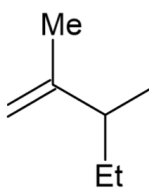


SECTION 1 (Maximum Marks:)

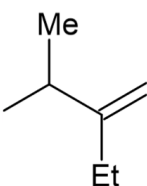
- This section contains **SIX (06)** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
 - Full Marks* : +4 If only (all) the correct option(s) is (are) chosen.
 - Partial Marks* : +3 If all the four options are correct but ONLY three options are chosen.
 - Partial Marks* : +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
 - Partial Marks* : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
 - Zero Marks* : 0 If none of the options is chosen (i.e. the question is unanswered).
 - Negative Marks* : -2 In all other cases.
- **For Example:** If first, third and fourth are the ONLY three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option) , without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

The maximum number of constitutional isomers of a compound having molecular formula $C_3H_2Cl_2$

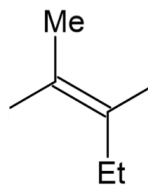
- A) 7
- B) 5
- C) 3
- D) 0



(I)



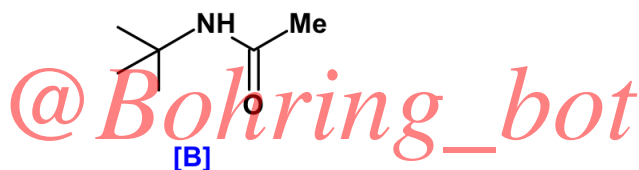
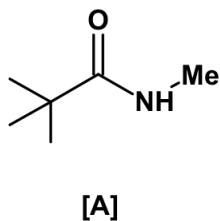
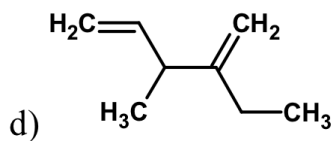
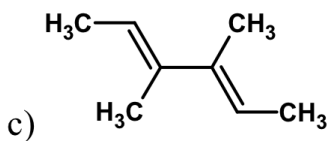
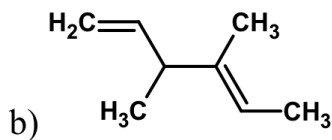
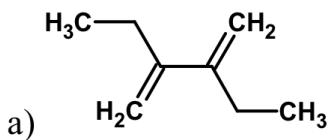
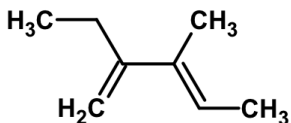
(II)



(III)

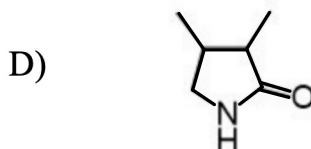
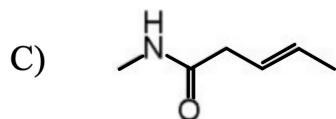
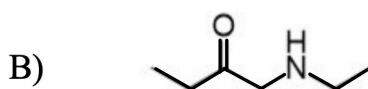
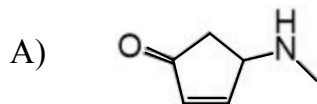
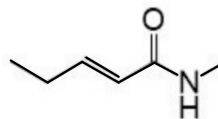
- A) I, II, III are constitutional isomers of each other
- B) I, II are chain isomers
- C) I, III are position isomers
- D) II, III are chain isomers

Which of the following is NOT the chain isomer of

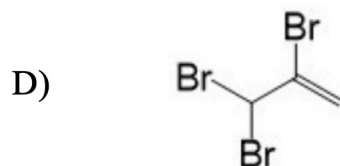
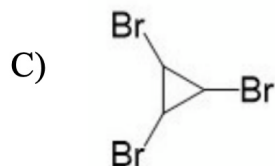
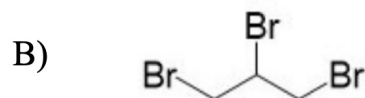
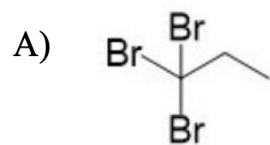
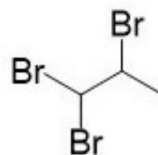


- A) They are metamers
- B) They are functional isomers
- c) They are constitutional isomers
- d) They are not isomers

Identify the constitutional isomers for molecule

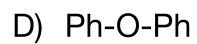
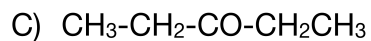
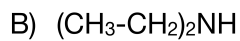
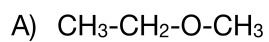


Identify the constitutional isomers for molecule



In the given structures which can NOT show matamerism

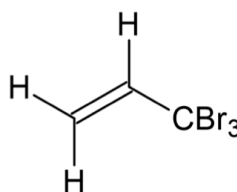
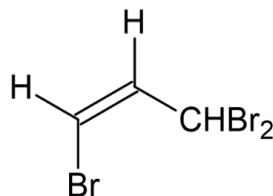
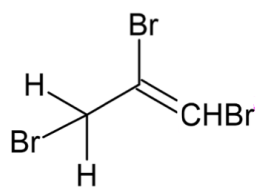
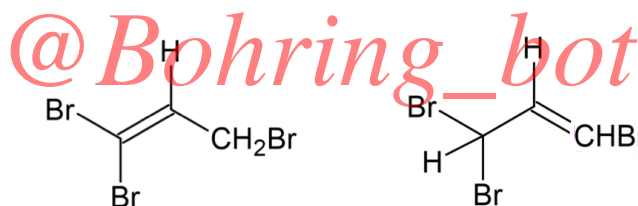
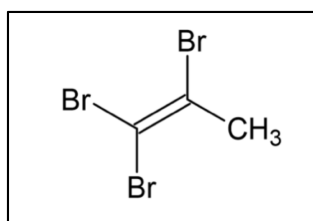
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SECTION 2 (Maximum Marks:)

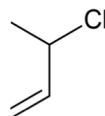
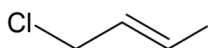
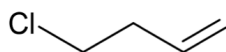
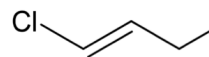
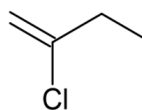
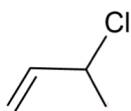
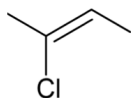
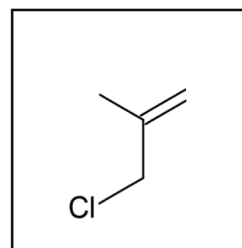
- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

In the given Position isomers of molecular formula $C_3H_3Br_3$, count the isomers in which position (locant) of double bond as per the IUPAC is changed With respect to given structure



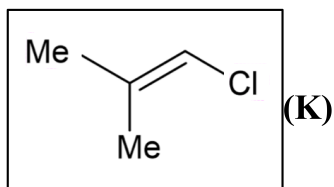
Ques 2

Identify the position isomers for the given molecule (X)

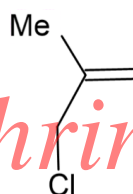
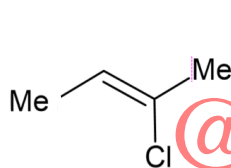
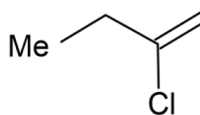
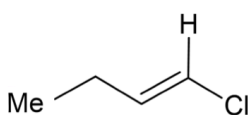


Calculate the cyclic constitutional isomers of $C_4H_6Cl_2$

In the given isomers of compound (K)



The count of isomers out of the following in which Locant (Position) of chlorine atom is changed

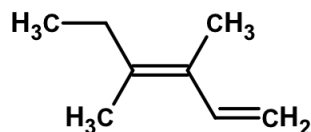
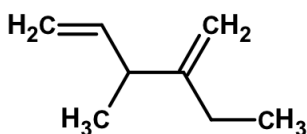
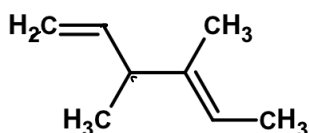
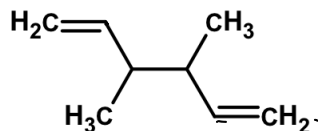
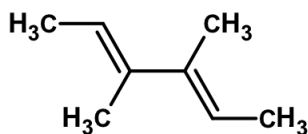
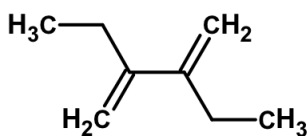
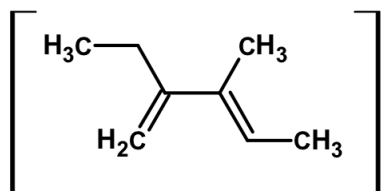


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How many chloroethenes are possible for molecular formula $C_2H_{4-n}Cl_n$, where 'n' can be an integer from 1 to 6

Total number of constitutional isomers (only position isomers) possible for trimethylcyclohexane is

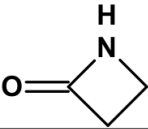
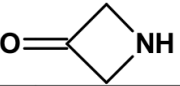
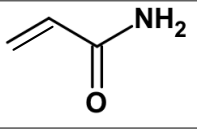
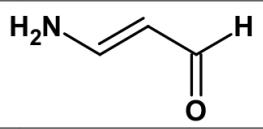
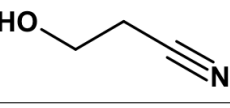
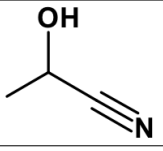
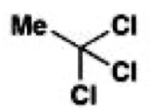
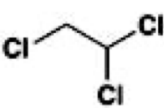
How many of the following are chain isomers of



SECTION 3 (Maximum Marks:)

- This section contains **FOUR (04)** questions.
- Each question has **TWO (02)** matching lists: **LIST-I** and **LIST-II**.
- **FOUR** options are given representing matching of elements from **LIST-I** and **LIST-II**. **ONLY ONE** of these four options corresponds to a correct matching.
- For each question, choose the option corresponding to the correct matching.
- For each question, marks will be awarded according to the following marking scheme:
Full Marks : +3 If **ONLY** the option corresponding to the correct matching is chosen.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.

Match the column

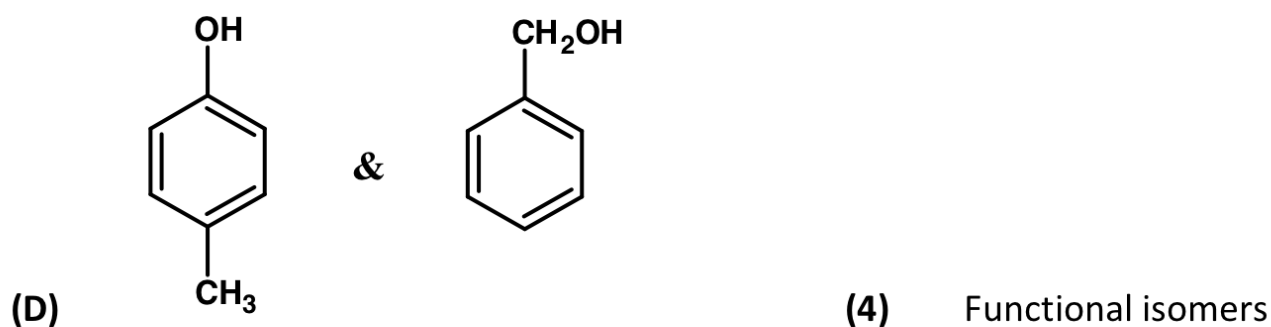
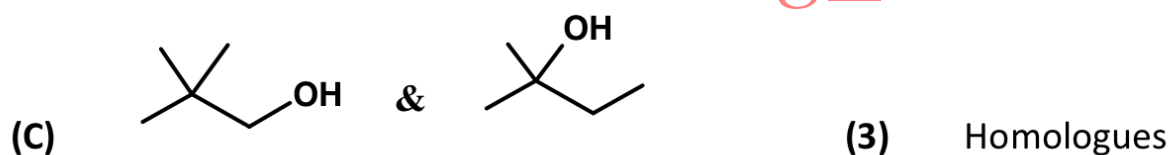
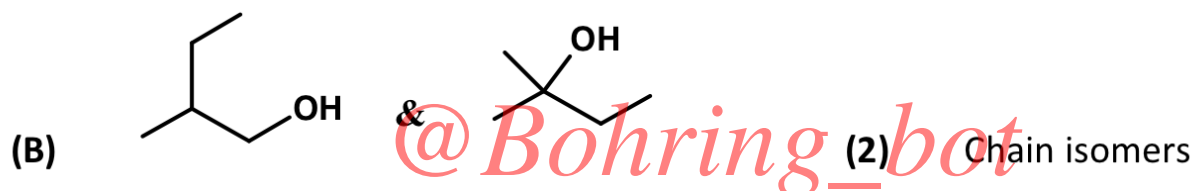
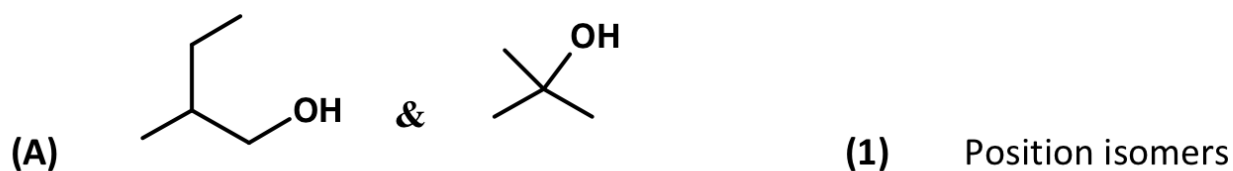
	Column-I		Column-II
A)	 	P)	Functional isomers
B)	 	Q)	Chain isomers
C)	 	R)	Position isomers
D)	 	S)	Metamers

- A) A-P B-P C-R D-R
- B) A-P B-R C-Q D-R
- C) A-P B-Q C-R D-S
- D) A-R B-S C-Q D-P

Match the following

(Compound)

(Type of isomerism)



A) A-3, B-1, C-2, D-4

B) A-2, B-1, C-3, D-4

C) A-2, B-1, C-4, D-3

D) A-2, B-3, C-4, D-1

Handout

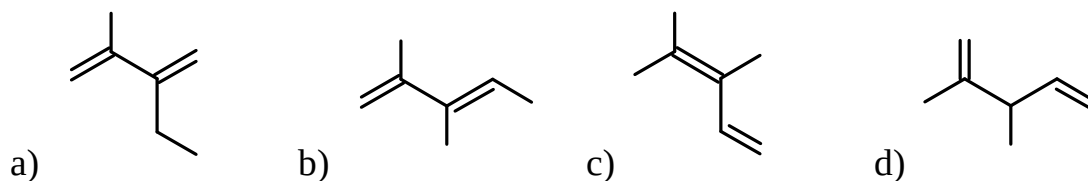
Geometrical

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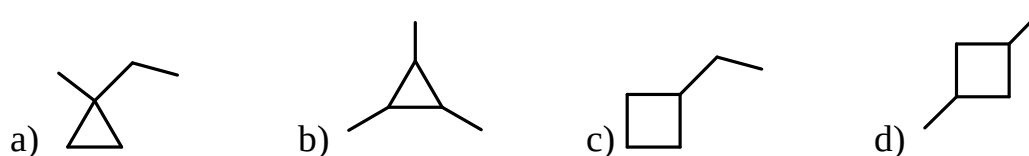
isomerism

ONE OR MORE THAN ONE ANSWER TYPE

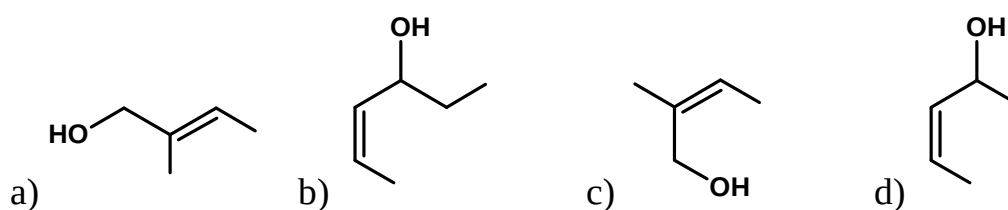
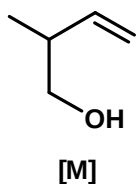
1. Which of the structural isomers given below can show GI



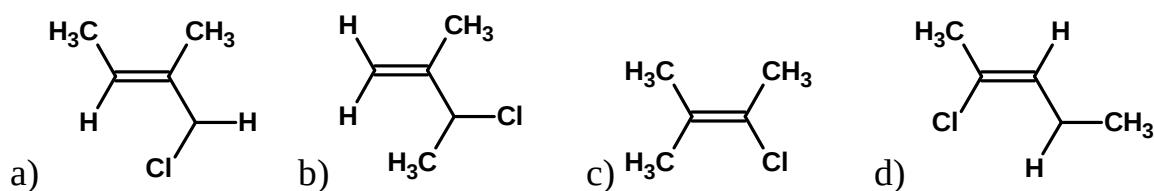
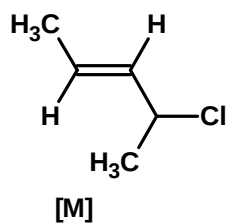
2. Identify the chain isomers of the given compound [K], which can show G.I.



3. Identify the position isomer(s) for the given molecule [M], which can show G.I.

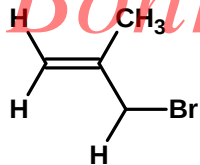
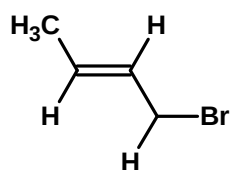
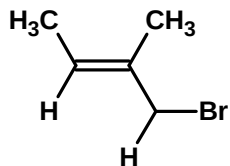
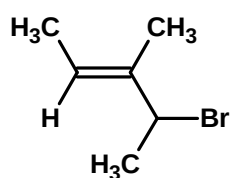
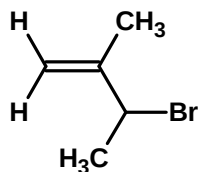
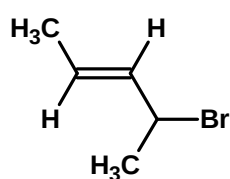


4. How many position isomers of given structure [M] can show geometrical isomerism?

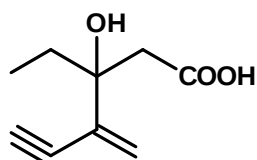


INTEGER ANSWER TYPE

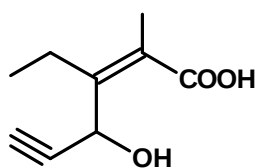
5. How many structural isomers of C_4H_8 can show G.I.?
6. Find the total structural isomers of C_5H_{10} can show G.I.?
7. How many isomers of C_5H_8 consist of $2sp^3$ carbons, $2sp^2$ carbons and 1 sp carbon atom which can show geometrical isomerism?
8. Among the given structure(s), identify those containing bromine at fourth position according to IUPAC system and shown geometrical isomerism



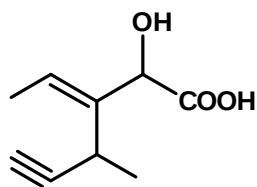
9. Structure(s) of the compound which is showing geometrical isomerism



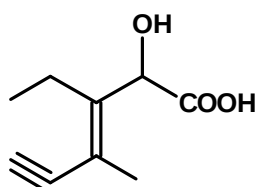
(1)



(2)

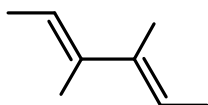


(3)

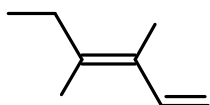


(4)

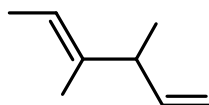
10. Consider the following FOUR structures I, II, III and IV



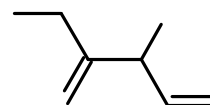
(I)



(II)



(III)



(IV)

G.I. of these I, II, III, IV are x, y, z and w. find sum of $x+y+z+w$.

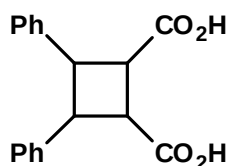
$$I = 3$$

$$II = 2$$

$$III = 2$$

$$IV = 0$$

11. Find the total G.I. possible for given compound

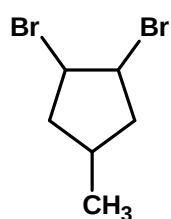


12. How many structural isomers of C_6H_{12} can show geometrical isomers?

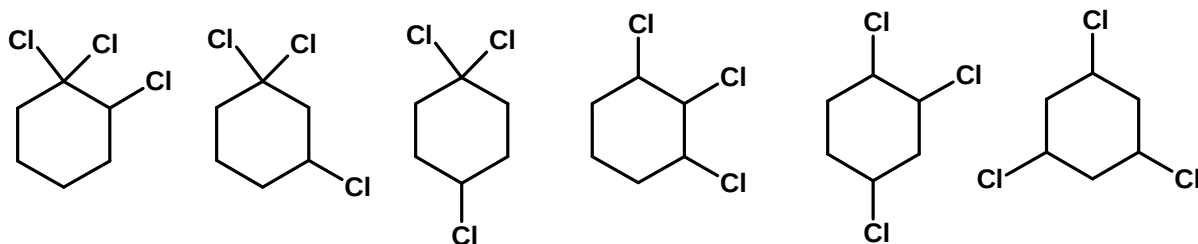
13. How many of the given can show G.I.?

$CH_2 = CH_2$, $CH_2 = CH.Cl$ 1-phenyl butene 2-methylbut-2-ene, 3-Ethylpent-2-ene

14. How many geometrical isomers possible for given structure [K]



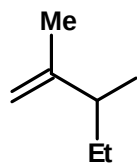
15. How many trichloro derivatives of cyclohexane can show geometrical isomerism



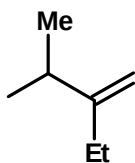
16. How many chloroderivatives of $C_2H_6 - nCl_n$ can show geometrical isomerism?

17. The maximum number of constitutional isomers of molecular formula $C_3H_2Cl_2$ which can show H.I..

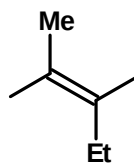
18. How many statement(s) is/are correct for given these three structures



(I)



(II)



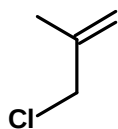
(III)

i) I is showing G.I.

ii) II is showing G.I.

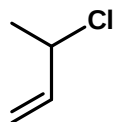
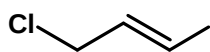
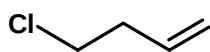
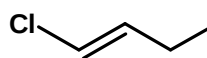
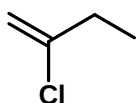
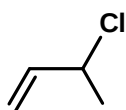
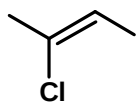
iii) III is showing G.I.

19. How many position isomers of [K] in given collection can show G.I.



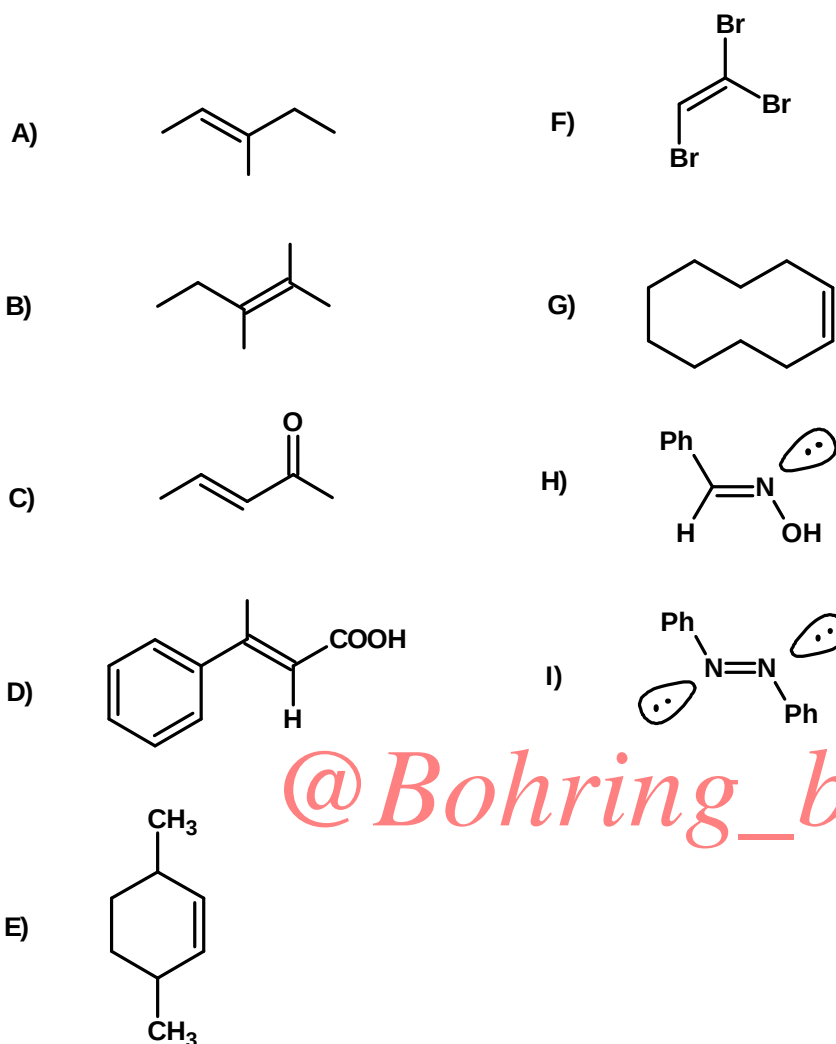
[K]

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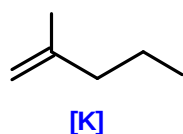
SUBJECTIVE QUESTIONS:

1. Which of the following show G.I. among the given

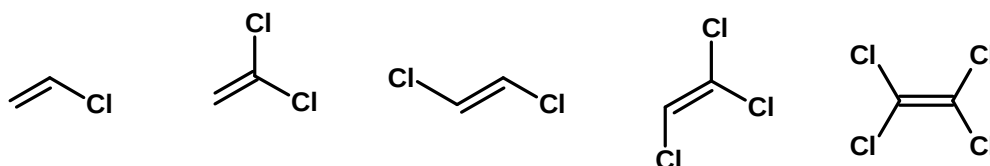


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2. How many position isomers of isohexene can show G.I. individually

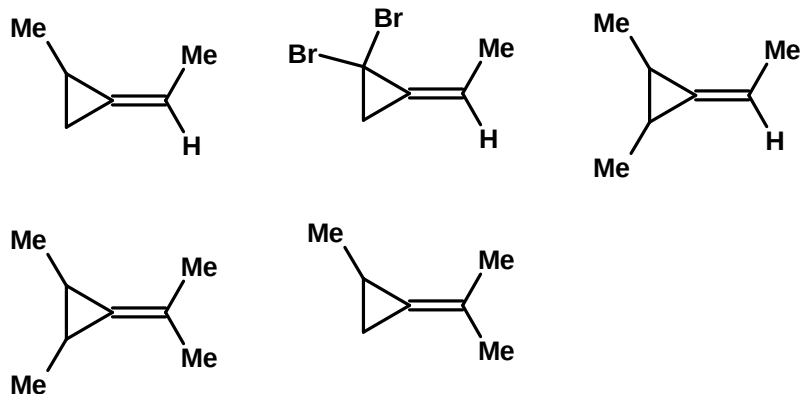


3. How many chloroderivatives of $C_2H_4 - nCl_n$ can show G.I., where n can be 1–4?

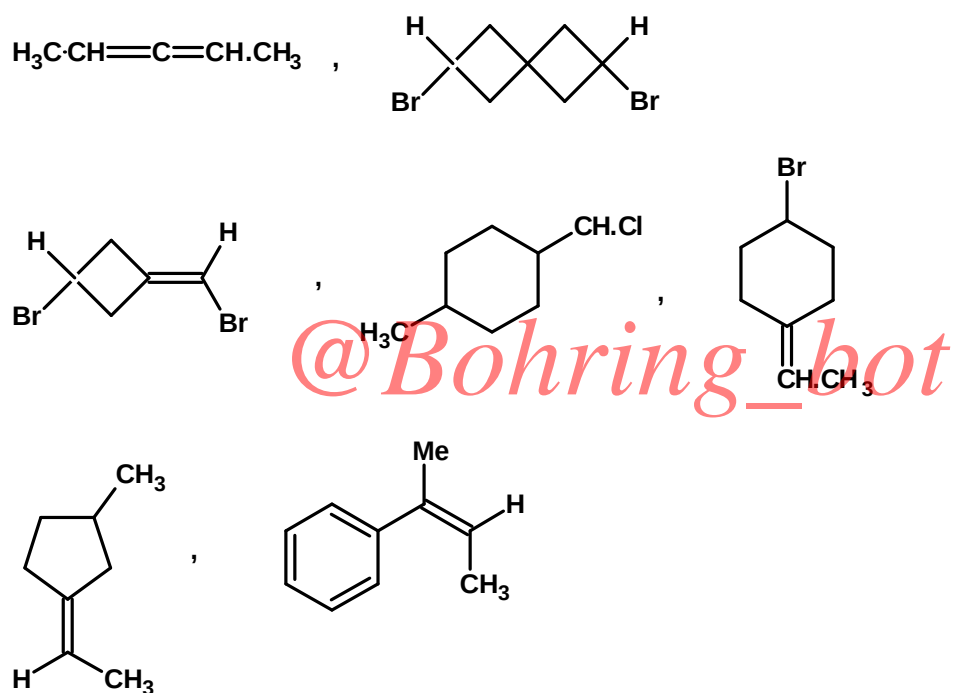


4. Find the total isomers possible for $C_2F.Cl.Br.I$. Including geometrical isomers.

5. How many of the given molecule can show G.I.?



6. How many of the given can "NOT" show geometrical isomerism



KEY SHEET

ONE OR MORE THAN ONE & INTEGER

1	2	3	4	5	6	7	8	9	10
B	BD		D	01	02	0	02	2,3,4	07
11	12	13	14	15	16	17	18	19	
06	08	01	03	03	0	00	0	0	

SUBJECTIVE

1. A,C,D,E,G,H,I
2. 02
3. 01
4. 06
5. 04
6. 02

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ORGANIC CHEMISTRY

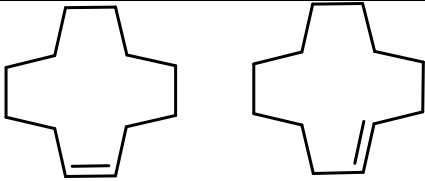
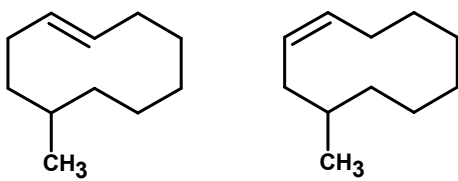
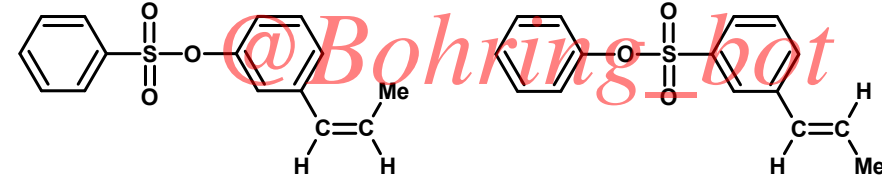
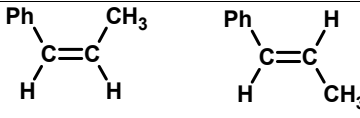
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GEOMETRICAL ISOMERISM

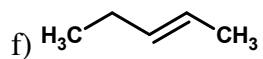
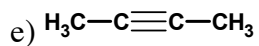
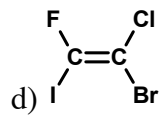
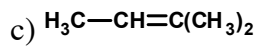
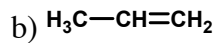
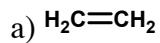
WORK SHEET

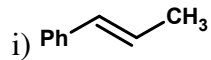
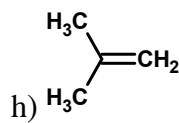
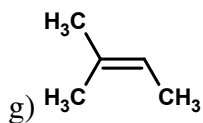
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EXERCISE-I

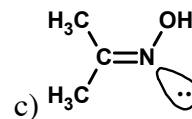
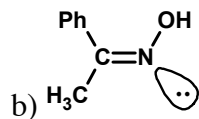
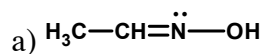
1.	Identify the relationship ?
a)	
b)	
c)	
d)	

2. Which of the given alkenes can show G.I. ?

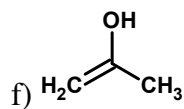
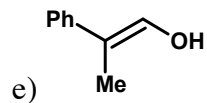
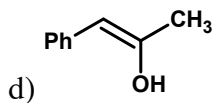
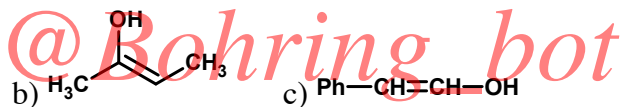
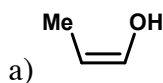




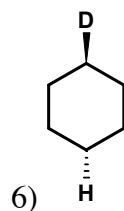
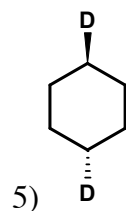
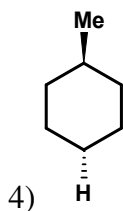
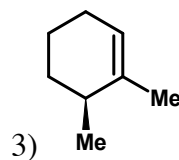
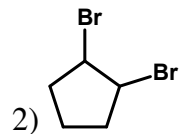
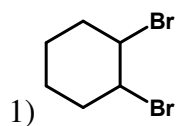
3. Which of the given oximes can show G.I. ?

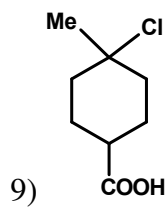
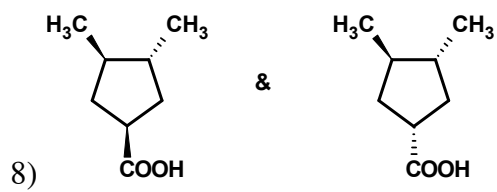
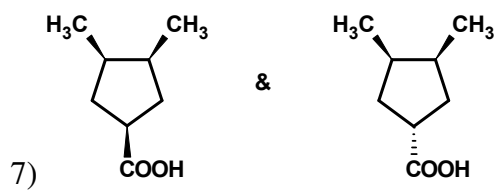


4. Which "ENOL" can show G.I. ?

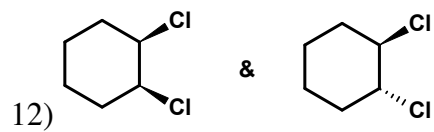
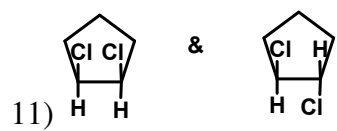
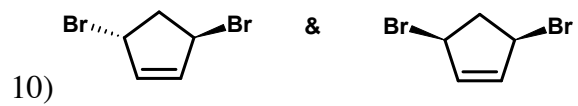


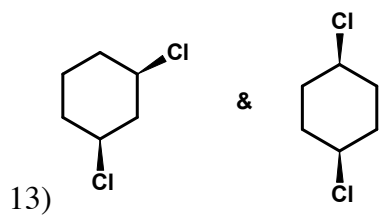
5. Which of the following cyclic structure can show G.I. ?



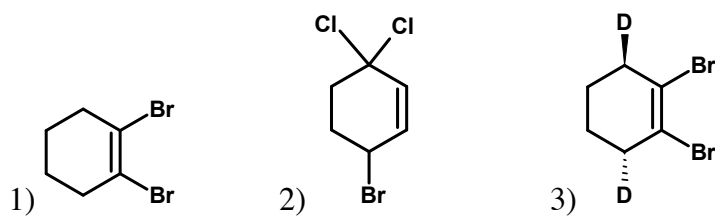


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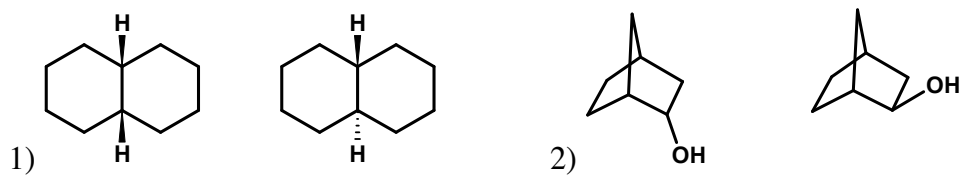




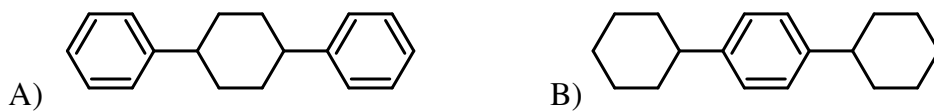
6. Which of given compound(s) can show G.I. ?

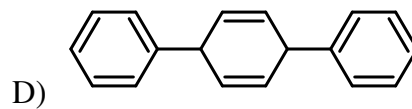
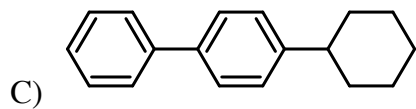


7. Given compound can show G.I. or NOT ?

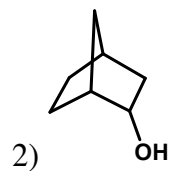
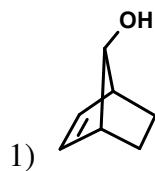


8. Which can show G.I. ?

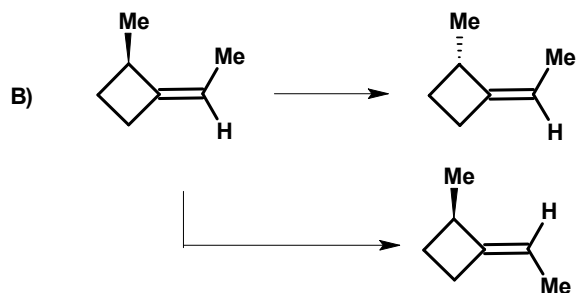
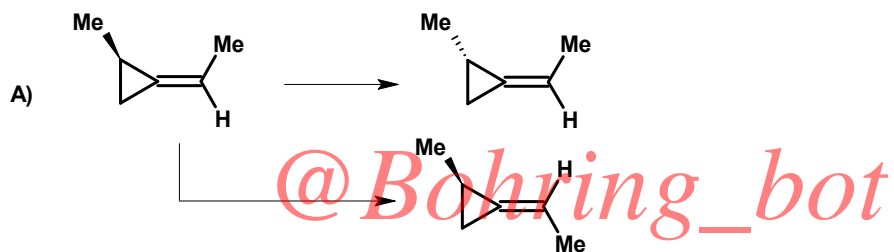




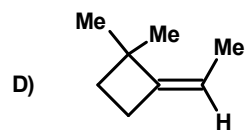
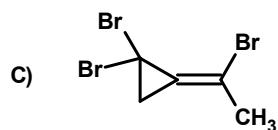
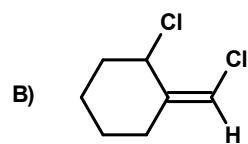
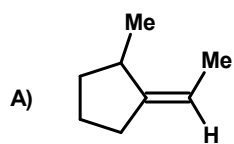
9. Which can show G.I. ?

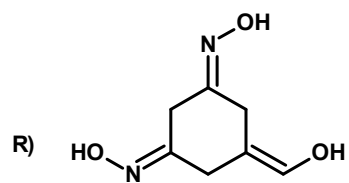
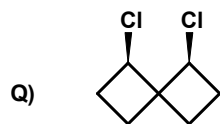
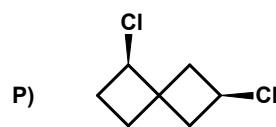
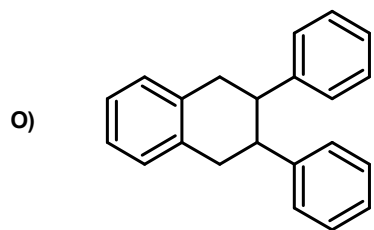
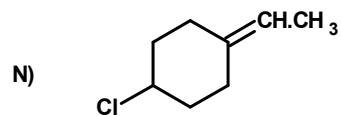
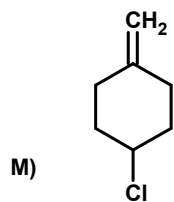
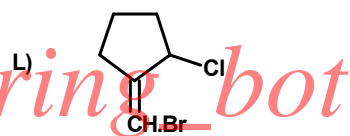
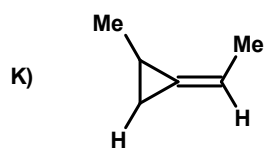
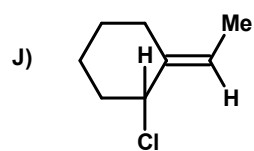
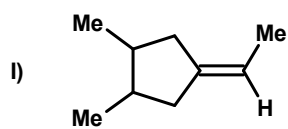
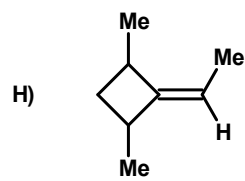
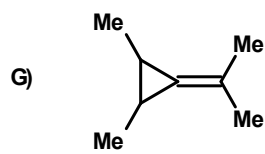
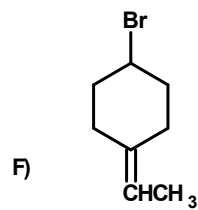
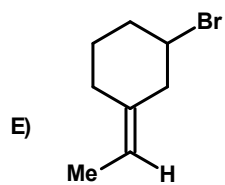


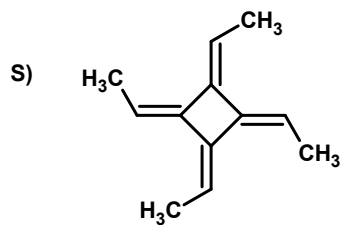
10. Which of the following can show G.I. ?



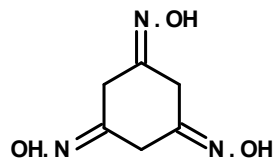
11. Which can show G.I. ?



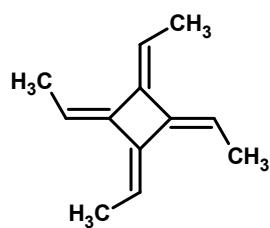




12. Find the no. of G.I ?

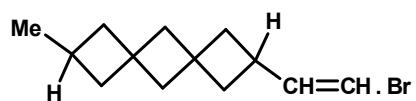


13. Find the no. of G.I. ?

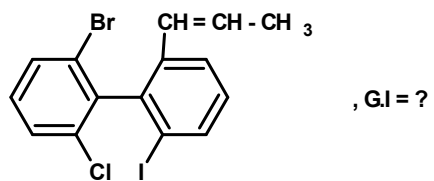


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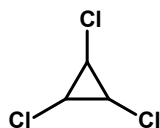
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15.

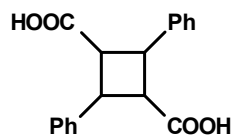


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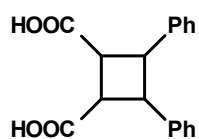
Find no of G.I. ?

17.



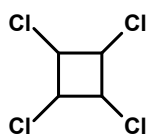
Find total no of G.I. ?

18.



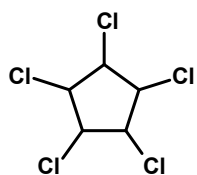
@Bohring_bot
Find total no of G.I. ?

19.



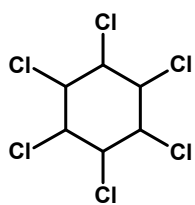
Total no of G.I. = ?

20.



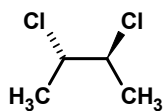
Total no of G.I. = ?

21.



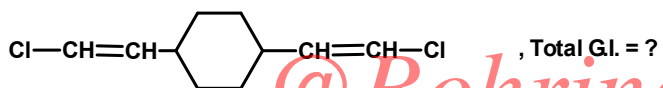
Total no of G.I. = ?

22.



Total no of G.I. = ?

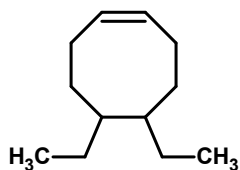
23.



, Total G.I. = ?

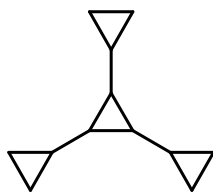
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24.



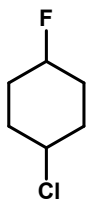
total no. of G. I ?

25.

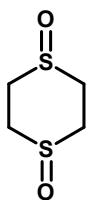
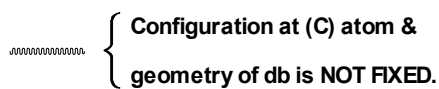
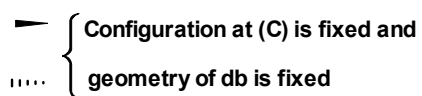
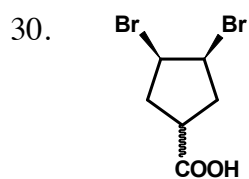
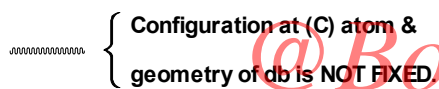
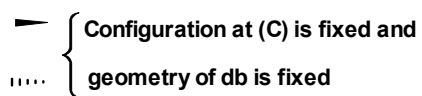
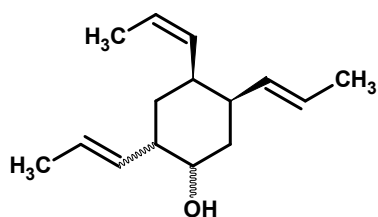
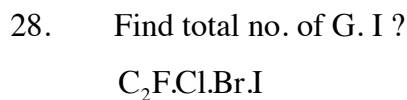
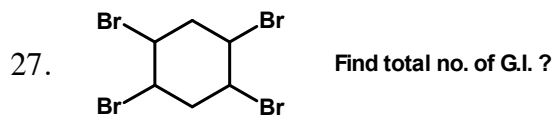


total no. of G. I ?

26.



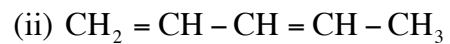
total no. of G. I ?



PREVIOUS YEAR'S JEE-MAINS QUESTIONS

1. Geometrical isomerism is not shown by
- a) 1,1-dichloro-1-pentene b) 1,2-dichloro-1-pentene
- c) 1,3-dichloro-2-pentene d) 1,4-dichloro-2-pentene
2. Which of the following does not show geometrical isomerism?
- a) $\text{CH}_3 - \text{CH} = \text{CH} - \text{CH}_3$ b) $\text{CH}_3 - \text{CH}_2 - \text{CH} = \text{CH}_2$
- c) $\begin{array}{c} \text{H}_3\text{C}-\text{C}=\text{CH}-\text{CH}_3 \\ | \\ \text{Cl} \end{array}$ d) $\begin{array}{c} \text{HC}=\text{CH}-\text{CH}_2-\text{CH}_3 \\ | \\ \text{Cl} \end{array}$
3. The alkene that exhibits geometrical isomerism is
- a) Propene b) 2-methyl propene
- c) 2-butene d) 2-methyl-2-butene
4. Maleic acid and fumaric acids are
- a) Chain isomers b) Geometrical isomers
- c) Tautomers d) functional isomers
5. Which of the following compounds will exhibit geometrical isomerism ?
- a) 1-phenyl-2-butene b) 3-phenyl-1-butene
- c) 2-phenyl-1-butene d) 1,1-diphenyl-1-propane

PREVIOUS YEAR'S JEE-ADVANCED QUESTIONS



The numbers of possible geometric isomers for the above compounds respectively are

a) 0,2,4

b) 2,2,4

c) 0,3,3

d) 0,2,3

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Conformers worksheet.

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HANDOUT

(CONFORMERS)

1. Conversion Fischer - Newmann's



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2. Which can show conformational isomerism.

- i) H_2O_2 ii) NH_2OH iii) CH_3Cl iv) NH_2OH
v) CH_3CHO

3. Which parameter will not change during conformation rotation?

- i) Bond angle ii) dihedral angle
iii) Dipole moment iv) Bond length

4. Calculate the rotation energy barrier on the basis of given data.

For $\theta = 0^\circ$

for $\theta = 60^\circ$

i) $\text{CH}_3, \text{CH}_3 = 4 \text{ kcal}$

i) $\text{CH}_3, \text{CH}_3 = 0.9 \text{ kcal}$

ii) $\text{CH}_3, \text{H} = 1.5 \text{ kcal}$

ii) remaining = negligible

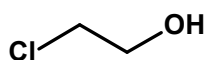
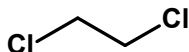
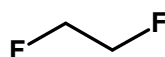
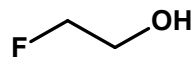
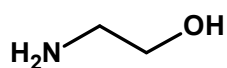
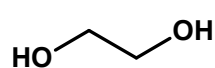
iii) $\text{H}, \text{H} = 1.0 \text{ kcal}$

Now calculate the rotation energy barrier for

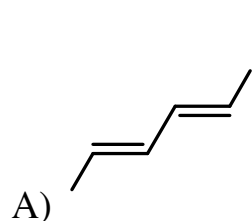
- 1) Propane 2) Neopentane 3) Isobutane 4) Ethane

5. Compare rotation energy barrier of $\text{CH}_3 - \text{CH}_3$ & $\text{CH}_3 - \text{SiH}_3$.

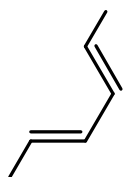
6. How many of given mostly exist as gauche conformer as stable conformer



7. Which of the following is not a correct conformer pair?



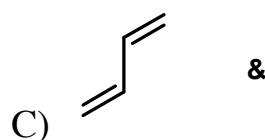
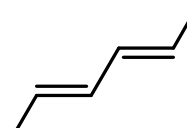
&



B)



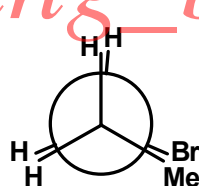
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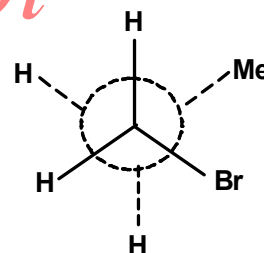
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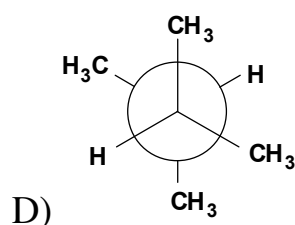
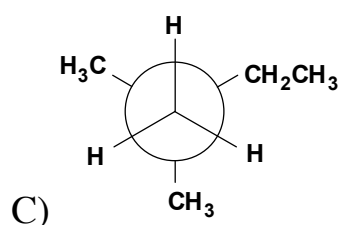
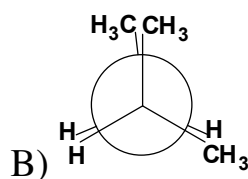
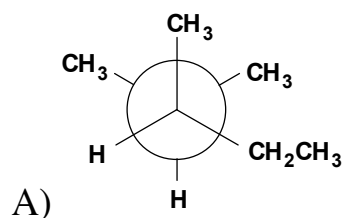
D)



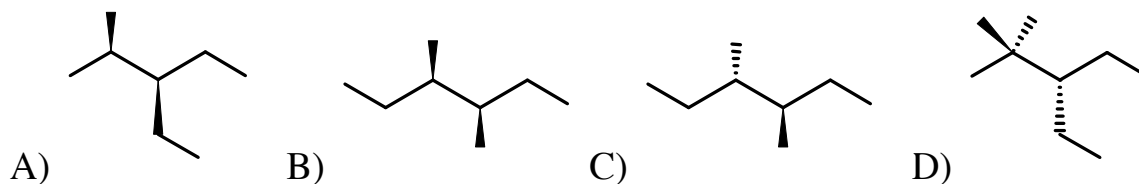
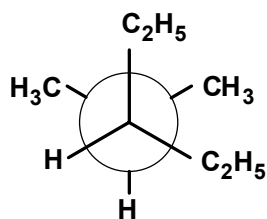
&



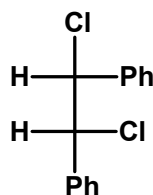
8. Which of the following Newmann's projections represent conformations of 2,3-dimethyl butane ?



9. The Newman projection shown is the same as

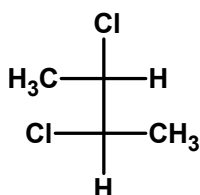


10. Find the number of stable conformers with non-zero dipole moment of given structure:

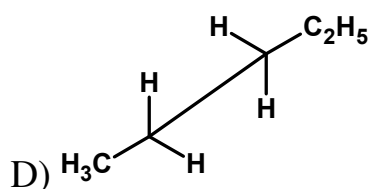
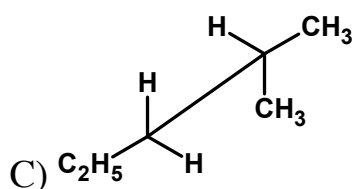
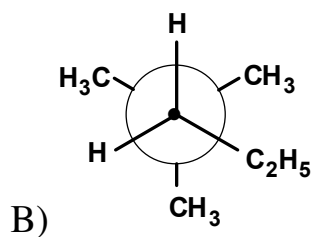
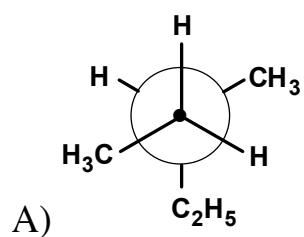


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11. Find the number of stable conformers with non-zero dipole moment of given structure:



12. Which of the following conformational structures is 2-methylpentane?



PASSAGE-1 (13-14)

The two major contributors of conformers of 1,2-dichloroethane are anti and gauche. At 32°C in gas phase, the measured dipole moment of 1,2-dichloroethane is 1.12 D. The dipole moment of a mixture of **X** and **Y** is given by the relationship

$$\mu^2 = N_X \mu_X^2 + N_Y \mu_Y^2$$

Here, N = mole fraction of each kind of molecule. From bond moment measurement, it has been estimated that gauche conformer of 1,2-dichloroethane should have a dipole moment of about 3.2 D.

13. What percent of conformers of 1,2-dichloroethane is anti, at 32°C ?
- A) 12% B) 36% C) 60% D) 89%
14. What happens if temperature is increased to 52°C ?
- A) Percentage of gauche conformers decreases and that of anti-conformers increases
- B) Percentage of both gauche and anti conformers increases
- C) No change in percentage of anti and gauche conformers occur
- D) Percentage of both anti and gauche conformers decreases

PASSAGE-2 (15-16)

Among conformers, the barrier to rotation originates mainly from torsional strain when atoms or groups eclipse one another. Following are the costs of energy required to cross the rotational barrier for the indicated pair of groups eclipsing

Groups eclipsing	Energy cost (kJ/mol)
H – H	4.00
CH ₃ – H	6.00
CH ₃ – CH ₃	11.00

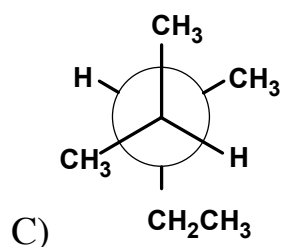
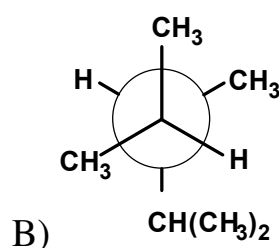
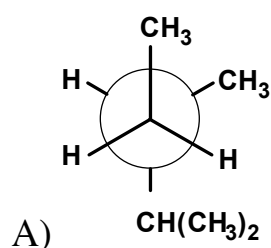
15. What is the energy barrier to rotation in neopentane?

- A) 12 kJ/mol B) 16 kJ/mol C) 18 kJ/mol D) 23 kJ/mol

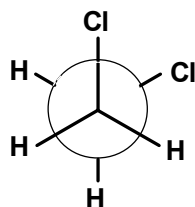
16. What is the energy barrier to rotation in 2-methyl propane?

- A) 10 kJ/mol B) kJ/mol C) 14 kJ/mol D) 16 kJ/mol

17. Which structure is of a compound different from the following?

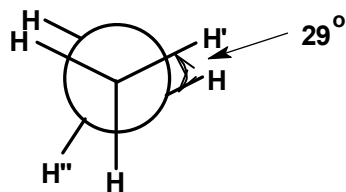


18. Conformation obtained on 60° clockwise rotation of C₁ and then 180° anticlockwise rotation of C₂ is



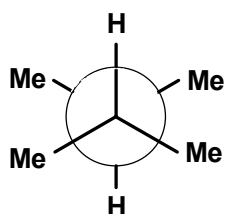
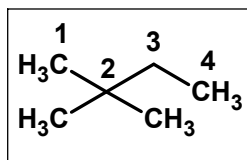
- A) Ellipse B) Gauche C) Anti D) Partially Ellipse

19. In the following skew conformation of ethane, H'-C-C-H" dihedral angle is:

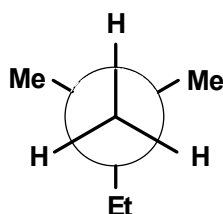


- A) 58° B) 149° C) 151° D) 120°

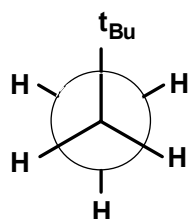
20. Identify the structure(s) that represent a conformation of 2,2-dimethylbutane sighting along any C-C bond.



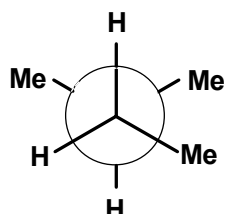
(P)



(Q)



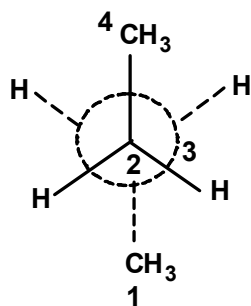
(R)



(S)

- A) P B) S C) Q *D) R

21. In the given conformation, if C_2 is rotated about $C_2 - C_3$ bond anticlockwise by an angle of 120° then conformation obtained is

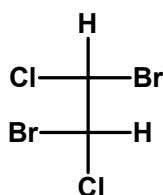


- A) Fully eclipsed B) Partially eclipsed
C) gauche D) Staggered

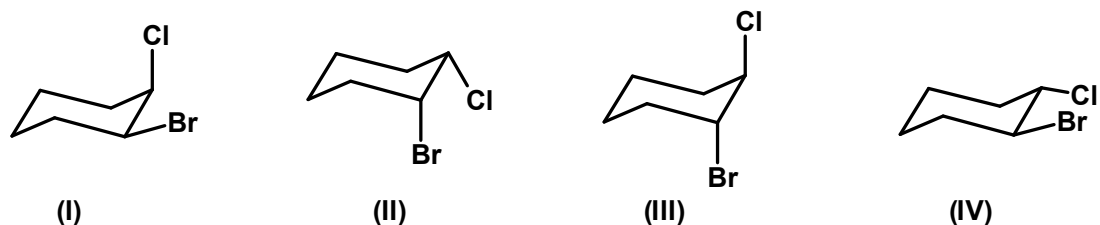
22. On increasing temperature, dipole moment of 1,2-dichloro ethane is

- A) increases B) decreases
C) remain constant D) None

23. Total number of stable conformers of having non-zero dipole moment are



24 Arrange the following dihalocyclohexanes derivatives in decreasing order of stability

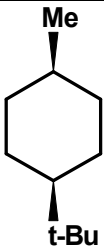
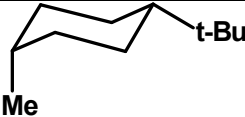
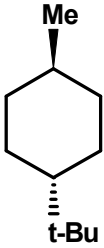
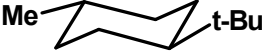
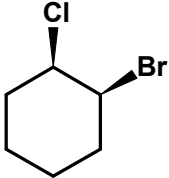
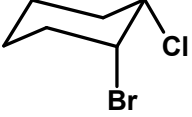
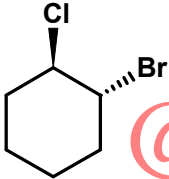


25.

	Column-I		Column-II
P)		1)	Posses more stability than R
Q)		2)	Possess more P.E. than Q
R)		3)	Lies between two R in P.E. diagram
S)		4)	All H-atoms are skew.

26

	Column-I		Column-II
P)	$\xrightleftharpoons{\text{Flip}}$	1)	$K > 1$
Q)	$\xrightleftharpoons{\text{Flip}}$	2)	$K < 1$
R)	$\xrightleftharpoons{\text{Flip}}$	3)	$K = 1$
S)	$\xrightleftharpoons{\text{Flip}}$	4)	$K = 0$

	Column-I		Column-II
P)		1)	
Q)		2)	
R)		3)	
S)		4)	

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Answer key:

2. 1),2),4),5)

3. 1 ,4

6. 5

7. B

8. D

9. A,D

10. 03

11. 03

12. C

13. D

14. D

15. C

16. D

17. B

18. C

19. B

20. C D

21. C

22. A

23. 02

24. $3 < 1 < 2 < 4$

25. P-1 Q-3 R-4 S-2

26. P-1 Q-2 R-3 S-4

27. P-1 Q-2 R-3 S-4

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Optical isomerism

Basic sheet

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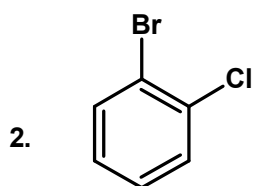
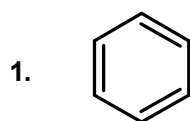
By Ashish mishra

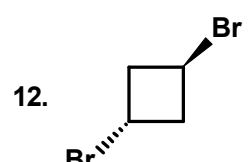
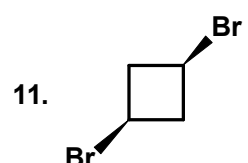
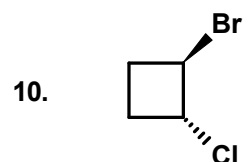
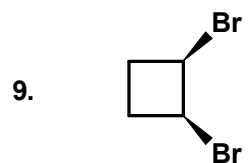
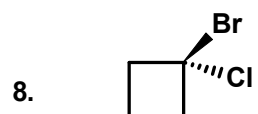
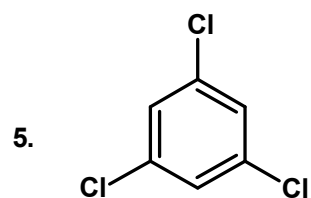
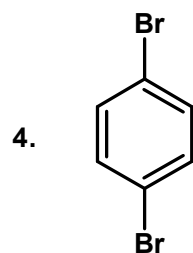
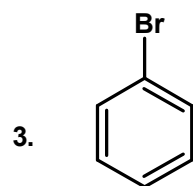
OPTICAL ISOMERISM HANDOUT

- 1) Interconversion of Fischer, Newman's, sawhorse and wedge-dash projection
- 2) Plane polarised light
- 3) Optically active
- 4) d(+) or dextro rotatory
- 5) $\ell(-)$ or leavo rotatory
- 6) Optically inactive
- 7) Optical isomerism $\longrightarrow$ definition
- 8) Chirality
- 9) Plane of symmetry
- 10) Centre of symmetry (or) centre of inversion.
- 11) Axis of symmetry
- 12) Alternate axis of symmetry
- 13) Meso
- 14) Racemic mixture
- 15) Enantiomers
- 16) Diastereomers
- 17) Resolution
- 18) D, L-nomenclature

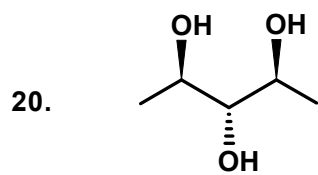
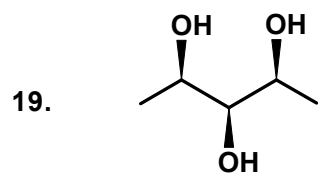
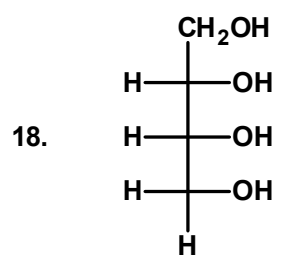
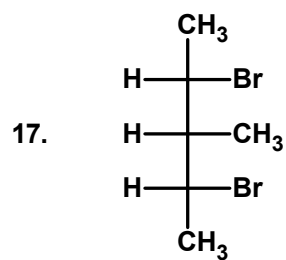
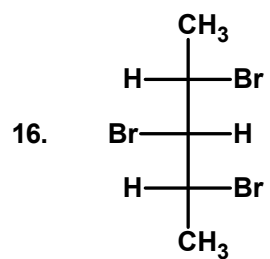
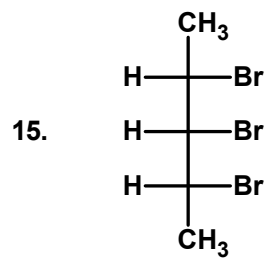
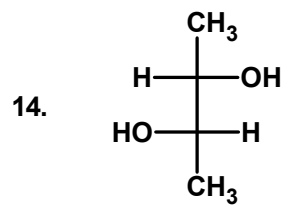
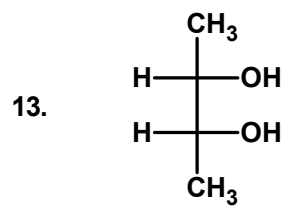
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Plane symmetry



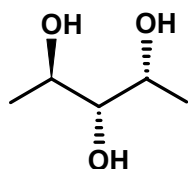


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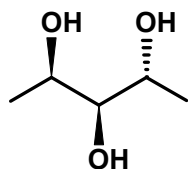


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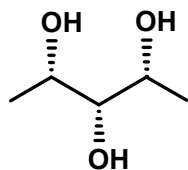
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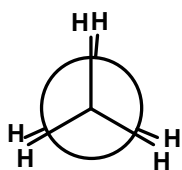
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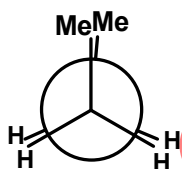
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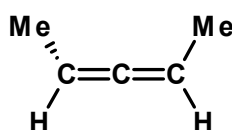
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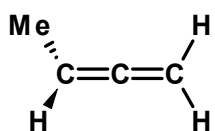
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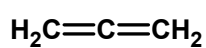
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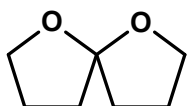
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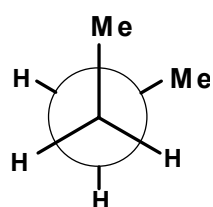
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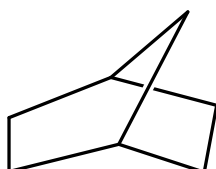
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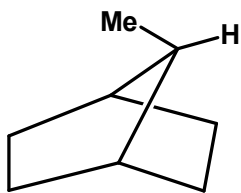


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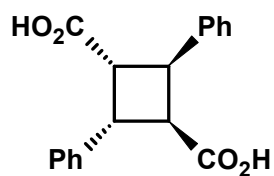
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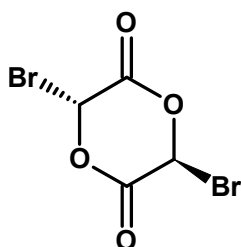


**** Centre of symmetry:**

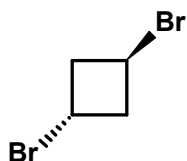
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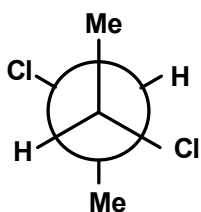
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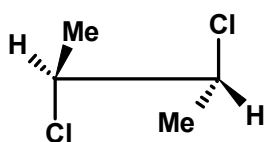
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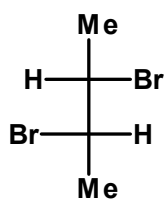
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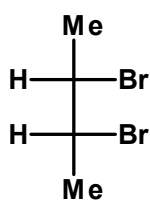
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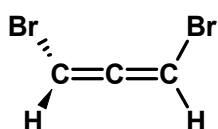
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7)



8)

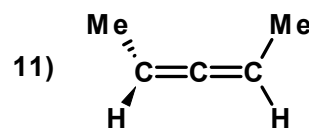
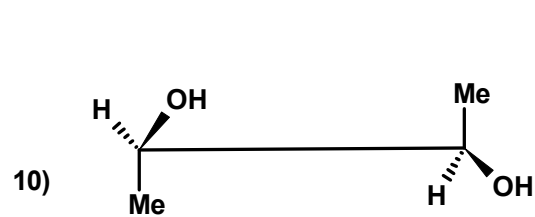
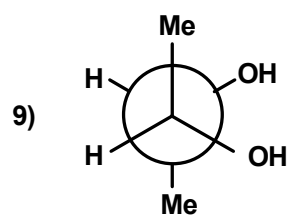
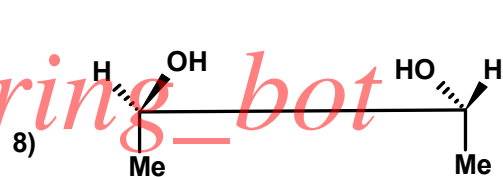
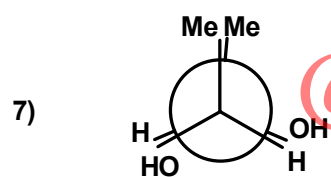
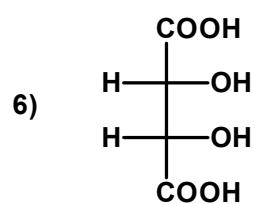
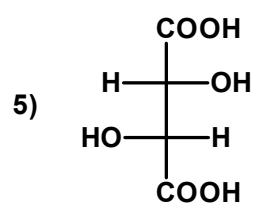
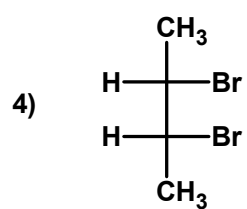
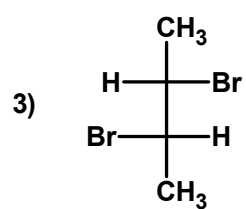


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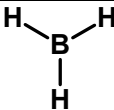
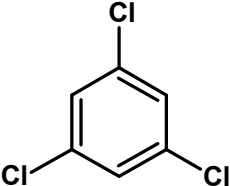
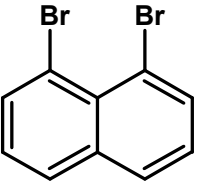
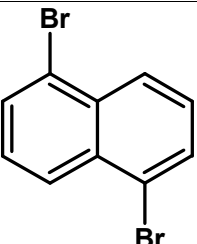
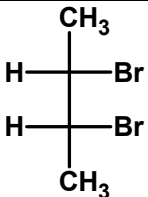
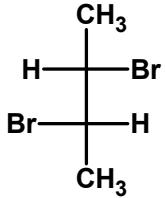
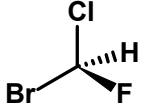
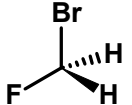
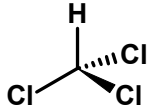
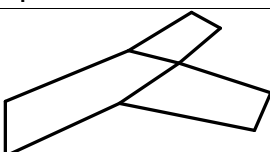
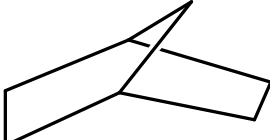
*** Axis of symmetry:**

1) CH_4

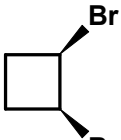
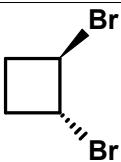
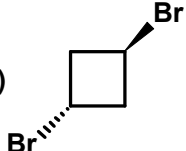
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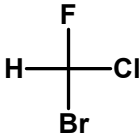
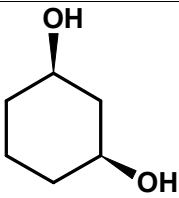
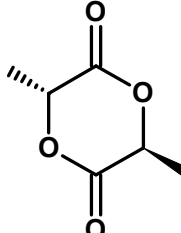
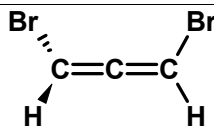
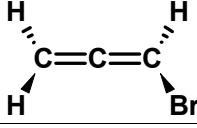
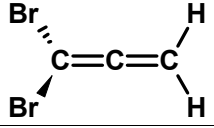
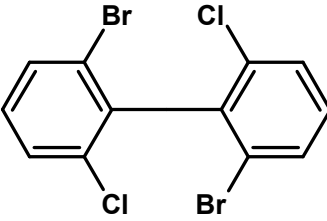
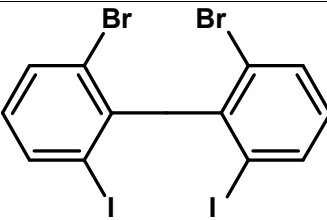
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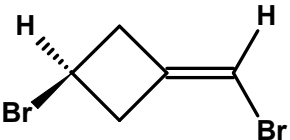
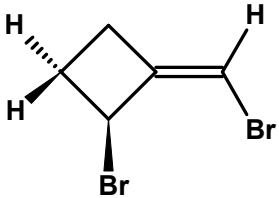
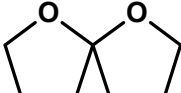
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1)				
2)				
3)				
4)				
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10)	CH ₄			
11)				
12)				

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13)				
14)				
15)				

Que:

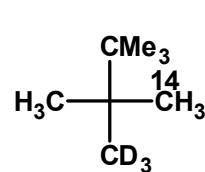
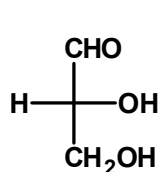
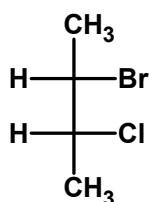
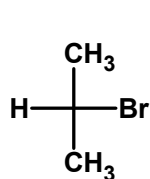
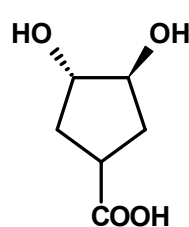
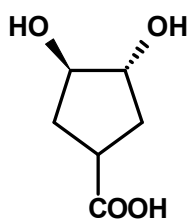
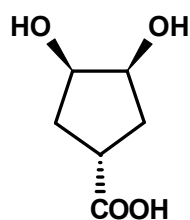
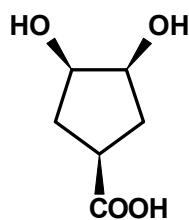
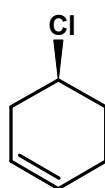
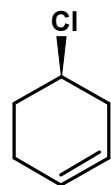
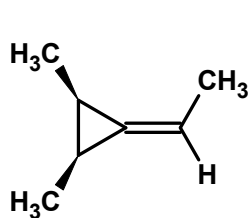
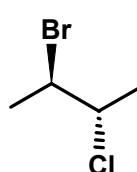
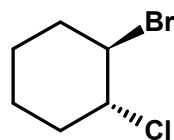
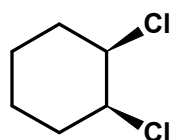
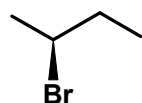
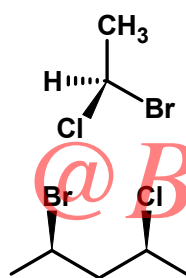
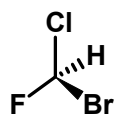
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1)				
2)				
3)				
4)				
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9)				
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11)				

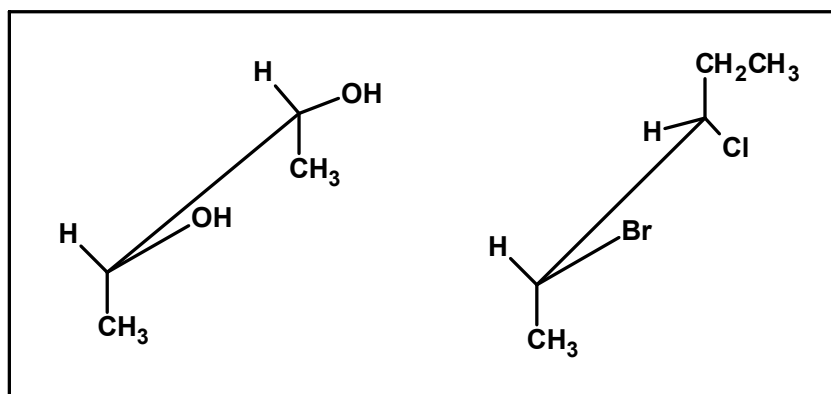
* Find the alternate axis of symmetry in CH_4

* Find the R, S configuration in following structures.

Ex:

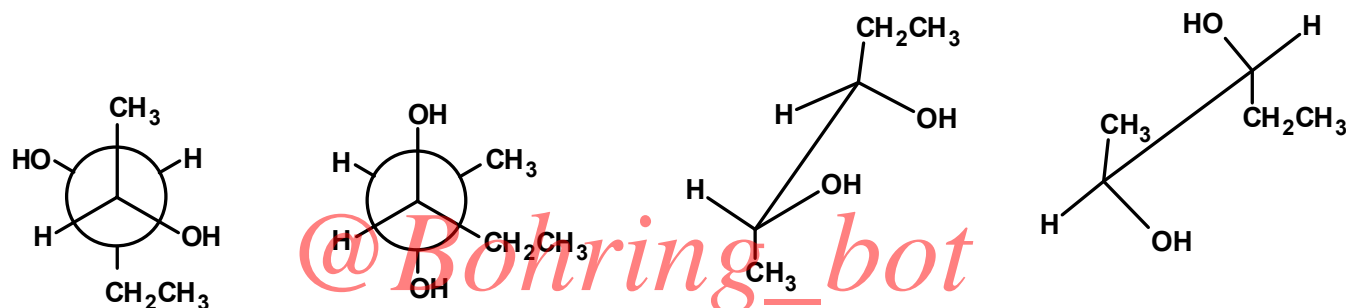


Q1.

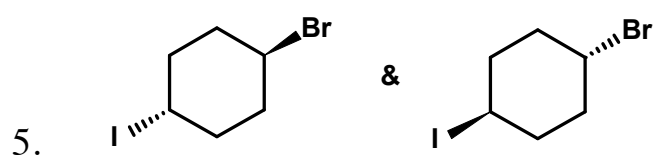
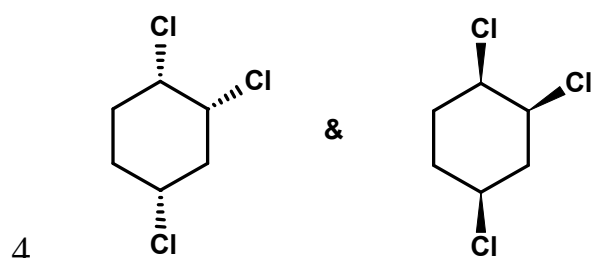
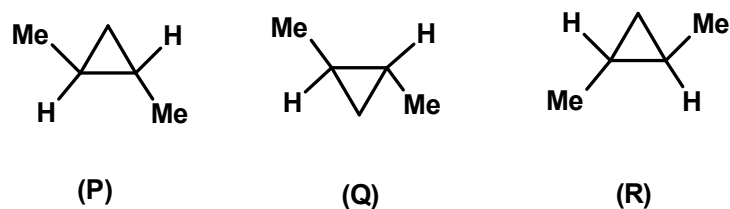


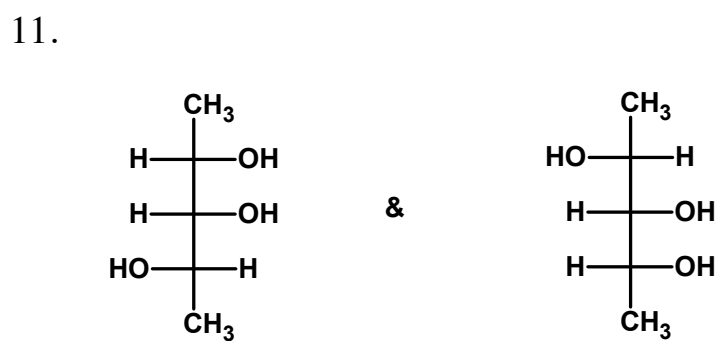
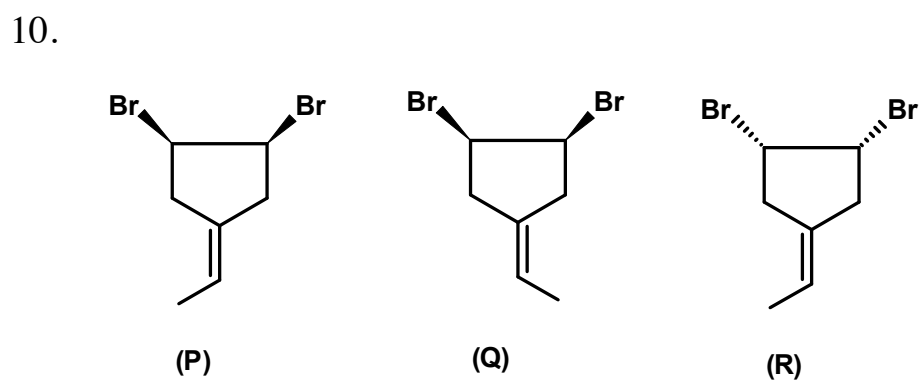
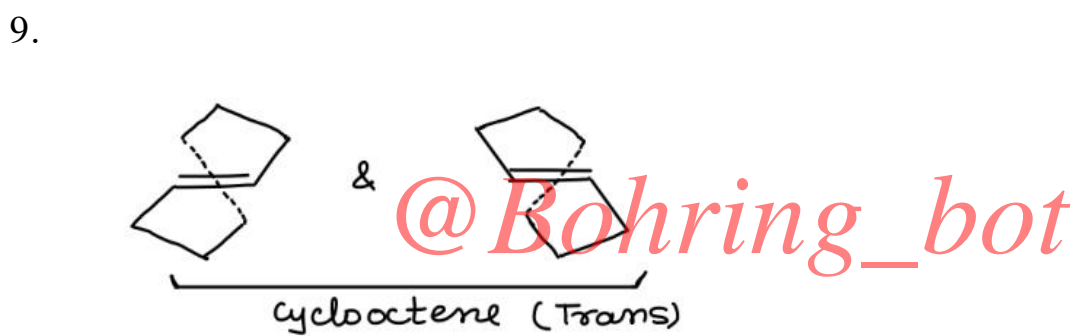
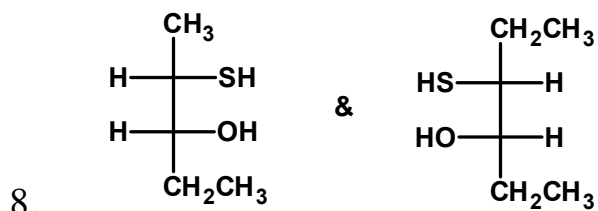
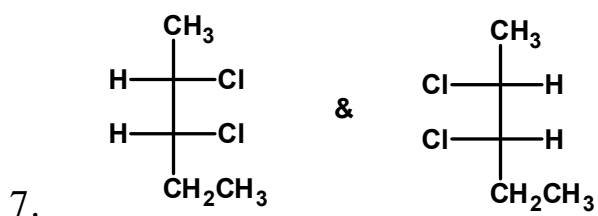
Find R, S configuration for both?

2. Find R, S configuration for given compound?



3. Find the relationship?

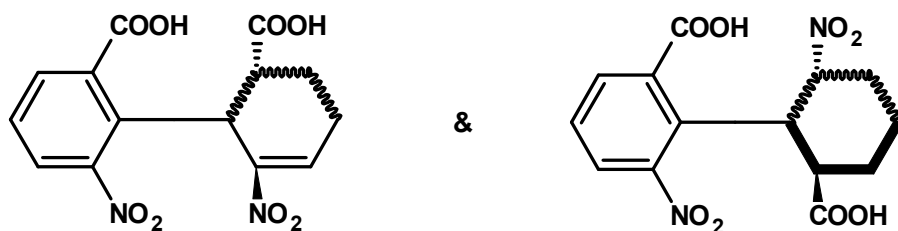




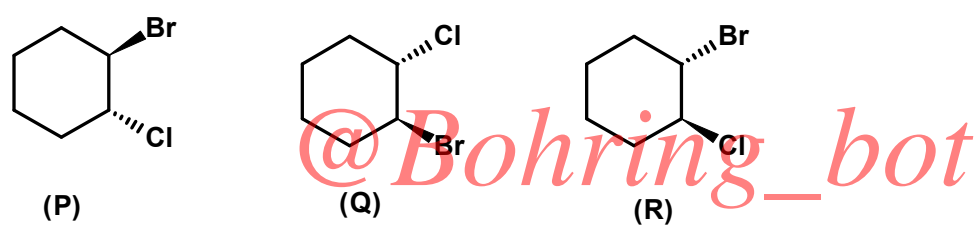
12.



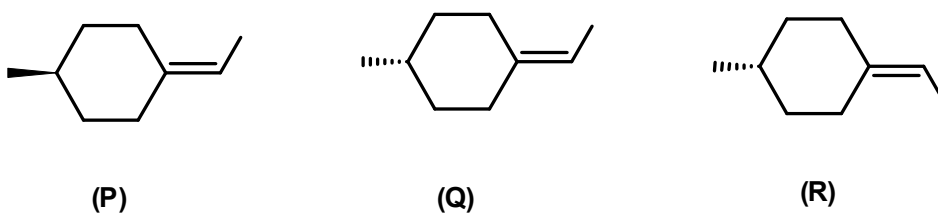
13.



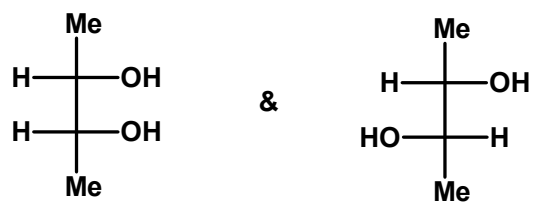
14.



15.



16.



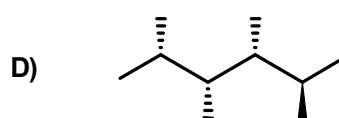
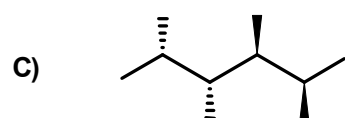
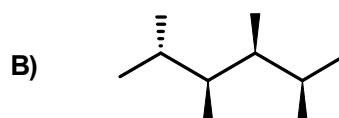
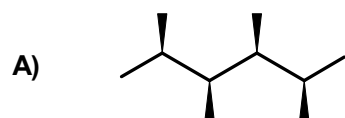
17.



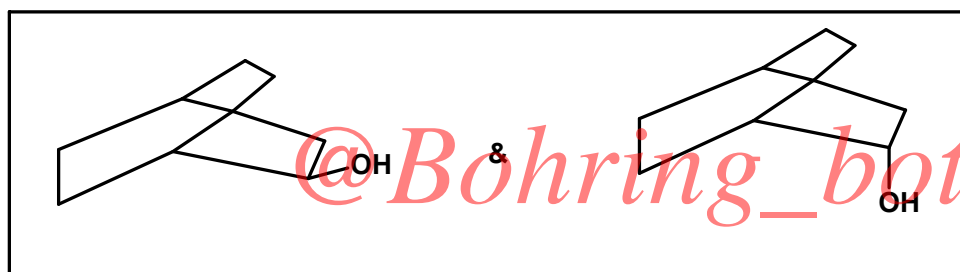
18.



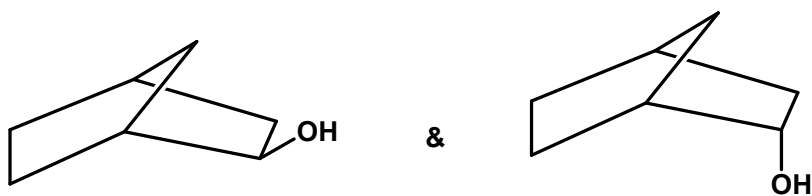
19.



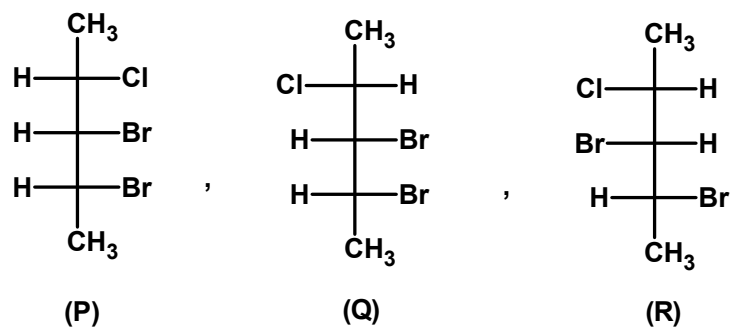
20.



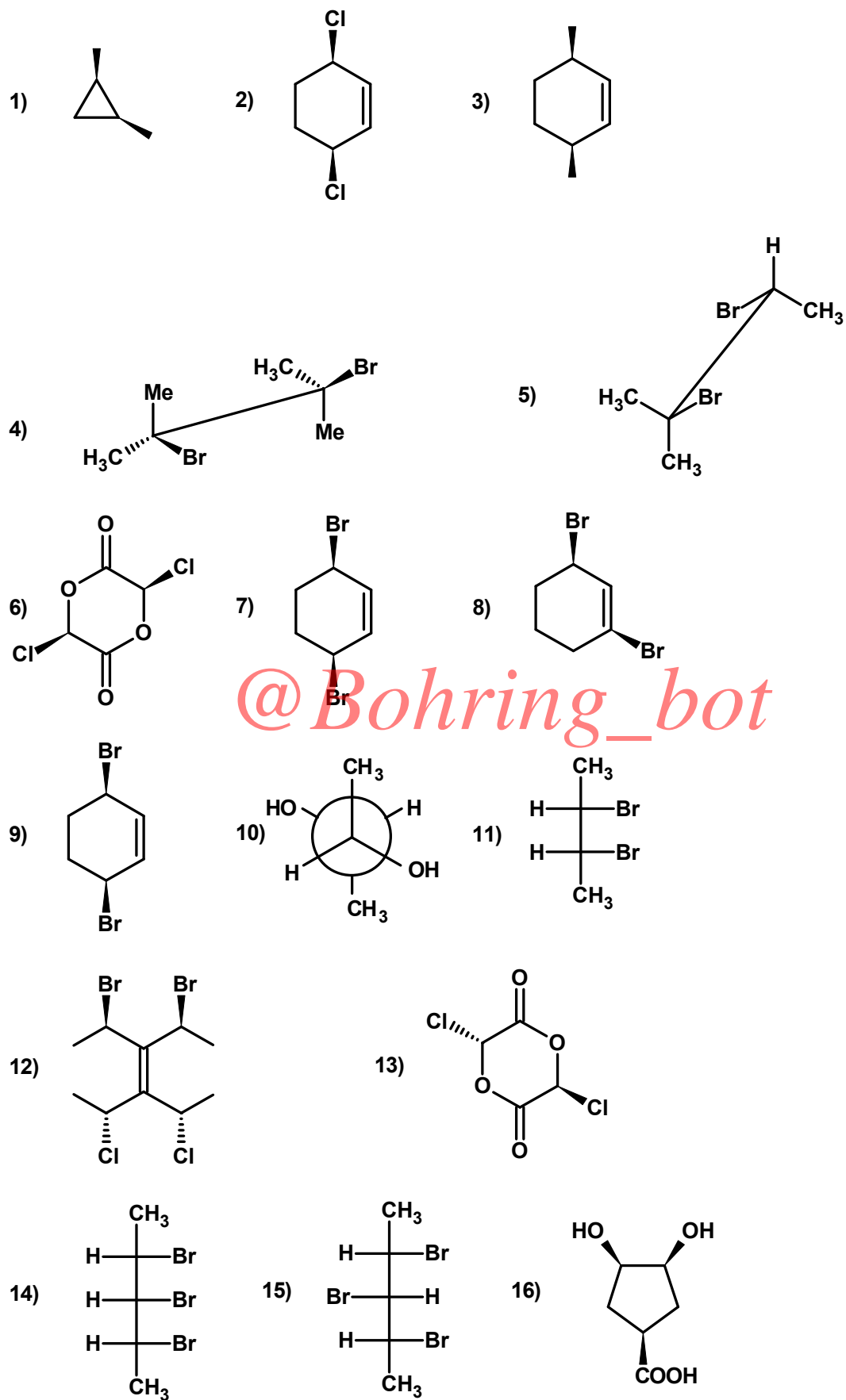
21.



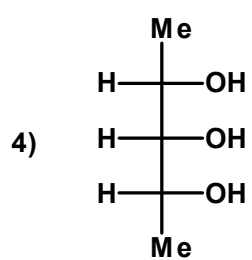
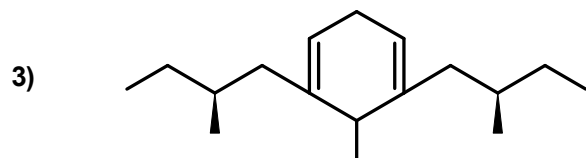
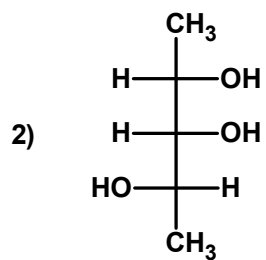
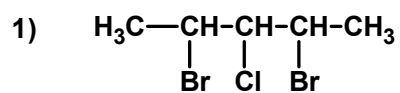
22.



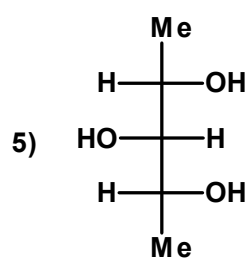
Q. Identify "Meso" compounds?



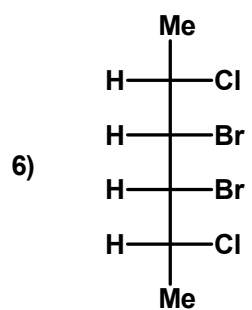
Q. Find the number of chiral centre?



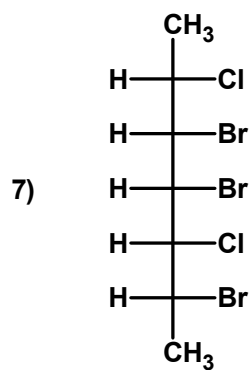
no. of C.C. =
no. of Dia =



no. of C.C. =
no. of Dia =

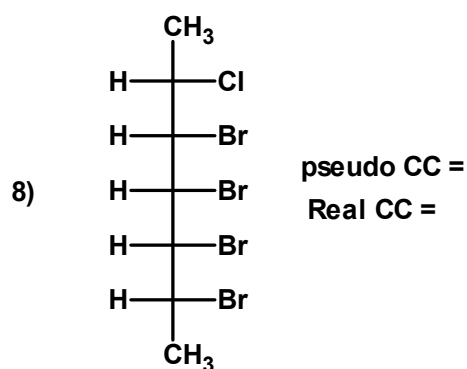


pseudo CC =
Real CC =



pseudo CC =
Real CC =

@Bohring_bot



Q. Find the number of acyclic diene of C_5H_8 ?

Q. Find the total S.I. of $\text{C}_6\text{H}_6\text{Cl}_6$? (Gamaxene)

1) Total SI

2) G.I.

3) O.I.

4) Opt. Active

5) Meso

@Bohring_bot

Q. Find total isomeric alkyne of $\text{C}_3\text{F.Cl.Br.I.}$? ((including stereo)

Q. Find total S.I. of $\text{CH}_3\text{CH}(\text{Br})\text{CH}(\text{Br})\text{CH}_3$?

Q. Total cyclic isomers of C_5H_{10} (including S.I.)

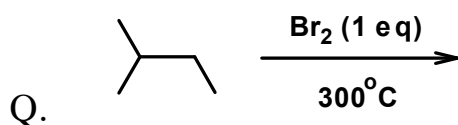
Q. Total cyclic isomers of dibromo cyclopropane? (including S.I.)

Q. Total cyclic isomers of bromochloro cyclopropane? (including S.I.)

Q. Total isomers of C_6H_{12} (only 4-membered ring) (including S.I.)

Q. Total isomers of $C_4H_6Br.Cl$. (only 4-member ring)

Q. Total isomers of $C_4H_6Br_2$ (only 4-member ring).

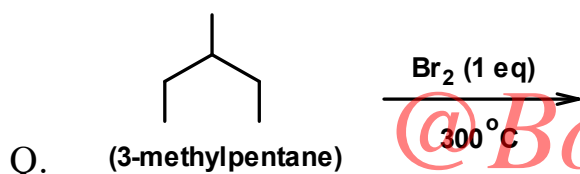


1) structural =

2) including stereo =

3) opt. active =

4) no. of fraction =

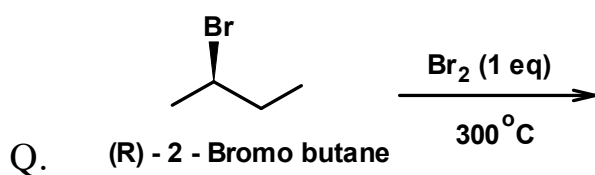


1) structural =

2) including stereo =

3) opt. active =

4) no. of fraction =



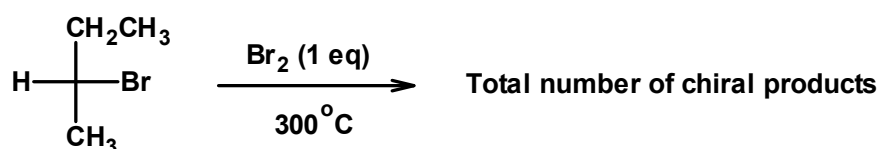
1) structural =

2) including stereo =

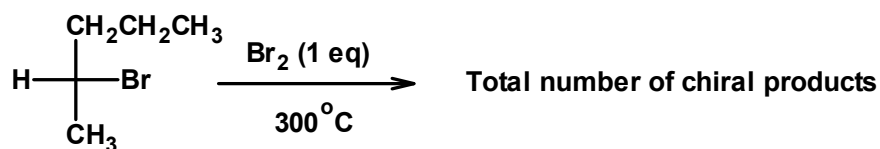
3) opt. active =

4) no. of fraction =

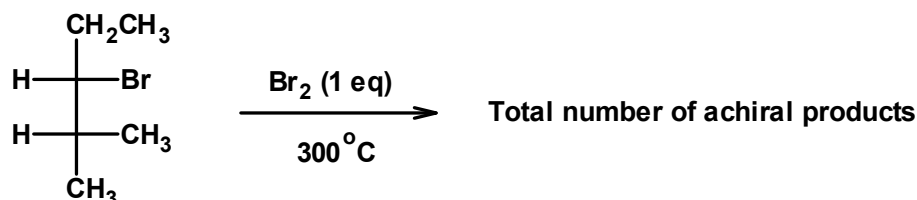
Q.



Q.



Q.



Level-2

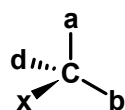
5. Dextrorotatory α - rinene has a specific rotation $[\alpha]_D^{20} = +51.3^\circ$. A sample of α - Rinene containing both the enantiomers was found to have a specific rotation $[\alpha]_D^{20} = +30.8^\circ$. The % of (+) & (-) enantiomer present in the sample respectively.

- | | (+) | (-) | | (+) | (-) |
|----|-----|-----|----|-----|-----|
| a) | 70% | 30% | b) | 20% | 80% |
| c) | 80% | 20% | d) | 60% | 40% |

6. (+) Mandelic acid has a specific rotation of 158° . What would be the observed specific rotation of a mixture of 25% (-) mandelic acid & 75 % (+) mandelic acid

- a) $+118.5^\circ$ b) -118.5° c) -79° d) $+79^\circ$

7. What is the correct Fischer for given wedge dash representation:



- | | | | | | | | |
|----|---|----|---|----|---|----|---|
| a) | $\begin{array}{c} \text{x} \\ \\ \text{d} - \text{C} - \text{b} \\ \\ \text{a} \end{array}$ | b) | $\begin{array}{c} \text{x} \\ \\ \text{a} - \text{C} - \text{b} \\ \\ \text{d} \end{array}$ | c) | $\begin{array}{c} \text{a} \\ \\ \text{x} - \text{C} - \text{b} \\ \\ \text{d} \end{array}$ | d) | $\begin{array}{c} \text{x} \\ \\ \text{b} - \text{C} - \text{d} \\ \\ \text{a} \end{array}$ |
|----|---|----|---|----|---|----|---|

8. $\text{C}_4\text{H}_6\text{O}_2$ does represent

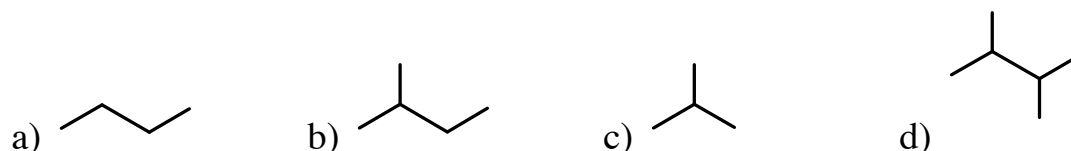
- | | |
|---------------------|---------------------------------|
| a) A diketone | b) A compound with two aldehyde |
| c) An alkenoic acid | d) An alkanoic acid |

9. Total possible isomers of dimethylcyclohexane

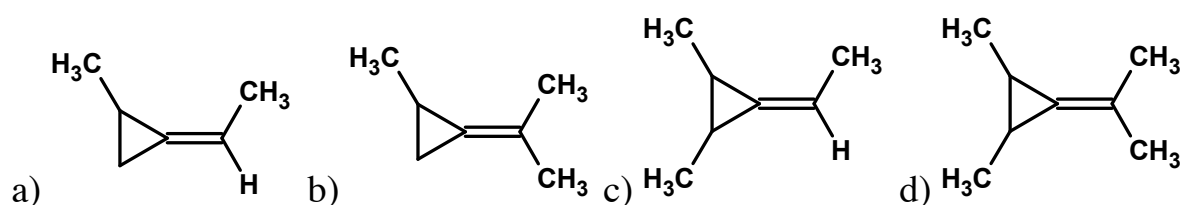
(6-member ring only), which are chiral?

- a) 4 b) 7 c) 3 d) 6

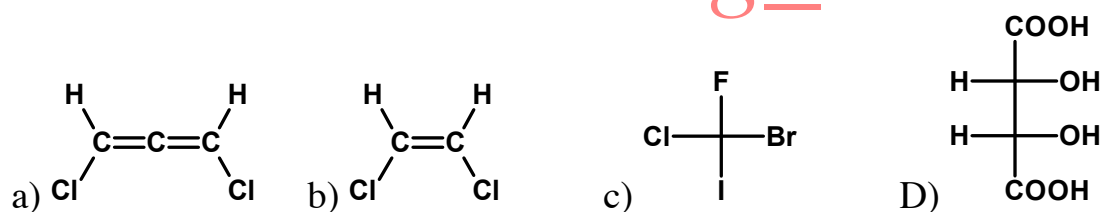
10. Which alkane does not give optically active product on monochlorination in presence of light.



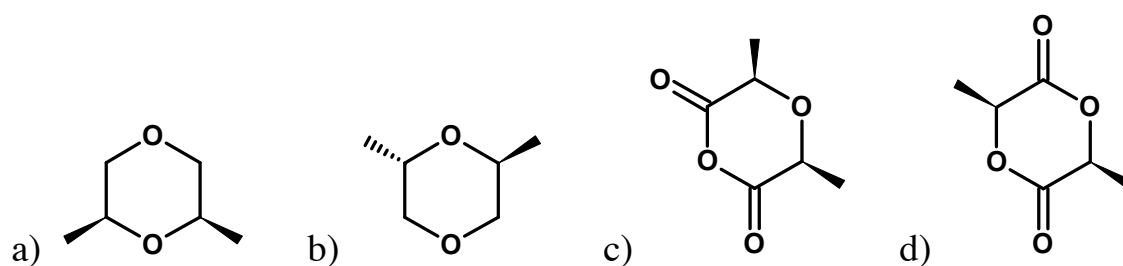
11. Among the following compound, which can show geometrical as well as optical isomerism.



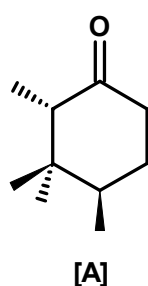
12. Which of the following can exist in diastereomeric form

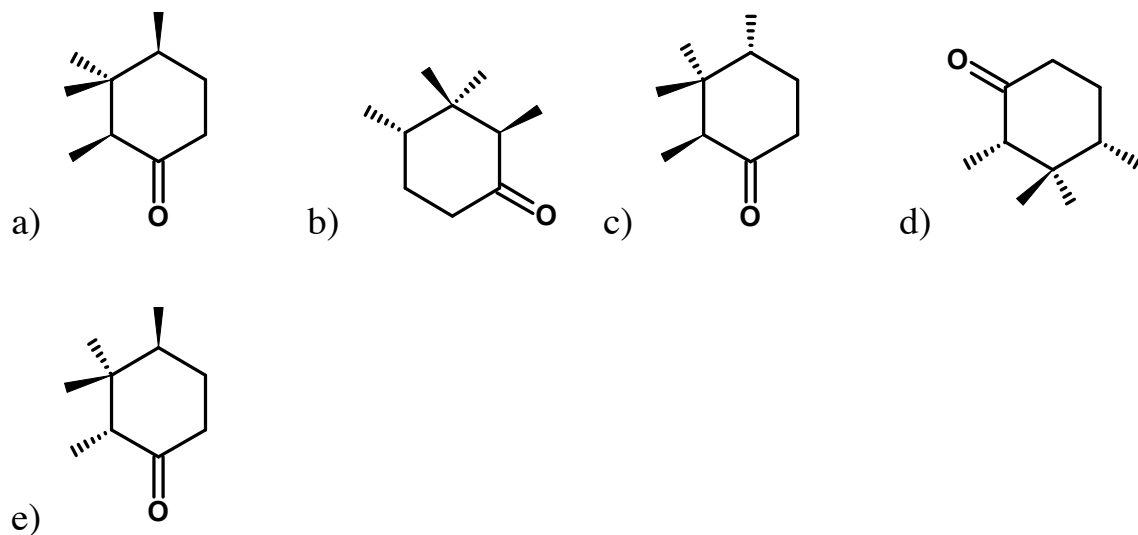


13. Identify the meso compound?

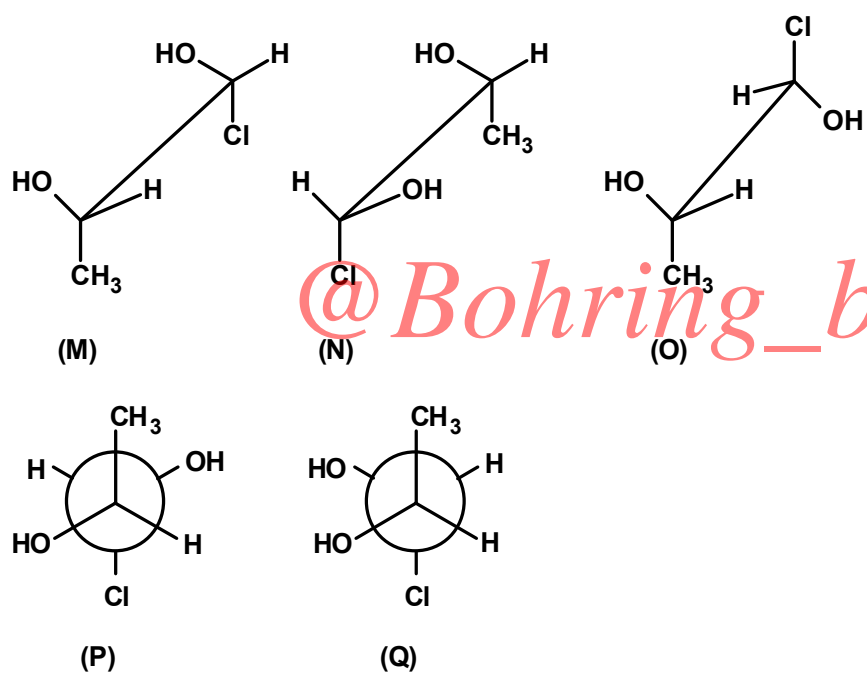


14. Which of the structure(s) is/are diastereomer of "A"?



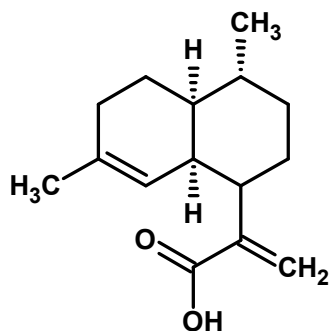


15. Which of the given statement(s) about N, O, P & Q with respect to M is/are correct.

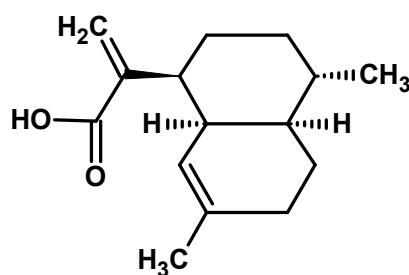


- a) M and N are non-mirror image stereoisomers.
- b) M and O are identical.
- c) M and P are enantiomers
- d) M and Q are identical

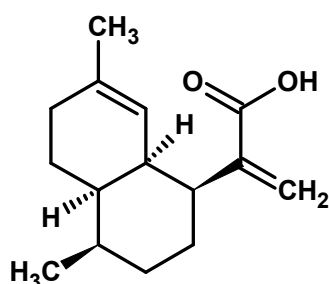
16. Identify which of the relationship is correct b/w the following pairs?



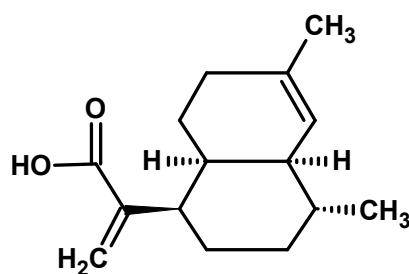
(P)



(Q)



(R)



(S)

a) Q, R are enantiomers of each other.

b) p, Q are enantiomers of each other.

c) P, Q are constitutional isomers of each other.

d) P, S are diastereomers of each other.

17. The correct statement(s) about the compound (X)



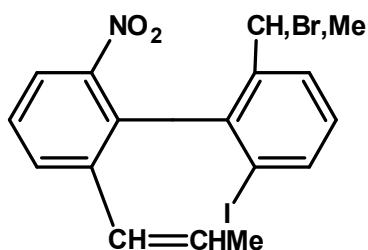
a) The total number of stereoisomers possible for [X] is 6.

b) The total number of diastereomers possible for [X] is 3.

c) If the stereochemistry about double bond in [X] is trans then number of enantiomers possible for [X] is cir.

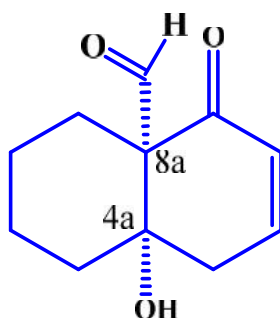
d) If the stereochemistry about double bond in [X] is cis, then number of enantiomers possible for [X] is 2.

18. How many stereoisomers possible for the following compound.



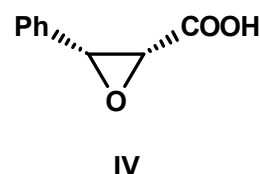
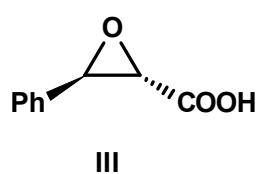
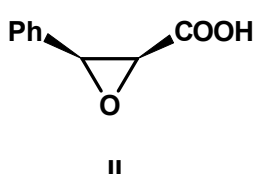
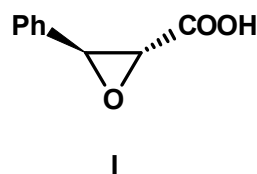
- a) 2 b) 4 c) 6 d) 8

19. Absolute stereochemistry of the given compound is?



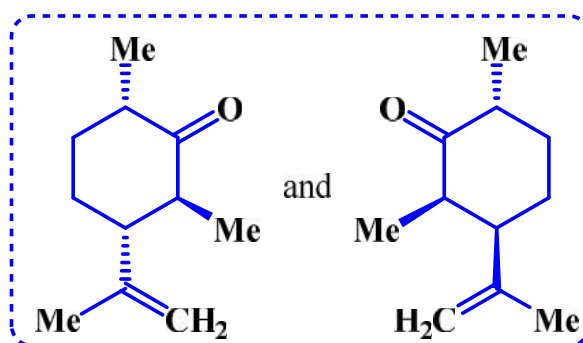
- a) 4aR, 8aS B) 4aR, 8aR C) 4aS, 8aS D) 4aS, 8aR

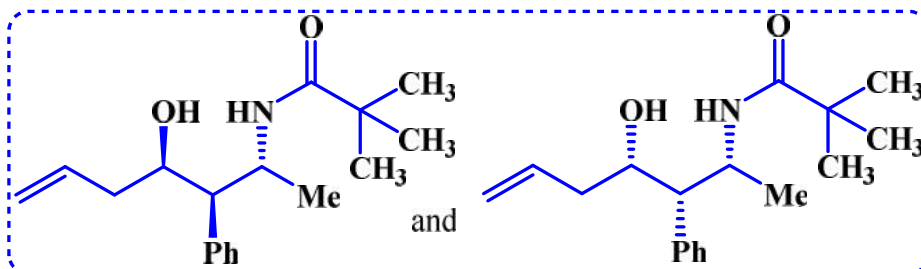
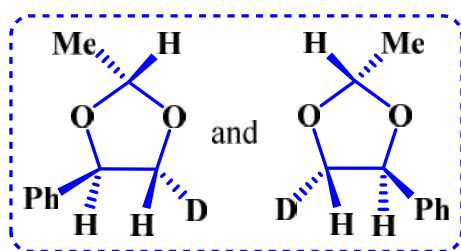
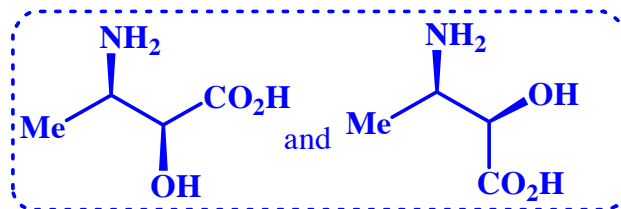
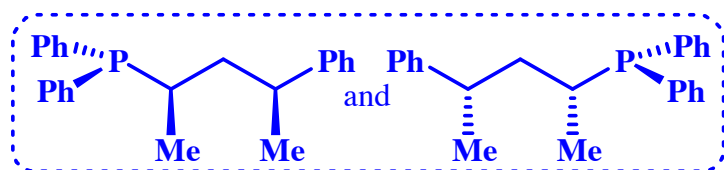
20. Identify relationship



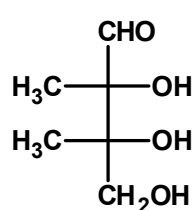
- A) (I, II) are diastereomers b) (II, IV) are enantiomers
c) (I, III) are enantiomers d) (III, IV) are homomers

31. Among the following the total number of set(s) of diastereomeric pair(s) is?

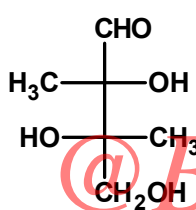




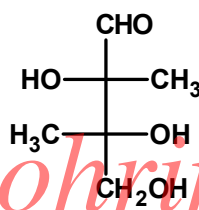
32. For the aldotetroses I-IV, the combination of TRUE statements, among P-T, is:



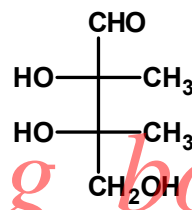
I



II



III



IV

P = I and II are diastereomers and II and III are enantiomers.

Q = I and IV are mesomers and are optically inactive.

R = I and III can be interconverted by a base catalysed isomerisation.

S = only I and IV are HIO_4 cleavable.

T = I and III are D-sugars and II and IV are L-sugars.

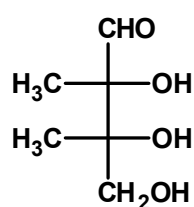
a) Q, R, T

b) P, R, T

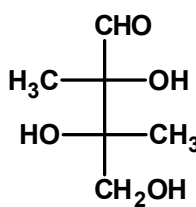
c) Q, S, T

d) P, Q, S

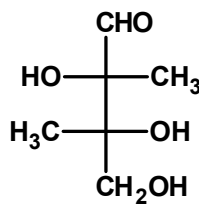
33. For the aldotetroses I-IV, the combination of TRUE statements, among P-S, is:



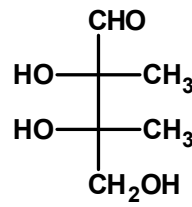
I



II



III



IV

P = I and II are diastereomers and II and III are enantiomers.

Q = I and IV are mesomers and are optically inactive.

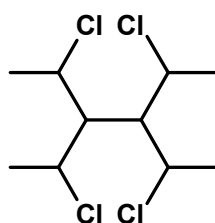
R = I and III can be interconverted by a base catalysed isomerisation.

S = only I and IV are HIO_4 cleavable.

T = I and III are D-sugars and II and IV are L-sugars.

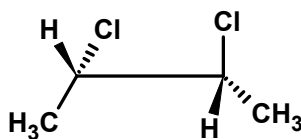
a) Q, R, T b) P, R, T c) Q, S, T d) P, Q, S

34. Total no. of possible stereoisomers for the given compound is



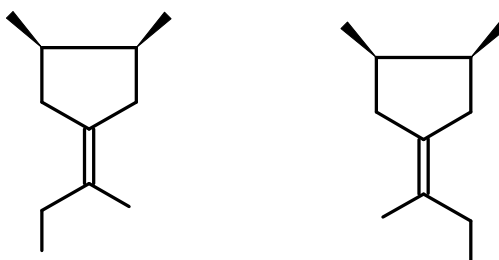
a) 2 b) 4 c) 6 d) 7

35. Find the total number of stable conformers with non zero dipolemoment for given compound



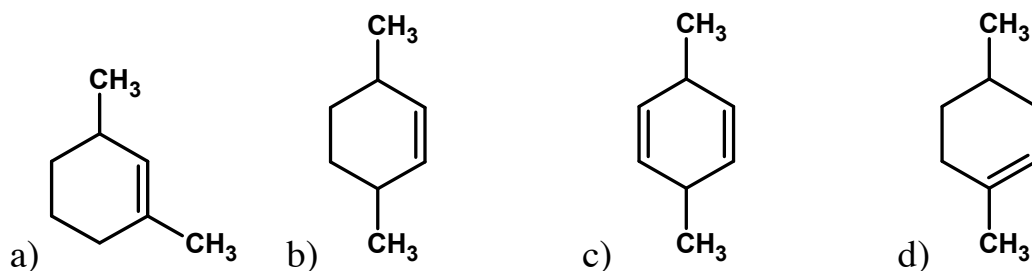
a) Zero b) 2 c) 3 d) 1

36. Both compounds are related as

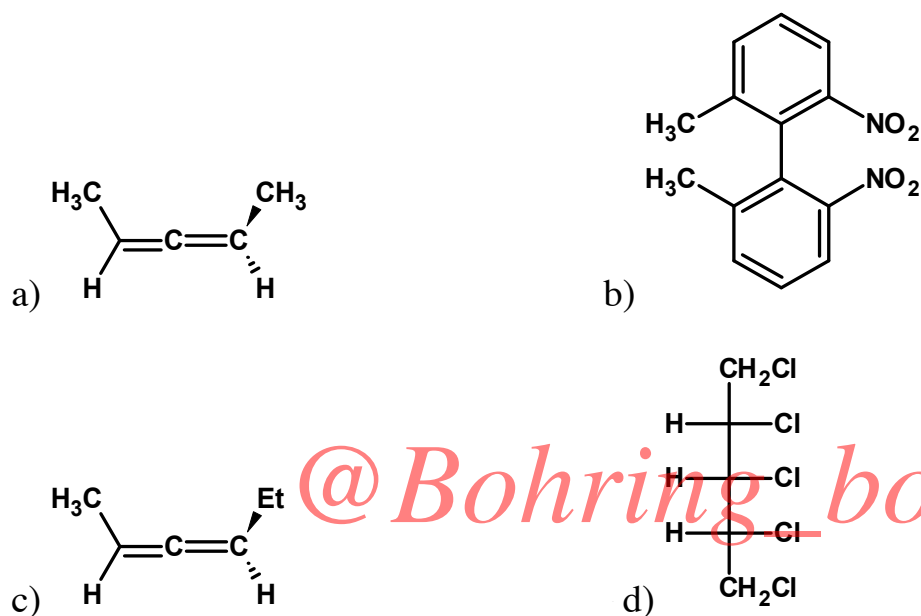


a) Identical b) Enantiomers c) Diastereomers d) Optical Isomers

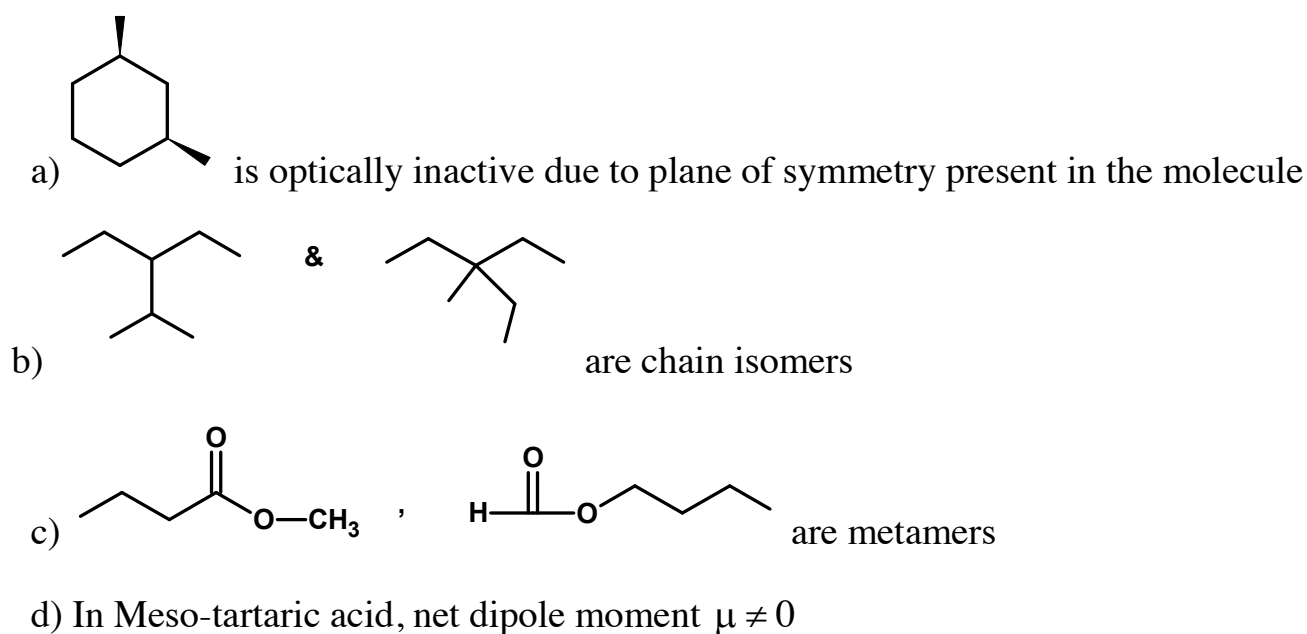
37. Compound which cannot show geometrical isomerism is/are?



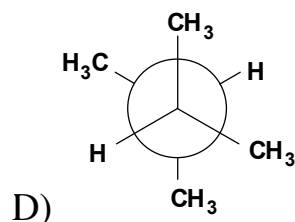
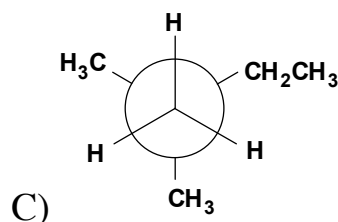
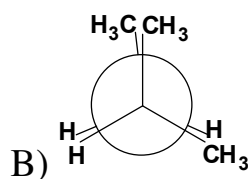
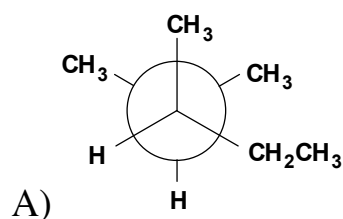
38. Which of the following is/'are achiral molecules



39. Which of the following is/are correct



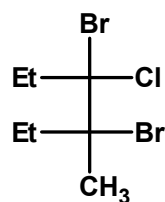
40. Which of the following Newmann's projections represent conformations of 2,3-dimethyl butane ?



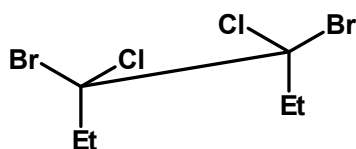
41. The specific rotation of optically pure (R)-2-bromo butane is -112.00° . A given sample of 2-bromobutane exhibited a specific rotation of -82.88° . The percentage of (S)-(+)-enantiomer present in the sample is

a) 74% b) 26% c) 13% d) 37%

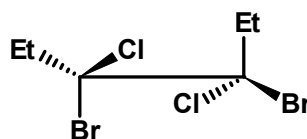
42. Which of the given statement(s) about N, O, P and Q with respect to M is/are correct?



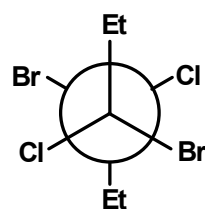
[M]



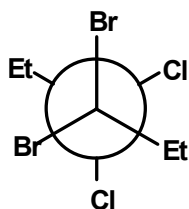
[N]



[O]



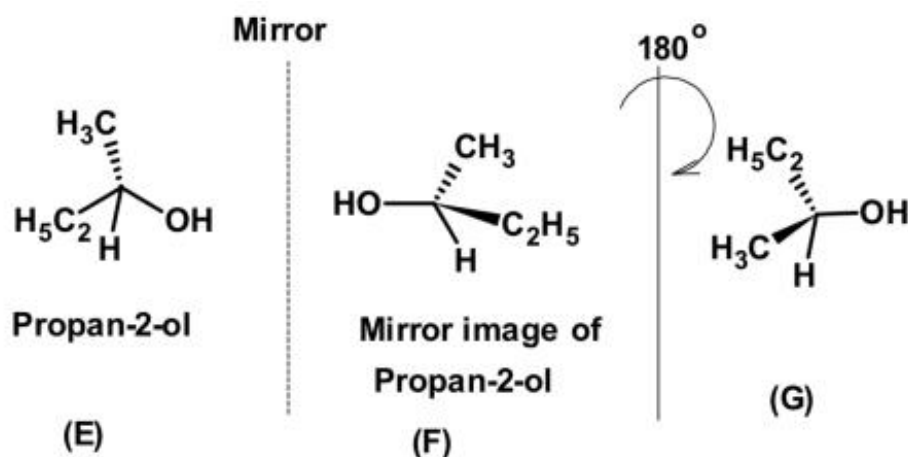
[P]



[Q]

- a) **M** and **N** are identical
 b) **M** and **O** are enantiomers
 c) **M** and **P** are non-mirror image stereoisomers
 d) **M** and **Q** are identical

43.

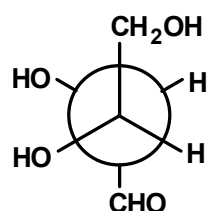


Identify the correct statement(s) is

- a) "G" is non-superimposable on
- b) "G" is superimposable on
- c) "G" is non-superimposable on "E"
- d) "G" is superimposable on "E".

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44. What is the correct Fischer Projection formula for the compound represented by the following Newmann Projection?

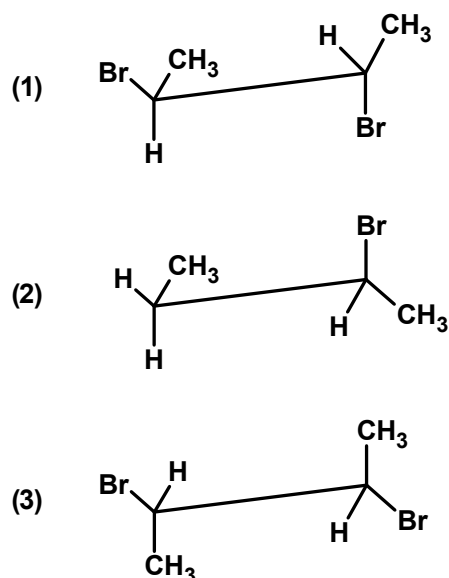


- a)
- b)
- c)
- d)

45. Which one of the following Newmann projection formulae correctly represents a meso structure?

- a)
- b)
- c)
- d)

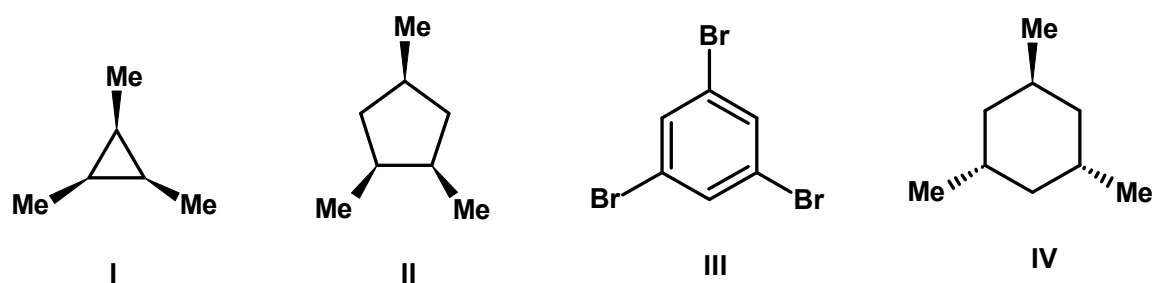
46. Consider the following configuration of 2,3-dibromobutane.



Which one of the following is the correct answer?

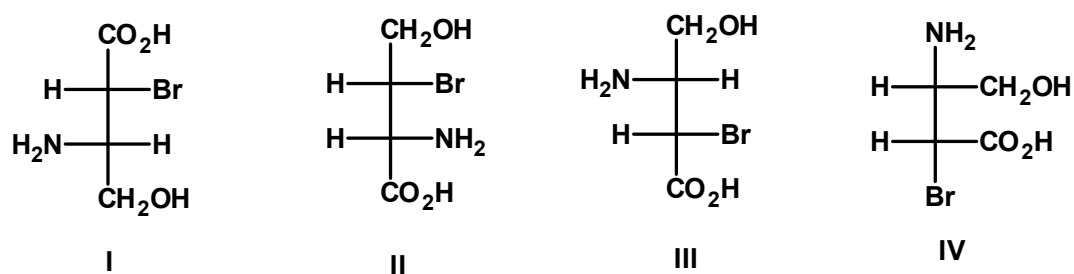
- a) Conformation (1) is the meso-form while (2) and (3) are an enantiomeric pair.
- b) Conformation (2) is the meso-form while (1) and (3) are an enantiomeric pair.
- c) Conformation (3) is the meso-form while (1) and (2) are an enantiomeric pair.
- d) Conformation (1) and (2) are identical and (2) is the meso-form..

47. The compounds having C_3 – axis of symmetry are



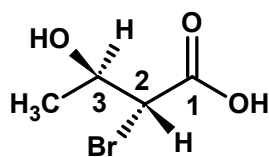
- a) I, III and IV b) I, II and III c) I and III d) III and IV

48. Among the following compounds, the pair of enantiomers is



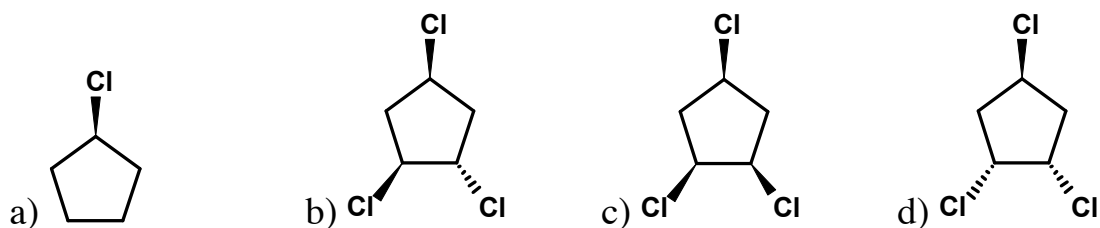
- a) I and IV b) I and III c) II and III d) III and IV

49. The absolute configuration of C₂ and C₃ in the following compound is



- a) 2R, 3S b) 2S, 3R c) 2S, 3S *d) 2R, 3R

50. Which of the following molecules is chiral?



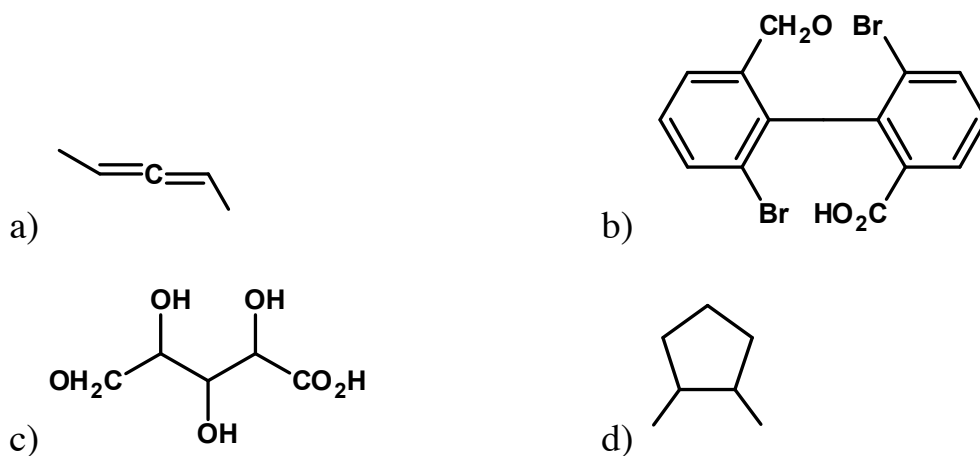
51. Allene has the structure $\text{H}_2\text{C}=\text{C}=\text{CH}_2$. What is the best description of the geometry of allene?

Geometry at central carbon

Positions of hydrogen atoms

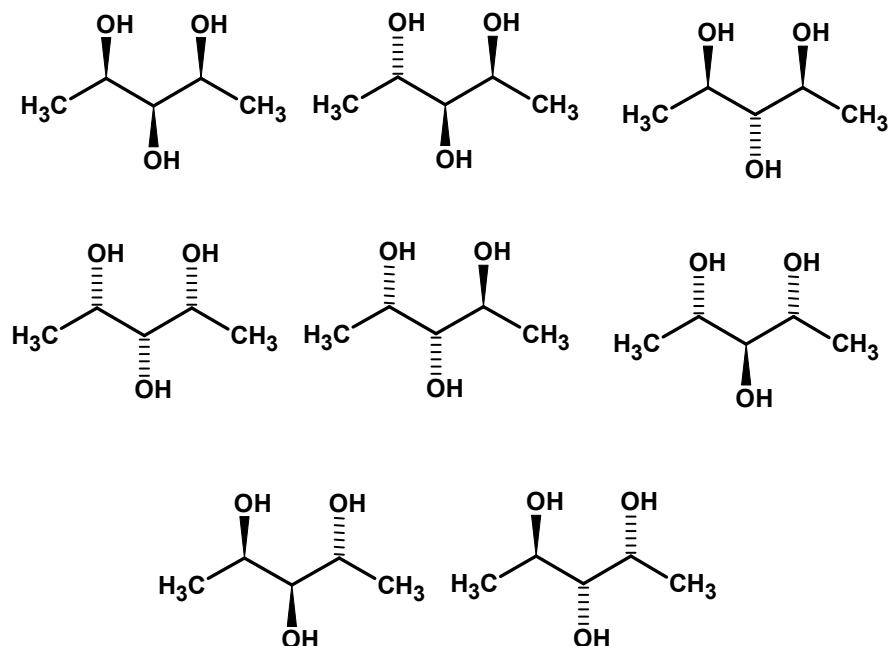
- | | | |
|----|--------|-----------------------------|
| A) | Linear | All in the same plane |
| B) | Linear | In two perpendicular planes |
| C) | Bent | All in the same plane |
| D) | Bent | In two perpendicular planes |

52. Achiral stereoisomer(s)



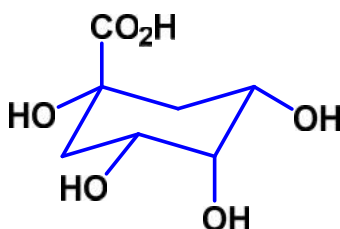
INTEGER

1. How many out of the given stereo isomers of pentane 2,3,4-triol show optical isomerism?



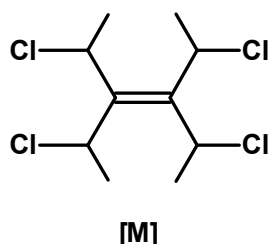
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2. (R)-2-methyl-1-butanol has a specific rotation of $+13.5^\circ$. The specific rotation of 2-methyl-1-butanol containing 15% of the (S)-enantiomer is__
3. Number of meso isomers possible with molecular formula $C_6H_{12}Cl_2$
4. The optical rotation of the given compound is?

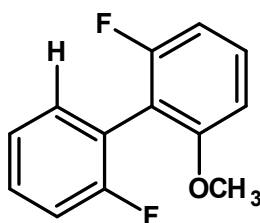
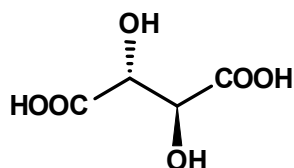
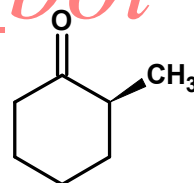
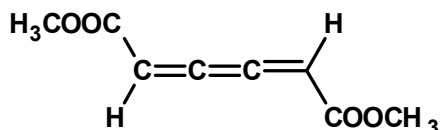
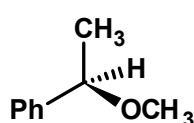


5. (–) mandelic acid has a specific rotation of -150° . What would be the specific rotation of solution which contains 40% (–) mandelic acid and 60% (+) mandelic acid.
6. The specific rotation of a pure substance is -5.90° . What is the percentage of this isomer in a mixture with an observed rotation of -2.95° ?

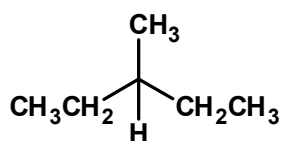
7. Find the total number of optically active stereoisomers of given compound [M].



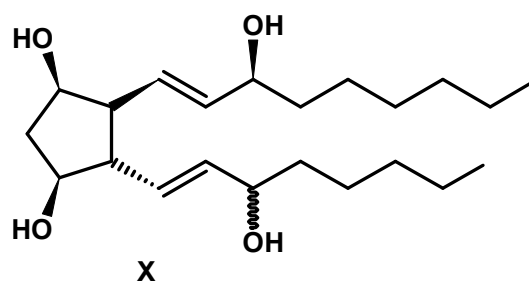
8. Specific rotation of the (R)-enantiomer of a chiral compound is 48. The specific rotation of a sample of this compound which contains 25% of (S)-enantiomer is__.
9. (R)-2-methyl-1-butanol has a specific rotation of $+13.5^\circ$. The specific rotation of 2-methyl-1-butanol containing 40% of the (S)-enantiomer is____ $^\circ$. (Round off to one decimal place)
10. Among the following compounds, the number of compounds that DO NOT exhibit optical activity at room temperature is_____.



11. The maximum number of isomers (including stereoisomers) that are possible on monochlorination of the following compound, is



12. For the given compound [X], the total number of optical active stereoisomers are



- This type of bond indicate that the configuration at the specific carbon and the geometry of the double bond is fixed
- ~~~~ This type of bond indicate that the configuration at the specific carbon and the geometry of the double bond is **NOT** fixed.

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ORGANIC CHEMISTRY

IIT JEE

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GOC-(A)

PREPARED BY: ASHISH SIR

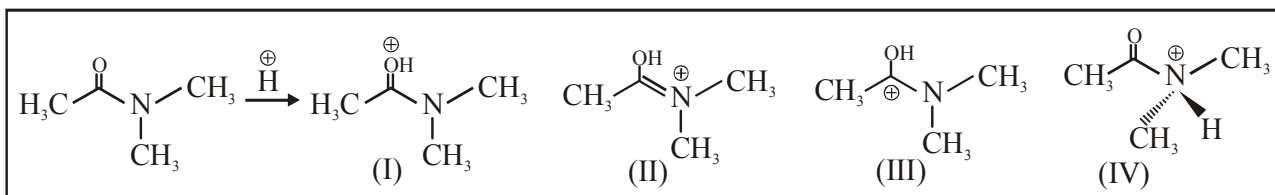
SYNOPSIS

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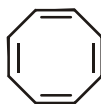
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One or more correct :

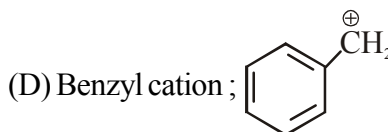
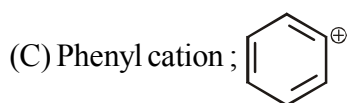
1. Which of the following resonating structure is most stable?
- (A) $\text{CH}_3 - \text{O} - \text{CH} = \text{CH} - \overset{\oplus}{\text{CH}}_2$ (B) $\text{CH}_3 - \text{O} - \overset{\oplus}{\text{CH}} - \text{CH} = \text{CH}_2$
- (C) $\text{CH}_3 - \overset{\oplus}{\text{O}} = \text{CH} - \text{CH} = \text{CH}_2$ (D) $\text{CH}_3 - \text{O} - \overset{\oplus}{\text{CH}} - \overset{\ominus}{\text{CH}} - \overset{\oplus}{\text{CH}}_2$
2. Shown below are three representations of protonated N, N-dimethylacetamide. Identify the incorrect statement(s) –



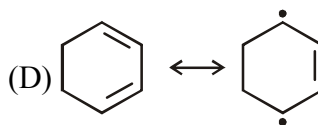
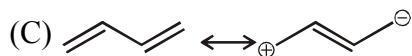
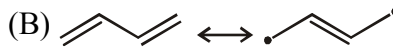
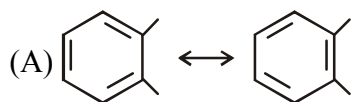
- (A) Structure (IV) is not a resonance form of the other.
- (B) Structure at the N-atom is predicted to be trigonal planar.
- (C) Structure (III) is the least important of the resonance structures shown.
- (D) The C – O Bond in protonated N, N-dimethyl acetamide would be shorter than the C–O bond in N,N-dimethyl acetamide.
3. Find the correct statement(s) regarding inductive effect?
- (A) Polarisation of σ -bond caused by the polarization of adjacent σ -bond is referred as inductive effect.
- (B) This effect is not passed on to the subsequent bonds.
- (C) During I-effect, $\bar{e}$ -does not leave the original orbital.
- (D) This effect increases rapidly as the number of intervening bonds increases.
4. Which of the following statement(s) is (are) correct?
- (A) In positive electromeric effect, $\pi-\bar{e}$ of multiple bonds are transferred to that atom to which the reagent gets attached.
- (B) In negative electromeric effect, $\pi-\bar{e}$ of multiple bond are transferred to that atom to which the reagent gets attached.
- (C) When inductive and electromeric effect operate in opposite directions, the electromeric effect predominates.
- (D) Electromeric effect is annulled as attacking reagent is removed from the domain of the reaction.
5. Which of the following statement is correct about cyclooctatetraene?



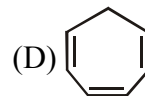
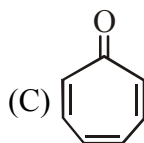
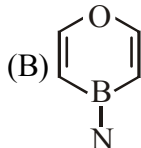
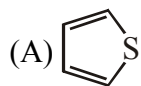
- (A) It is non conjugated system. (B) It has "Tub" like conformation.
- (C) It is a cyclic planar compound. (D) Chemically, it behaves like an alkene, not like benzene.
6. Which of the following cation is(are) not resonance stabilized?



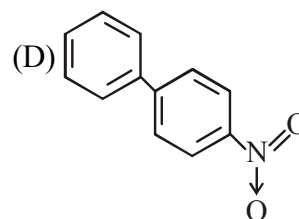
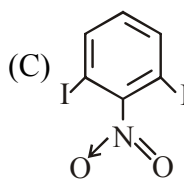
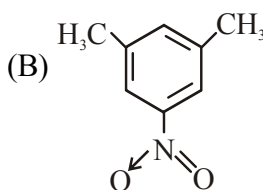
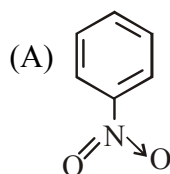
7. Which of the following pairs are resonance structures of each other?



8. Which of the following is not aromatic compound?

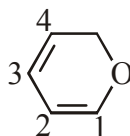


9. In which of the following molecules, nitro group is not coplanar with phenyl ring?



10. In given molecule, π - e^- density is more on –

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(A) 1 & 3

(B) 2 & 4

(C) 2 & 3

(D) 1 & 4

11. Among these compounds, the correct order of resonance energy is :-



(I)

(II)

(III)

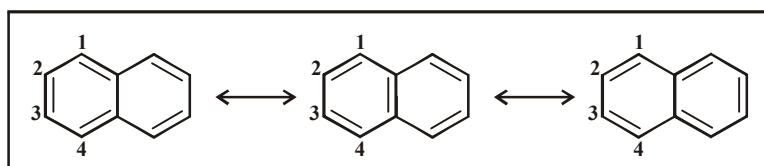
(A) I > II > III

(B) III > II > I

(C) II > III > I

(D) II > I > III

12. There are three canonical structures of naphthalene, Examine them and find correct statement among the following :-



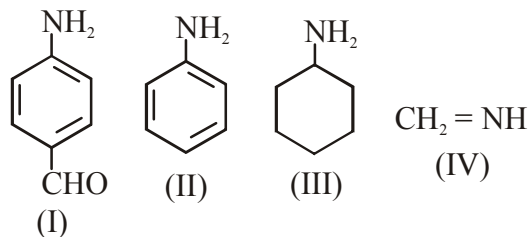
(A) All C – C bonds are of same length.

(B) $\text{C}_1 - \text{C}_2$ bond is shorter than $\text{C}_2 - \text{C}_3$ bond

(C) $\text{C}_1 - \text{C}_2$ bond is longer than $\text{C}_2 - \text{C}_3$ bond

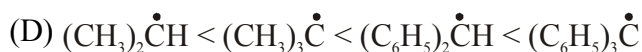
(D) None

13. Among these compounds, the correct order of C – N bond length is :-



- (A) IV > I > II > III (B) III > I > II > IV (C) III > II > I > IV (D) III > I > IV > II

14. The increasing order of stability of the following free radicals is :-



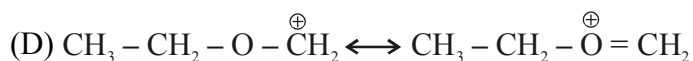
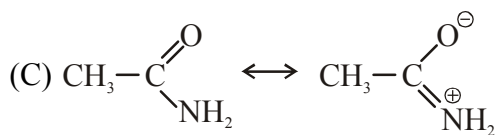
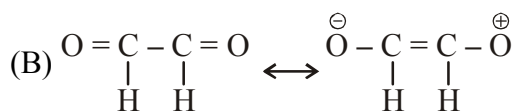
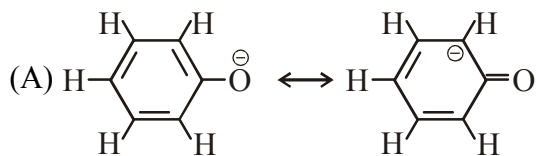
15. Arrange the carbanions, in order of their decreasing stability –



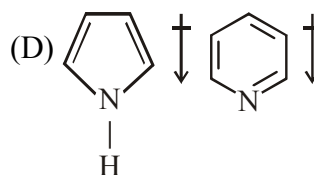
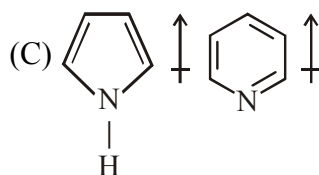
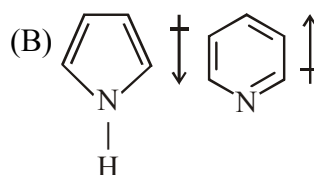
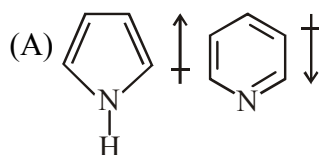
16. For 1-methoxy-1,3-butadiene, which of the following structure is the least stable?



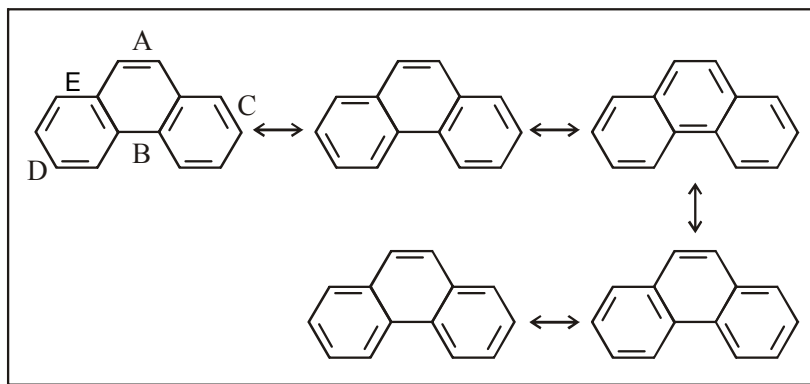
17. In which of the following case, resonance contribute LEAST toward stabilisation :-



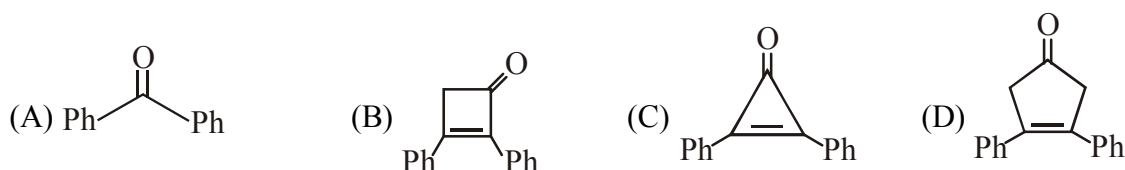
18. The correct orientation of dipoles in pyrrole and pyridine is :-



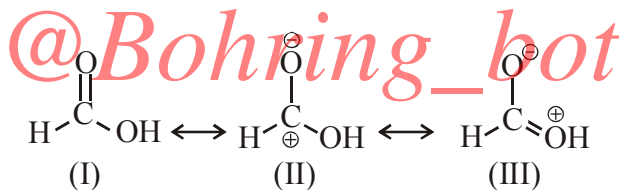
19. Assuming that each resonance structure contributes equally to the final resonance hybrid of phenanthrene, Identify the longest bond and shortest bond in phenanthrene.



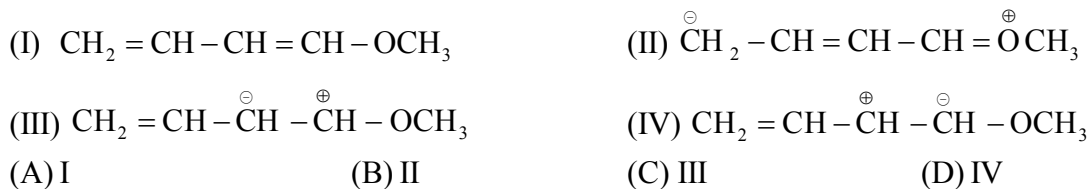
- (A) A shortest & B longest
(B) C shortest D longest
(C) E shortest & A longest
(D) B shortest & C longest
20. Molecule with the highest dipole moment among the following is :-



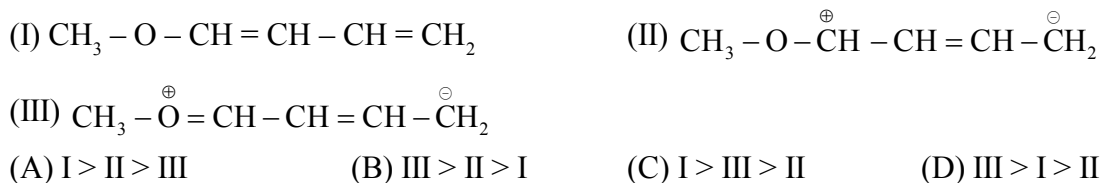
21. Among these cononical structure, the correct order of stability is :-



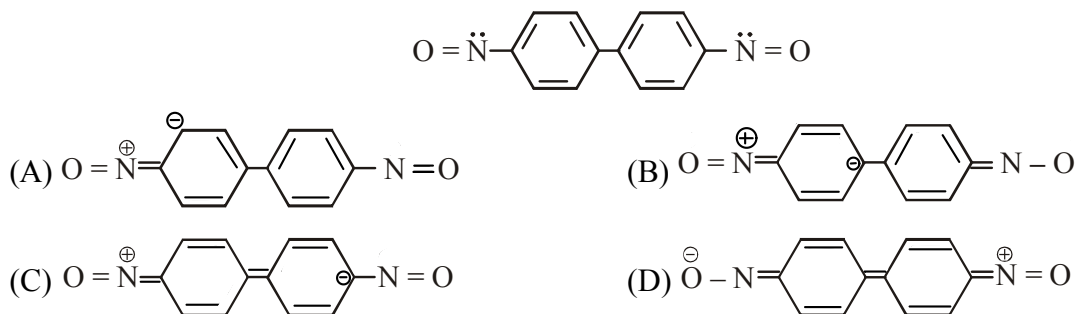
- (A) I > II > III (B) III > II > I (C) I > III > II (D) II > I > III
22. Among these canonical structures, the least stable is :-



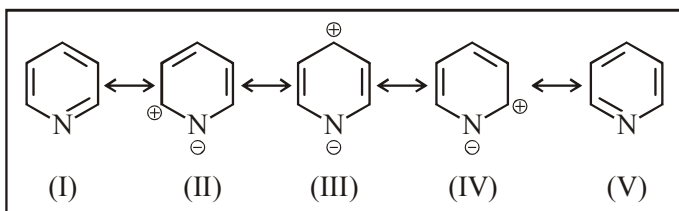
23. Among these 3 canonical structures (through more are possible) what would be their contribution in the "HYBRID"?



24. The most stable resonating structure of following compound is :-



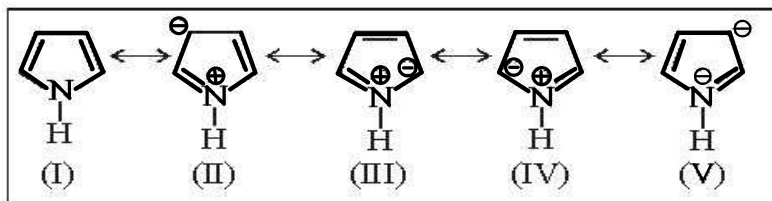
25.



Among these canonical structures of pyridine, the correct stability order is :-

(A) $(I = V) > (II = IV) > III$ (B) $(II = IV) > (I = V) > III$ (C) $(I = V) > III > (II = IV)$ (D) $III > (II = IV) > (I = V)$

26.

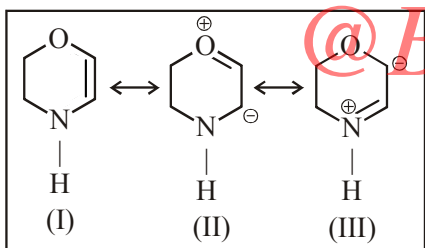


Among these canonical structures of "Pyrrole".

The correct stability order is :-

(A) $(III = IV) > (II = V) > I$ (B) $I > (II = V) > (III = IV)$ (C) $I > (III = IV) > (II = V)$ (D) $(II = V) > (III = V) > I$

27.



The least stable canonical structure among these –

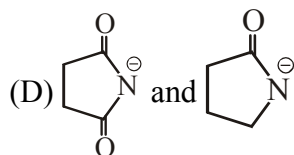
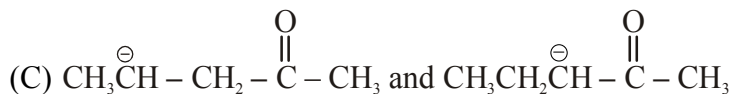
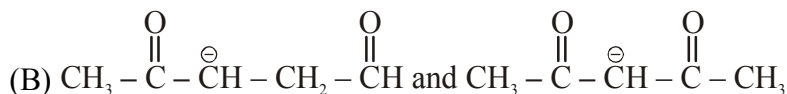
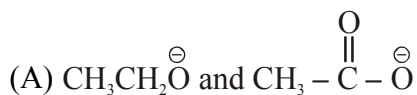
(A) I

(B) II

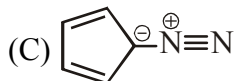
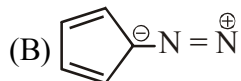
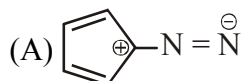
(C) III

(D) All are equally stable

28. In which of the following pairs, first species is more stable than second?



29.



(D) All are equally stable

ORGANIC CHEMISTRY

IIT JEE

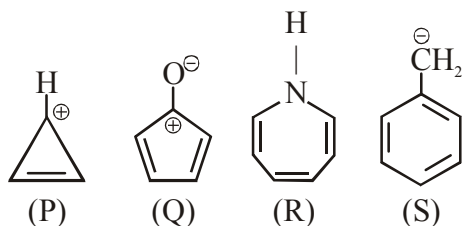
@Bohring_bot

GOC-(B)

PREPARED BY: ASHISH SIR

EXERCISE-B

1. The species that are aromatic among the following is(are) :-



- (A) R (B) P (C) Q (D) S

2. The compound is

- (A) Aromatic and has high dipole moment. (B) Aromatic and has no dipole moment
(C) non-aromatic and has high dipole moment (D) Anti-aromatic and has no dipole-moment

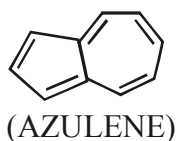
3. The compound that is "NOT AROMATIC" is :-



4. Cycloocta tetraene appears to be a conjugated annulene. In addition to the planar configuration, three non planar structures may be considered, all of which are shown below, which of these is most likely to be the favoured conformation.

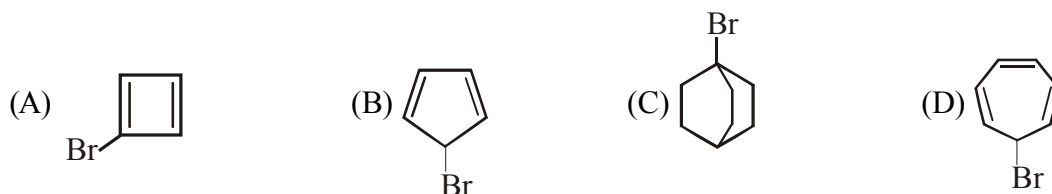


5. Which of the following statement is are true for AZULENE :



- (A) Azulene is an ionic compound
(B) By charge transfer from Biggerring to smaller ring
(C) Azulene is non benzenoid system
(D) Azulene has greater dipole moment

6. Organic compounds sometimes adjust their electronic as well as steric structures to attain stability. Among the following, the compound having highest dipole moment is :

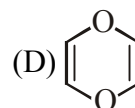
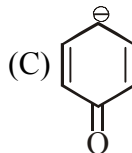
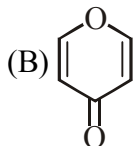
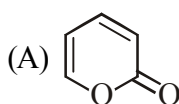


7. cyclooctatetraene is :

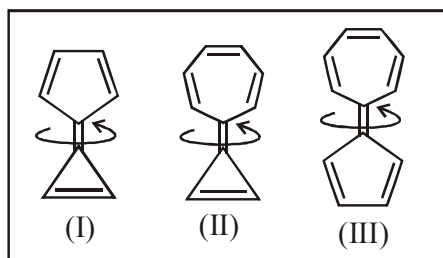


- (A) Aromatic because it is planar and follow Huckle's rule
 (B) Anti aromatic because it is not planar and it does not follow Huckle's rule
 (C) antiaromatic because it is planar and it does not follow Huckel's Rule
 (D) Non aromatic because it is not planar and it does not follow Huckel's rule

8. Which of the following compound is "NOT AROMATIC".



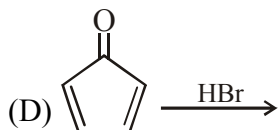
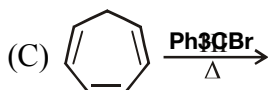
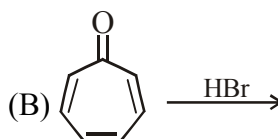
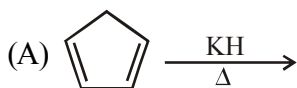
9.



The barrier for rotation about the indicated bonds will be maximum in which of these three compounds

- (A) I (B) II (C) III (D) Same in all three

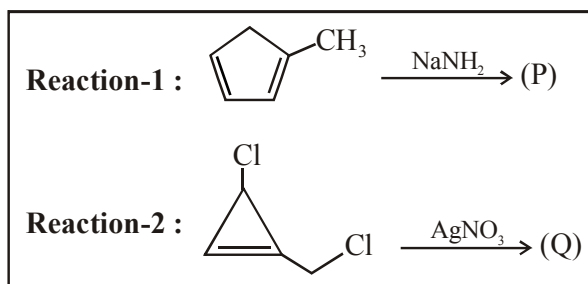
10. Which of the following reactions give aromatic compound ?

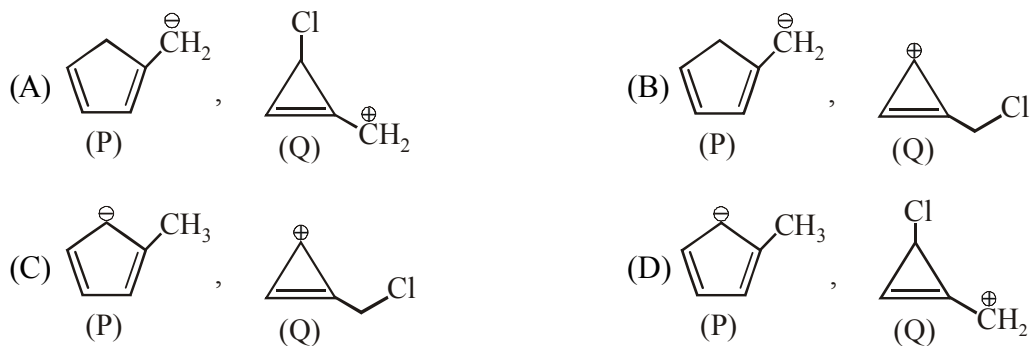


11. (P)

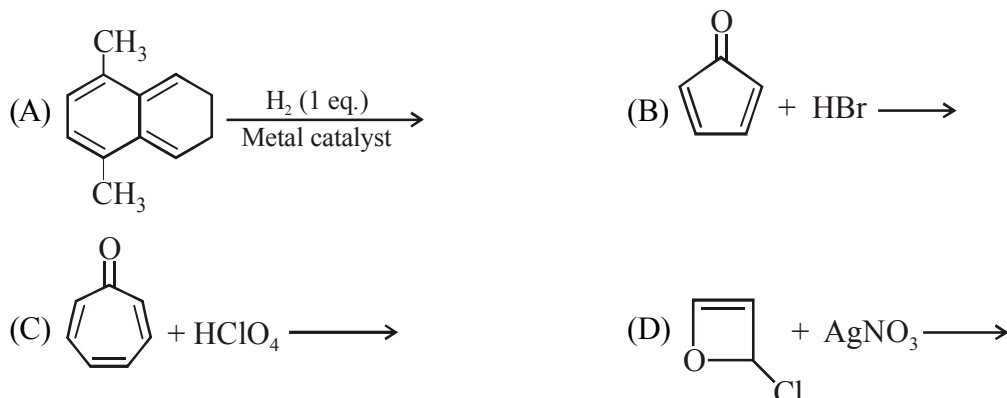
- (A) , ClO_4^- (B) , Ag^+ (C) mixture of (A) & (B) (D) None

12. The products P and Q in the following reaction respectively are

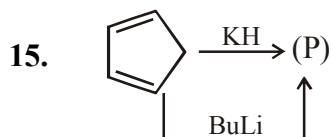
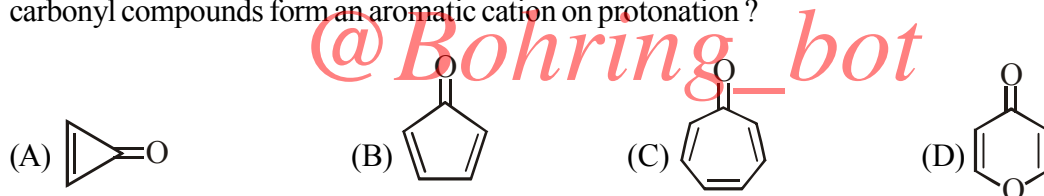




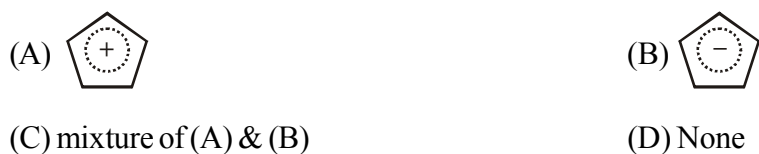
13. How many of the following compounds can form "AROMATIC" product on the given reactions ?



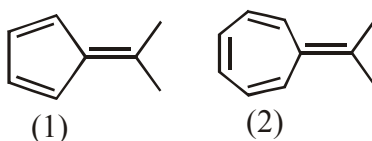
14. Protonation of a carbonyl group form a cation with significant charge on carbon atom which of the following carbonyl compounds form an aromatic cation on protonation ?



(P) will be :

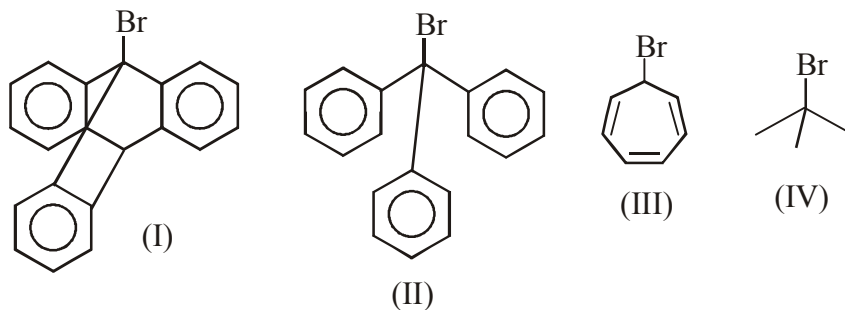


16. Which one of the following statement(s) is(are) true :

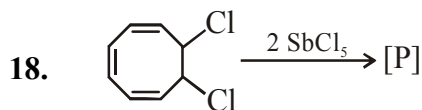


- (A) PhLi^+ react readily with both compound
 (B) PhLi^+ does not react to either of the compound
 (C) PhLi^+ react readily with (1) but does not react to (2)
 (D) PhLi^+ react readily with (2) but does not react to (1)

17. Correct order of rate of hydrolysis or rate of reaction toward AgNO_3 aq for given compounds-



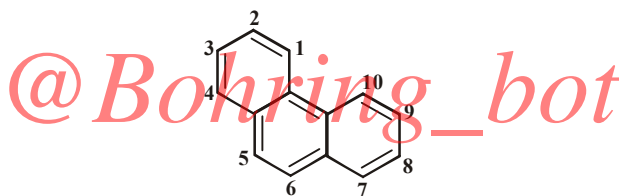
- (A) $\text{III} > \text{II} > \text{IV} > \text{I}$ (B) $\text{I} > \text{II} > \text{III} > \text{IV}$ (C) $\text{III} > \text{I} > \text{II} > \text{IV}$ (D) $\text{III} > \text{II} > \text{I} > \text{IV}$



Product [P] will be -

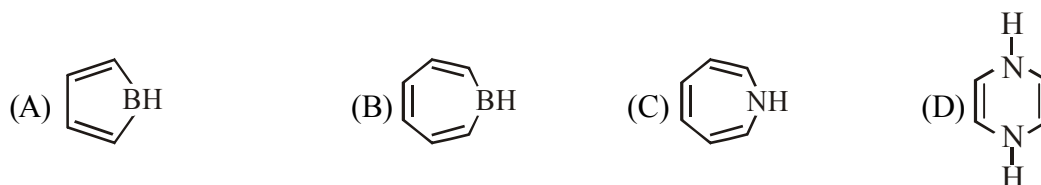
- (A) (B) $\cdot 2\text{SbCl}_6^-$ (C) (D) mixture of (A) & (B)

19. In the case of dibromo derivatives of the following compound, the derivative having highest energy has the bromo substituents in positions

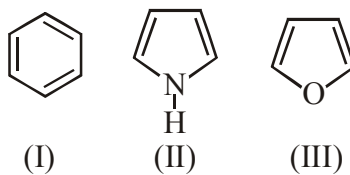


- (A) 1, 2 (B) 2, 3 (C) 4, 5 (D) 1, 10

20. Which of the following compounds is/are aromatic ?

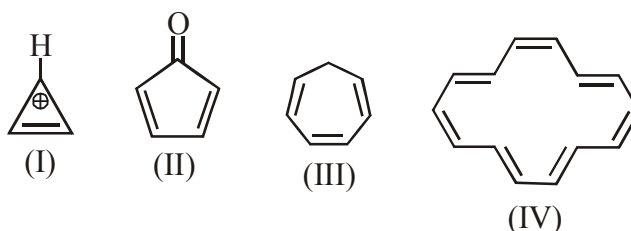


21. The correct order of resonance energy (in KJ/mol) of the following aromatic compounds is :



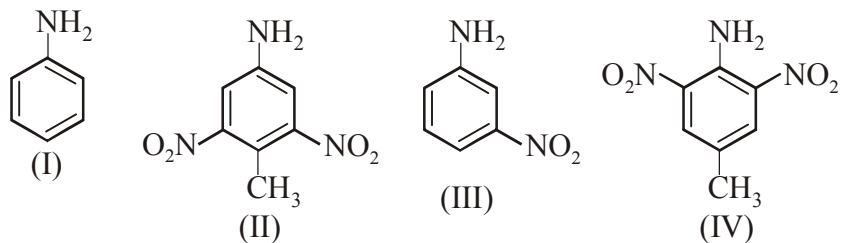
- (A) $\text{I} > \text{II} > \text{III}$ (B) $\text{II} > \text{I} > \text{III}$ (C) $\text{III} > \text{II} > \text{I}$ (D) $\text{II} > \text{III} > \text{I}$

22. The aromatic systems among the following are :



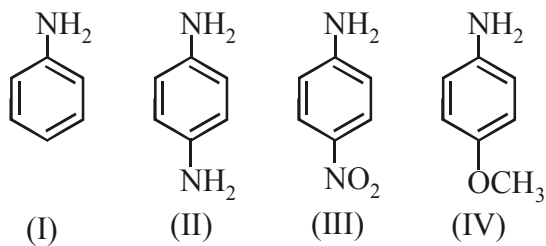
- (A) III and IV (B) I and II (C) II and IV (D) I and IV

23. The order of basicities of the following substituted anilines is -



- (A) I > II > III > IV (B) IV > I > III > II (C) IV > II > I > III (D) IV > III > I > II

24. The basicities (pK_b) of the following series of compounds (I - IV),



will vary as :

- (A) IV > II > I > III (B) III > I > IV > II (C) I > II > III > IV (D) IV > III > II > I

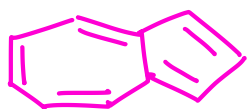
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GOC-(C)



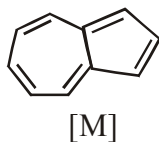
PREPARED BY: ASHISH SIR



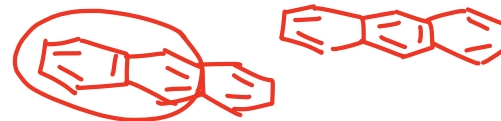
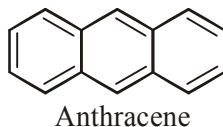
EXERCISE-C

Integer Type :

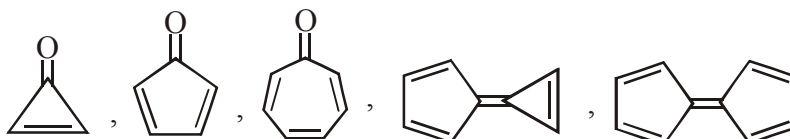
1. For the given structure [M] find the total number of Resonating structure (Uncharged).



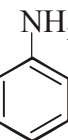
2. Find the total number of Benzenoid rings present in all possible resonating structures of anthracene -



3. How many of the following structures will have high contribution of ionic or polarized structures in their resonance hybrid



4. M = number of Resonance structures contributed in Resonance hybrid of



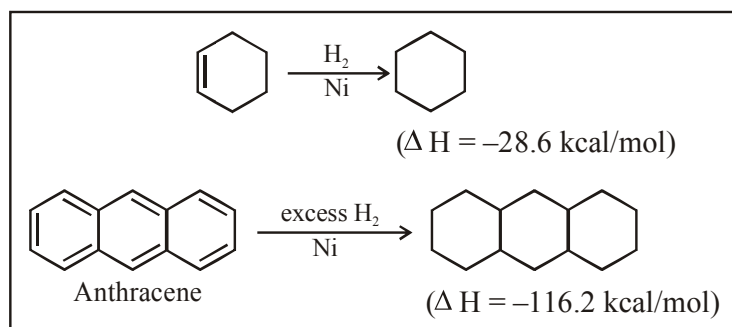
@Bohring_bot

N = number of Resonance structures contributed in Resonance hybrid of



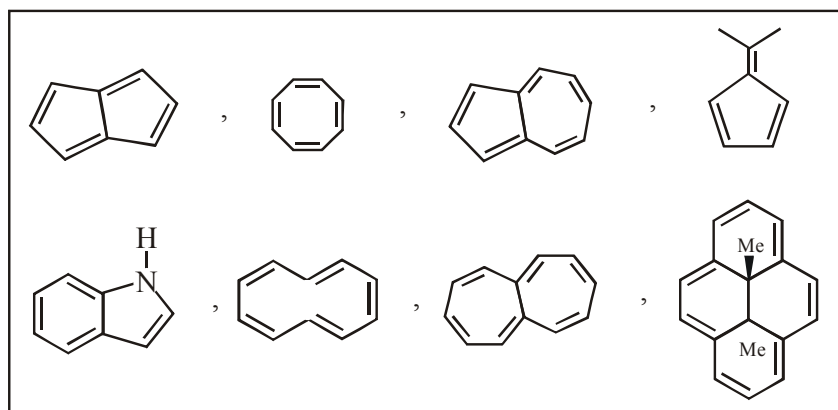
Find the sum of (M + N) ?

5. Use the following data to answer the question below

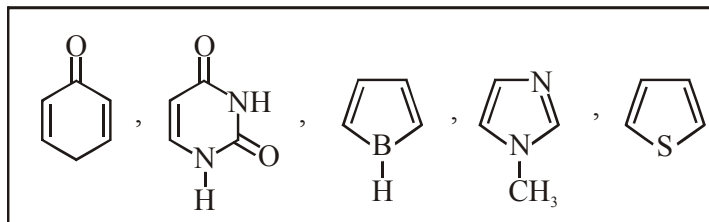


Calculate the resonance energy of anthracene in kcal / mole :

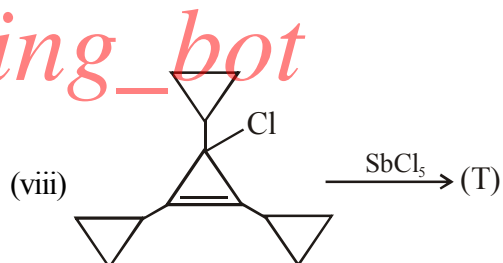
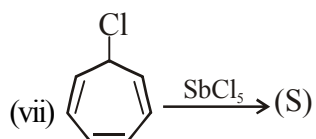
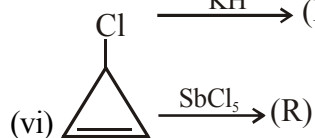
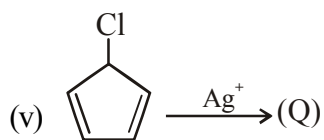
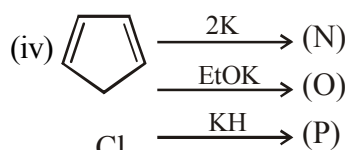
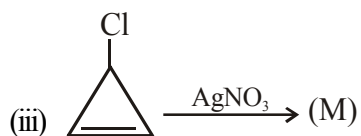
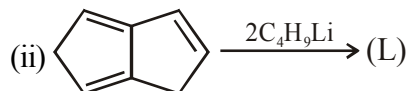
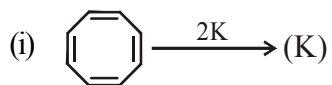
6. Among the following the number of molecules that are aromatic as they are drawn -



7. Among the following how many are aromatic compound(s) -



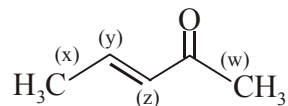
8. How many of the given reactions product can be aromatic ?



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EXERCISE-D

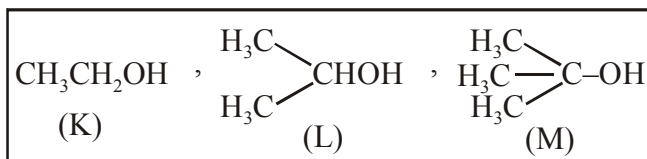
1. The abstraction of proton will be fastest in which carbon in the following compound -



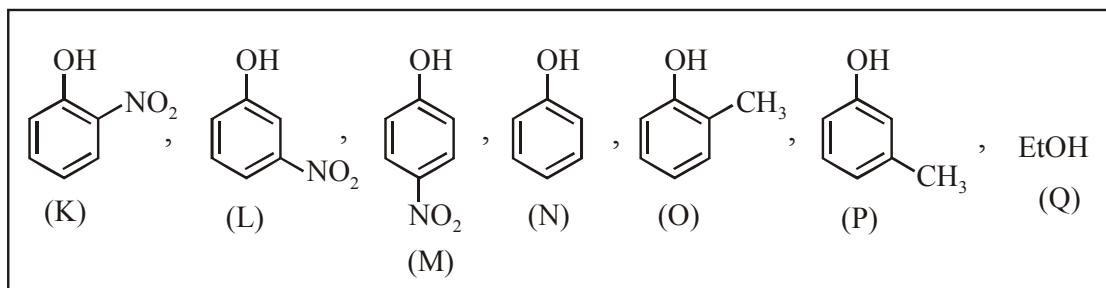
- (A) X (B) Y (C) Z (D) W
2. Consider the acidity of the carboxylic acids -
- PhCOOH (P) $\text{o-NO}_2\text{C}_6\text{H}_4\text{COOH}$ (Q) $\text{p-NO}_2\text{C}_6\text{H}_4\text{COOH}$ (R) $\text{m-NO}_2\text{C}_6\text{H}_4\text{COOH}$ (S)

which of the following order is correct

- (A) $\text{P} > \text{Q} > \text{R} > \text{S}$ (B) $\text{Q} > \text{S} > \text{R} > \text{P}$
 (C) $\text{Q} > \text{S} > \text{P} > \text{R}$ (D) $\text{Q} > \text{R} > \text{S} > \text{P}$
3. Find the correct acidic strength order -



- (A) $\text{K} > \text{L} > \text{M}$ (B) $\text{M} > \text{L} > \text{K}$ (C) $\text{K} > \text{M} > \text{L}$ (D) $\text{M} > \text{K} > \text{L}$
4. Which order of acidic strength is correct for given structures -

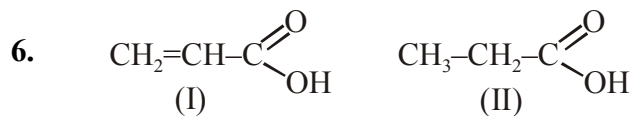


- (A) $\text{M} > \text{K} > \text{L} > \text{N}$ (B) $\text{N} > \text{P} > \text{O} > \text{Q}$
 (C) $\text{L} > \text{N} > \text{P} > \text{O}$ (D) $\text{M} > \text{K} > \text{P} > \text{Q}$
5. Consider the acidic strength of given structures

CF_3COOH , Cl_3COOH , CHCl_2COOH , CH_2ClCOOH , FCH_2COOH , BrCH_2COOH , $\text{NO}_2\text{CH}_2\text{COOH}$,
 $\text{NC-CH}_2\text{COOH}$, HCOOH , $\text{Cl-CH}_2\text{CH}_2\text{COOH}$, $\text{C}_6\text{H}_5\text{COOH}$, $\text{C}_6\text{H}_5\text{CH}_2\text{COOH}$, CH_3COOH ,
 $\text{CH}_3\text{CH}_2\text{COOH}$

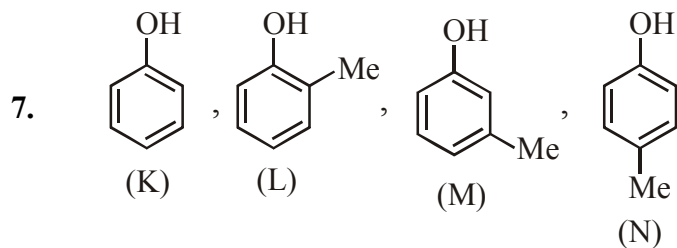
Find the correct order of acidic strength -

- (A) $\text{CCl}_3\text{COOH} > \text{CHCl}_2\text{COOH} > \text{NO}_2\text{CH}_2\text{COOH} > \text{NC-CH}_2\text{COOH}$
 (B) $\text{NO}_2\text{CH}_2\text{COOH} > \text{F-CH}_2\text{COOH} > \text{Cl-CH}_2\text{COOH} > \text{HCOOH}$
 (C) $\text{HCOOH} > \text{C}_6\text{H}_5\text{COOH} > \text{C}_6\text{H}_5\text{CH}_2\text{COOH} > \text{CH}_3\text{COOH}$
 (D) $\text{CF}_3\text{COOH} > \text{NO}_2\text{-CH}_2\text{COOH} > \text{NC-CH}_2\text{COOH} > \text{CHCl}_2\text{COOH}$



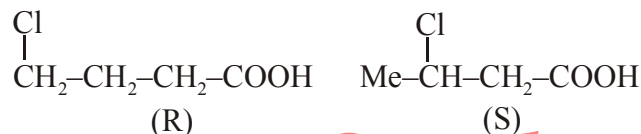
Find the correct statement(s) -

- (A) $\text{pK}_a = \text{I} > \text{II}$ (B) $\text{pK}_a = \text{II} > \text{I}$ (C) $\text{K}_a = \text{I} > \text{II}$ (D) $\text{K}_a = \text{II} > \text{I}$



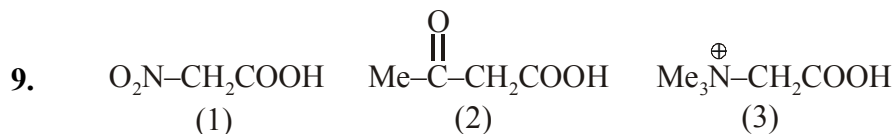
Acidic strength order -

- (A) $\text{K} > \text{M} > \text{N} > \text{L}$ (B) $\text{N} > \text{L} > \text{M} > \text{K}$ (C) $\text{K} > \text{N} > \text{L} > \text{M}$ (D) $\text{M} > \text{L} > \text{N} > \text{K}$



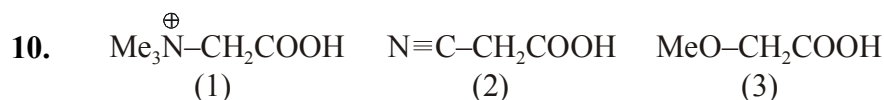
Correct acidic strength is -

- (A) $\text{Q} > \text{S} > \text{R} > \text{P}$ (B) $\text{S} > \text{Q} > \text{R} > \text{P}$ (C) $\text{P} > \text{Q} > \text{R} > \text{S}$ (D) $\text{Q} > \text{P} > \text{S} > \text{R}$



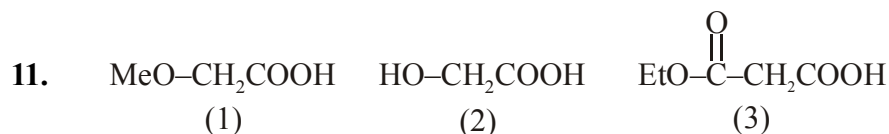
Correct acidic strength order is -

- (A) $1 > 3 > 2$ (B) $3 > 1 > 2$ (C) $1 > 2 > 3$ (D) $3 > 2 > 1$



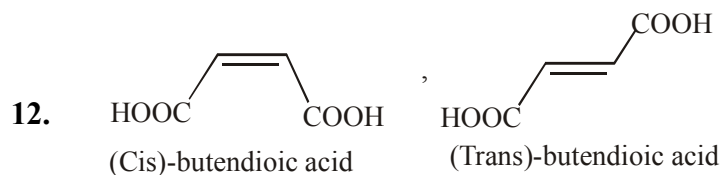
Correct acidic strength order is -

- (A) $1 > 2 > 3$ (B) $2 > 1 > 3$ (C) $2 > 3 > 1$ (D) $1 > 3 > 2$



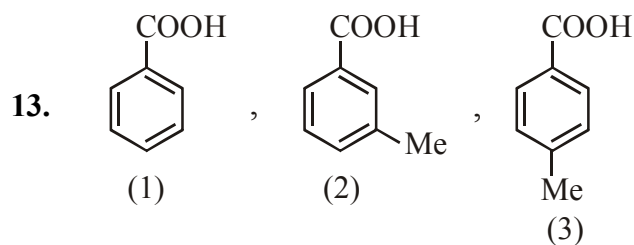
Correct acidic strength order is -

- (A) $1 > 2 > 3$ (B) $3 > 1 > 2$ (C) $3 > 2 > 1$ (D) $2 > 3 > 1$



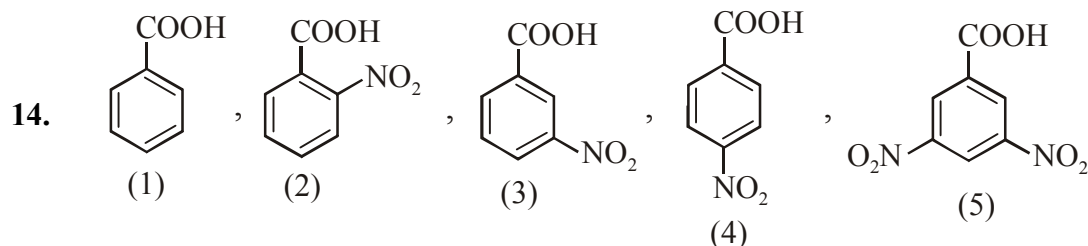
Find the correct statement(s)-

- (A) $\text{pK}_a^1 = \text{Cis} > \text{Trans}$ (B) $\text{pK}_a^1 = \text{Trans} > \text{cis}$
(C) $\text{pK}_a^2 = \text{cis} > \text{Trans}$ (D) $\text{pK}_a^2 = \text{Trans} > \text{cis}$



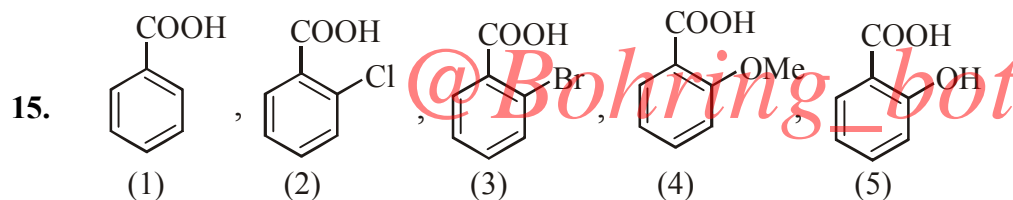
Correct acidic strength order is -

- (A) $1 > 2 > 3$ (B) $3 > 2 > 1$ (C) $2 > 3 > 1$ (D) $1 > 2 > 3$



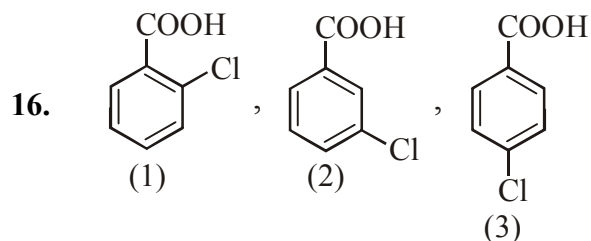
Correct acidic strength order is -

- (A) $2 > 5 > 4 > 3 > 1$ (B) $2 > 5 > 3 > 4 > 1$
(C) $5 > 2 > 4 > 3 > 1$ (D) $5 > 2 > 3 > 4 > 1$



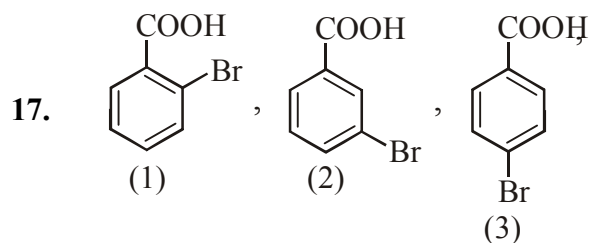
Arrange them in decreasing order of acidic strength -

- (A) $-\text{Br} > -\text{Cl} > -\text{OH} > -\text{OMe} > -\text{H}$ (B) $-\text{OH} > -\text{OMe} > -\text{Cl} > -\text{Br} > -\text{H}$
(C) $-\text{Br} > -\text{Cl} > -\text{OMe} > -\text{OH} > -\text{H}$ (D) $-\text{OMe} > -\text{OH} > -\text{Br} > -\text{Cl} > -\text{H}$



Correct order of acidic strength -

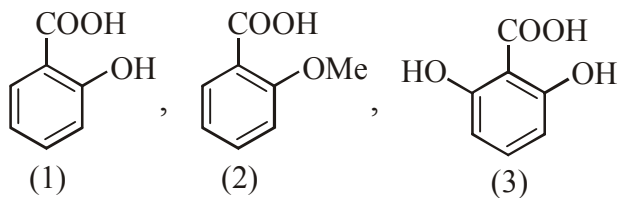
- (A) $1 > 2 > 3$ (B) $1 > 3 > 2$ (C) $3 > 1 > 2$ (D) $3 > 2 > 1$



Correct order of acidic strength -

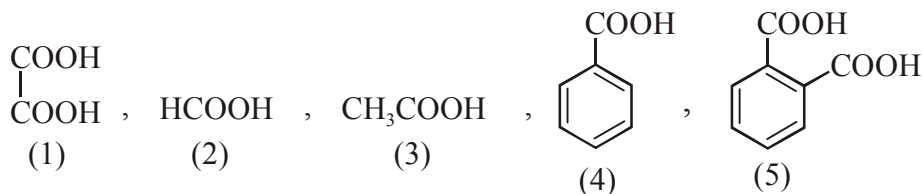
- (A) $1 > 2 > 3$ (B) $1 > 3 > 2$ (C) $3 > 1 > 2$ (D) $2 > 1 > 3$

18. Compare acidic strength



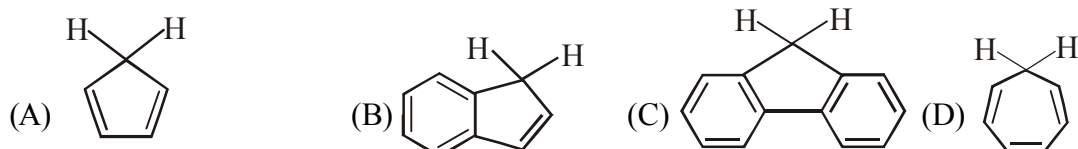
- (A) $3 > 1 > 2$ (B) $3 > 2 > 1$ (C) $1 > 3 > 2$ (D) $1 > 2 > 3$

19. Consider the given dicarboxylic acid -



- (A) $1 > 5 > 2 > 4 > 3$ (B) $1 > 5 > 4 > 2 > 3$
 (C) $5 > 1 > 4 > 2 > 3$ (D) $5 > 1 > 2 > 4 > 3$

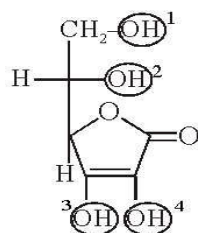
20. Which of the following is most acidic -



21. Which is the correct method for separating a mixture of benzoic acid, p-methyl aniline and phenol

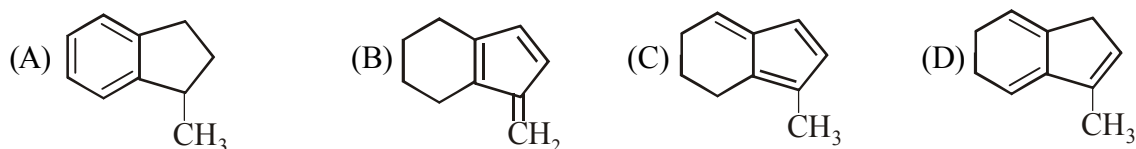
- (A) $\xrightarrow{\text{NaHCO}_3} \xrightarrow{\text{aq. NaOH}}$ (B) $\xrightarrow{\text{aq. HCl}} \xrightarrow{\text{aq. NaHCO}_3}$
 (C) $\xrightarrow{\text{aq. NaOH}} \xrightarrow{\text{aq. NaHCO}_3}$ (D) $\xrightarrow{\text{aq. NaOH}} \xrightarrow{\text{aq. HCl}}$

22. The structural formula for vitamin C is shown below out of the four hydroxyl group identified by circles which is most acidic

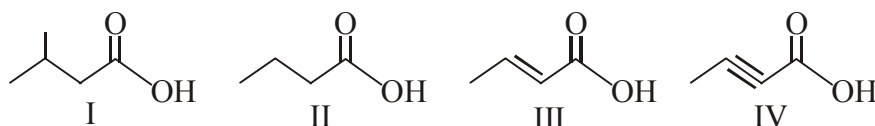


- (A) 1 (B) 2 (C) 3 (D) 4

23. The most acidic isomer hydrocarbon among the following is.-



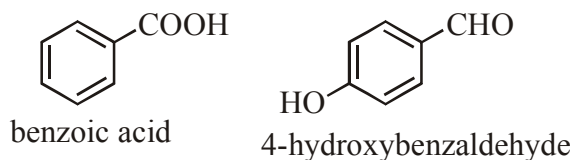
23. Among the following given compounds



The decreasing order of their acidity is :

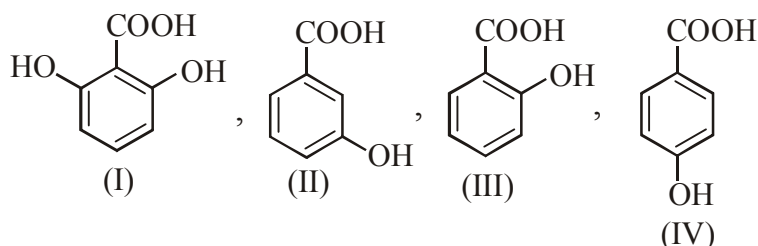
- (A) $I > II > III > IV$ (B) $II > I > IV > III$ (C) $III > IV > I > II$ (D) $IV > III > II > I$

24. A chemist wishes to separate benzoic acid from 4-hydroxybenzaldehyde. Which is the best method to achieve this separation ?



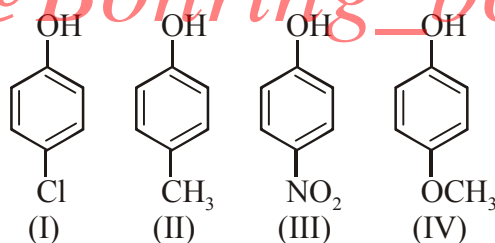
- (A) Partitioning the mixture between diethyl ether and water
 (B) Partitioning the mixture between diethyl ether and 1 M aqueous NaHCO_3
 (C) Partitioning the mixture between diethyl ether and 1 M aqueous NaOH
 (D) Partitioning the mixture between diethyl ether and 1 M aqueous HCl

25. The correct order of acidity for the following compounds is -



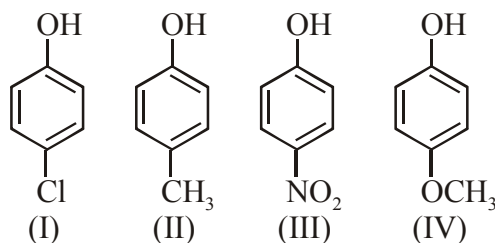
- (A) $\text{I} > \text{II} > \text{III} > \text{IV}$ (B) $\text{III} > \text{I} > \text{II} > \text{IV}$ (C) $\text{III} > \text{IV} > \text{II} > \text{I}$ (D) $\text{I} > \text{III} > \text{II} > \text{IV}$

26. Arrange the following compounds in order of decreasing acidity -



- (A) $\text{II} > \text{IV} > \text{I} > \text{III}$ (B) $\text{I} > \text{II} > \text{III} > \text{IV}$ (C) $\text{III} > \text{I} > \text{II} > \text{IV}$ (D) $\text{IV} > \text{III} > \text{I} > \text{II}$

27. The correct acidity order of the following is -

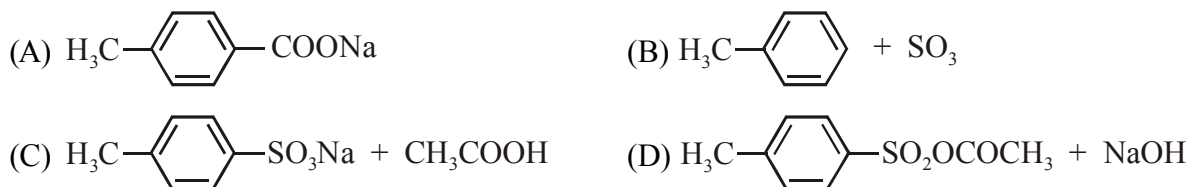


- (A) $\text{III} > \text{IV} > \text{II} > \text{I}$ (B) $\text{IV} > \text{III} > \text{I} > \text{II}$ (C) $\text{III} > \text{II} > \text{I} > \text{IV}$ (D) $\text{II} > \text{III} > \text{IV} > \text{I}$

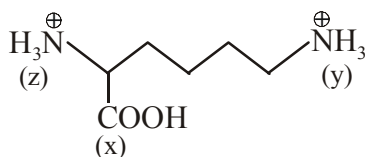
28. When benzene sulphonic acid and p nitrophenol are treated with NaHCO_3 the gases evolve respectively are

- (A) SO_2 , NO_2 (B) SO_2 , NO (C) SO_2 , CO_2 (D) CO_2 , CO_2

29. Which of the following is obtained when 4 methylbenzene sulphonic acid is hydrolysed with excess of sodium acetate -

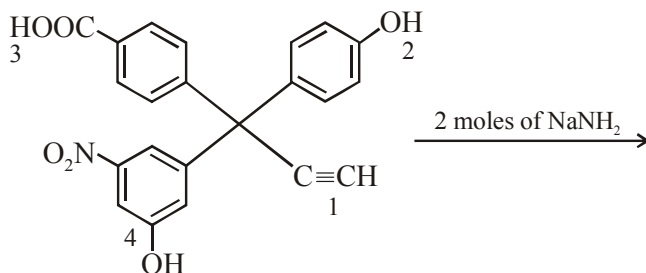


30. Arrange in order of increasing acidic strength -



- (A) $X > Z > Y$ (B) $Z < X < Y$ (C) $X > Y > Z$ (D) $Z > X > Y$

31.



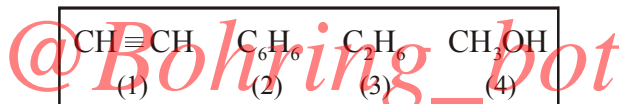
Proton will be removed from -

- (A) 1 (B) 2 (C) 3 (D) 4

32. which of the following has the most acidic hydrogen is

- (A) 3 Hexanone (B) 2,4 hexanedione (C) 2,5 hexandione (D) 2,3 hexanedione

33. Among the following find the correct acidic strength order



- (A) $1 > 4 > 2 > 3$ (B) $4 > 1 > 3 > 2$ (C) $4 > 1 > 2 > 3$ (D) $1 > 4 > 3 > 2$

Note : Answer the Q no 34, 35,36 by appropriately matching the information given in the three columns of the following table

Column-I	Column-II (pK_a)	Column-III
(1)	(P) 16	(K) Give effervescence with NaHCO_3
(2)	(Q) 15.3	(L) React with NaOH
(3)	(R) -65	(M) More acidic than water
(4) CH_3OH	(S) 4.2	(N) Gives H_2 gas on reaction with Na metal

34. Which option is correct

- (A) 1-R-K (B) 2-Q-K (C) 3-S-L (D) 4-P-K

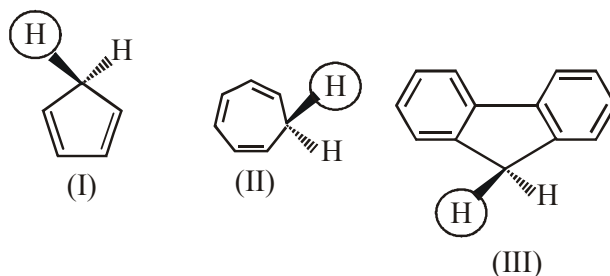
35. Which option is incorrect

- (A) 1-R-M (B) 2-P-N (C) 3-P-N (D) 4-Q-M

36. Which of the following option is correct

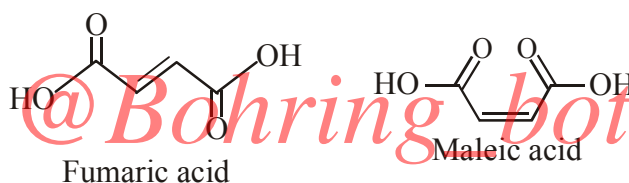
- (A) 1-Q-L (B) 2-R-M (C) 3-P-N (D) 4-Q-K

37. Arrange the following compounds in the increasing order of the acidity of circled hydrogens



- (A) III < II < I (B) II < I < III (C) I < II < III (D) II < III < I

38. The two compounds shown below, fumaric and maleic acid, are geometric (E/Z) isomers of each other. Interestingly, maleic acid is more acidic with a pK_a of 1.9 as compared to fumaric acid's pK_a of 3.0. This is because.



- (A) maleic acid has more strain and is thus more acidic
- (B) Maleic acid has a larger molecular dipole and dissolves better in water
- (C) the conjugate base of fumaric acid is better stabilized through the pi-orbitals
- (D) the conjugate base of maleic acid has an intramolecular hydrogen bond

ORGANIC CHEMISTRY

IIT JEE

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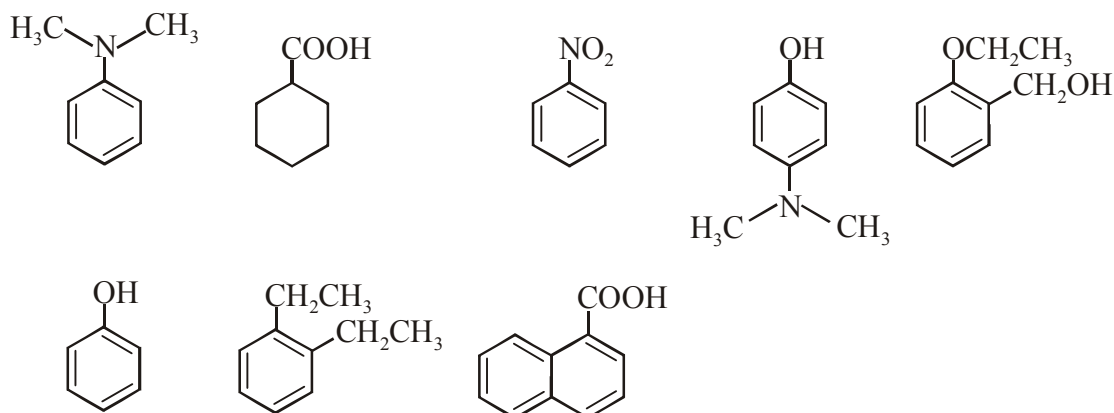
GOC-(D)

PREPARED BY: ASHISH SIR

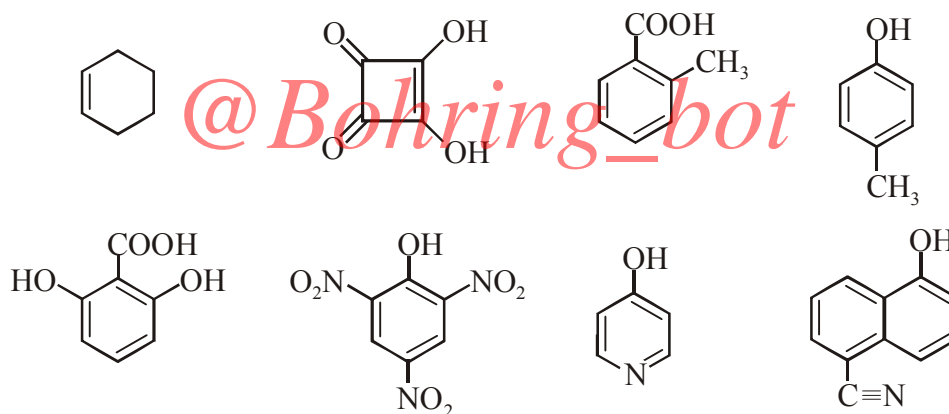
EXERCISE-E

Integer Type :

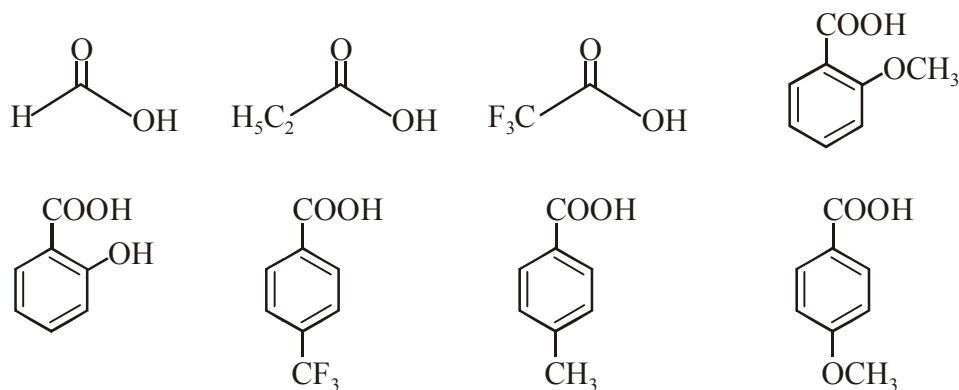
1. Among the following the total number of compounds soluble in aq NaOH is



2. How many compounds are soluble in aqueous NaOH ?



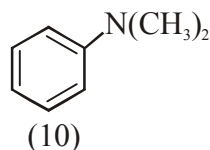
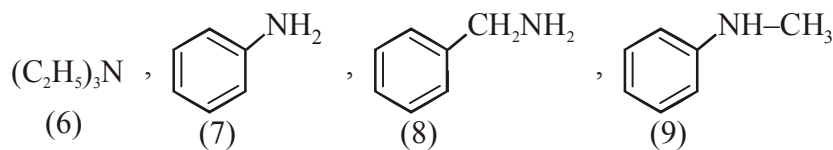
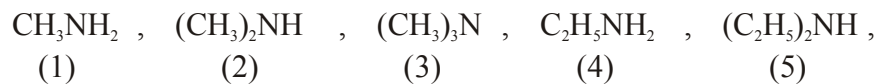
3. How many of the following acids are more acidic than benzoic acid ?



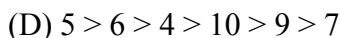
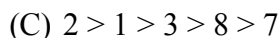
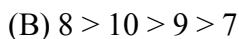
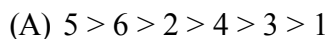
EXERCISE-F

One or More than Correct :

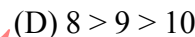
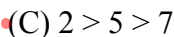
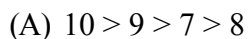
NOTE : Consider the basic strength of following given compounder and answer the Q No 1, 2, 3 & 4.



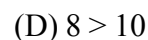
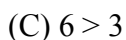
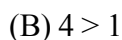
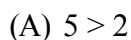
1. Find the correct basic strength order



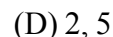
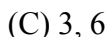
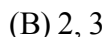
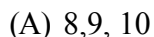
2. Find the incorrect order of basic strength



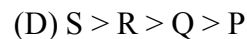
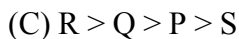
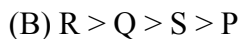
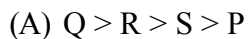
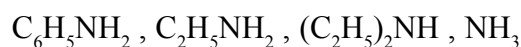
3. Find the correct order -



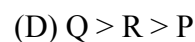
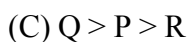
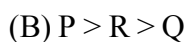
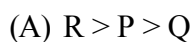
4. Which of the following are homologous -



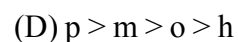
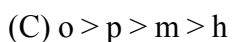
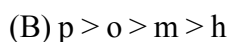
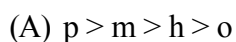
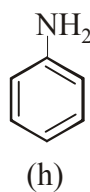
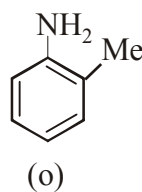
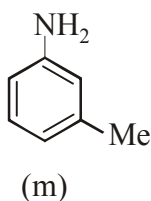
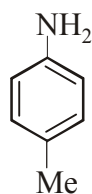
5. Arrange the following in decreasing order of basic strength order -

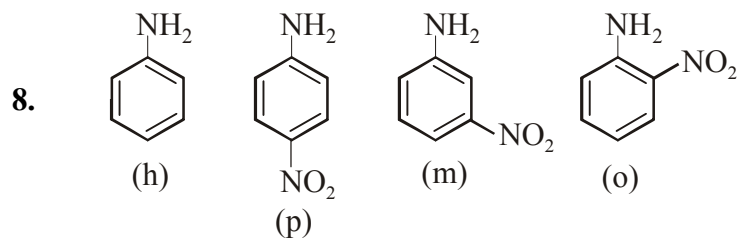


6. Arrange the following in increasing order of their basic strength -



7. Compare the basic strength of following -

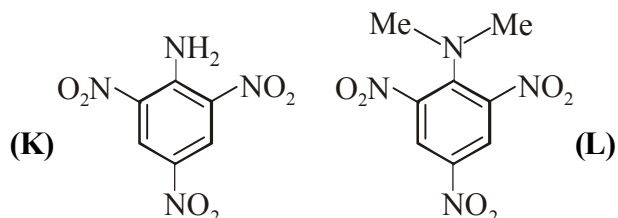




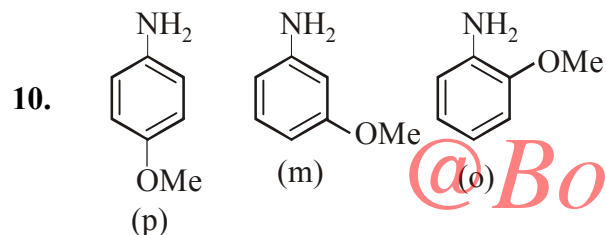
Find the correct order of basic strength -

- (A) $h > m > p > o$ (B) $h > p > o > m$ (C) $h > o > m > p$ (D) $h > o > p > m$

9. Find the correct statements regarding the given compounds -

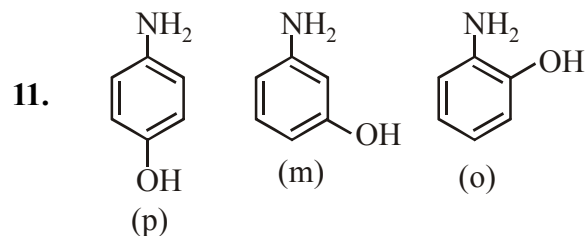


- (A) $pK_b = K > L$ (B) $pK_b = K < L$ (C) $K_a = K > L$ (D) $K_a = K < L$



Correct order of Basic strength -

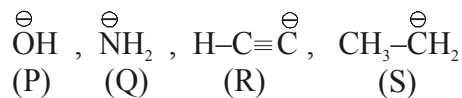
- (A) $p > o > m$ (B) $p > m > o$ (C) $o > p > m$ (D) $o > m > p$



Correct order of Basic strength -

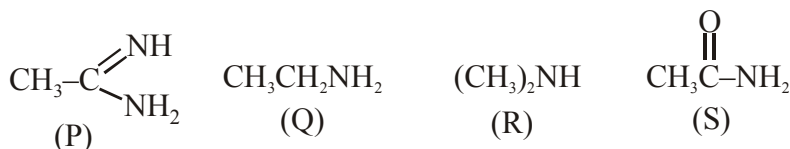
- (A) $p > o > m$ (B) $p > m > o$ (C) $o > p > m$ (D) $o > m > p$

12. What is the decreasing order of strength of the base -



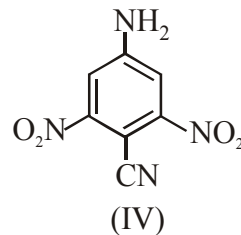
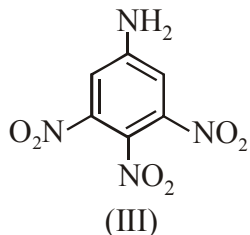
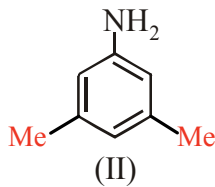
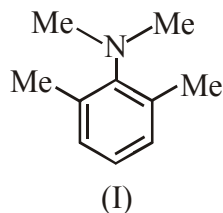
- (A) $S > Q > R > P$ (B) $R > S > Q > P$ (C) $P > Q > R > S$ (D) $Q > R > P > S$

13. The correct order of basicities of the following compound is -



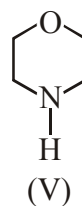
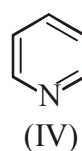
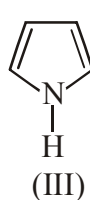
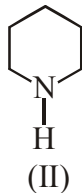
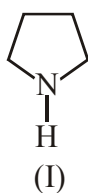
- (A) $P > R > Q > S$ (B) $P > Q > R > S$ (C) $R > P > Q > S$ (D) $R > Q > P > S$

14. The correct order of basic strength of following amines is :



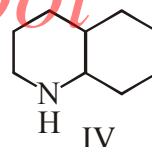
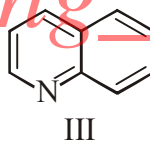
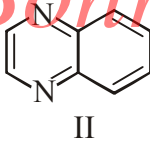
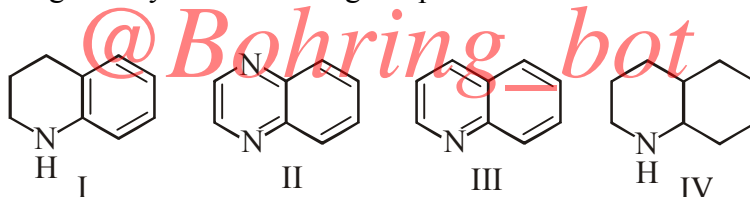
- (A) I > II > III > IV (B) I < II < III < IV (C) II < I > IV > III (D) I > II > IV > III

15. The reactivity of following compounds with proton will be in the order of -



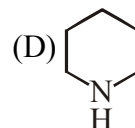
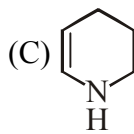
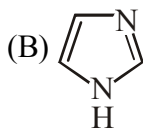
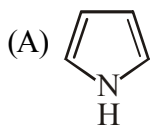
- (A) I > II > V > IV > III (B) II > I > V > IV > III
(C) I > II > IV > V > III (D) III > IV > V > II > I

16. The order of decreasing basicity of the following compounds is :

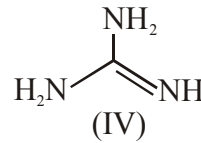
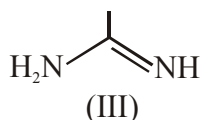
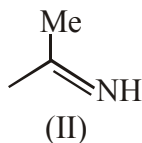
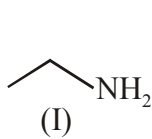


- (A) II > III > I > IV (B) III > II > I > IV
(C) IV > I > II > III (D) IV > I > III > II

17. The least basic amine among the following is -



18. The increasing order of basicity of the following compounds is :



- (A) I < II < III < IV (B) I < II < IV < III (C) II < I < III < IV (D) IV < III < II < I

ORGANIC CHEMISTRY

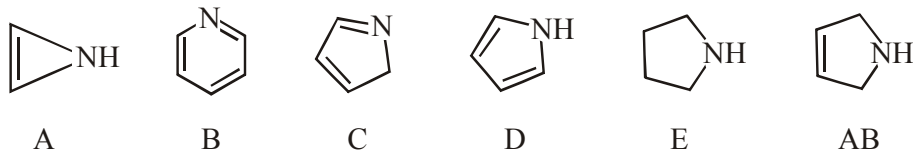
IIT JEE

@Bohring_bot

GOC-(E)

PREPARED BY: ASHISH SIR

EXERCISE - G



Answer question 1-5 by selecting a **SINGLE** compound from SET - 1 above :

- Select a **SINGLE** compound that has an sp^2 N and is non - aromatic as drawn.
- Select the **SINGLE** compound that when protonated on N has the lowest pK_a .
- Select a **SINGLE** compound that is anti - aromatic as drawn.
- Select a **SINGLE** compound that is aromatic as drawn and has an aromatic conjugate acid .
- Select a **SINGLE** compound that is non - aromatic as drawn but has an aromatic tautomer .
- The electron - donating nature of the ethyl group accounts for the difference in base strength of ethylamine and ammonia.

Which one of the following correctly shows the effect of the ethyl group ?

effect on strength of $C_2H_5NH_2$ as a base relative to NH_3

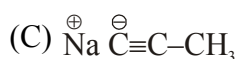
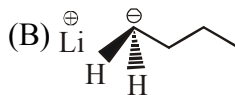
effect on strength of $C_2H_5NH_3^+$ as an acid relative to NH_4^+

- | | |
|--------------|----------|
| (A) increase | increase |
| (B) increase | decrease |
| (C) decrease | decrease |
| (D) decrease | increase |

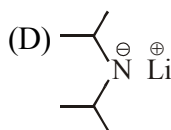
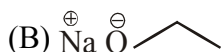
7. Which sequence gives the correct order of basic strength (strongest first) ?

	most basic $\longrightarrow$ least basic		
(A).	$C_2H_5NH_2$	NH_3	$(C_2H_5)_2NH$
(B).	$C_2H_5NH_2$	$(C_2H_5)_2NH$	$C_6H_5NH_2$
(C).	$(C_2H_5)_2NH$	NH_3	$C_6H_5NH_2$
(D).	$C_6H_5NH_2$	NH_3	$C_2H_5NH_2$

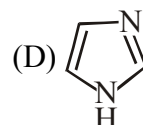
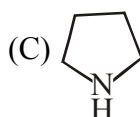
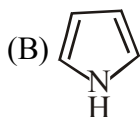
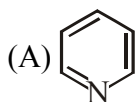
8. Choose the weakest base :-



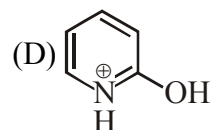
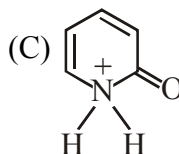
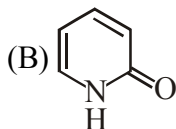
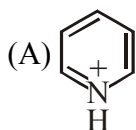
9. Choose the species that could not be used as the base in the following reaction :-



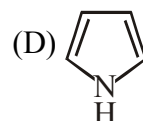
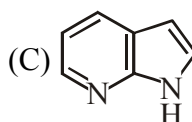
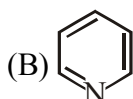
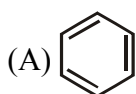
10. Which compound is most basic ?



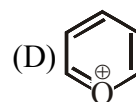
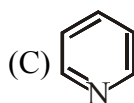
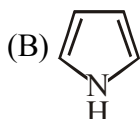
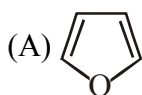
11. Which of (a) - (d) is not aromatic ?



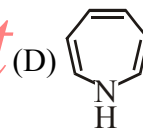
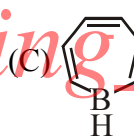
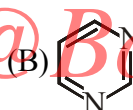
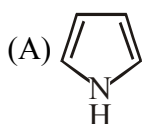
12. Which is least reactive towards an electrophile ?



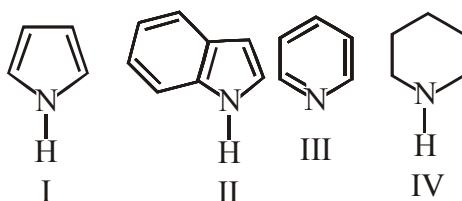
13. Which is most stabilized by electron delocalization (resonance) ?



14. Which of (a) - (d) is not aromatic ?

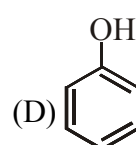
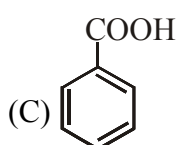
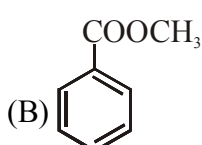
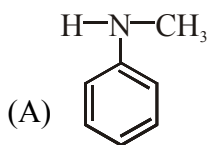


15. The correct order of the basicity of the following compounds is :-

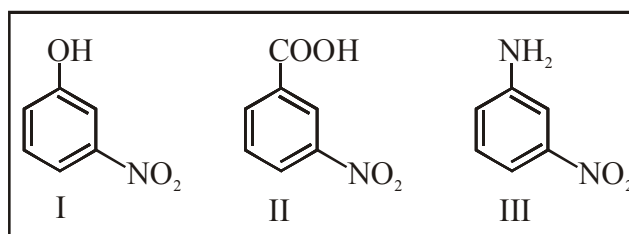


(A) IV > III > II > I (B) III > IV > I > II (C) IV > III > I > II (D) III > IV > II > I

16. Which of the following compounds form salt with NaHCO_3 :-



17. The order in which the reagent must be used to separate the compound I - III is -



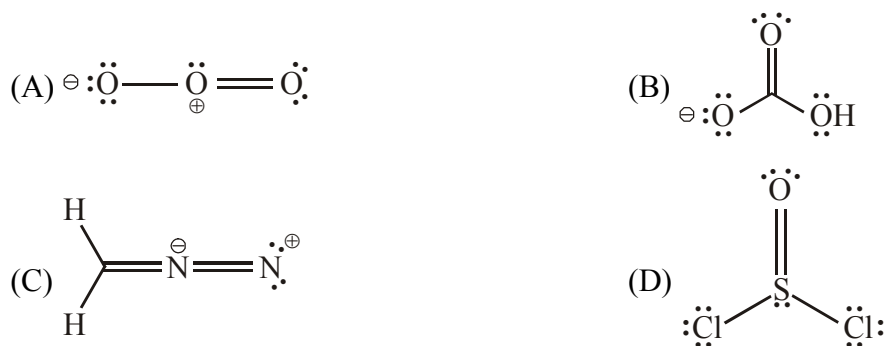
(A) NaOH , NaHCO_3 , HCl

(B) HCl , NaOH , NaHCO_3

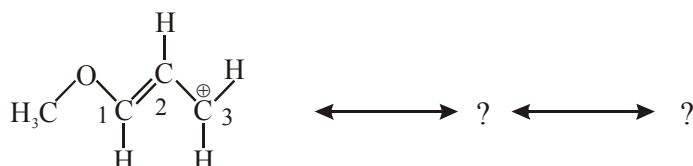
(C) NaHCO_3 , NaOH , HCl

(D) NaOH , HCl , NaHCO_3

18. Which one of the following Lewis structure is not correct :-



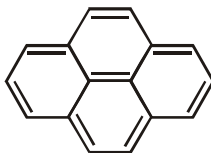
19. Draw the two most important additional contributing structures for the carbocation shown below :-



Use your structures as a guide to determine which one of the following statements about this carbocation is incorrect?

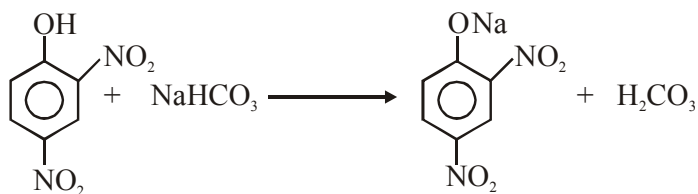
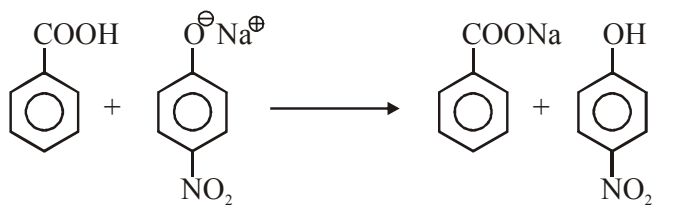
- (A) There is a significant build up of positive charge on the oxygen atom
- (B) The C-O bond between the methoxy group and the allyl group has partial double bond character
- (C) Except for the methyl hydrogen atoms, you would expect this cation to be planar.
- (D) There is a significant build up to positive charge on C2 atom of the allyl group.

20. The value of 'n' for the following molecule according to Huckel's rule is :-

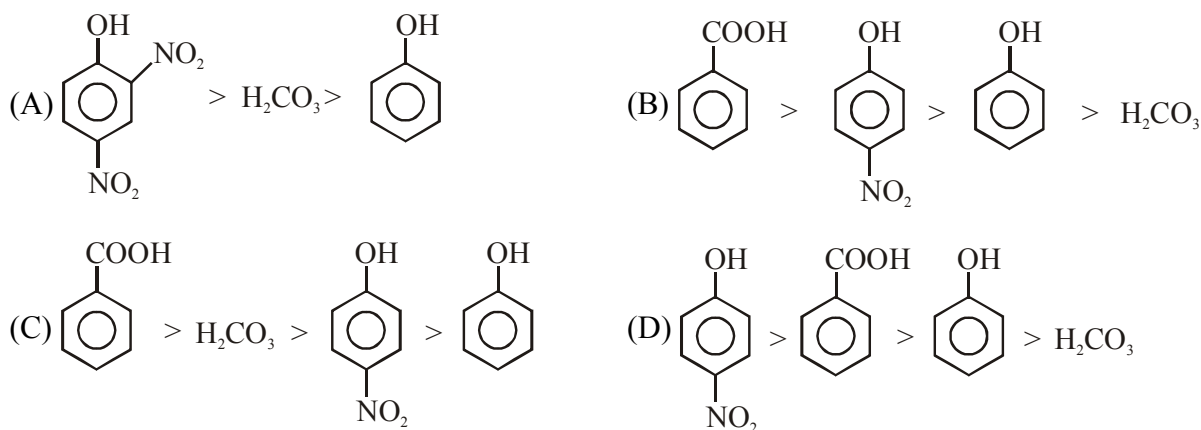


- (A) 16 (B) 4 (C) 3 (D) 14

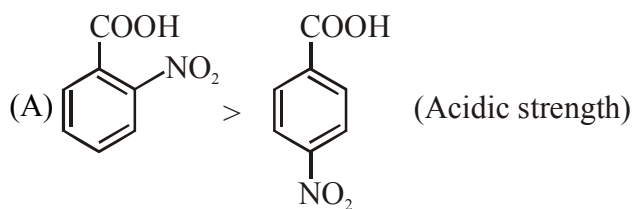
21. Observe the following reaction :-



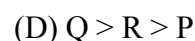
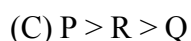
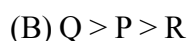
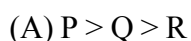
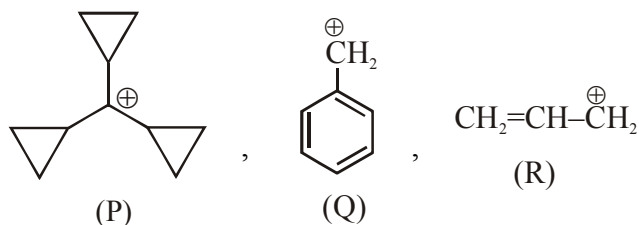
Which of the following is / are incorrect order of acid strength :



22. Which of the following statements are correct ?



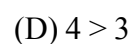
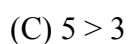
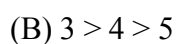
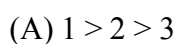
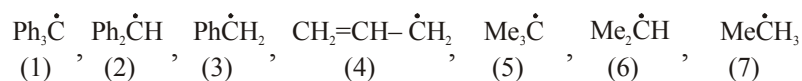
23. Which of the following order of cation stability is correct ?



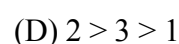
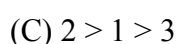
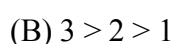
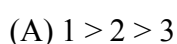
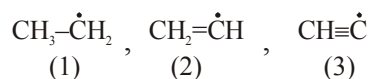
24. A solution of (+)-1-Chloro-1-phenylethane in toluene racemise slowly in presence of small amount of SbCl_5 due to formation of :-



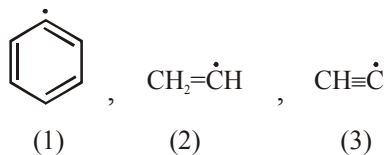
25. Compare the stability of the following free radicals :-



26. Compare the stability of following free radicals :-

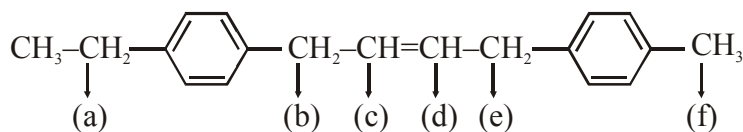


27. Compare the stability :-



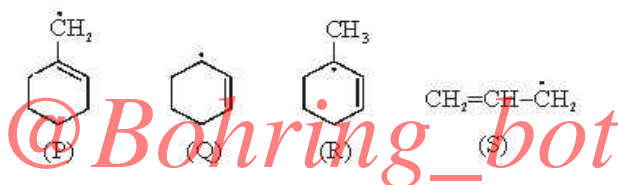
- (A) $1 > 2 > 3$ (B) $2 > 1 > 3$ (C) $3 > 2 > 1$ (D) $2 > 3 > 1$

28. Compare the bond energy of indicated C-H Bond :-



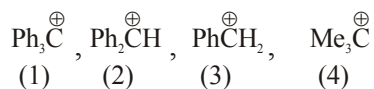
- (A) $e > b > a > f$ (B) $c = d$
 (C) $c = d > f > a > b > e$ (D) $e > b > a > f > c = d$

29. Compare the stability of the following free Radical :-



- (A) $P > Q > R > S$ (B) $R > P > Q > S$
 (C) $R > Q > P > S$ (D) $R > Q > S > P$

30. Compare the Stability of following cations :-



- (A) $1 > 2 > 3 > 4$ (B) $1 > 2 > 4 > 3$
 (C) $1 > 4 > 2 > 3$ (D) $3 > 4 > 2 > 1$

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GOC-(F)

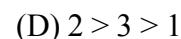
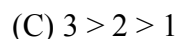
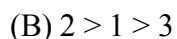
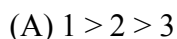
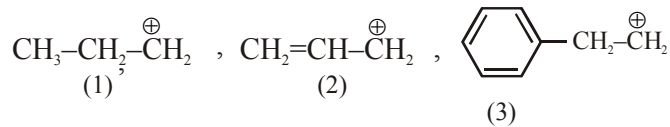
PREPARED BY: ASHISH SIR

EXERCISE - H

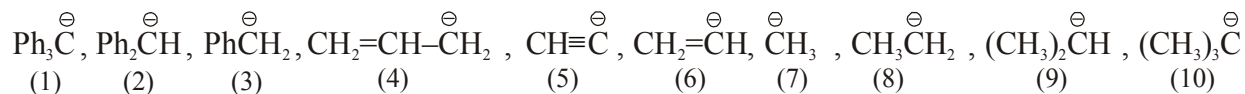
1. Identify the correct order of stability of given :-



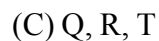
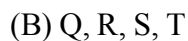
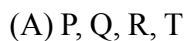
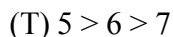
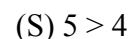
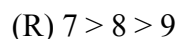
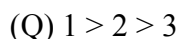
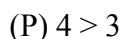
2. Compare the stability of given cations :-



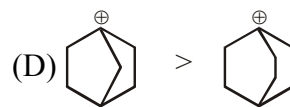
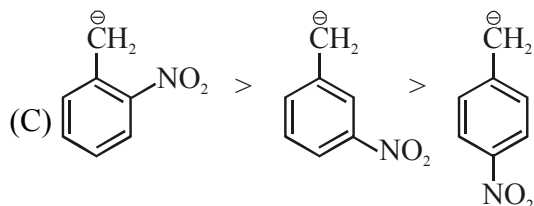
3. Consider the following carbanions :-



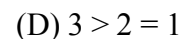
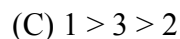
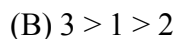
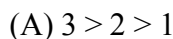
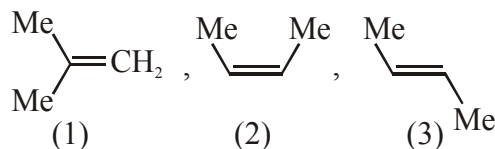
Identify how many stability order is (are) correct :-



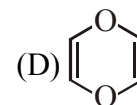
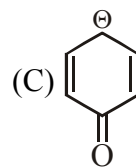
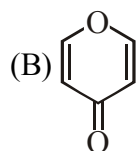
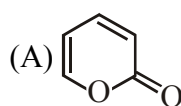
4. Identify the correct stability order :-



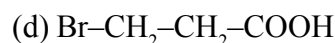
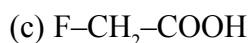
5. Identify the correct stability order :-



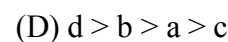
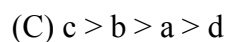
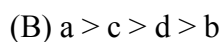
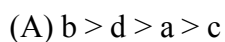
6. Which of the following compounds is not aromatic :-



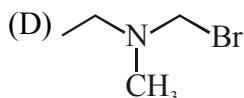
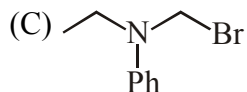
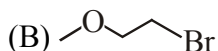
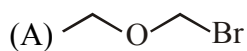
7. Arrange in decreasing K_a :-



Correct answer is :



8. Ease of ionization to produce carbocation and bromide ion under the treatment of $\text{Ag}^{\oplus}$ will be maximum in which of the following compounds ?

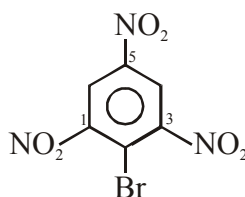


9. (I) (II) (III)

Which of the following choice is the correct order of resonance energy of these molecules :-

- (A) $\text{I} > \text{II} > \text{III}$ (B) $\text{II} > \text{I} > \text{III}$ (C) $\text{III} > \text{II} > \text{I}$ (D) $\text{III} > \text{I} > \text{II}$

10. Which of the following statements would be true about this compound :-



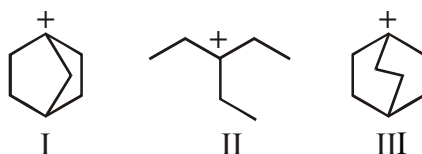
- (A) All three C – N bonds are of same length
 (B) C1 – N and C3 – N bonds are of same length but shorter than C5 – N bond
 (C) C1 – N and C3 – N bonds are of same length but longer than C5 – N bond
 (D) C1 – N and C3 – N bonds are of different length but both are longer than C5 – N bond.

11. I II III IV

Which of the following statement/s is / are correct regarding above compounds ?

- (A) I & II are aromatic and have equal basic strength
 (B) I is aromatic, II is antiaromatic but II is stronger base than I
 (C) Basic strength of the above compounds is $\text{I} < \text{II} < \text{III} < \text{IV}$
 (D) Conjugate acid of IV is more stabilized than conjugate acid of II

12. The **CORRECT** order of stability of the following carbonium ions is :-



- (A) $\text{II} > \text{I} > \text{III}$ (B) $\text{III} > \text{II} > \text{I}$
 (C) $\text{I} > \text{III} > \text{II}$ (D) $\text{II} > \text{III} > \text{I}$

13. What is the increasing order of stability of following carbocations (give least stable first) ?

- (I) Tropylium cation (II) $\text{CH}_2=\text{CH}-\text{CH}_2^+$ (III) $(\text{C}_6\text{H}_5)_2\text{C}^+$ (IV) CH_3^+
 (A) $\text{III} < \text{I} < \text{II} < \text{IV}$ (B) $\text{IV} < \text{II} < \text{III} < \text{I}$
 (C) $\text{I} < \text{III} < \text{II} < \text{IV}$ (D) $\text{IV} < \text{III} < \text{II} < \text{I}$

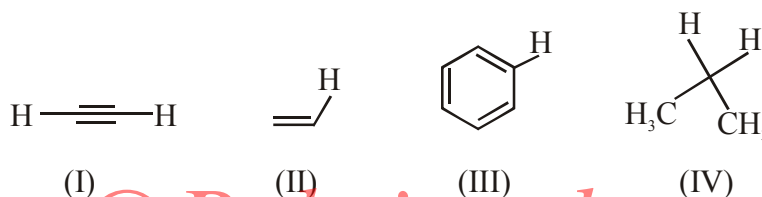
14. Which one of the following is not aromatic ?



15. The least basic amine among the following is :-

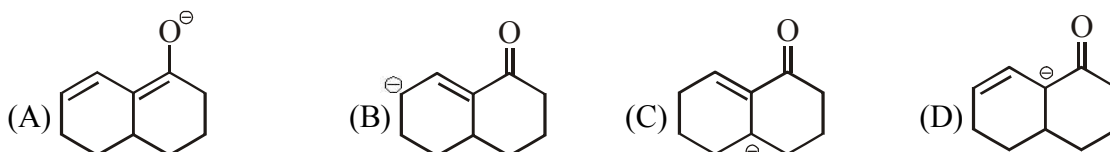


16. Arrange the following molecules in the increasing order of C-H bond dissociation energy (kJ mol^{-1}). :-



- (A) $\text{IV} < \text{II} < \text{III} < \text{I}$ (B) $\text{I} < \text{III} < \text{IV} < \text{II}$
 (C) $\text{III} < \text{II} < \text{IV} < \text{I}$ (D) $\text{II} < \text{IV} < \text{I} < \text{III}$

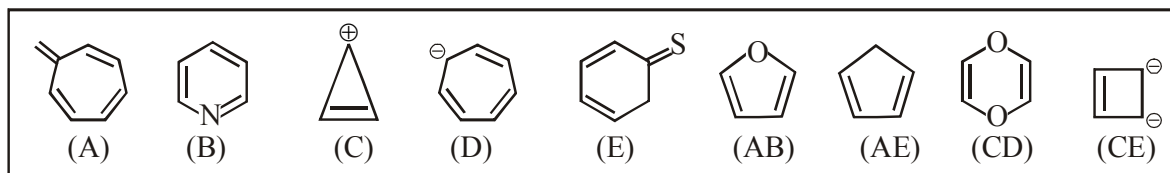
17. The structure which is not a resonance structure of others in the following is :-



18. Among the following the compound which would not be considered aromatic in its behavior is :-



19. Answer the Following questions :-



- An ionic molecule that is aromatic as drawn with 6 p-electrons.
- A non-aromatic, uncharged heterocycle.
- A hydrocarbon molecule that is non-aromatic as drawn, but reacts readily with H^+ to give an aromatic carbocation.
- Not aromatic as drawn, but has an aromatic tautomer.
- Of the aromatic heterocycles, which would be the strongest base ?

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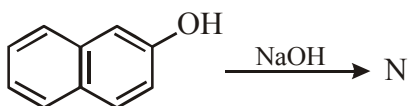
PREPARED BY: ASHISH SIR

EXERCISE - I

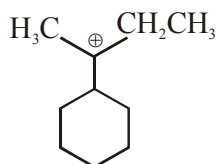
1. Match the column :-

Column - I(Acids)	Column - II(K_a)
1. Benzoic acid	(P) 4.2×10^{-5}
2. p-nitrobenzoic acid	(Q) 3.3×10^{-5}
3. p-chlorobenzoic acid	(R) 6.4×10^{-5}
4. p-methylbenzoic acid	(S) 36.2×10^{-5}
5. p-methoxybenzoic acid	(T) 10.2×10^{-5}

2. The number of resonance structures for N is :-



3. The total number of contributing structures showing hyperconjugation (involving C-H bonds) for the following carbocation is

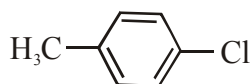


4. Among the following, the number of aromatic compound (s) is



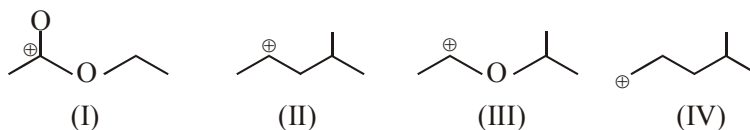
One or More Correct :

5. The IUPAC name (s) of the following compound is (are) :-



- (A) 1-chloro-4-methylbenzene (B) 4-chlorotoluene
(C) 4-methylchlorobenzene (D) 1-methyl-4-chlorobenzene
6. The compound that does NOT liberate CO_2 , On treatment with aqueous sodium bicarbonate solution, is
(A) Benzoic acid (B) Benzenesulphonic acid
(C) Salicylic acid (D) Carboic acid (Phenol)
7. The hyperconjugative stabilities of tert-butyl cation and 2-butene, respectively, are due to
(A) $\sigma \rightarrow p$ (empty) and $\sigma \rightarrow \pi^*$ electron delocalisations.
(B) $\sigma \rightarrow \sigma$ and $\sigma \rightarrow \pi$ electron delocalisations.
(C) $\sigma \rightarrow p$ (filled) and $\sigma \rightarrow \pi$ electron delocalisation.
(D) p (filled) $\rightarrow \sigma$ and $\sigma \rightarrow \pi$ electron delocalisation.

8. The correct stability order for the following species is :-



(A) (II) > (IV) > (I) > (III)

(B) (I) > (II) > (III) > (IV)

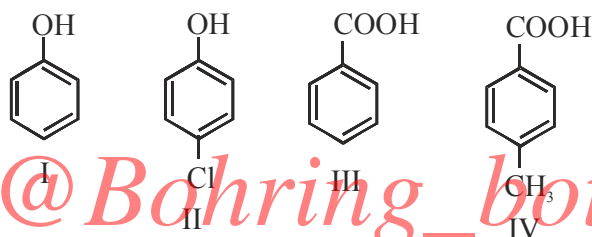
(C) (II) > (I) > (IV) > (III)

(D) (I) > (III) > (II) > (IV)

9. Among the following the least stable resonance structure is :-



10. The correct acidity order of the following is :-



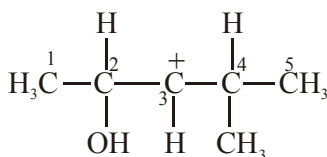
(A) (III) > (IV) > (II) > (I)

(B) (IV) > (III) > (I) > (II)

(C) (III) > (II) > (I) > (IV)

(D) (II) > (III) > (IV) > (I)

11. In the following carbocation, H/CH₃ that is most likely to migrate to the positively charged carbon is :-



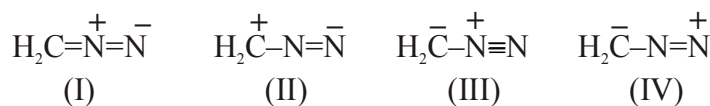
(A) CH₃ at C-4

(B) H at C-4

(C) CH₃ at C-2

(D) H at C-2

12. The correct stability order of the following resonance structures is



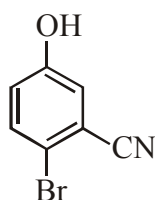
(A) I > II > IV > III

(B) I > III > II > IV

(C) II > I > III > IV

(D) III > I > IV > II

13. The IUPAC name of the following compound is



(A) 4-Bromo-3-cyanophenol

(B) 2-Bromo-5-hydroxybenzonitrile

(C) 2-Cyano-4-hydroxybromobenzene

(D) 6-Bromo-3-hydroxybenzonitrile

ANSWER KEY

EXERCISE-A

1. (C)	2. (D)	3. (A,C)	4. (A,C,D)	5. (A,B,D)
6. (A,C)	7. (A,C)	8. (D)	9. (C)	10. (B)
11. (C)	12. (B)	13. (C)	14. (D)	15. (B)
16. (C)	17. (B)	18. (A)	19. (A)	20. (C)
21. (C)	22. (D)	23. (C)	24. (D)	25. (A)
26. (C)	27. (B)	28. (D)	29. (C)	

EXERCISE-B

1. (B,D)	2. (A)	3. (C)	4. (A)	5. (A,B,C,D)
6. (D)	7. (D)	8. (D)	9. (B)	10. (A,B,C)
11. (A)	12. (C)	13. (A,C)	14. (A,C,D)	15. (B)
16. (C)	17. (A)	18. (B)	19. (D)	20. (B)
21. (A)	22. (D)	23. (A)	24. (B)	

EXERCISE-C

1. (10)	2. (6)	3. (3)	4. (7)	5. (84)
6. (3)	7. (3)	8. (9)		

EXERCISE-D

1. (A)	2. (D)	3. (A)	4. (A,B,C,D)	5. (A,B,C)
6. (B,C)	7. (A)	8. (A)	9. (A)	10. (A)
11. (B)	12. (B,C)	13. (A)	14. (A)	15. (A)
16. (A)	17. (A)	18. (A)	19. (A)	20. (A)
21. (A,B)	22. (C)	23. (C)	23. (D)	24. (B)
25. (D)	26. (C)	27. DEL	28. (D)	29. (C)
30. (A)	31. (C,D)	32. (B)	33. (C)	34. (A)
35. (B)	36. (C)	37. (D)	38. (A & D)	

EXERCISE-E

1. (4)	2. (7)	3. (5)		
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EXERCISE-F

1. (B,C,D)	2. (A,C,D)	3. (A,B,C,D)	4. (C,D)	5. (B)
6. (A)	7. (A)	8. (A)	9. (A,C)	10. (A)
11. (A)	12. (A)	13. (A)	14. (A)	15. (A)
16. (D)	17. (A)	18. (C)		

EXERCISE - G

1. (C)	2. (D)	3. (A)	4. (B)	5. (C)
6. (B)	7. (C)	8. (A)	9. (B)	10. (C)
11. (C)	12. (B)	13. (C)	14. (D)	15. (C)
16. (C)	17. (C)	18. (C)	19. (D)	20. (C)
21. (B,D)	22. (A,B,D)	23. (A)	24. (D)	25. (A,D)
26. (A)	27. (B)	28. (B,C)	29. (C)	30. (B)

EXERCISE - H

-
- | | | | | |
|--------------|---------|---------|---|---------|
| 1. (A,B,C,D) | 2. (B) | 3. (B) | 4. (B,C) | 5. (C) |
| 6. (D) | 7. (C) | 8. (D) | 9. (C) | 10. (C) |
| 11. (C,D) | 12. (D) | 13. (B) | 14. (C) | 15. (A) |
| 16. (A) | 17. (C) | 18. (B) | 19. (i)-CE ; (ii)-CD ; (iii)-A ; (iv)-E ; (v)-B | |

EXERCISE - I

-
- | | | |
|--|----------|---------|
| 1. (1)→R ; (2)→S ; (3)→T ; (4)→P ; (5)→Q | 2. (9) | 3. (6) |
| 4. (5) | 5. (A,B) | 6. (D) |
| 9. (A) | 10. (A) | 11. (D) |
| | 12. (B) | 13. (B) |

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21. The compound that does **NOT** liberate CO_2 , on treatment with aqueous sodium bicarbonate solution, is

- (A) Benzoic acid (B) Benzenesulphonic acid
(C) Salicylic acid (D) Carboic acid (Phenol)

33. The hyperconjugative stabilities of tert-butyl cation and 2-butene, respectively, are due to

(A) $\sigma \rightarrow p$ (empty) and $\sigma \rightarrow \pi^*$ electron delocalisations.

(B) $\sigma \rightarrow \sigma^*$ and $\sigma \rightarrow \pi$ electron delocalisations.

(C) $\sigma \rightarrow p$ (filled) and $\sigma \rightarrow \pi$ electron delocalisations.

(D) p (filled) $\rightarrow \sigma^*$ and $\sigma \rightarrow \pi^*$ electron delocalisations.

21. The correct combination of names for isomeric alcohols with molecular formula $\text{C}_4\text{H}_{10}\text{O}$ is/are

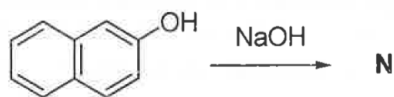
(A) *tert*-butanol and 2-methylpropan-2-ol

(B) *tert*-butanol and 1, 1-dimethylethan-1-ol

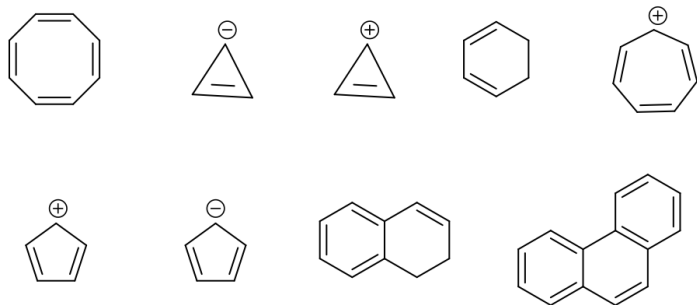
(C) *n*-butanol and butan-1-ol

(D) isobutyl alcohol and 2-methylpropan-1-ol

Q.22 The number of resonance structures for N is

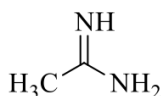


Q.30 Among the following, the number of aromatic compound(s) is

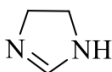


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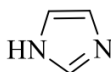
Q.25 The order of basicity among the following compounds is



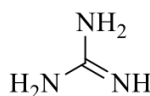
I



II



III



IV

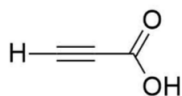
[A] II > I > IV > III

[B] IV > II > III > I

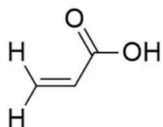
[C] IV > I > II > III

[D] I > IV > III > II

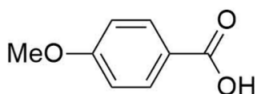
Q.4 The correct order of acid strength of the following carboxylic acids is



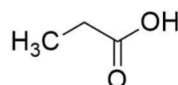
I



II



III



IV

(A) III > II > I > IV

(B) I > II > III > IV

(C) I > III > II > IV

(D) II > I > IV > III

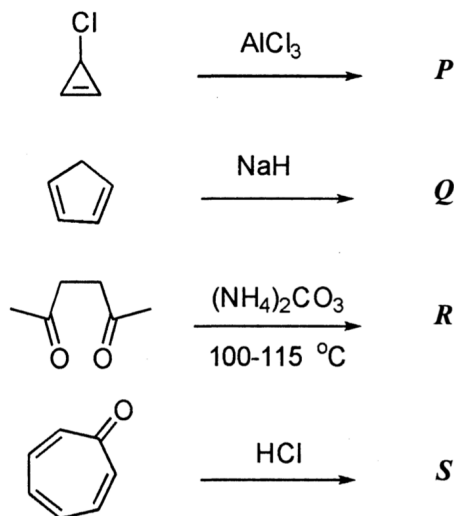
32. Among *P*, *Q*, *R* and *S*, the aromatic compound(s) is/are

(A) *P*

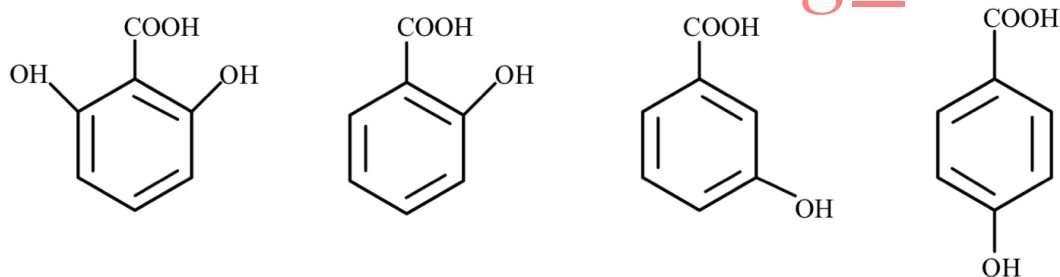
(B) *Q*

(C) *R*

(D) *S*

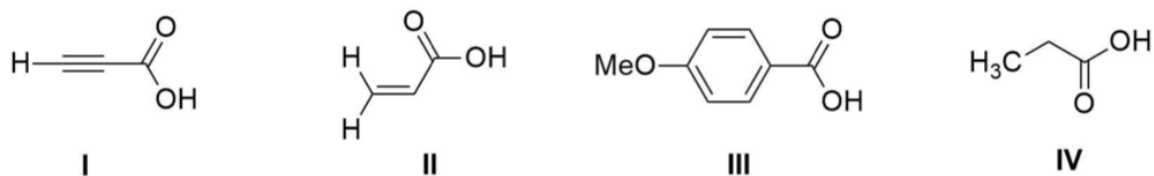


The correct order of acidic strength for the following compounds



(A) 1>2>3>4. (B) 3>1>2>4. (C) 3>4>2>1 (D) 1>3>4>2

Q.4 The correct order of acid strength of the following carboxylic acids is



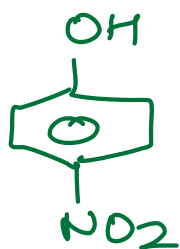
(A) III > II > I > IV

(B) I > II > III > IV

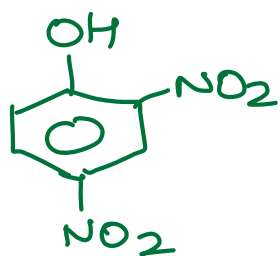
(C) I > III > II > IV

(D) II > I > IV > III

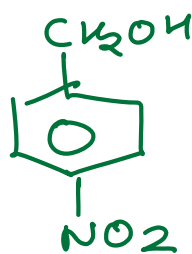
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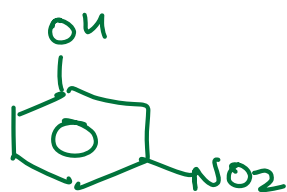
(I)



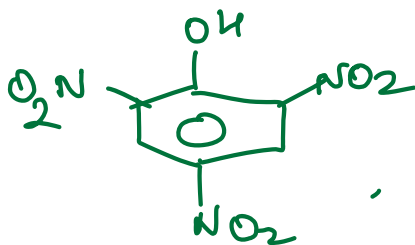
(II)



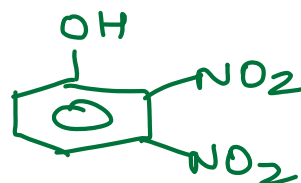
(III)



(IV)



(V)



(VI)

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35. the ^{correct} order of K_a is -

A) I > II > III

B) II > III > IV

C) IV > VI > V

☒ D) V > VI > IV

36) which of the following gives effervescence with NaHCO_3 ?

A) IV

B) III

☒ C) V

D) All

ALKANES HANDOUT

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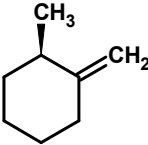
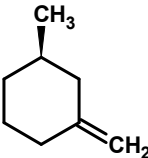
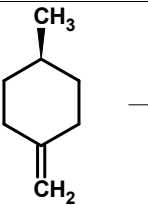
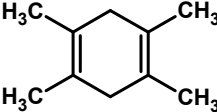
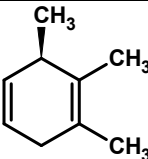
SYNOPSIS

REAGENT	NAME REACTION	TEST	CONCEPTUAL POINTS
i) H_2 / Ni	i) Sabatier - Sandaren's reaction		i) $HBO \rightarrow BH_3$ attach to side having more hydrogen & less steric site.
ii) Raney Nickel Ni-Al alloy (NaOH)	ii) Hydroboration acidolysis		ii) Don't use clemenson reduction for compound having acid sensitive group
iii) $H_2 / Pd - C$	iii) clemenson reduction		iii) don't use Wolf Kishner reduction compound having base sensitive group
iv) $H_2 / Pd(or)Pt$	iv) Wolf Kishner reduction		iv) Odd c-alkane is not prepared in good yield by Wurtz reaction.
v) H_2 / PtO_2 (Adom's catalyst)	v) Kolbe electrolysis		v) Methane = Marsh gas $Co + 3H_2 \xrightarrow{Ni/300^\circ C} CH_4 + H_2O$
vi) $N_2H_4 + H_2O_2$ (Di-imide)	vi) Soda-lime decarboxylation		vi) Alkanes $\rightarrow$ $C_1 - C_4 \Rightarrow$ gases $C_5 - C_{17} \Rightarrow$ liquid $\geq C_{18} \Rightarrow$ solids (waxy)
vii) $[(ph_3P)_3 Rh]^+ Cl^-$ (Wilkinson's catalyst)	vii) Wurtz reaction		vii) $\frac{1}{\text{branching}} \propto BP(\text{alkanes}) \propto \text{no. of carbon } \Delta H_{\text{combustion}}$
viii) 1) B_2H_6 / THF 2) CH_3COOH	viii) Fitting reaction		viii) Reactivity in F.R. substitution $F_2 > Cl_2 > Br_2 > I_2$ & Selectivity $\rightarrow Br_2 > Cl_2 > I_2$ & Reactivity of (H) $\rightarrow 3^\circ > 2^\circ > 1^\circ$
xi) Red P + HI	ix) Wurtz-Fitting reaction		
x) $Zn - Hg / HCl$	x) Correy-house synthesis		
xi) N_2H_4 / KOH , glycol	xi) Allylic substitution		
xii) Zn/HCl	xii) Aromatisation		
xiii) $LiAlH_4$	xiii) oxidation (alkanes)		
xiv) $NaBH_4$	xiv) Nitration		
xv) $NaOH + Cao \Delta$	xv) Sulphonation		
xvi) Na/dry ether	xvi) Isomerisation		
xvii) Grignard reagent	xvii) Free radical substitution		
xviii) R_2CuLi			
xix) $Br_2 / h\nu$			
xx) $NBS / h\nu$			
xxi) SO_2Cl_2			
xxii) conc. H_2SO_4 , Δ			

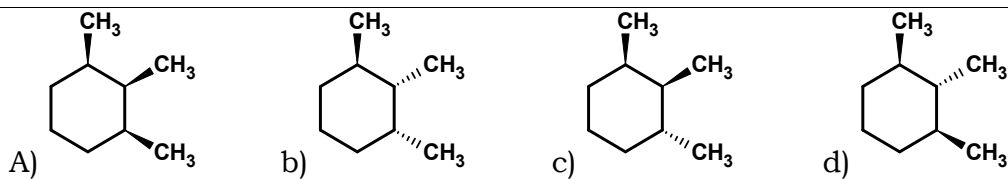
i) $\text{Cr}_2\text{O}_3^- \text{Al}_2\text{O}_3$ (or) V_2O_5 (or) $\text{Mo}_2\text{O}_3, \Delta$ ii) $\text{Cu} / 528 \text{ K } 100 \text{ atm}$ iii) Mo_2O_3 iv) $\text{Mn}(\text{OAc})_2$ v) KMnO_4 vi) HNO_3, Δ vii) Anhyd AlCl_3 / Δ			<u>*) Stero-selective reaction:</u> A reaction which yields stereoisomers irrespective of reactant <u>*) Sterospecific reaction:</u> A reaction in which a set of stereoisomers through same mechanism gives a set of stereoisomers
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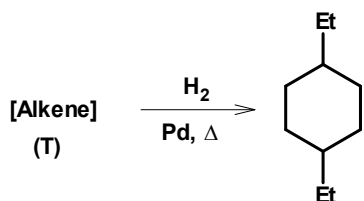
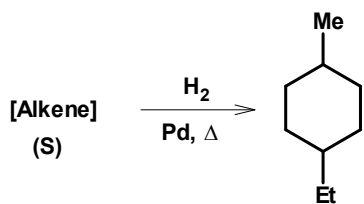
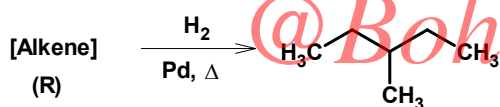
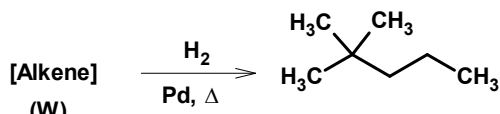
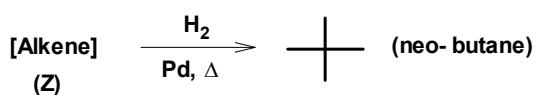
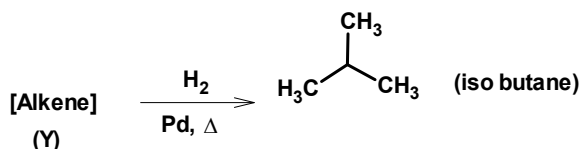
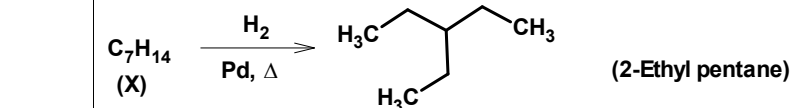
1. HYDROGENATION

1.	cis - 2 - Pentene $\xrightarrow[\text{Ni}]{\text{D}_2}$ [X]
2.	Trans - 2 - Pentene $\xrightarrow[\text{Ni}]{\text{D}_2}$ [Y]
3.	 $\xrightarrow[\text{Ni}]{\text{H}_2}$ [K] + (L)
4.	 $\xrightarrow[\text{Ni}]{\text{H}_2}$ (M) + (N)
5.	 $\xrightarrow[\text{Ni}]{\text{H}_2}$ (R) + (S)
6.	[W] $\xrightarrow[\text{Ni}]{\text{H}_2}$ 2-Methyl butane (iso pentane) isomeric alkene Find the no. of alkene [W], including stereoisomers, which can give 2-methyl butane on hydrogenation
7.	C_4H_8 $\xrightarrow[\text{Ni}]{\text{D}_2}$ [M] (all isomeric alkene) Find the no. of [M] product, including stereo.
8.	 $\xrightarrow[\text{Ni}]{\text{H}_2}$ [X]
9.	 $\xrightarrow[\text{Ni}]{\text{H}_2}$ Products (s)
	Which of the product(s) are possible ?

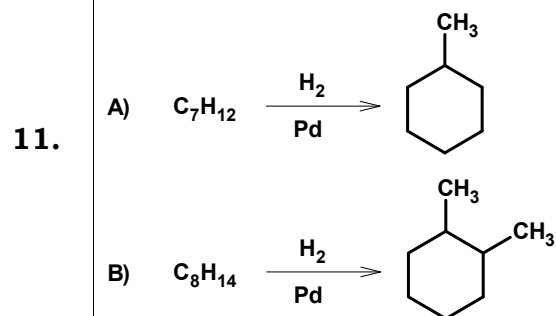
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How many alkene (including stereo) can give following product on catalyst hydrogenation ?



Find the number of reactant(s) possible including stereoisomers

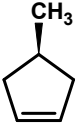
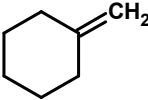
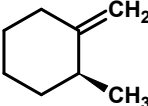


	C) $C_8H_{14} \xrightarrow[\text{Pd}]{H_2} \text{1,3-dimethylcyclohexane}$ D) $C_8H_{14} \xrightarrow[\text{Pd}]{H_2} \text{1,4-dimethylcyclohexane}$
12.	$\text{1-methylcyclopentene} \xrightarrow[\text{Pd}]{H_2} \text{Number of product (s) ?}$
13.	$C_{10}H_{18} \xrightarrow[\text{Pd}]{H_2} \text{1-isopropyl-3-methylcyclohexane}$ Find the number of reactant(s) including stereomers
14.	$\begin{array}{c} \text{CH}_3 \\ \\ \text{H}-\text{C}-\text{D} \\ \\ \text{CH} \\ \\ \text{CH} \\ \\ \text{D}-\text{C}-\text{H} \\ \\ \text{CH}_3 \end{array} \xrightarrow[\text{Pd}]{D_2} \text{Number of product (s) ?}$ (cis) <i>@Bohring_bot</i>
15.	$\text{1-(cyclopent-1-en-1-yl)ethyne} \xrightarrow[\text{Pd}]{H_2 (1 \text{ eq})}$
16.	$\begin{array}{c} \text{H}_3\text{C}-\text{C}=\text{CH}_2 \\ \\ \text{Cyclohexane ring} \\ \\ \text{CH}=\text{CH}_2 \end{array} \xrightarrow[\text{Pd}]{H_2 (1 \text{ eq})}$
17.	$R-\text{NO}_2 \xrightarrow[\text{Pd}]{H_2}$
18.	$R-\text{C}\equiv\text{N} \xrightarrow[\text{Pd}]{H_2}$
19.	$R-\text{CH}=\text{O} \xrightarrow[\text{Pd}]{H_2}$
20.	$R-\overset{\text{O}}{\parallel}{\text{C}}-R^1 \xrightarrow[\text{Pd}]{H_2}$

21.	Identify [K] and [L]
22.	Number of product (s) & their relationship.

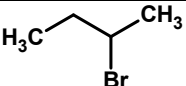
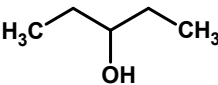
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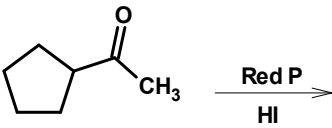
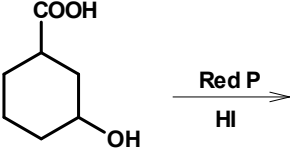
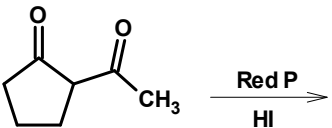
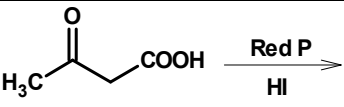
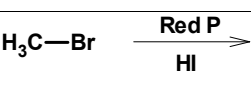
2. HYDROBORATION - ACIDOLYSIS

1.	$\text{H}_2\text{C}=\text{CH}_2 \xrightarrow[2. \text{CH}_3\text{COOH}]{1. (\text{H-BH}_2), \text{THF}} [\text{A}]$
2.	$\text{H}_3\text{C}-\text{CH}=\text{CH}_2 \xrightarrow[2. \text{CH}_3\text{COOH}]{1. (\text{H-BH}_2), \text{THF}} [\text{B}]$ (Trans)
3.	 $\xrightarrow[2. \text{CH}_3\text{COOH}]{1. (\text{H-BH}_2), \text{THF}} [\text{C}]$
4.	 $\xrightarrow[2. \text{AcOH}]{1. \text{BD}_3, \text{THF}} [\text{D}]$
5.	 $\xrightarrow[2. \text{AcOH}]{1. \text{BD}_3, \text{THF}} [\text{E}]$

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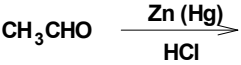
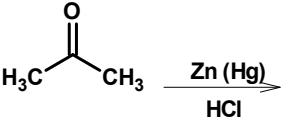
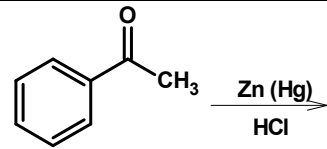
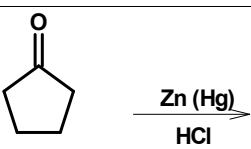
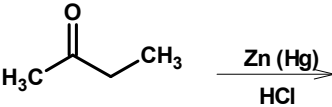
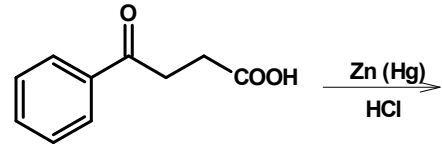
3. Red P + HI

1.	$\text{H}_3\text{C}-\text{CH}_2-\text{CH}_2-\text{Cl} \xrightarrow[\text{HI}]{\text{Red P}}$
2.	 $\xrightarrow[\text{HI}]{\text{Red P}}$
3.	 $\xrightarrow[\text{HI}]{\text{Red P}}$
4.	$\begin{array}{c} \text{CH}_2-\text{OH} \\ \\ \text{CH}-\text{OH} \\ \\ \text{CH}_2-\text{OH} \end{array} \xrightarrow[\text{HI}]{\text{Red P}}$ (glycerol)
5.	$\begin{array}{c} \text{CHO} \\ \\ \text{H}-\text{C}-\text{OH} \\ \\ \text{HO}-\text{C}-\text{OH} \\ \\ \text{H}-\text{C}-\text{OH} \\ \\ \text{H}-\text{C}-\text{OH} \\ \\ \text{CH}_2\text{OH} \end{array} \xrightarrow[\text{HI}]{\text{Red P}}$

6.	
7.	
8.	
9.	
10.	

4. Clemenson - Reduction

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1.	
2.	
3.	
4.	
5.	
6.	

7.	$\text{OHC}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{C}(=\text{O})\text{OCH}_3 \xrightarrow[\text{HCl}]{\text{Zn (Hg)}}$
8.	$\begin{array}{c} \text{H}_3\text{C} \\ \\ \text{H}_3\text{C}-\text{C}-\text{CHO} \\ \\ \text{H}_3\text{C} \end{array} \xrightarrow[\text{HCl}]{\text{Zn (Hg)}}$ (Pivaldehyde)
9.	$\begin{array}{c} \text{H}_3\text{C} \\ \\ \text{CH}-\text{CHO} \\ \\ \text{H}_3\text{C} \end{array} \xrightarrow[\text{HCl}]{\text{Zn (Hg)}}$
10.	$\text{HO}-\text{C}_4\text{H}_7-\text{C}(=\text{O})\text{CH}_3 \xrightarrow[\text{HCl}]{\text{Zn (Hg)}}$

5. Wolf - Kishner reduction

1.	$\text{1,4-naphthoquinone} \xrightarrow{\text{H}_2\text{N}-\text{NH}_2} \xrightarrow[\text{ethylene glycol}]{\text{KOH}, \Delta}$
2.	$\text{Cyclopentanone} \xrightarrow{\text{H}_2\text{N}-\text{NH}_2} \xrightarrow[\text{ethylene glycol}]{\text{KOH}, \Delta}$
3.	$\text{2-nitrobenzaldehyde} \xrightarrow[\text{HCl}]{\text{Zn (Hg)}} \xrightarrow[\text{ethylene glycol}]{\text{H}_2\text{N}-\text{NH}_2, \text{KOH}, \Delta}$
4.	$\text{2-(2-oxocyclohexyl)propanal} \xrightarrow[\text{ethylene glycol}]{\text{H}_2\text{N}-\text{NH}_2, \text{KOH}, \Delta}$
5.	$\text{C}_5\text{H}_{10}\text{O} \xrightarrow[\text{ethylene glycol}]{\text{H}_2\text{N}-\text{NH}_2, \text{KOH}, \Delta} \text{H}_3\text{C}-\text{CH}_2-\text{CH}(\text{CH}_3)_2$ (isopentane)

	How many carbonyl compounds of mol. Formula $C_5H_{10}O$ can give isopentane on reduction with $NH_2 - NH_2$, KOH, Δ ? $\begin{array}{c} \text{H}_3\text{C}-\text{CH}_2-\text{CH}_2-\text{CHO} \\ 2 \end{array}$ $\begin{array}{c} \text{O} \\ \\ \text{H}_3\text{C}-\text{C}-\text{CH}_2-\text{CH}_3 \\ 1 \end{array}$ $\begin{array}{c} \text{OHC}-\text{CH}_2-\text{CH}_2-\text{CH}_3 \\ 1 \end{array}$
6.	$C_6H_{12}O$ Carbonyl compound [M] Find the number of [M] ? $\begin{array}{c} \xrightarrow[\text{ethylene glycol}]{\text{H}_2\text{N}-\text{NH}_2, \text{KOH}, \Delta} \\ \text{H}_3\text{C}-\text{CH}_2-\text{CH}(\text{CH}_3)-\text{CH}_2-\text{CH}_3 \end{array}$

6. Kolbe Electrolysis

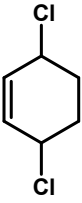
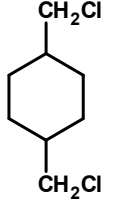
1.	$\begin{array}{c} \text{CH}_2\text{COOK} \\ \\ \text{C}_6\text{H}_{10} \\ \\ \text{CH}_2\text{COOK (aq)} \end{array}$ $\xrightarrow{\text{Electrolysis}}$
2.	$\begin{array}{c} \text{CH}_2\text{COOK} \\ \\ \text{KOOCH}_2\text{C}-\text{CH}_2\text{COOK} \\ \\ \text{CH}_2\text{COOK} \\ \text{(aq)} \end{array}$ $\xrightarrow{\text{Electrolysis}}$
3.	$\begin{array}{c} \text{CH}_2\text{COOK} \\ \\ \text{KOOCH}_2\text{C}-\text{CH}_2-\text{C}-\text{CH}_2\text{COOK} \\ \quad \\ \text{KOOCH}_2\text{C} \quad \text{CH}_2\text{COOK} \\ \text{(aq)} \end{array}$ $\xrightarrow{\text{Electrolysis}}$
4.	$\begin{array}{c} \text{COOK} \quad \text{COOK} \\ \quad \\ \text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2 \end{array}$ $\xrightarrow{\text{Electrolysis}}$

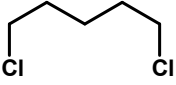
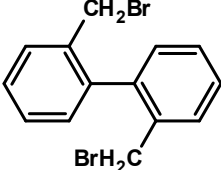
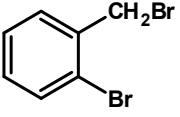
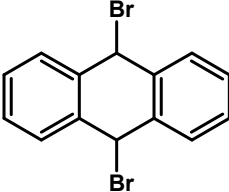
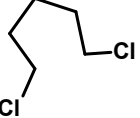
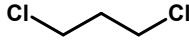
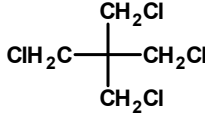
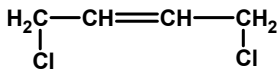
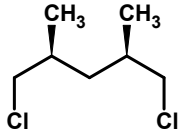
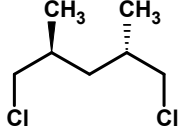
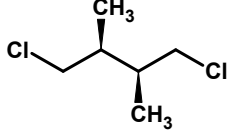
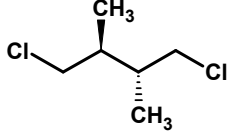
7. Sodalime decarboxylation

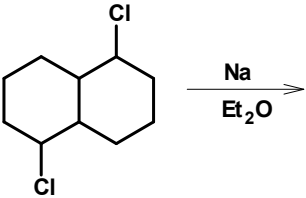
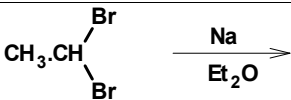
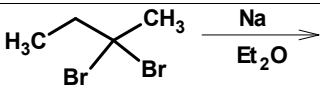
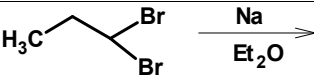
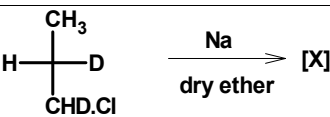
1.	$\begin{array}{c} \text{COOH} \\ \\ \text{C}_5\text{H}_7 \\ 14 \end{array}$ $\xrightarrow[\Delta]{\text{NaOH, CaO}}$ [X] number of product(s) ?
2.	$\begin{array}{c} \text{COOH} \\ \\ \text{C}_6\text{H}_5 \end{array}$ $\xrightarrow[\Delta]{\text{NaOH, CaO}}$ [M] number of product(s) ?
3.	How many carboxylic acid on sodalime decarboxylation give isopentane?

	$\text{C}_6\text{H}_{12}\text{O}_2 \xrightarrow[\Delta]{\text{NaOH, CaO}} \text{H}_3\text{C}-\text{CH}(\text{CH}_3)-\text{CH}_2-\text{CH}_3$
4.	How many carboxylic acid of molecular formula $\text{C}_7\text{H}_{12}\text{O}_2$ give methylcyclopentane on sodalime decarboxylation ? $\text{C}_6\text{H}_{12}\text{O}_2 \xrightarrow[\Delta]{\text{NaOH, CaO}} \text{Cyclopentane ring}-\text{CH}_3$
5.	Find number of carboxylic acid, which on sodalime decarboxylation give following products ? [A] $\xrightarrow[\Delta]{\text{NaOH, CaO}} \text{H}_3\text{C}-\text{CH}_2-\text{CH}_2-\text{CH}_3$ [B] $\xrightarrow[\Delta]{\text{NaOH, CaO}} \text{H}_3\text{C}-\text{CH}(\text{CH}_3)-\text{CH}_2-\text{CH}_3$ [C] $\xrightarrow[\Delta]{\text{NaOH, CaO}} \text{H}_3\text{C}-\text{C}(\text{CH}_3)_3$ [D] $\xrightarrow[\Delta]{\text{NaOH, CaO}} \text{H}_3\text{C}-\text{CH}(\text{CH}_3)-\text{CH}(\text{CH}_3)-\text{CH}_3$ [E] $\xrightarrow[\Delta]{\text{NaOH, CaO}} \text{Cyclohexane ring}-\text{CH}_3$

8. Wurtz reaction

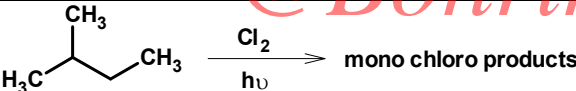
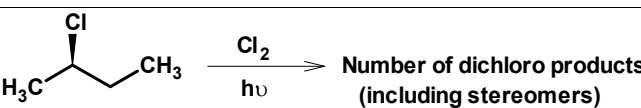
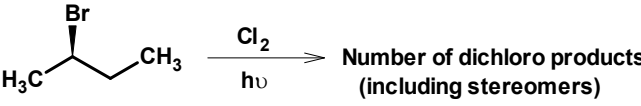
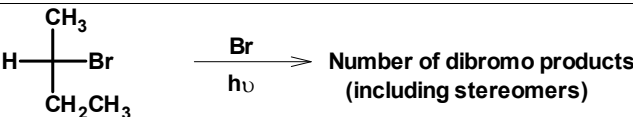
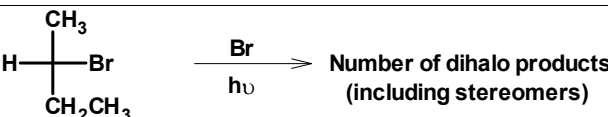
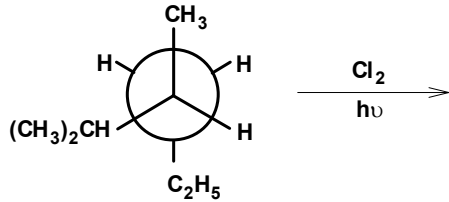
1.	 $\xrightarrow[\text{dry ether}]{\text{Na}}$
2.	 $\xrightarrow[\text{dry ether}]{\text{Na}}$
3.	 $\xrightarrow[\text{dry ether}]{\text{Na}}$

4.	 $\xrightarrow[\text{dry ether}]{\text{Na}}$
5.	 $\xrightarrow[\text{dry ether}]{\text{Na}}$
6.	 $\xrightarrow[\text{dry ether}]{\text{Na}}$ [A] + [B]
7.	 $\xrightarrow[\text{dry ether}]{\text{Na}}$
8.	$\text{H}_2\text{C}=\text{CH}-\text{CH}_2\text{Br}$ $\xrightarrow[\text{dry ether}]{\text{Na}}$ [A] + [B] + [C]
9.	 $\xrightarrow[\text{Et}_2\text{O}]{\text{Na}}$
10.	 $\xrightarrow[\text{Et}_2\text{O}]{\text{Na}}$
11.	 $\xrightarrow[\text{Et}_2\text{O}]{\text{Na}}$
12.	 $\xrightarrow[\text{Et}_2\text{O}]{\text{Na}}$
13.	 $\xrightarrow[\text{Et}_2\text{O}]{\text{Na}}$ number of product(s) (Opt. Act + Opt Inact)
14.	 $\xrightarrow[\text{Et}_2\text{O}]{\text{Na}}$ number of product(s) (Opt. Act + Opt Inact)
15.	 $\xrightarrow[\text{Et}_2\text{O}]{\text{Na}}$ number of product(s) (chiral + Achiral)
16.	 $\xrightarrow[\text{Et}_2\text{O}]{\text{Na}}$ number of product(s) (chiral + Achiral)

17.	
18.	 (1,1- dibromo ethane)
19.	 (2,2- dibromo buane)
20.	
21.	

9. Halogenation

@Bohring_bot

1.	 Find the number of monochloro product(s), including stereomer
2.	 Number of dichloro products (including stereomers)
3.	 Number of dichloro products (including stereomers)
4.	 Number of dibromo products (including stereomers)
5.	 Number of dihalo products (including stereomers)
6.	Find the number of monochloro derivatives ? A) 

B)	$ \begin{array}{c} \text{CH}_3 \\ \\ \text{H} - \text{C} - \text{Cl} \\ \\ \text{H} - \text{C} - \text{Cl} \\ \\ \text{CH}_2\text{CH}_3 \end{array} \xrightarrow[h\nu]{\text{Cl}_2} $
C)	$ \begin{array}{c} \text{CH}_3 \\ \\ \text{Cyclohexane ring} \end{array} \xrightarrow[h\nu]{\text{Cl}_2} $
D)	$ \begin{array}{c} \text{CH}_3 \\ \\ \text{Cyclopentane ring} \end{array} \xrightarrow[h\nu]{\text{Cl}_2} $
E)	$ \begin{array}{c} \text{H}_3\text{C} \quad \text{CH}_3 \\ \quad \\ \text{Cyclopentane ring} \end{array} \xrightarrow[h\nu]{\text{Cl}_2} $

NCERT

1.	$ \text{H}_3\text{C}-\text{C}\equiv\text{CH} + 2\text{H}_2 \xrightarrow{\text{Pt/Pd/Ni}} $
2.	$ \text{H}_3\text{C}-\text{Cl} \xrightarrow{\text{Zn, HCl}} $
3.	$ \text{H}_2\text{CH}_2\text{CH}_2\text{CH}_2-\text{Cl} \xrightarrow{\text{Zn, H}^+} $
4.	$ \text{CH}_3\text{Br} + 2\text{Na} + \text{CH}_3\text{Br} \xrightarrow{\text{dry ether}} $
5.	$ \text{CH}_3\text{COONa} + \text{NaOH} \xrightarrow[\Delta]{\text{CaO}} $
6.	$ 2\text{CH}_3\text{COONa} + 2\text{H}_2\text{O} \xrightarrow{\text{Electrolysis}} $
7.	$ \begin{array}{l} 2\text{CH}_4 + \text{O}_2 \xrightarrow{\text{Cu / 523 K / 100 atm}} 2\text{CH}_3\text{OH} \\ \text{CH}_4 + \text{O}_2 \xrightarrow{\text{Mn}_2\text{O}_3} \text{HCHO} \\ 2\text{CH}_3\text{CH}_3 + 3\text{O}_2 \xrightarrow[\Delta]{(\text{CH}_3\text{COO})_2\text{Mn}} 2\text{CH}_3\text{COOH} \end{array} $
8.	$ (\text{CH}_3)_3\text{CH} \xrightarrow[\text{Oxidation}]{\text{KMnO}_4} $
9.	$ \text{H}_3\text{C}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_3 \xrightarrow[\text{HCl}]{\text{Anhyd. AlCl}_3} \text{H}_3\text{C}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_3 + \text{H}_3\text{C}-\text{CH}_2-\text{CH}(\text{CH}_3)-\text{CH}_2-\text{CH}_2-\text{CH}_3 $

10.	$ \begin{array}{c} \text{H}_2\text{C} \quad \text{CH}_2 \\ \diagup \quad \diagdown \\ \text{CH}_2 \quad \text{CH}_2 \\ \diagdown \quad \diagup \\ \text{H}_2\text{C} \quad \text{CH}_2 \end{array} \xrightarrow[\substack{\text{V}_2\text{O}_5 \text{ (or)} \\ \text{Mo}_2\text{O}_3, 773 \text{ K} \\ 10 - 20 \text{ atm}}]{\text{Cr}_2\text{O}_3 \text{ (or)}} $
11.	$ \text{CH}_4 + \text{H}_2\text{O} \xrightarrow[\Delta]{\text{Ni}} \text{CO} + 3\text{H}_2 $

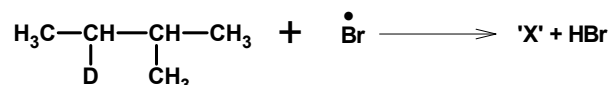
@Bohring_bot

TRUE/FALSE

1. Photobromination of 2-methylpropane gives a mixture of 1-bromo-2-methylpropane and 2-bromo-2-methylpropane in the ratio of 9 : 1

MCQS WITH ONE CORRECT ANSWER

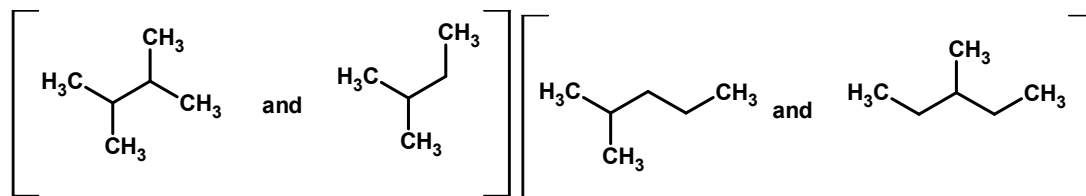
1. marsh gas mainly contains
a) C_2H_2 b) CH_4 c) H_2S d) CO
3. The compound with the highest boiling point is
a) n-hexane b) n-pentane c) 2,2-dimethylpropane d) 2-methylbutane
10. The highest boiling point is expected for :
a) iso-octane b) n-octane
c) 2,2,3,3-tetramethylbutane d) n-butane
12. When cyclohexane is poured on water, it floats, because :
a) cyclohexane is in 'boat' form
b) cyclohexane is in 'chair' form
c) cyclohexane is in 'crown' form
d) cyclohexane is less dense than water
20. Consider the following reaction



Identify the structure of the major product 'X'

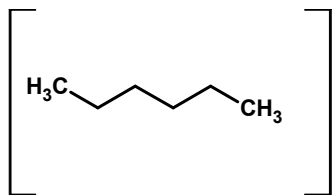
- a) $\begin{array}{c} H_3C-CH-CH-\dot{C}H_2 \\ | \quad | \\ D \quad CH_3 \end{array}$ B) $\begin{array}{c} H_3C-CH-\dot{C}-CH_3 \\ | \quad | \\ D \quad CH_3 \end{array}$
- C) $\begin{array}{c} H_3C-\dot{C}-CH-CH_3 \\ | \quad | \\ D \quad CH_3 \end{array}$ D) $\begin{array}{c} H_3C-\dot{C}H-CH-CH_3 \\ | \quad | \\ \quad CH_3 \end{array}$

24. On monochlorination of 2-methylbutane, the total number of chiral compounds formed is
a) 2 b) 4 c) 6 d) 8
31. Isomers of hexane, based on their branching, can be divided into three distinct classes as shown in the figure.



(I)

(II)



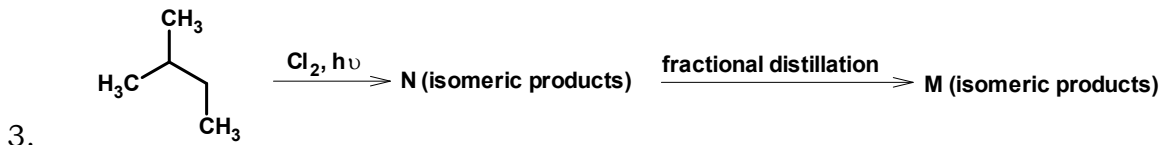
(III)

The correct order of their boiling point is

- a) I>II>III b) III>II>I c) II>III>I d) III>I>II

1. Which one of the following has the smallest heat of hydrogenation per mole ?

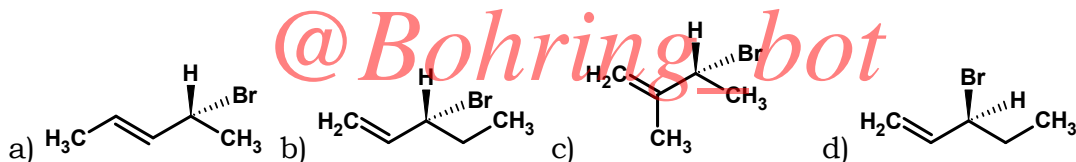
- a) 1-butene b) trans-2-butene c) cis-2-butene d) 1,3-butadiene



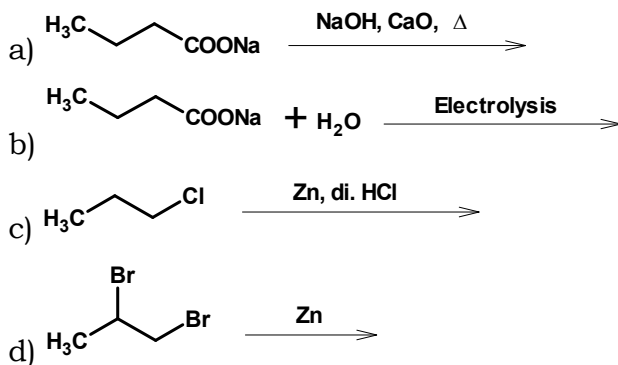
Identify N and M

- a) 6, 4 b) 6, 6 c) 4, 4 d) 3, 3

6. Compound(s) that on hydrogenation product(s) optically inactive compound(s) is (are) :



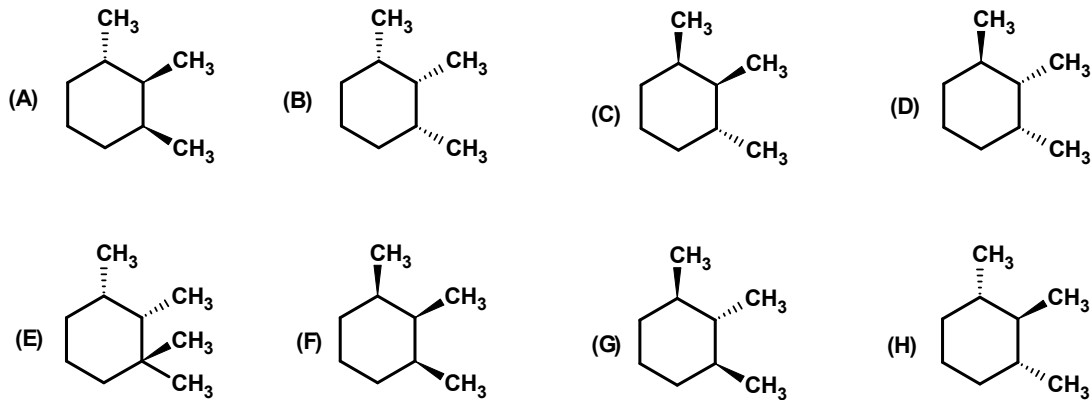
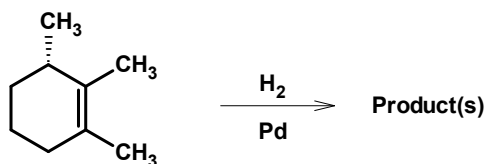
9. Which of the following reactions produce(s) propane as a major product ?



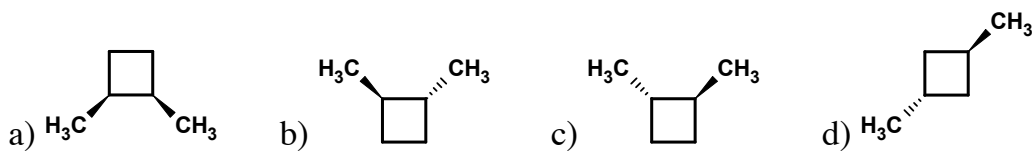
3. On mixing a certain alkane with chlorine and irradiating it with ultraviolet light, it forms only one monochloroalkane. This alkane could be
 a) pentane B) isopentane c) neopentane d) propane
4. 2-Methylbutane on reacting with bromine in the presence of sunlight gives mainly
 a) 1-bromo-3-methylbutane b) 2-bromo-3-methylbutane
 c) 2-bromo-2-methylbutane d) 1-bromo-2-methylbutane
5. Butene-1 may be converted to butane by reaction with
 a) Sn-HCl b) Zn-Hg c) Pd / H₂ d) Zn-HCl
7. Of the five isomeric hexanes, the isomer which can give two monochlorinated compound is
 a) 2-methylpentane b) 2,2-dimethylbutane
 c) 2,3-dimethylbutane d) n-hexane
9. Which types of isomerism is shown by 2,3-dichlorobutane ?
 a) structural b) Geometrical c) Optical d) diastereo
16. The treatment of CH₃MgX with CH₃C≡C-H produces
 a) CH₃-CH≡CH₂ b) CH₃C≡C-CH₃
 c) $\begin{array}{c} \text{H} \quad \text{H} \\ | \quad | \\ \text{H}_3\text{C}-\text{C}=\text{C}-\text{CH}_3 \end{array}$ d) CH₄
19. Which branched chain isomer of the hydrocarbon with molecular mass 72u gives only one isomer of mono substituted alkyl halide ?
 a) Tertiary butyl chloride b) Neopentane
 c) Isohexane d) Neohexane

TEST

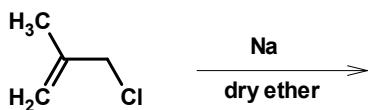
2. Identify the product(s) obtained after hydrogenation



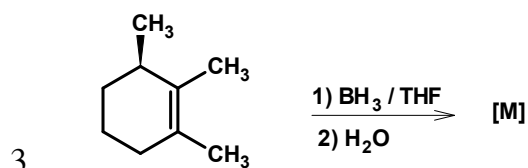
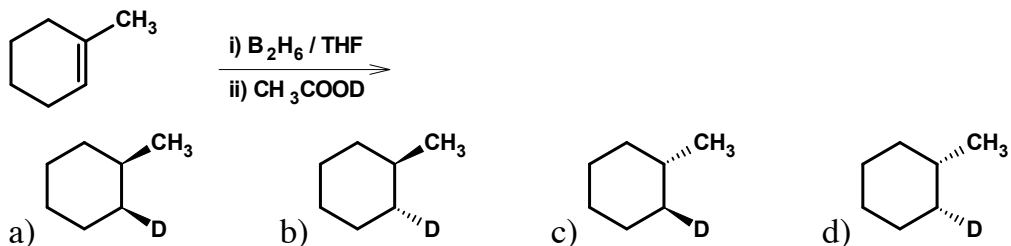
2. The major product obtained in the following reaction is



7. Find the number of product formed in given reaction

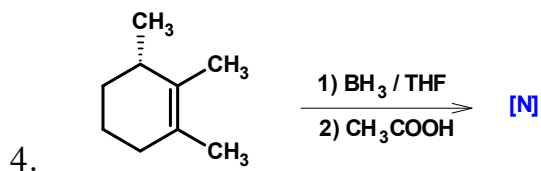
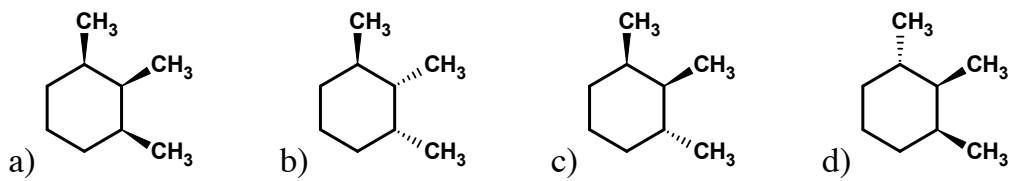


2. The major product obtained in the following reaction is

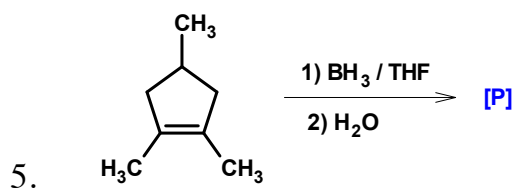
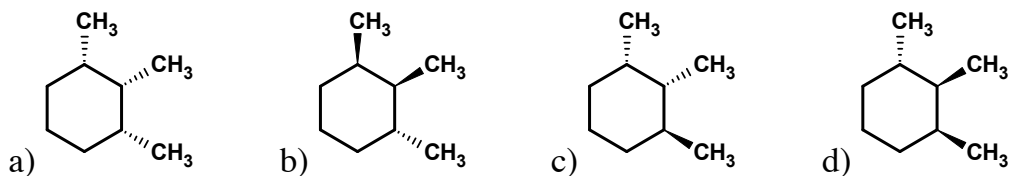


- 3.

[M] will be



[N] will be

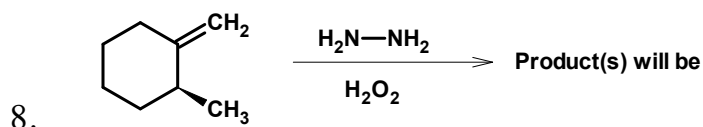


[P] will be

a) Diastereomeric pair

b) Racemic mixture

c) Pure enantiomer d) Meso Compounds

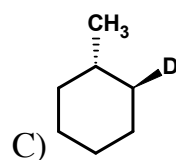
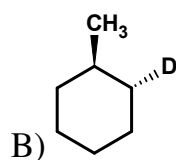
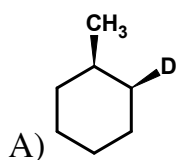
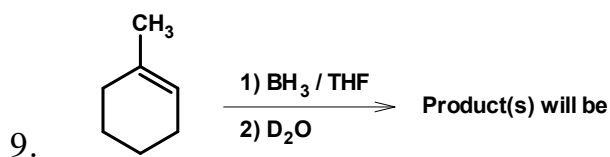


a) Racemic mixture

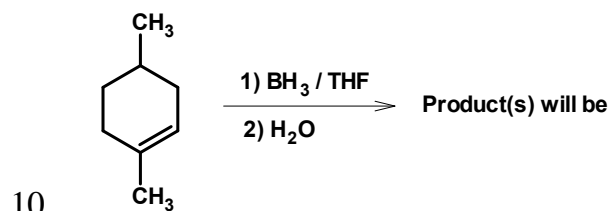
b) Diastereomeric

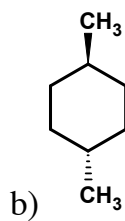
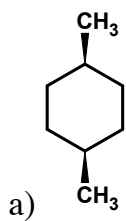
c) Meso

d) Constitutional isomers



D) All correct

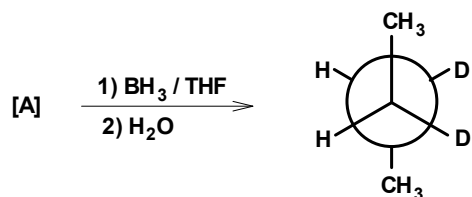




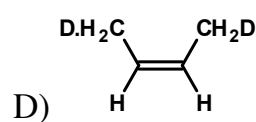
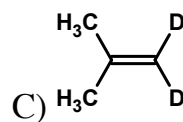
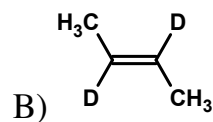
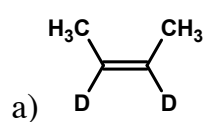
c) Achiral products

d) Diastereomers of each other

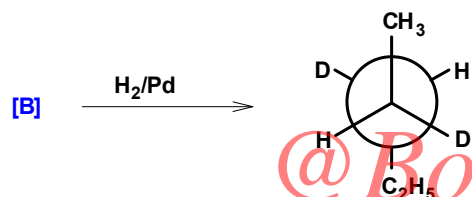
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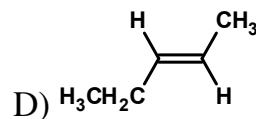
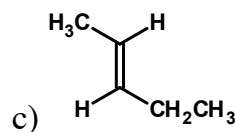
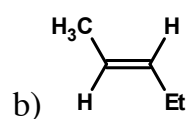
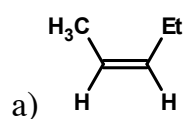
[A] is



19.



[B] is



8.



[X] and [Y] are

a) 6,5

b) 6,4

c) 5,4

d) 6,3

ORGANIC CHEMISTRY

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ALKENE-1

WORK SHEET

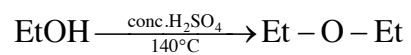
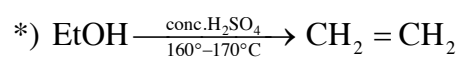
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SYNOPSIS

REAGENT	NAME REACTION	TEST	CONCEPTUAL POINTS
i) conc. H_2SO_4 , Δ	*) dyhydraton		$\text{E}_2 \rightarrow$ strong $\bar{\text{B}}$, good L.G., single T.S., sytzeff product
ii) conc. H_3PO_4 , Δ	*) Dehalogenation		
iii) Al_2O_3 , Δ	*) Dehydrohalogenation		$\rightarrow$ strong bulky $\bar{\text{B}}$, good L.G., single T.S. hoffman product.
iv) Al_2O_3 – pyridine	*) Pyrolysis of ester		
v) ThO_2 , Δ	*) Pyrolysis of Xanthate		$\rightarrow$ strong $\bar{\text{B}}$, bad L.G., single T.S., hoffman product, +ve nucleofuse
vi) CHCl_3 (excess) AgOH / Δ	*) Cope elimination *) Ammonolysis		$\text{E}_{1\text{CB}} \rightarrow$ strong $\bar{\text{B}}$, bad L.G., acidic $\alpha - \text{H}$, [anion](2-step), Hoffman (general)
vii) H_2 / Ni(1eq)	*) Hoffmann Exhaustive Elimination (HEE)		$\text{E}_1 \rightarrow$ poor B^- , Nu^- , good L.G. [cation], 2-step, Sytzeff
viii) H_2 / Pd – BaSO_4 (Lindlar catalyst)	*) Hydrogenation		$\Rightarrow$ Rigid structure = "Syn" elimination
ix) H_2 / pd – CaCO_3 Poisoned with quinoline/boiling sulphur/xylene	*) Birch reduction *) Witting reaction *)		$\Rightarrow \text{NaI/Acetone } \Delta$ - first SN^2 at 1°C then E_2
x) Na / liq NH_3			*) $\text{E}_{1\text{CB}} \rightarrow$ pseudo first order = "second order"
xi) $\text{Ph}_3\text{P}^+ - \text{CH}_2^-$ $\text{Ph}_3\text{P} = \text{CH}_2$			$\text{E}_{1\text{CB}} \rightarrow$ EWG- PhSO_2 , NO_2 , COOH , dihalogen, $-\text{CN}$
xii) Zn / Δ			$\text{E}_{1\text{CB}} \rightarrow$ Bad L.G, $-\text{OH}$, $-\text{OR}$, $-\text{CN}$, $-\text{F}$, $-\text{OMe}$
xiii) NaI / Acetone, Δ			*) $\text{E}_{1\text{CB}} \rightarrow$ Anion attacking i.e. removal of bad L.G. is RDS. (Deuterium exchange by anion)
xiv) alc.KOH, Δ			*) $\text{E}_1 \rightarrow$ first order reaction formation of cation is RDS H_2SO_4 , H_3PO_4 , Al_2O_3 , Al_2O_3 – py, ThO_2 Generate carbocation sytzeff without rearrange Hoffman

*) Kinetic isotopci effect $\rightarrow E_1, E_2, E_{1CB}$

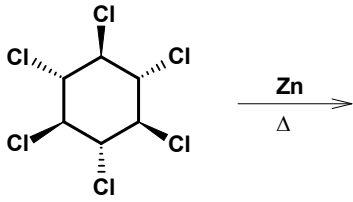
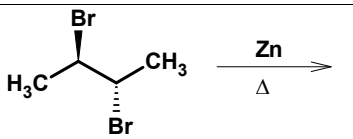
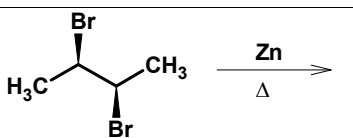
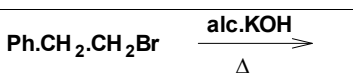
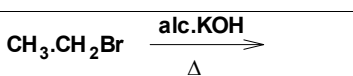
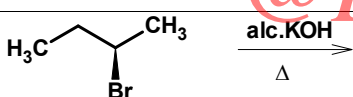
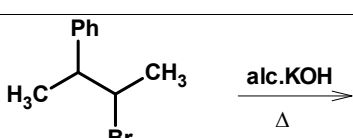
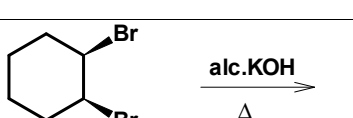
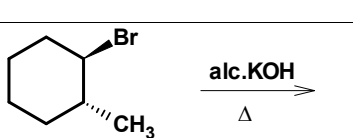
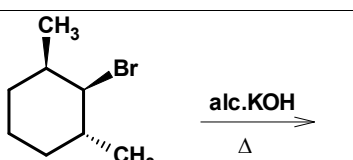
*) Solvent isotopic effect $\rightarrow E_1, E_2, E_{1CB}$

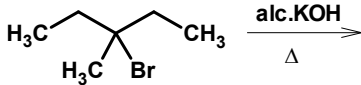
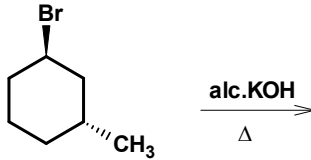
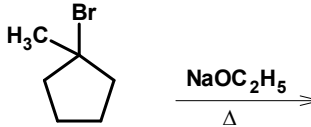
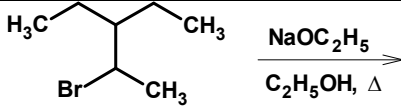
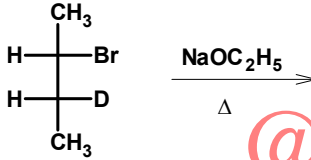
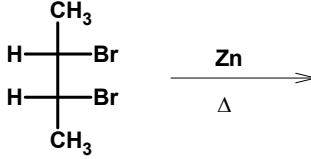
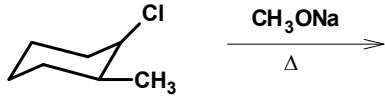
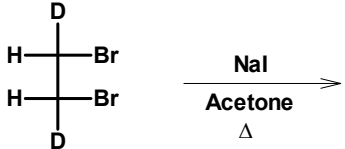


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E2 (Sytzeff)

1.	$\begin{array}{c} \text{H}_2\text{C}-\text{CH}_2 \\ \quad \\ \text{Br} \quad \text{Br} \end{array} \xrightarrow[\Delta]{\text{Zn}}$
2.	$\begin{array}{c} \text{H}_2\text{C}-\text{CH}_2 \\ \quad \\ \text{Br} \quad \text{Br} \end{array} \xrightarrow[\text{Acetone}]{\text{NaI}}$ (Ethylene di bromide)
3.	$\begin{array}{c} \text{Br} \\ \\ \text{H}_3\text{C}-\text{CH}-\text{CH}_3 \\ \\ \text{Br} \end{array} \xrightarrow[\text{Acetone}]{\text{NaI}}$
4.	$\begin{array}{c} \text{H} \\ \\ \text{D}-\text{C}-\text{Br} \\ \\ \text{Br}-\text{C}-\text{D} \\ \\ \text{H} \end{array} \xrightarrow[\text{Acetone}]{\text{NaI}}$
5.	$\begin{array}{c} \text{CH}_3 \\ \\ \text{H}-\text{C}-\text{Br} \\ \\ \text{H}-\text{C}-\text{Br} \\ \\ \text{CH}_3 \end{array} \xrightarrow[\Delta]{\text{Zn}}$
6.	$\begin{array}{c} \text{CH}_3 \\ \\ \text{H}-\text{C}-\text{Br} \\ \\ \text{Br}-\text{C}-\text{H} \\ \\ \text{CH}_3 \end{array} \xrightarrow[\Delta]{\text{Zn}}$
7.	$\begin{array}{c} \text{CH}_3 \\ \\ \text{H}-\text{C}-\text{Br} \\ \\ \text{Br}-\text{C}-\text{H} \\ \\ \text{Ph} \end{array} \xrightarrow[\Delta]{\text{Zn}}$
8.	$\begin{array}{c} \text{Br} \\ \\ \text{C}_6\text{H}_{10} \\ \\ \text{Br} \end{array} \xrightarrow[\Delta]{\text{Zn}}$

9.	 <chem>ClC1(Cl)C(Cl)C(Cl)C(Cl)C(Cl)C1>>C1=CC=CC=C1</chem>
10.	 <chem>Br[C@H](C)[C@@H](C)Br>>C/C=C/C</chem>
11.	 <chem>Br[C@H](C)[C@H](C)Br>>C/C=C\C</chem>
12.	 <chem>PhCH2CH2Br>>PhCH=CH2</chem>
13.	 <chem>CH3CH2Br>>CH2=CH2</chem>
14.	 <chem>CC[C@H](C)Br>>C/C=C/C</chem>
15.	 <chem>CC(=C)C[C@H](C)Br>>CC(=C)C=C</chem>
16.	 <chem>BrC1CCC(CC1)Br>>C1=CCCCC1</chem>
17.	 <chem>BrC1CCC(CC1)C>>CC1=CCCCC1</chem>
18.	 <chem>CC1CCC(CC1)C(Br)C>>CC1=CCC(CC1)C</chem>

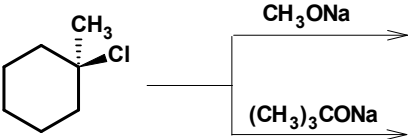
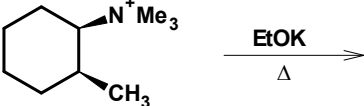
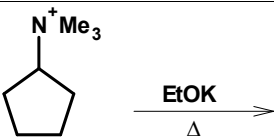
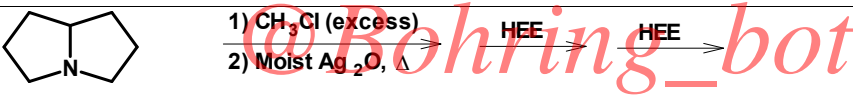
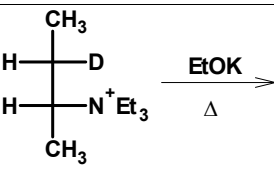
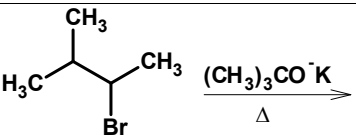
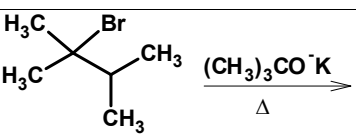
19.	 $\xrightarrow[\Delta]{\text{alc.KOH}}$
20.	 $\xrightarrow[\Delta]{\text{alc.KOH}}$
21.	 $\xrightarrow[\Delta]{\text{NaOC}_2\text{H}_5}$
22.	 $\xrightarrow[\text{C}_2\text{H}_5\text{OH}, \Delta]{\text{NaOC}_2\text{H}_5}$
23.	 $\xrightarrow[\Delta]{\text{NaOC}_2\text{H}_5}$
24.	 $\xrightarrow[\Delta]{\text{Zn}}$ $\xrightarrow[\Delta]{\text{Alc. KOH}}$ [X] But-2-ene (Trans) + [Y] But-2-ene (cis) + [Z] But-1-ene (enantic) + [W]
25.	 $\xrightarrow[\Delta]{\text{CH}_3\text{ONa}}$
26.	 $\xrightarrow[\Delta]{\text{NaI, Acetone}}$
27.	 $\xrightarrow[\Delta]{\text{NaI, Acetone}}$

28.	$\begin{array}{c} \text{Ph} \quad \text{H} \\ \diagdown \quad \diagup \\ \text{C}=\text{C} \\ \diagup \quad \diagdown \\ \text{Ph} \quad \text{Br} \end{array} \xrightarrow{\text{NaNH}_2} \quad \Rightarrow$
29.	$\left[\begin{array}{c} \text{CH}_3 \quad \text{CH}_3 \\ \quad \\ \text{H}_3\text{C}-\text{C}-\text{N}^+-\text{CH}_2\text{CH}_3 \\ \quad \\ \text{CH}_3 \quad \text{CH}_3 \end{array} \right] \cdot \text{OH}^- \xrightarrow{\Delta} \quad \Rightarrow$
30.	$\text{Cyclohexane ring with } \text{Br} \text{ (wedge) and } \text{COOH} \text{ (dash)} \xrightarrow[\Delta]{\text{K}_2\text{CO}_3} \text{Alkene ?}$
31.	$\begin{array}{c} \text{Ph} \\ \\ \text{H}-\text{C}-\text{Cl} \\ \\ \text{H}-\text{C}-\text{Cl} \\ \\ \text{CH}_3 \end{array} \xrightarrow[\text{Acetone}]{\text{NaI}} \quad \Rightarrow$
32.	$\text{Cyclohexane ring with } \text{Br} \text{ (wedge) and } \text{Br} \text{ (dash)} \xrightarrow[\text{Acetone}]{\text{NaI}} \quad \Rightarrow$
33.	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{Cl} \\ / \quad \backslash \\ \text{CH}_3\text{H}_2\text{C} \quad \text{C}-\text{Cl} \\ \backslash \quad / \\ \text{CH}_3 \end{array} \xrightarrow[\text{Acetone}]{\text{NaI}} \quad \Rightarrow$
34.	$\text{Cyclohexane ring with } \text{CH}_3 \text{ (wedge) and } \text{Cl} \text{ (wedge)} \xrightarrow[\Delta]{\text{alc.KOH}} \quad \Rightarrow$

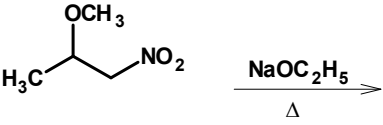
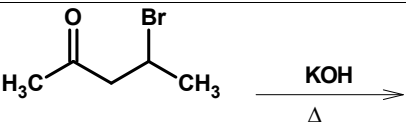
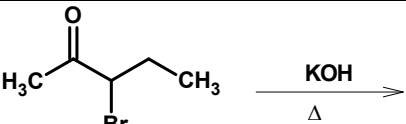
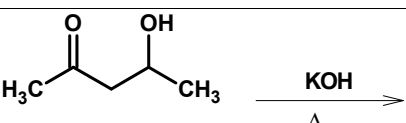
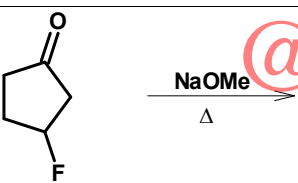
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E2 (Hoffmann)

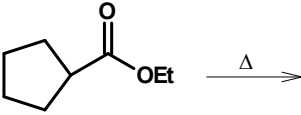
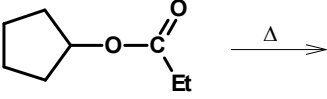
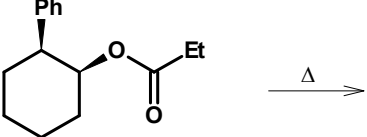
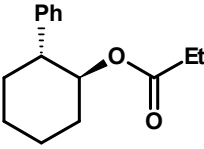
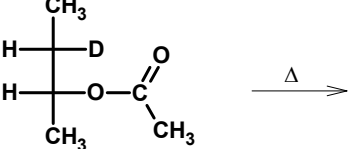
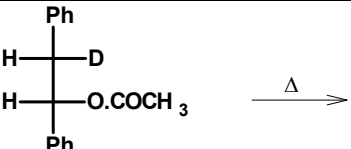
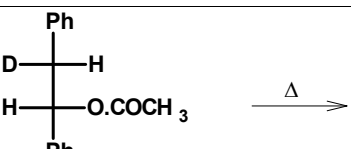
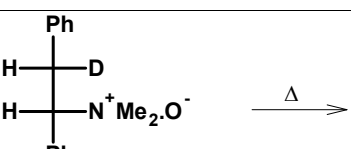
35.	$\text{H}_3\text{C}-\text{CH}_2-\text{CH}(\text{F})-\text{CH}_3 \xrightarrow[\text{C}_2\text{H}_5\text{OH}, \Delta]{\text{KOH}} \longrightarrow$
36.	$\text{H}_3\text{C}-\text{CH}_2-\text{CH}(\text{NMe}_3^+)-\text{CH}_3 \xrightarrow[\Delta]{\text{alc.KOH}} \longrightarrow$
37.	$\text{H}_3\text{C}-\text{CH}_2-\text{CH}(\text{SMe}_2^+)-\text{CH}_3 \xrightarrow[\Delta]{\text{alc.KOH}} \longrightarrow$
38.	$\text{CH}_3\text{CH}_2-\text{N}^+(\text{CH}_3)_2-\text{CH}_2\text{CH}_2\text{CH}_3 \xrightarrow[\Delta]{\text{EtOK}} \longrightarrow$
39.	$\text{CH}_3\text{CH}_2-\text{NH}-\text{CH}(\text{CH}_3)_2 \xrightarrow[2) \text{C}_2\text{H}_5\text{OH}, \Delta]{1) \text{CH}_3-\text{Cl}} \longrightarrow$
40.	$\text{H}_3\text{C}-\text{CH}_2-\text{CH}_2-\text{N}^+(\text{Me}_3) \xrightarrow[\Delta]{\text{alc.KOH}} \longrightarrow$
41.	$\text{Cyclohexyl}-\text{N}^+(\text{Me}_2)(\text{Et})(\text{n-Bu}) \cdot \text{OH}^- \xrightarrow{\Delta} \longrightarrow$
42.	$\text{Ph}-\text{CH}(\text{D})-\text{CH}_2\text{CH}_3 \cdot \text{I}^+ (\text{CH}_2\text{CH}_3)_3 \xrightarrow[\text{ii) } \Delta]{\text{i) Moist Ag}_2\text{O}} \text{Major product}$
43.	$\left[\text{H}_3\text{C}-\text{C}(\text{CH}_3)_2-\text{N}^+(\text{CH}_3)_2-\text{CH}_2\text{Ph} \right] \cdot \text{OH}^- \xrightarrow{\Delta} \longrightarrow$

44.	
45.	
46.	
47.	$\text{CH}_3\text{CH}_2\text{NH}_2 \xrightarrow[1) \text{ Moist Ag}_2\text{O}, \Delta]{1) \text{ CH}_3\text{-I (excess)}} \rightarrow$
48.	$\text{H}_3\text{C}-\text{CH}_2-\text{NH}-\text{CH}_2\text{CH}_3 \xrightarrow[2) \text{ Moist Ag}_2\text{O}, \Delta]{1) \text{ CH}_3\text{Cl (excess)}} \rightarrow$
49.	
50.	
51.	
52.	

E1 CB

53.	 <chem>COC(C)C[N+](=O)[O-]>[Na]OCC>C=C[N+](=O)[O-]</chem>
54.	 <chem>CC(=O)C(Br)C(C)C>[K]OH>CC(=O)C=C(C)C</chem>
55.	 <chem>CC(=O)C(Br)CC>[K]OH>CC(=O)C=CC</chem>
56.	 <chem>CC(=O)C(O)C(C)C>[K]OH>CC(=O)C=C(C)C</chem>
57.	 <chem>O=C1CCCC(F)1>[Na]OC>C=C1CCCC1=O</chem>

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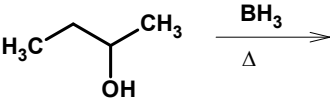
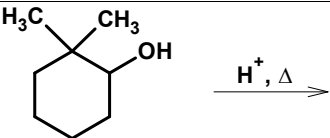
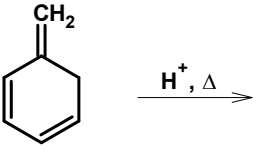
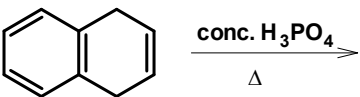
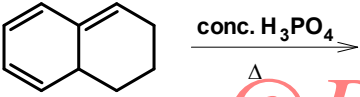
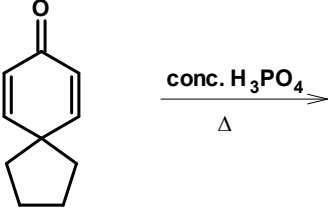
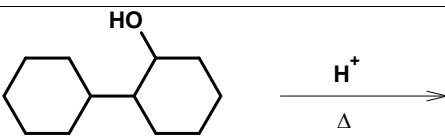
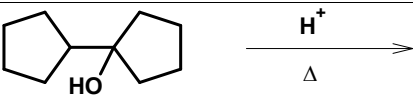
58.	$\text{H}_3\text{C}-\overset{\text{O}}{\parallel}{\text{C}}-\text{OCH}_2\text{CH}_3 \xrightarrow{\Delta}$
59.	 $\xrightarrow{\Delta}$
60.	 $\xrightarrow{\Delta}$
61.	 $\xrightarrow{\Delta}$
62.	 $\xrightarrow{\Delta}$
63.	 $\xrightarrow{\Delta}$
64.	 $\xrightarrow{\Delta}$
65.	 $\xrightarrow{\Delta}$
66.	 $\xrightarrow{\Delta}$

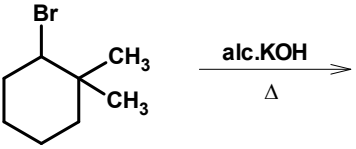
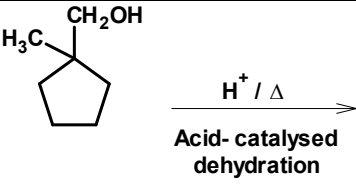
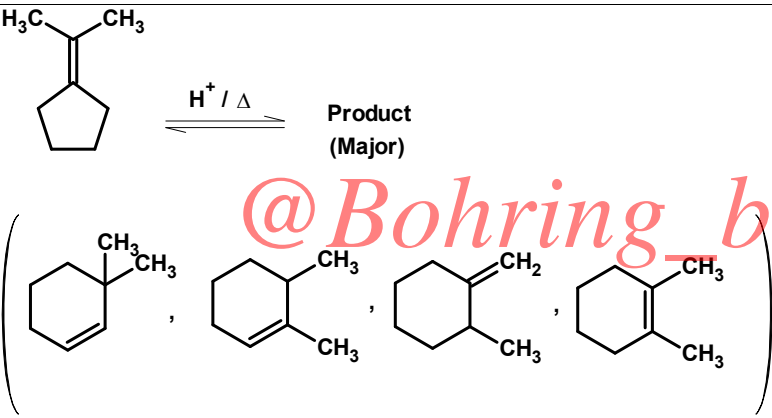
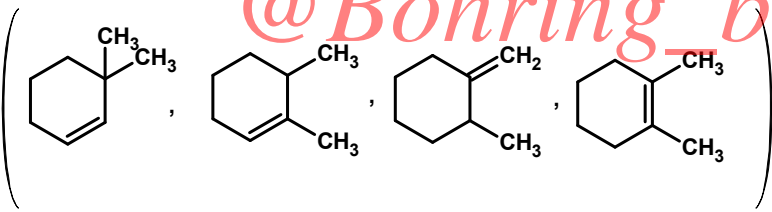
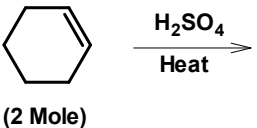
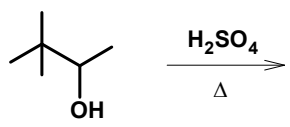
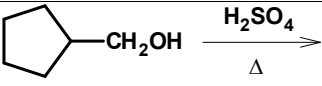
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67.	$ \begin{array}{c} \text{Ph} \\ \\ \text{D} - \text{C} - \text{H} \\ \\ \text{H} - \text{C} - \text{N}^+ \text{Me}_2 \text{O}^- \\ \\ \text{Ph} \end{array} \xrightarrow{\Delta} $
68.	$ \begin{array}{c} \text{Ph} \\ \\ \text{H} - \text{C} - \text{D} \\ \\ \text{H} - \text{C} - \text{COOC}_2\text{H}_5 \\ \\ \text{Ph} \end{array} \xrightarrow{\Delta} $
69.	$ \begin{array}{c} \text{Ph} \\ \\ \text{D} - \text{C} - \text{H} \\ \\ \text{H} - \text{C} - \text{COOC}_2\text{H}_5 \\ \\ \text{Ph} \end{array} \xrightarrow{\Delta} $
70.	$ (\text{CH}_3\text{CH}_2)_3\ddot{\text{N}} \xrightarrow[\Delta]{\text{H}_2\text{O}_2} $
71.	$ \begin{array}{c} \text{CH}_3 \\ \\ \text{Cyclopentane ring} - \text{O} - \text{C}(=\text{O}) - \text{CH}_3 \end{array} \xrightarrow{\Delta} $
72.	$ \text{H}_3\text{C} - \overset{\text{O}}{\parallel}{\text{C}} - \text{O} - \underset{\text{CH}_3}{\text{CH}} - \text{CH}_2\text{CH}_3 \xrightarrow{\Delta} $

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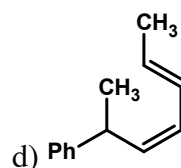
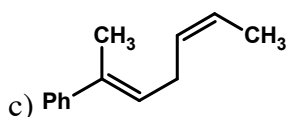
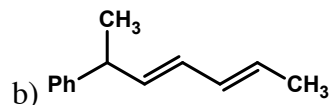
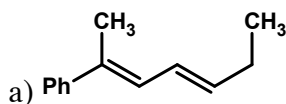
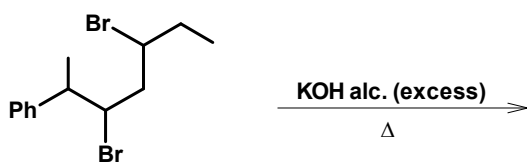
73.	$\begin{array}{c} \text{CH}_3 \\ \\ \text{H}_3\text{C}-\text{C}-\text{Br} \\ \\ \text{CH}_3 \end{array} \xrightarrow[\Delta]{\text{EtOH}}$
74.	$\text{Cyclohexene} \xrightarrow{\text{conc. H}_2\text{SO}_4}$ 2 Moles
75.	$\begin{array}{c} \text{CH}_3 \\ \\ \text{H}_3\text{C}-\text{C}-\text{CH}_2\text{OH} \\ \\ \text{CH}_3 \end{array} \xrightarrow[\Delta]{\text{BF}_3}$
76.	$\begin{array}{c} \text{O} \\ \\ \text{Cyclohex-2-en-1-one} \\ \quad \\ \text{H}_3\text{C} \quad \text{CH}_3 \end{array} \xrightarrow[\Delta]{\text{conc. H}_2\text{SO}_4}$
77.	Find the total number of possible products formed in following dehydration reaction $\begin{array}{c} \text{CH}_3 \\ \\ \text{H}_3\text{C}-\text{CH}-\text{CH}-\text{CH}_3 \\ \quad \\ \text{CH}_3 \quad \text{OH} \end{array} \xrightarrow[\Delta]{\text{conc. H}_2\text{SO}_4} \quad [\text{M}]$
78.	$\begin{array}{c} \text{H}_3\text{C} \\ \\ \text{H}_3\text{C}-\text{C}-\text{CH}_2\text{OH} \\ \\ \text{H}_3\text{C} \end{array} \xrightarrow[\Delta]{\text{H}^+}$
79.	$\text{Ph}-\text{CH}_2-\underset{\text{OH}}{\text{CH}}-\text{CH}_2-\text{Ph}-\text{CH}_3 \xrightarrow[\Delta]{\text{conc. H}_2\text{SO}_4} \quad \text{Number of products.}$
80.	$\text{C}_2\text{H}_5\text{OH} \begin{cases} \xrightarrow[140^\circ]{\text{conc. H}_2\text{SO}_4} \text{H}_5\text{C}_2-\text{O}-\text{C}_2\text{H}_5 \\ \xrightarrow[160^\circ - 170^\circ]{\text{conc. H}_2\text{SO}_4} \text{C}_2\text{H}_4 \end{cases}$

81.	
82.	
83.	
84.	
85.	
86.	
87.	
88.	
89.	
90.	$\text{R}-\text{Br} \xrightarrow[\text{C}_2\text{H}_5\text{OH}]{\text{C}_2\text{H}_5\text{OK}} [\text{Alkene}]_{\text{single only}} \xrightarrow[\text{Pd}]{\text{H}_2} \text{H}_3\text{C}-\text{CH}(\text{CH}_3)-\text{CH}_2-\text{CH}_3$ (2 - Methyl butane) Identify R-Br ?

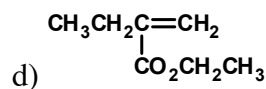
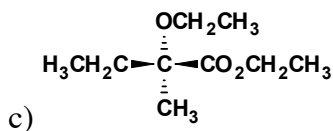
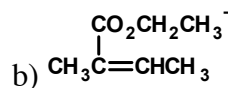
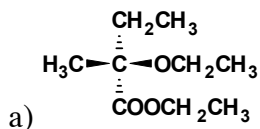
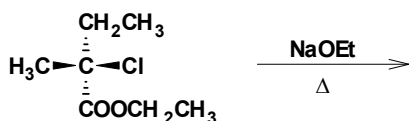
91.	 <chem>CC1(C)CCCCC1Br >> alc.KOH, \Delta</chem>
92.	C_4H_9Cl [A] $\xrightarrow[\text{(dehydro halogenation)}]{alc.KOH / \Delta}$ 3 isomeric alkene Identify [A] ?
93.	 <chem>CC1(CCCCC1)CO >> H^+, \Delta</chem> Acid-catalysed dehydration
94.	 <chem>CC(=C)C1CCCC1 >> H^+, \Delta</chem> Product (Major) 
95.	 <chem>C1=CCCCC1 >> H_2SO_4, \text{Heat}</chem> (2 Mole)
96.	 <chem>CC(C)C(C)O >> H_2SO_4, \Delta</chem>
97.	 <chem>C1CCCC1CO >> H_2SO_4, \Delta</chem>

PREVIOUS YEAR'S JEE-MAINS QUESTIONS

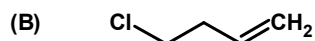
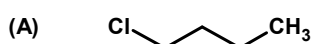
1. The major product of the following reaction is

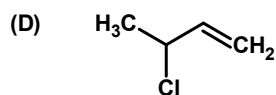
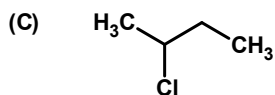


2. The major product of the following reaction is



3. The decreasing order of reactivity towards dehydrohalogenation (E_1) reaction of the following compounds is :





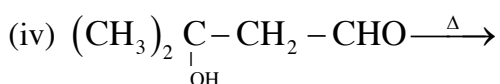
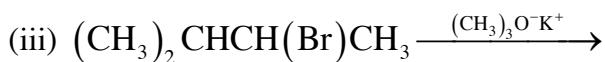
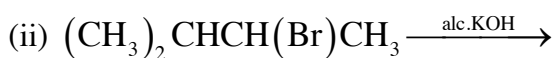
a) $B > A > D > C$

b) $B > D > A > C$

c) $D > B > C > A$

d) $B > D > C > A$

4. Consider the following reactions :



Which of these reaction(s) will not produce Saytzeff product ?

a) iii only

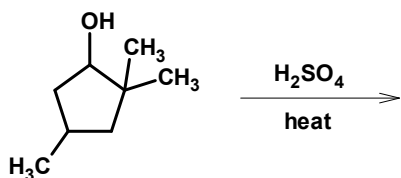
b) iv only

c) ii and iv

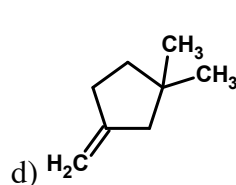
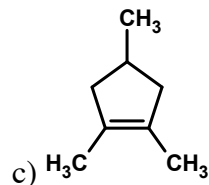
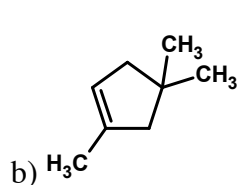
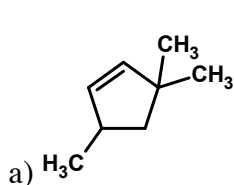
d) i, iii and iv

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5. The major product formed in the following reaction is :



Major product ?



PREVIOUS YEAR'S JEE-ADVANCED QUESTIONS

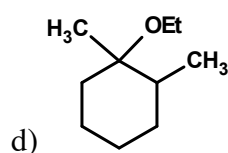
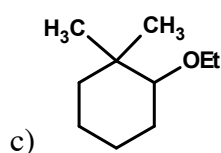
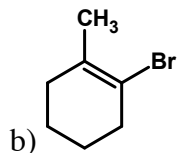
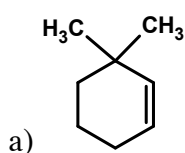
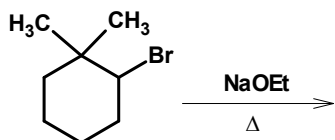
1. Match the following

	Column-I		Column-II
1)	$\text{O}_2\text{N}-\text{C}_6\text{H}_4-\text{CH}_2-\underset{\text{CH}_3}{\text{CH}}-\text{CH}_3 \xrightarrow[\text{C}_2\text{H}_5\text{OD}]{\text{High conc. C}_2\text{H}_5\text{O}^-} \text{O}_2\text{N}-\text{C}_6\text{H}_4-\text{CH}=\text{CH}-\text{CH}_3$	P)	Rearrangement
2)	$\text{H}_3\text{CO}-\text{C}_6\text{H}_4-\underset{\text{Br}}{\text{CH}}-\text{CH}_2-\text{CH}_3 \xrightarrow[\text{C}_2\text{H}_5\text{OD}]{\text{very - very low conc. C}_2\text{H}_5\text{O}^-}$ $\text{H}_3\text{CO}-\text{C}_6\text{H}_4-\text{CH}=\text{CH}-\text{CH}_3$	Q)	E ₁
3)	$\text{O}_2\text{N}-\text{C}_6\text{H}_4-\text{CH}_2-\underset{\text{Br}}{\text{CH}}-\text{CH}_3 \xrightarrow[\text{C}_2\text{H}_5\text{OD}]{\text{High conc. C}_2\text{H}_5\text{O}^-}$ $\text{O}_2\text{N}-\text{C}_6\text{H}_4-\text{CD}=\text{CH}-\text{CH}_3$	R)	E ₂
4)	$\text{H}_3\text{CO}-\text{C}_6\text{H}_4-\text{CHD}-\underset{\text{Br}}{\text{CH}}-\text{CH}_3 \xrightarrow[\text{C}_2\text{H}_5\text{OD}]{\text{very - very low conc. C}_2\text{H}_5\text{O}^-}$ $\text{H}_3\text{CO}-\text{C}_6\text{H}_4-\text{CH}=\text{CD}-\text{CH}_3$	S)	E ₁ CB

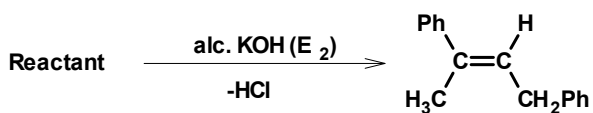
- | | | | | | | | | | |
|----|---|---|---|---|----|---|---|---|---|
| | P | Q | R | S | | P | Q | R | S |
| a) | 1 | 2 | 3 | 4 | b) | 1 | 3 | 2 | 4 |
| c) | 4 | 1 | 2 | 3 | d) | 4 | 2 | 1 | 3 |

TEST

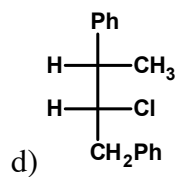
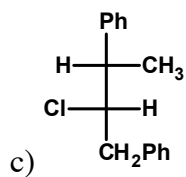
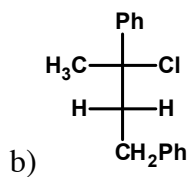
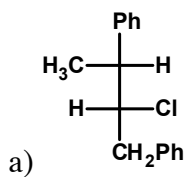
1. The major product of the following reaction is :



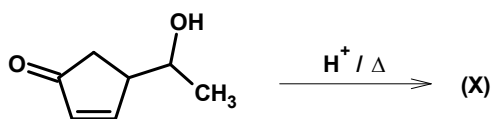
2.



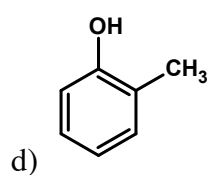
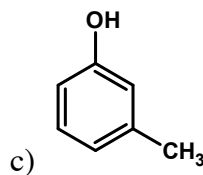
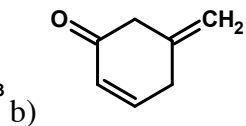
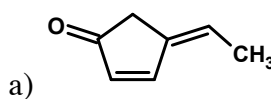
Which of the following molecule is NOT suitable reactant?

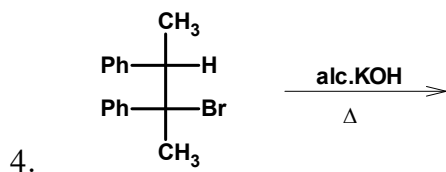


3.

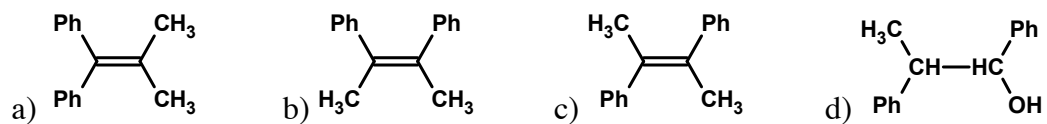


The major product (X) is



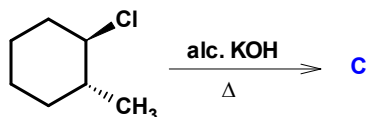
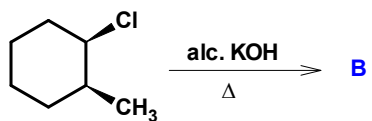
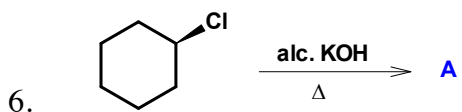


Major product of the reaction is :



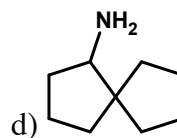
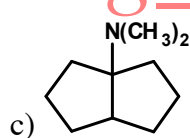
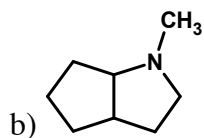
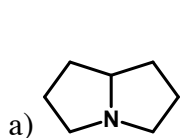
5.

	List-I (Reaction)		List-II (Major product formed by)
P)		1)	Syn addition
Q)		2)	Syn eliminatoin
R)		3)	Anti addition
S)		4)	Anti elimination
	P Q R S		P Q R S
A)	3 2 4 1	B)	3 1 4 2
C)	1 4 2 3	D)	3 4 2 1

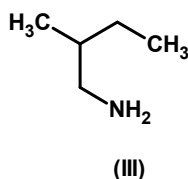
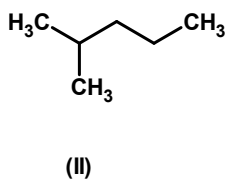
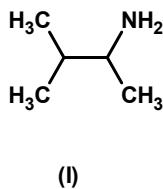


A, B, C are major product of the reactions and let x, y, z are the total number of allylic hydrogen present in A, B, C respectively then find $x + y - z$.

7. In how many of the given compound nitrogen atom may be removed as trimethyl amine by repeated Hofmann eliminations (exhaustive methylation followed by heating with AgOH) ?

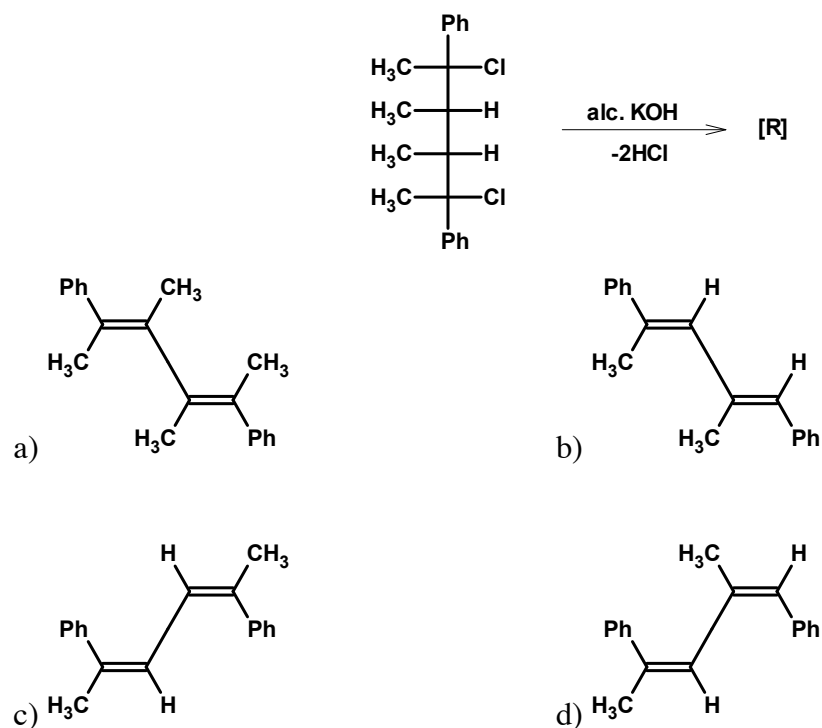


8. Which of the following compound(s) on exhaustive methylation following by elimination gives 3-methyl but-1-ene ?

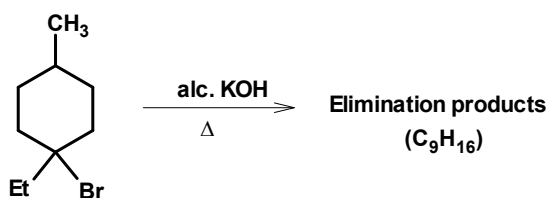


- a) only I b) I & II c) I, II & III d) only III

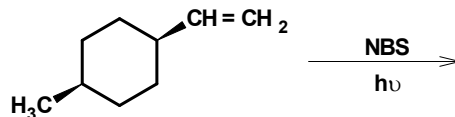
9. Find the major product formed in the following reaction



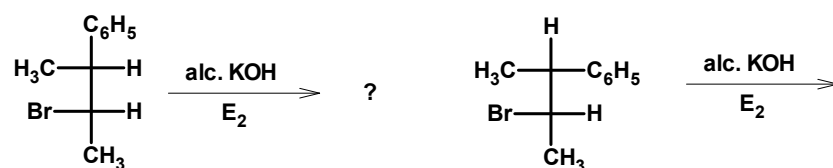
10. How many product(s) will be formed ?



11. Find the total number of monobromo products obtained during the reaction



12. Identify the correct statement(s) ?



- a) same products in both reaction
- b) both are constitutional isomers
- c) both are diastereomers
- d) both are meso compounds

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SYNOPSIS

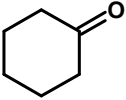
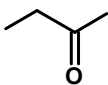
REAGENT	NAME REACTION	TEST	CONCEPTUAL POINTS
xii) HX / CCl ₄	*) Addition of HX	*) Bromine decoloration test	*) Kharash effect only for HBr*)
xiii) HX / peroxide (R ₂ O ₂)	*) Free radical addition	(Unsaturation test)	*) OMDM– Markonikov addition of H ₂ O without C ⁺ rearrangement
xiv) H ₃ O ⁺ or H ⁺ / H ₂ O	*) Hydration	*) Decoloration of KMnO ₄ (unsaturation test)	*) HBO– Anti-Markonikov addition of H ₂ O without C ⁺ rearrangement (syn)
xv) ROH / H ⁺	*) Addition of ROH		*) For addition of X ₂ / CCl ₄ formation of cyclic bromonium ion is RDS
*) i) Hg(OAc) ₂ , H ₂ O	*) Oxymercuration Demercuration (OMDM)		*) For rxn involving, non classical carbocation, the electronic effect runs through bonds (Inductive effect) until resonance stabilised C ⁺ is formed.
ii) NaBH ₄	*) Hydroboration oxidation (HBO)		
*) i) SiA ₂ BH or BH ₃ – THF	*) Additoin of conc. H ₂ SO ₄		
ii) H ₂ O ₂ , NaOH	*) dimerisation		
*) cold. conc H ₂ SO ₄	*) isomerisation		
*) H ⁺ / Δ	*) Addition of X ₂		
*) X ₂ / CCl ₄ (X = Cl ₂ , Br)	*) Syn hydroxylation		
*) X ₂ / H ₂ O	*) Anti hydroxylation.		
*) NBS / Δ or Br ₂ / Δ			
*) cold, dil, aq.sol. of KMnO ₄ (1% alkaline)			
*) 1) m-CPBA			
2) H ₂ O / H ⁺			
*) OsO ₄ – NaHSO ₃ / H ₂ O			

*) 1) $\text{Ag}_2\text{O} / \Delta$

2) H_2O

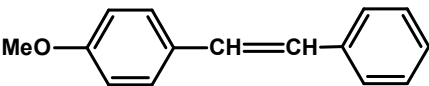
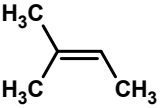
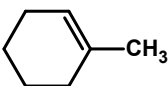
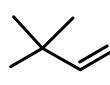
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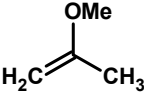
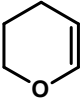
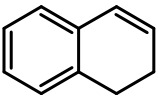
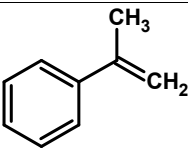
1. Wittig reaction

1.	$\text{HCH}_3\text{C=O} + \text{PPh}_3\text{-}\ddot{\text{C}}\text{H}_2 \longrightarrow$
2.	$\text{H}_3\text{C}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_3 + \text{PPh}_3\text{-}\ddot{\text{C}}\text{H}_2 \longrightarrow$
3.	 + $\text{PPh}_3\text{-}\ddot{\text{C}}\text{H}_2 \longrightarrow$
4.	 + $\text{PPh}_3\text{-}\ddot{\text{C}}\text{H}_2 \longrightarrow$
5.	 + $\text{PPh}_3\text{-}\ddot{\text{C}}\text{H}_2 \longrightarrow$
6.	$\text{Ph}_3\text{P} \xrightarrow[2) \text{ BuLi}]{1) \text{ CH}_3\text{-I}}$ 3) 

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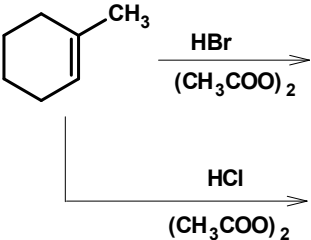
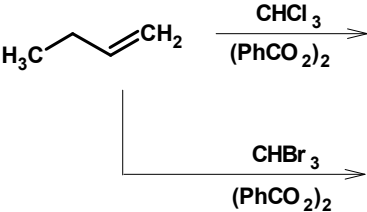
2. Addition of H-X

7.	 $\xrightarrow[\text{CCl}_4]{\text{HBr}}$
8.	 $\xrightarrow[\text{CCl}_4]{\text{HBr}}$
9.	 $\xrightarrow[\text{CCl}_4]{\text{HBr}}$
10.	 $\xrightarrow[\text{CCl}_4]{\text{HBr}}$

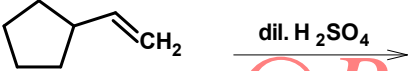
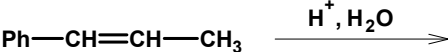
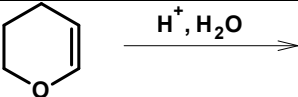
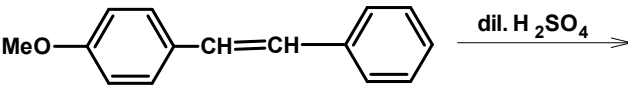
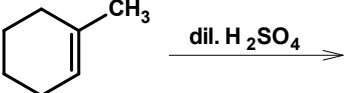
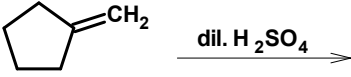
11.	$\text{Ph} - \text{CH} = \text{CH}_2$ (styrene) $\xrightarrow[\text{CCl}_4]{\text{HBr}}$
12.	$\text{Ph} - \text{CH} = \text{CH} - \text{Ph}$ (stilbene) $\xrightarrow[\text{CCl}_4]{\text{HBr}}$
13.	 $\xrightarrow[\text{CCl}_4]{\text{HBr}}$
14.	 $\xrightarrow[\text{CCl}_4]{\text{HBr}}$
15.	 $\xrightarrow[\text{CCl}_4]{\text{HBr}}$
16.	 $\xrightarrow[\text{CCl}_4]{\text{HBr}}$
17.	$\text{H}_3\text{C}-\text{O}-\text{CH}=\text{CH}_2$ $\xrightarrow[\text{CCl}_4]{\text{HBr}}$
18.	$\text{H}_2\text{C}=\text{CH}-\underset{\text{Br}}{\text{CH}_2}$ $\xrightarrow[\text{CCl}_4]{\text{HBr}}$
19.	$\text{H}_2\text{C}=\text{CH}-\text{CF}_3$ $\xrightarrow[\text{CCl}_4]{\text{HBr}}$

3. Addition of H-X/Peroxide

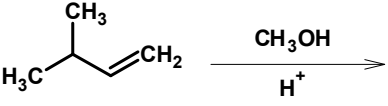
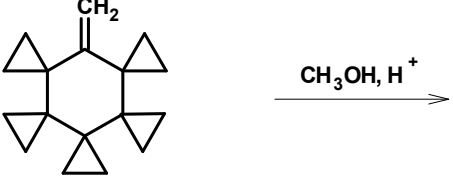
20.	$\text{H}_3\text{C}-\text{CH}=\text{CH}_2$ $\xrightarrow[\text{(PhCOO)}_2]{\text{HCl}}$
21.	$\text{H}_3\text{C}-\text{CH}=\text{CH}_2$ $\xrightarrow[\text{(PhCOO)}_2]{\text{HBr}}$

22.	
23.	

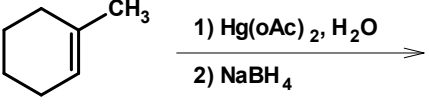
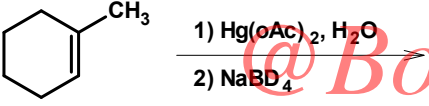
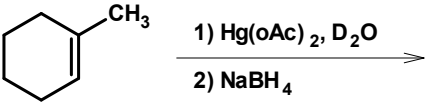
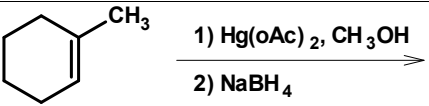
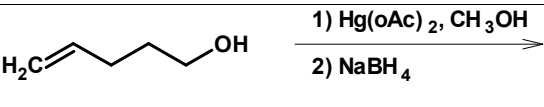
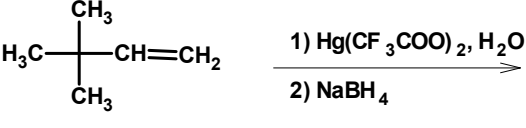
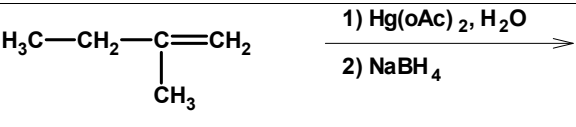
4. Hydration

24.	
25.	
26.	
27.	
28.	
29.	

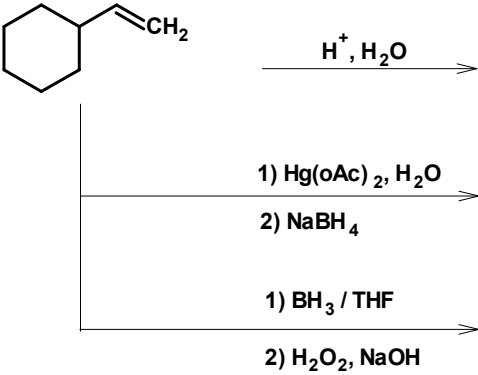
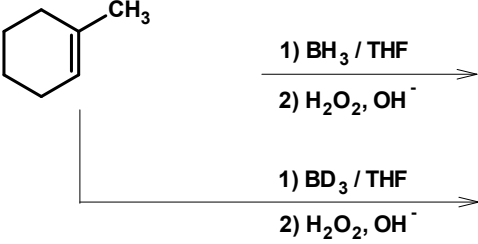
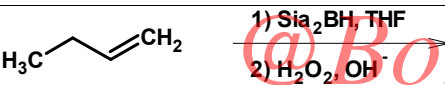
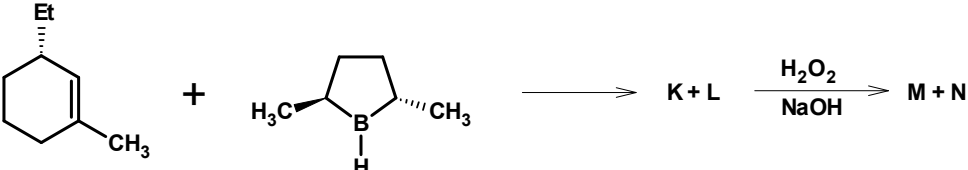
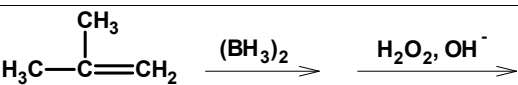
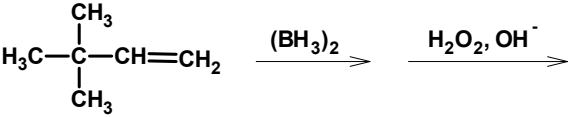
5. Addition of R-OH

30.	 <chem>CC(C)=CC.O>>CC(C)(O)CC</chem>
31.	 <chem>CC1=CCCCC1.CO>>CC1(O)CCCCC1</chem>

6. Oxymercuration - demercuration

32.	 <chem>CC1=CCCCC1.O=C(O)C(=O)O.O>>CC1(O)CCCCC1</chem>
33.	 <chem>CC1=CCCCC1.O=C(O)C(=O)O.O>>CC1(O)CCCCC1</chem>
34.	 <chem>CC1=CCCCC1.O=C(O)C(=O)O.O>>CC1(O)CCCCC1</chem>
35.	 <chem>CC1=CCCCC1.O=C(O)C(=O)O.CO>>CC1(O)CCCCC1</chem>
36.	 <chem>OCC=CCC.O=C(O)C(=O)O.CO>>OCC(O)CCC</chem>
37.	 <chem>CC(C)(C)=CC.O=C(O)C(F)(F)F(=O)O.O>>CC(C)(O)CC</chem>
38.	 <chem>CC(C)=CCC.O=C(O)C(=O)O.O>>CC(C)(O)CCC</chem>

7. Hydroboration - oxidation

39.	
40.	
41.	
42.	
43.	
44.	

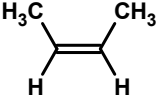
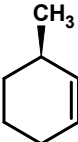
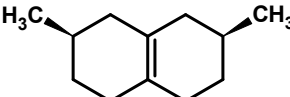
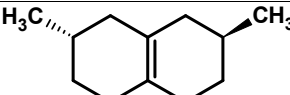
8. Addition of Br₂

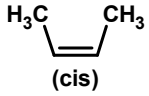
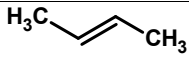
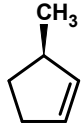
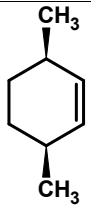


53.	$ \begin{array}{c} \text{CH}_3 \\ \\ \text{H}-\text{C}-\text{Br} \\ \\ \text{CH} \\ \\ \text{CH} \\ \\ \text{H}-\text{C}-\text{Br} \\ \\ \text{CH}_3 \end{array} \xrightarrow[\text{CCl}_4]{\text{Br}_2} $ (Trans)
54.	$ \text{C}_4\text{H}_8 \xrightarrow[\text{CCl}_4]{\text{Br}_2} \text{[M], number of products} $ (all isomeric alkene)
55.	$ \text{Cyclohexene with a methyl group} \xrightarrow[\text{CHCl}_3]{\text{Br}_2} \text{K + L} $
56.	$ \begin{array}{l} \text{H}_3\text{C} \\ \diagup \\ \text{C}=\text{CH}_2 \\ \diagdown \\ \text{H}_3\text{C} \end{array} \xrightarrow[\text{CHCl}_3]{\text{Br}_2} $ $\begin{array}{l} \xrightarrow{\text{Br}_2} \\ \xrightarrow{\text{H}_2\text{O}} \\ \xrightarrow{\text{Br} - \text{Cl}} \\ \xrightarrow{\text{I} - \text{Cl}} \\ \xrightarrow{\text{HOCl}} \end{array}$ @Bohring_bot
57.	$ \begin{array}{l} \text{H}_2\text{C}=\text{CH}-\text{CH}_2 \\ \\ \text{Br} \end{array} \xrightarrow[\text{H}_2\text{O}]{\text{Br}_2^{82}} $ $\begin{array}{l} \xrightarrow[\text{CHCl}_3]{\text{HBr}^{82}} \\ \xrightarrow[\text{H}_2\text{O}]{\text{H}^+} \end{array}$
58.	$ \text{H}_3\text{C}-\text{CH}=\text{CH}_2 \xrightarrow[\text{AlCl}_3 (\text{Anhyd})]{\text{R}-\text{Cl}} $
59.	$ \text{H}_3\text{C}-\text{CH}=\text{CH}_2 \xrightarrow[\text{AlCl}_3 (\text{Anhyd})]{\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{Cl}} $

60.	$\text{H}_3\text{C}-\text{CH}=\text{CH}_2 \xrightarrow[\text{ZnCl}_2]{\text{I}-\text{Cl}}$
61.	 $\xrightarrow[\text{CCl}_4]{\text{Br}_2}$

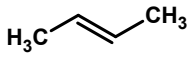
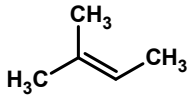
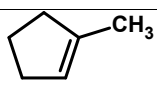
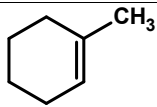
9. cold, dilute & aq. KMnO_4

62.	 $\xrightarrow[\text{KMnO}_4, 273 \text{ K}]{\text{cold, dilute}}$
63.	$\text{H}_3\text{C}-\text{CH}=\text{CH}_2 \xrightarrow[\text{KMnO}_4, 273 \text{ K}]{\text{cold, dilute}}$
64.	 $\xrightarrow[\text{KMnO}_4, 273 \text{ K}]{\text{cold, dilute}}$
65.	$\text{cis - 2 - pentene} \xrightarrow[\text{KMnO}_4, 273 \text{ K}]{\text{cold, dilute}}$
66.	$\text{trans - 2 - pentene} \xrightarrow[\text{KMnO}_4, 273 \text{ K}]{\text{cold, dilute}}$
67.	$\text{H}_3\text{C}-\text{CH}=\text{CH}-\text{CH}_3 \xrightarrow[2) \text{ NaHSO}_3, \text{H}_2\text{O}, \text{H}_2\text{O}_2]{1) \text{ OsO}_4}$
68.	 $\xrightarrow[2) \text{ NaHSO}_3, \text{H}_2\text{O}_2]{1) \text{ OsO}_4}$
69.	 $\xrightarrow{\text{KMnO}_4, 273 \text{ K}}$
70.	 $\xrightarrow{\text{KMnO}_4, 273 \text{ K}}$

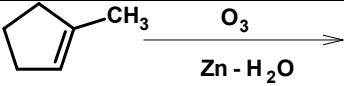
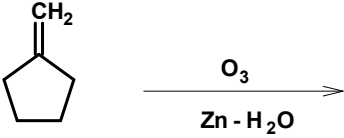
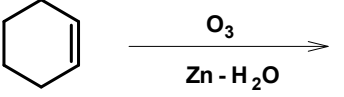
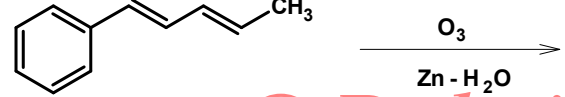
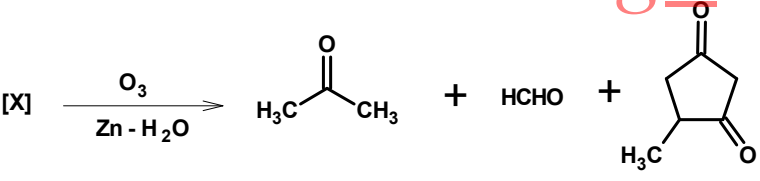
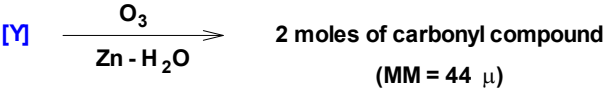
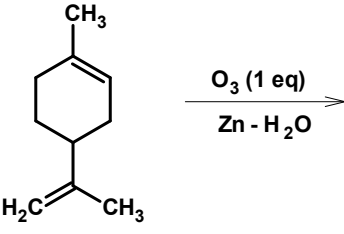
71.	 $\xrightarrow[2) \text{H}_3\text{O}^+]{1) \text{m - CPBA}}$
72.	 $\xrightarrow[2) \text{H}_3\text{O}^+]{1) \text{m - CPBA}}$
73.	 $\xrightarrow[2) \text{H}_3\text{O}^+]{1) \text{Ag}_2\text{O} / \Delta}$
74.	 $\xrightarrow[2) \text{H}_3\text{O}^+]{1) \text{m - CPBA}}$

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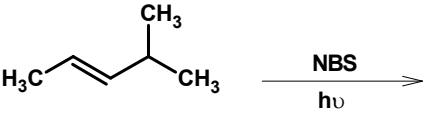
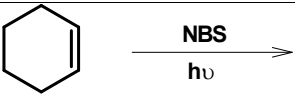
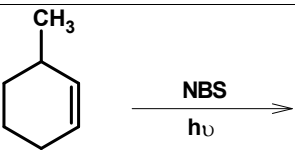
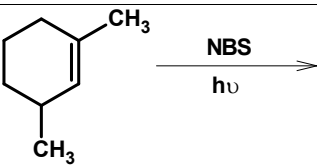
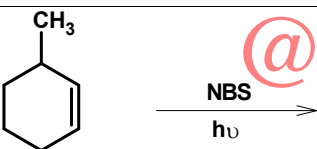
10. Hot & conc. KMnO_4

75.	 $\xrightarrow[\text{H}^+, \text{Heat}]{\text{KMnO}_4, 273 \text{ K}}$
76.	 $\xrightarrow[\text{H}^+, \text{Heat}]{\text{KMnO}_4}$
77.	 $\xrightarrow[\text{KOH, Heat}]{\text{KMnO}_4}$
78.	 $\xrightarrow[\text{KOH, Heat}]{\text{KMnO}_4}$
79.	$\text{Alkene} \xrightarrow[\text{KOH, Heat}]{\text{KMnO}_4} \xrightarrow{\text{H}^+} \text{HOOC}-(\text{CH}_2)_4-\text{COOH}$ (Adipic acid)

11. Ozonolysis

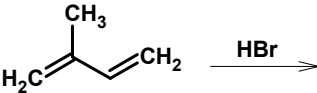
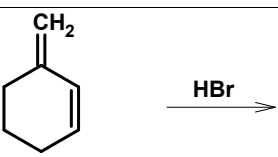
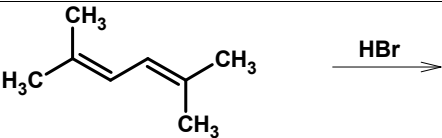
80.	 <chem>CC1=CCCC1.O=[O]>>[Zn-H2O]</chem>
81.	 <chem>C=C1CCCC1.O=[O]>>[Zn-H2O]</chem>
82.	 <chem>C1=CCCCC1.O=[O]>>[Zn-H2O]</chem>
83.	 <chem>C1=CCCC2CCCC2C1.O=[O]>>[Zn-H2O]</chem>
84.	 <chem>CC=CC=Cc1ccccc1.O=[O]>>[Zn-H2O]</chem>
85.	 <chem>[X].O=[O]>>[Zn-H2O]CC(=O)C.C=O.CC1CCC(=O)C1=O</chem>
86.	 <chem>[Y].O=[O]>>[Zn-H2O]2 C4H8O (MM = 44)</chem>
87.	 <chem>CC(=C)C1=CC(C)CCC1.O=[O]>>[Zn-H2O]</chem>

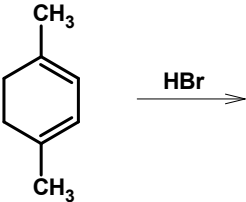
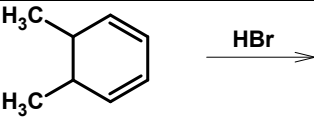
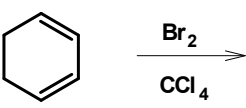
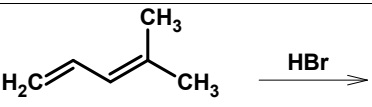
12. NBS

88.	 <chem>CC(=C)C(C)C</chem> $\xrightarrow[h\nu]{\text{NBS}}$
89.	 <chem>C1=CCCCC1</chem> $\xrightarrow[h\nu]{\text{NBS}}$
90.	 <chem>CC1=CCCCC1</chem> $\xrightarrow[h\nu]{\text{NBS}}$
91.	 <chem>CC1=CC(C)CCC1</chem> $\xrightarrow[h\nu]{\text{NBS}}$
92.	 <chem>CC1=CCCCC1</chem> $\xrightarrow[h\nu]{\text{NBS}}$

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13) 1,2 & 1,4 - addition

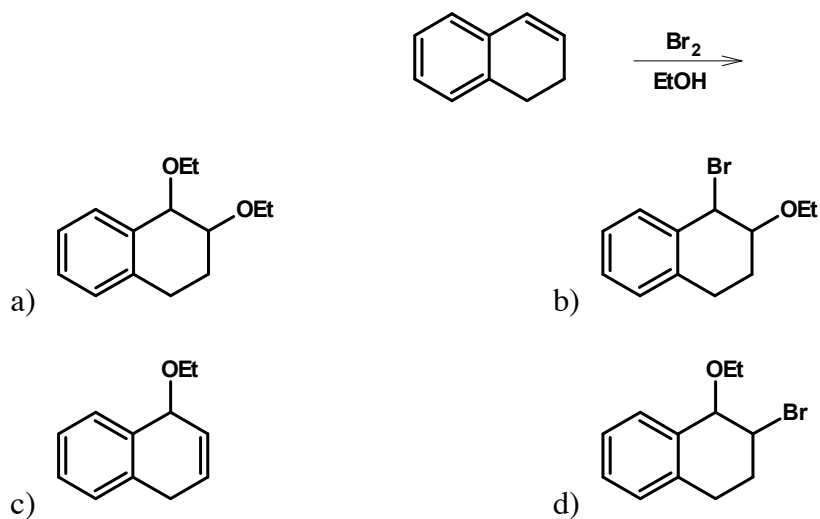
93.	 <chem>CC(=C)C=C</chem> $\xrightarrow{\text{HBr}}$
94.	 <chem>C=C1CCCCC1</chem> $\xrightarrow{\text{HBr}}$
95.	 <chem>CC(C)=CC(=C)C(C)C</chem> $\xrightarrow{\text{HBr}}$

96.	 Reaction of 1,4-dimethylbenzene (p-xylene) with HBr.
97.	 Reaction of 1,2-dimethylbenzene (o-xylene) with HBr.
98.	 Reaction of benzene with Br_2 in CCl_4.
99.	 Reaction of 2-methyl-2-pentene with HBr.

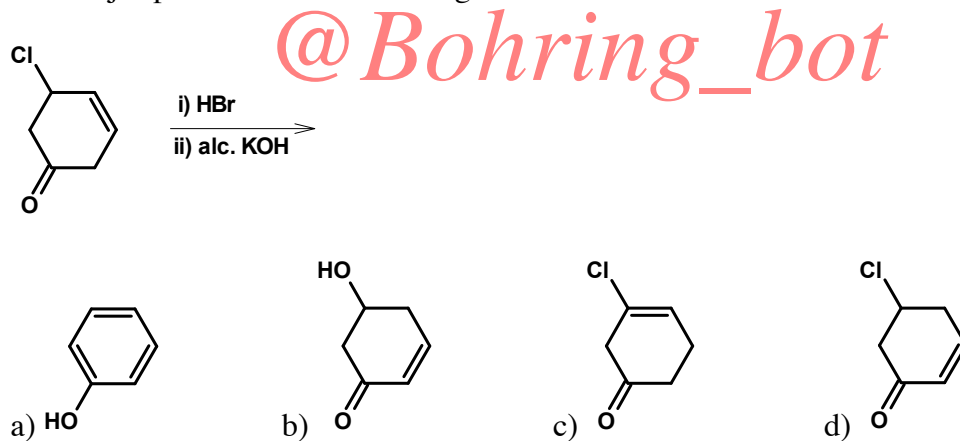
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PREVIOUS YEAR'S JEE-MAINS QUESTIONS

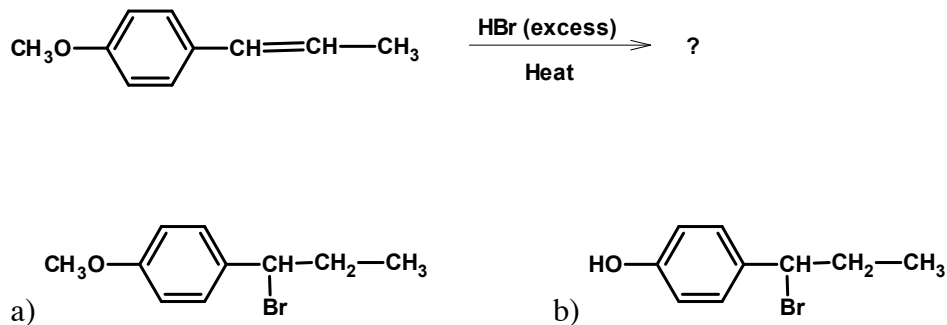
1. The major product of the following reaction is

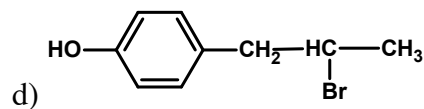
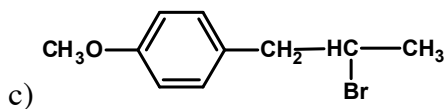


2. The major product of the following reaction is

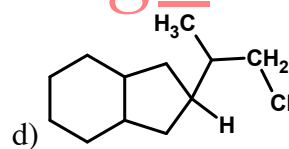
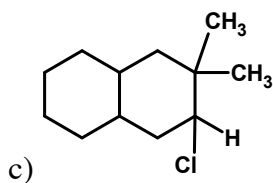
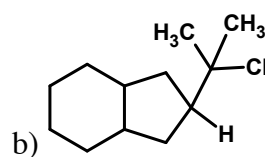
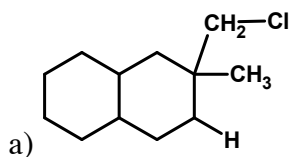
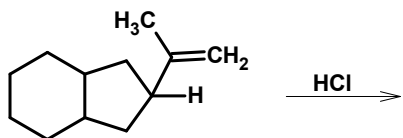


3. The major product of the following conversion is



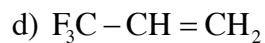
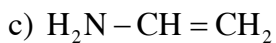
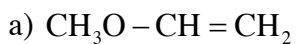


4. The major product of the following reaction is

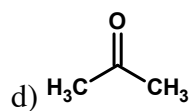
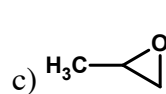
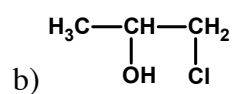
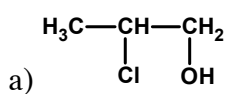
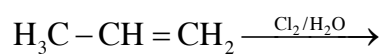


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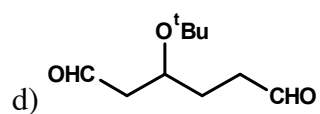
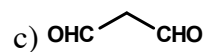
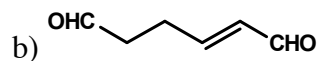
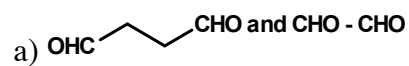
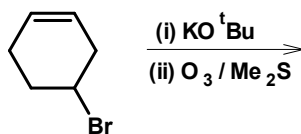
5. Which one of the following alkenes when treated with HCl yields majorly an anti Markovnikov product ?



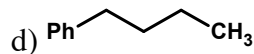
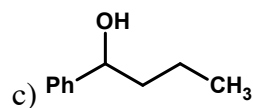
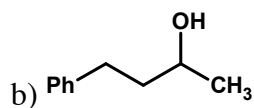
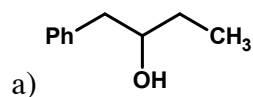
6. The major product of the following addition reaction is



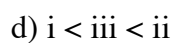
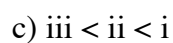
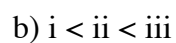
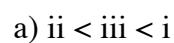
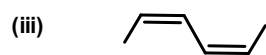
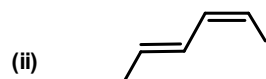
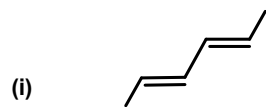
7. The major product(s) obtained in the following reaction is/are :



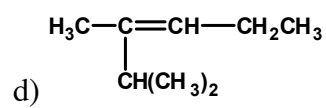
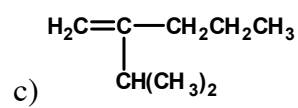
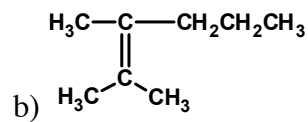
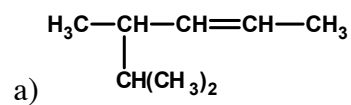
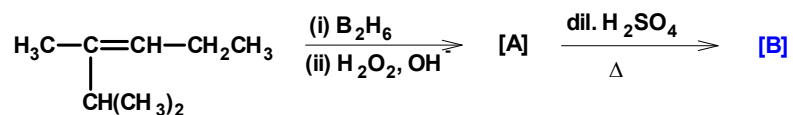
8. Heating of 2-chloro-1-phenylbutane with EtOK/EtOH gives X as the major product. Reaction of X with $\text{Hg}(\text{OAc})_2 / \text{H}_2\text{O}$ followed by NaBH_4 gives Y as the major product. Y is :



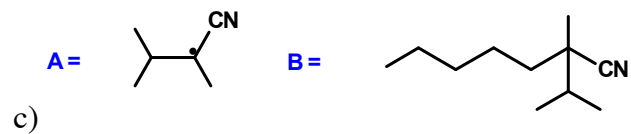
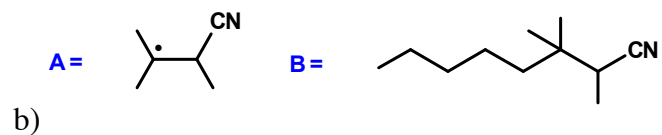
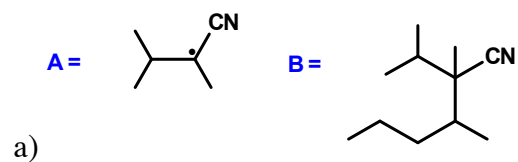
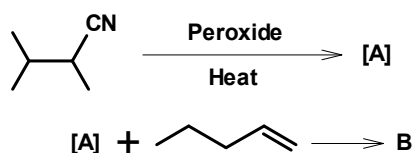
9. The correct order heat of combustion for following alkadienes is :

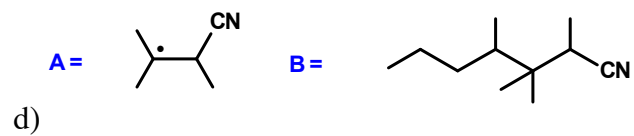


10. The major product [B] in the following sequence of reactions is :

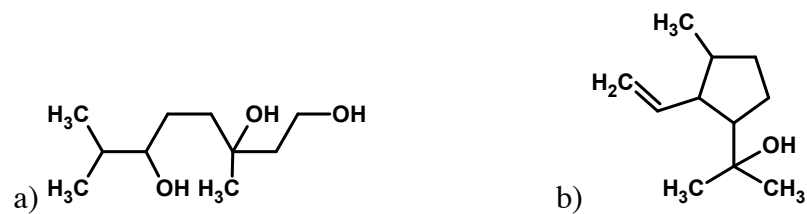
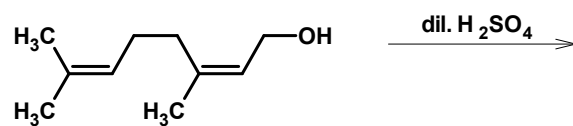


11. The major products A and B in the following reactions are :

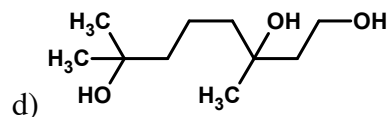
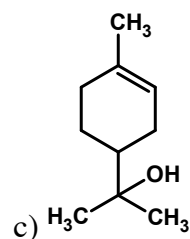




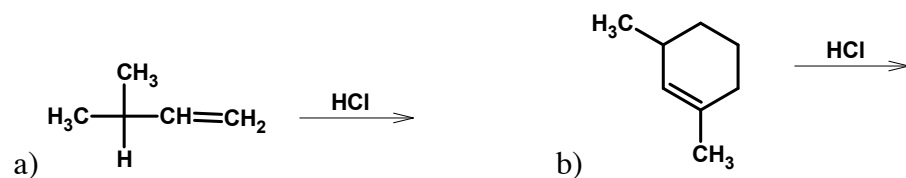
12. The major product of the following reaction is :



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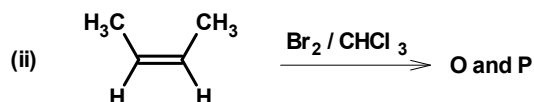
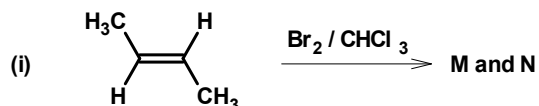


13. Which of the following reactions will not produce a racemic product ?

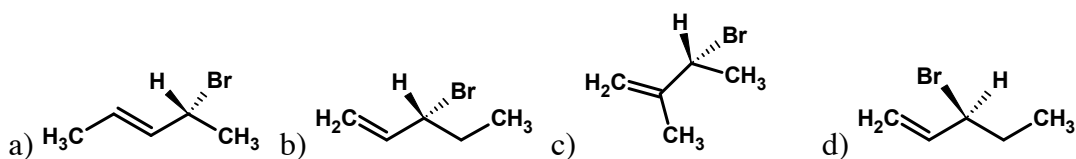


PREVIOUS YEAR'S JEE-ADVANCED QUESTIONS

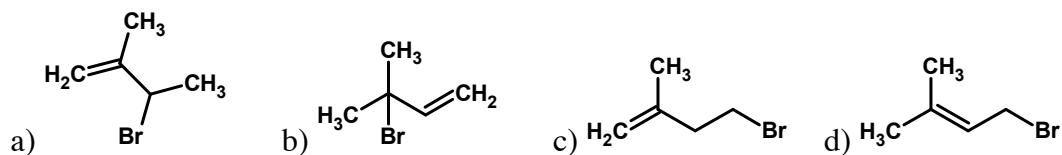
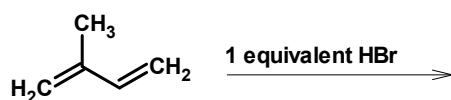
1. The correct statement(s) for the following addition reactions is (are)



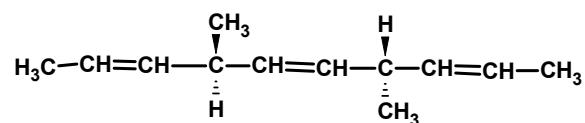
- a) **O** and **P** are identical molecules
- b) (**M** and **O**) and (**N** and **P**) are two pairs of diastereomers
- c) (**M** and **O**) and (**N** and **P**) are two pairs of enantiomers
- d) Bromination proceeds through trans-addition in both the reactions
2. Compound(s) that on hydrogenation produce(s) optically inactive compound(s) is (are)



3. In the following reaction, the major product is

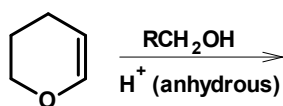


4. The number of optically active products obtained from the complete ozonolysis of the given compound is



- a) 0 b) 1 c) 2 d) 4

5. The major product of the following reaction is



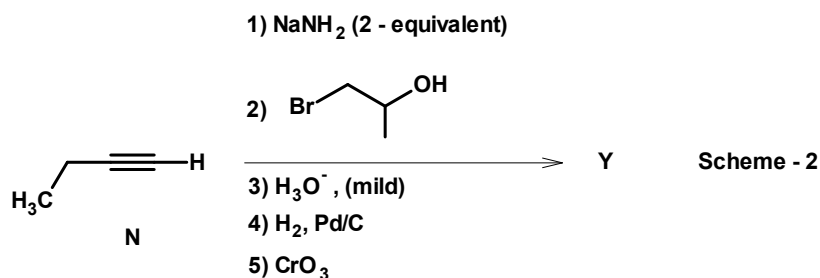
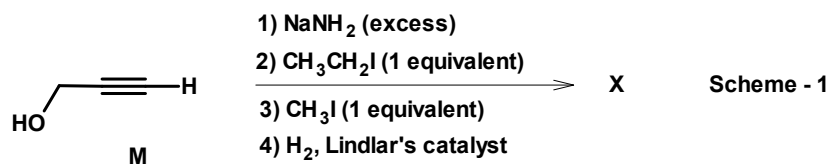
- a) a hemiacetal b) an acetal c) an ether d) an ester

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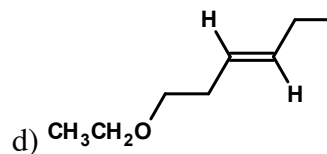
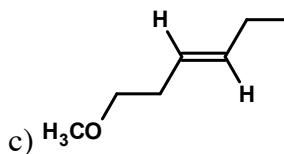
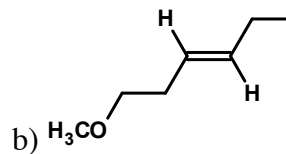
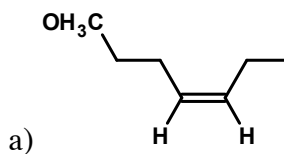
PARAGRAPH

PASSAGE-1

Schemes 1 and 2 describe sequential transformation of alkynes M and N. Consider only the major products formed in each step for both the schemes.



1. The product 'X' is

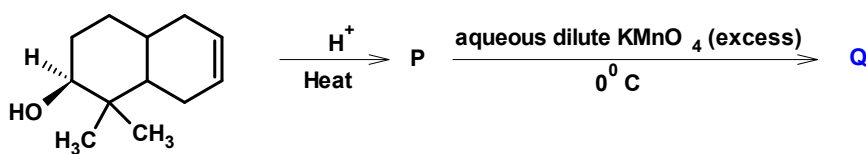


2. The correct statement with respect to product 'Y' is

- a) It gives a positive Tollens test and is a functional isomer of **X**.
- b) It gives a positive Tollens test and is a geometrical isomer of **X**.
- c) It gives a positive iodoform test and is a functional isomer of **X**.
- d) It gives a positive iodoform test and is a geometrical isomer of **X**.

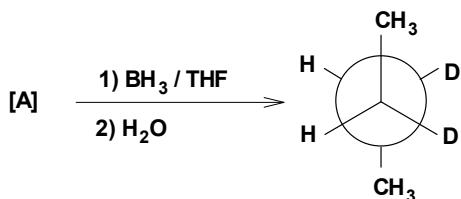
INTEGER

1. The number of hydroxyl group(s) in **Q** is

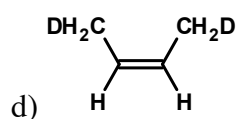
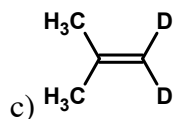
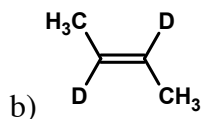
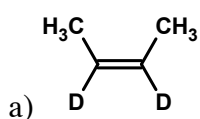


TEST

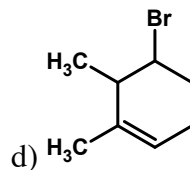
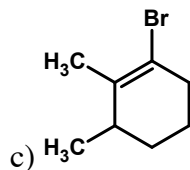
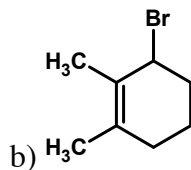
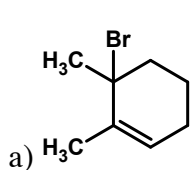
1.



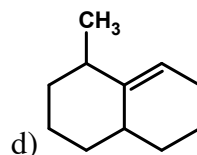
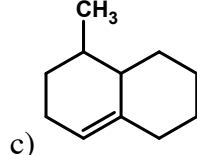
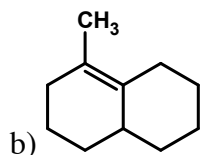
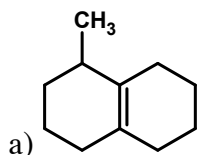
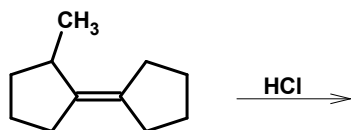
[A] is



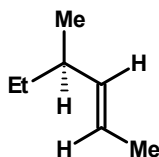
2. Addition of HBr to 2,3-dimethyl-1,3-cyclohexadiene may occur in the absence or presence of peroxides. In each case two isomeric $\text{C}_8\text{H}_{13}\text{Br}$ products are obtained. Which of the following is a common product from both reactions ?



4. The major product of the following reaction is :

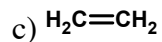
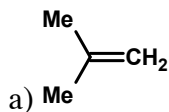


5. Correct statement(s) regarding addition of Br_2 to the following compound is/are

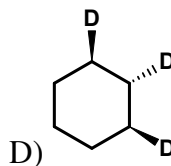
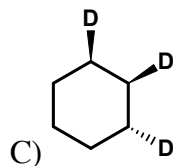
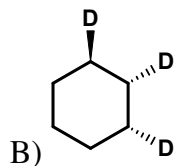
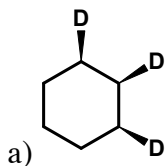
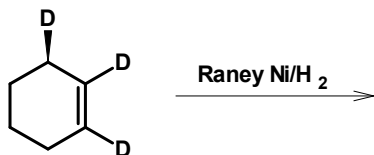


- a) Br_2 undergoes anti addition
 b) two diastereometric products are formed
 c) cyclic bromonium ion is formed as the intermediate
 d) addition proceeds via syn stereochemistry resulting in enantiomeric products.
6. When an aqueous solution of MeCOOK and $\text{Me}_3\text{C.COOK}$ is electrolysed, possible products is/are

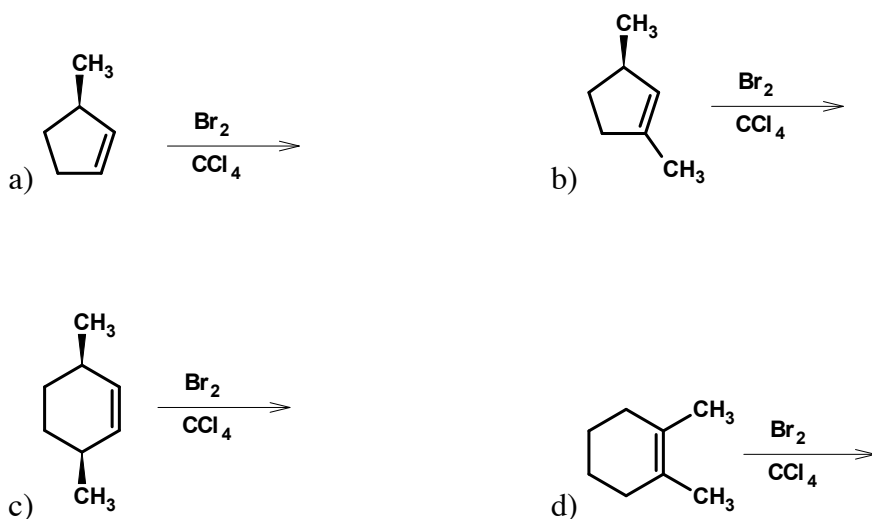
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7. Identify correct products which will be obtained on hydrogenation of



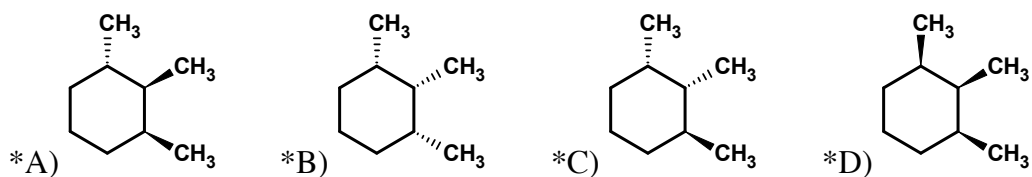
8. The reaction in which Diastereomeric product obtained is/are



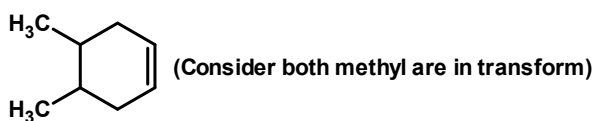
9.



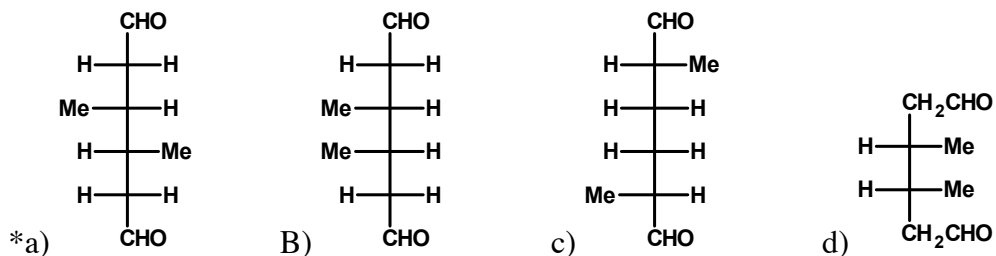
[M] will be



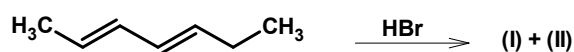
10. Consider the ozonolysis of trans-4,5-dimethyl cyclohexene



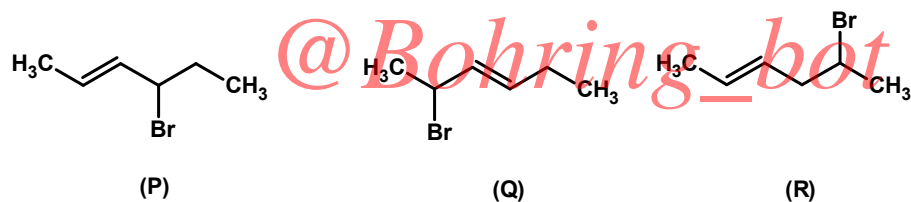
Which product(s) is/are formed in the above reaction ?



11. In the addition of HBr to 2,4-hexadiene, two major products (I & II) are formed and almost none of compound III is formed, identify I & II.

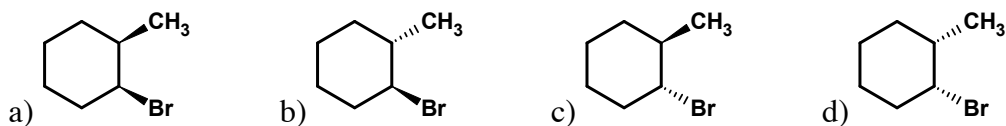
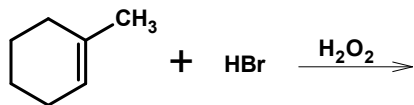


Find I & II

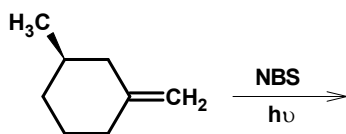


- A) P = I B) Q = II C) R = I D) P = I, R = II

12. Which of the following can be the product(s) for given reaction



13. Observe the following reaction :



Pick the correct statement :

- *a) Six products, including stereoisomers would be formed
- b) All products formed in reaction, are showing geometrical isomerism
- c) Only two constitutional isomers formed
- *d) All products are chiral

Matching type

1.

	List-I (Reaction)		List-II (Major product formed by)
P)		1)	Syn addition
Q)		2)	Syn elimination
R)		3)	Anti addition
S)		4)	Anti elimination

	P	Q	R	S		P	Q	R	S
A)	3	2	4	1	B)	3	1	4	2
C)	1	4	2	3	*D)	3	4	2	1

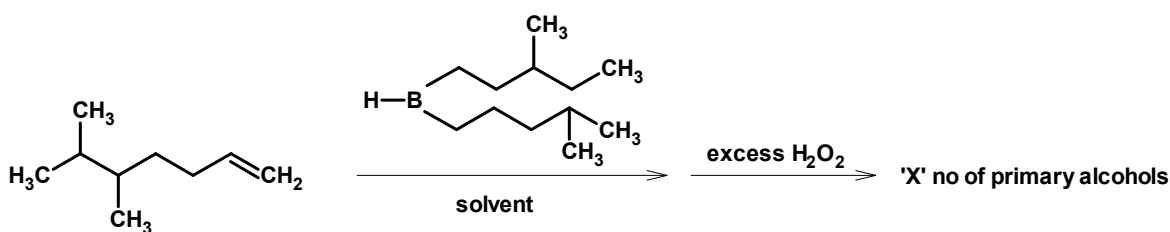
2. Matching type

	List-I (Reaction)		List-II (Major product formed by)
P)	$\text{H}_2\text{C}=\underset{\text{Br}}{\text{CH}}-\text{CH}_2 \xrightarrow[\text{H}_2\text{O}]{\text{Br}_2} \text{H}_2\underset{\text{OH}}{\text{C}}-\underset{\text{Br}}{\text{CH}}-\text{CH}_2$	1)	Major product
Q)	$\text{H}_2\text{C}=\underset{\text{Br}}{\text{CH}}-\text{CH}_2 \xrightarrow[\text{H}_2\text{O}]{\text{Br}_2} \text{H}_2\underset{\text{Br}}{\text{C}}-\underset{\text{OH}}{\text{CH}}-\underset{\text{Br}}{\text{CH}_2}$	2)	Minor product
R)	$\text{H}_2\text{C}=\text{CH}-\text{Cl} \xrightarrow[\text{CCl}_4]{\text{HBr}} \text{H}_3\text{C}-\underset{\text{Br}}{\text{CH}}-\text{Cl}$	3)	Trans addition
S)	$\text{H}_2\text{C}=\text{CH}-\text{CH}_3 \xrightarrow[2) \text{H}_2\text{O}, \text{NaOH}]{1) \text{BH}_3 / \text{THF}} \text{H}_2\underset{\text{OH}}{\text{C}}-\text{CH}_2-\text{CH}_3$	4)	Syn addition

Ans: A-PR, B-QR, C-P, D-PS

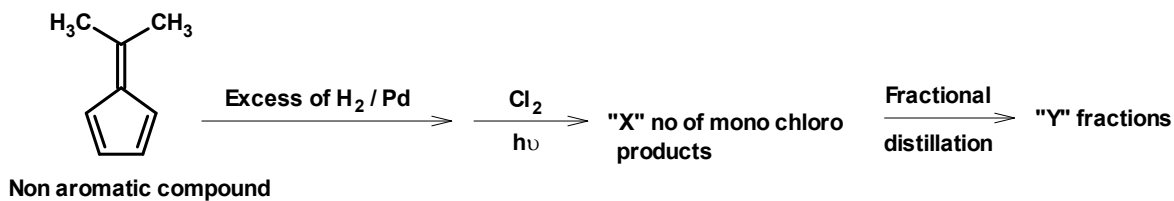
INTEGER

1.



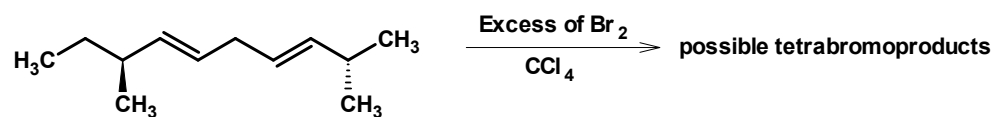
Value of "X" is.....

2.

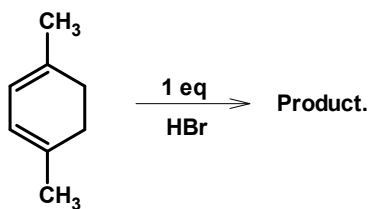


Then $|X - Y|$ value is...

3. No of meso products formed in the following reaction is....



4.



No of allylic hydrogens in thermodynamic product is "X"

No of vinylic hydrogens atom in kinetic product is "Y"

The value of $|X - Y|$ is....

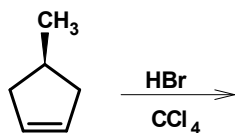
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5.



Then, $x + y$ is

6. Find the total number of product(s) in the following reaction



ORGANIC CHEMISTRY

HYDROCARBON

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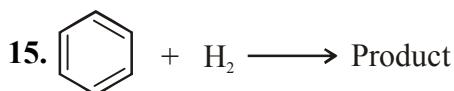
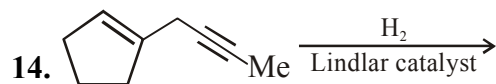
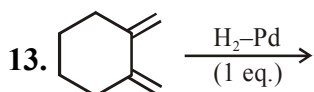
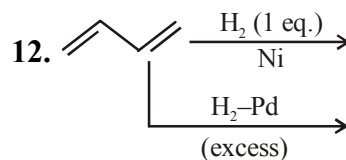
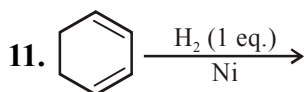
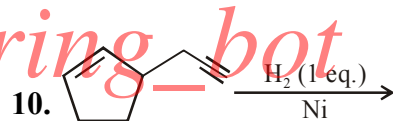
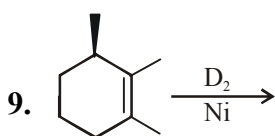
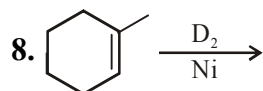
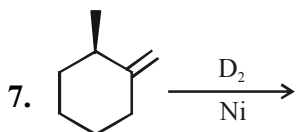
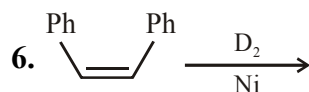
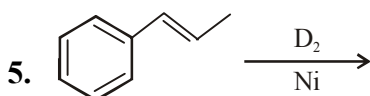
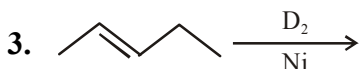
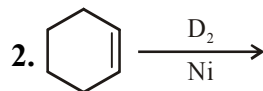
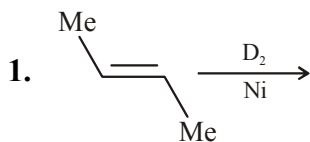
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HYDROCARBON

EXERCISE -A

Q. What would be the major product in the following reaction ?



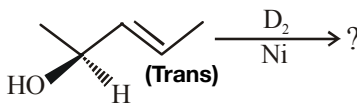
(1 : 3) (L)

(1 : 1) (K)

EXERCISE-B

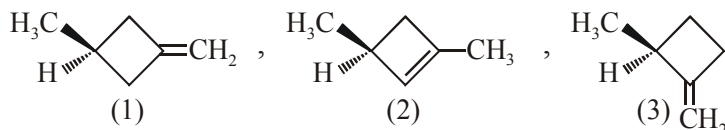
One or more correct

1. Hydrogenation of the given molecule produces



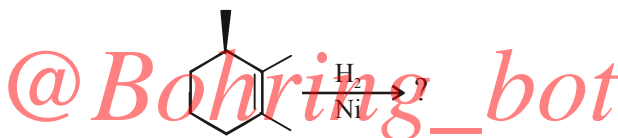
- (A) a single enantiomer (B) a pair of enantiomer
(C) a pair of diastereomer (D) a racemic mixture

2. Compound [X] is chiral and on hydrogenation yields a mixture of two stereoisomers one of which is chiral and other is achiral. Identify [X] ?

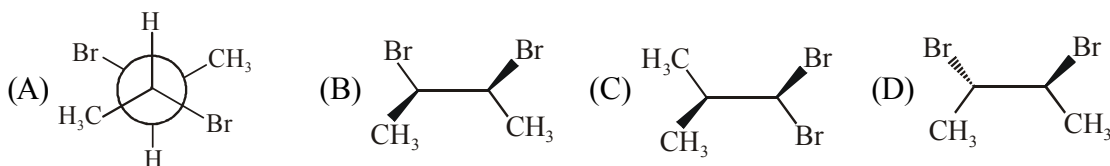


- (A) Compound [X] can be (1) (B) Compound [X] can be (3)
(C) Compound [X] can be (1) and (2) (D) Compound [X] can be (2) and (3)

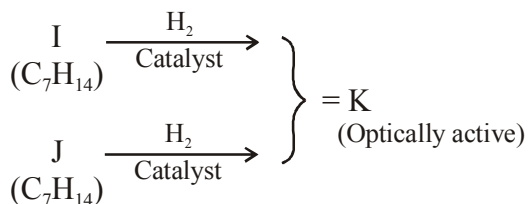
3. Identify the product



4. $\text{CH}_3\text{--C}\equiv\text{C--CH}_3 \xrightarrow[2. \text{Br}_2 / \text{CCl}_4]{1. \text{H}_2 / \text{Pd}}$



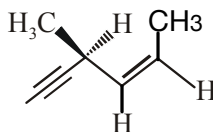
5. I and J both have molecular formula C_7H_{14} . I and J both are optically active both rotate plane polarised light in the same direction.



On catalytic hydrogenation I and J yield the same compound $\text{K}(\text{C}_7\text{H}_{16})$. Compound K is optically active. Find the correct statement -

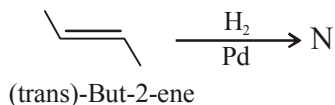
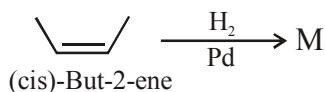
- (A) I and J are diastereomers (B) I and J are enantiomers
(C) I on addition with $\xrightarrow[\text{CHCl}_3]{\text{Br}_2}$ give diastereomeric mixture
(D) J on addition with $\xrightarrow[\text{CHCl}_3]{\text{Br}_2}$ give a racemic mixture

6. Hydrogenation of the following compound in the presence of poisoned palladium catalyst does not give -

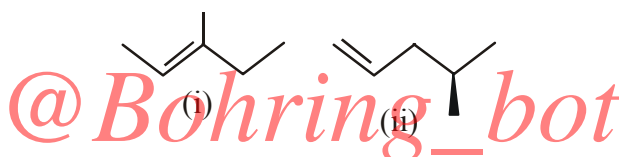


- (A) an optically active compound
(B) an optically inactive compound
(C) a racemic mixture
(D) a diastereomeric mixture

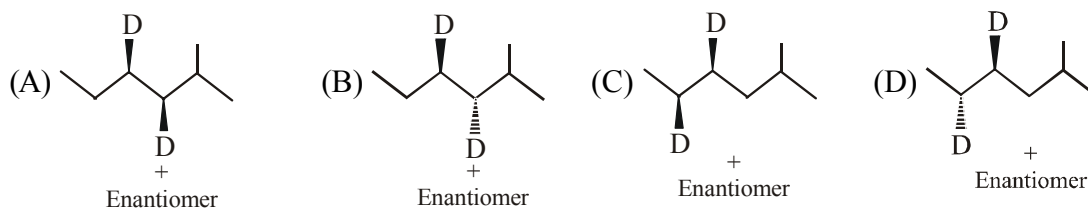
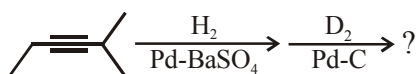
7.



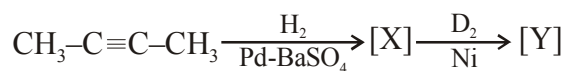
- (A) Hydrogenation of but-2-ene is stereospecific
(B) Hydrogenation of but-2-ene is stereoselective
(C) It is syn addition
(D) Rate of formation of M is faster than N.
8. Choose the incorrect statement about the following two alkenes -



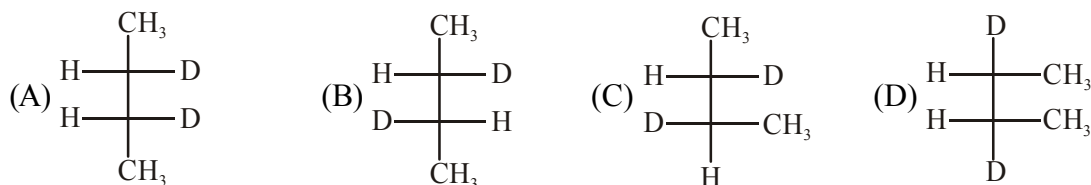
- (A) Alkene (i) is more stable than alkene (ii)
(B) Hydrogenation of achiral alkene (i), will give chiral products
(C) Hydrogenation of chiral alkene, (ii) will give a chiral product
(D) The two alkenes (i) and (ii) are configurational isomers
9. Select the major product from the following reaction sequence :



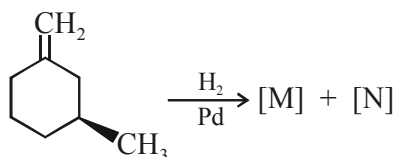
10.



Identify the structure of [Y] :



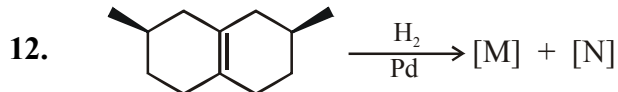
11. For the following given reaction-



[M] & [N] are -

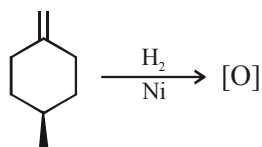
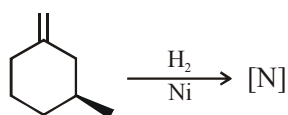
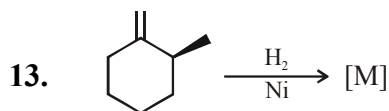
- (A) Enantiomers of each other
(C) Both are meso compounds

- (B) Diastereomers of each other
(D) Constitutional isomer of each other



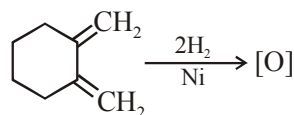
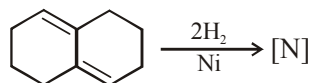
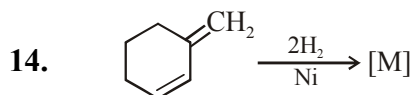
- (A) M & N are diastereomers
(C) M & O are diastereomers

- (B) O & P are identical
(D) N & O enantiomers



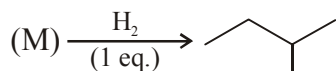
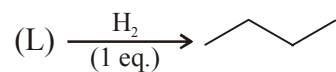
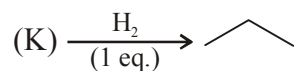
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Find the number of optically active product(s) out of M, N & O.

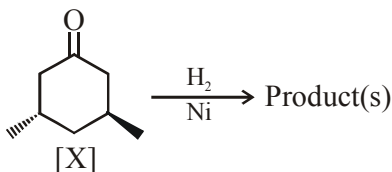


Find the sum of (M + N + O) ?

15. Find the number of possible alkenes :

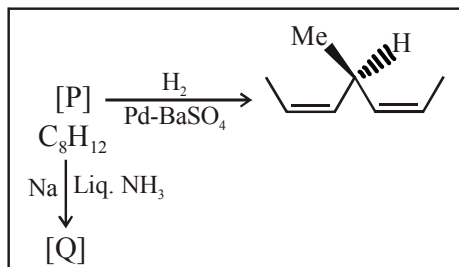


16. Correct statement(s) regarding the compound [X] and product(s) is/are :



- (A) [X] has one optically active diastereomer (B) Two products are obtained
(C) Single optically active product is formed (D) Diastereomeric mixture is obtained

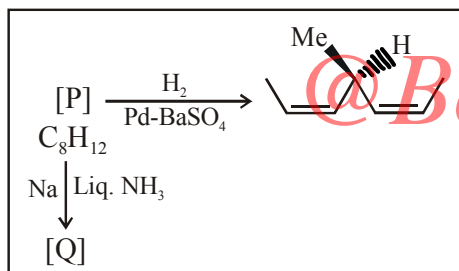
17.



Product (Q) is -

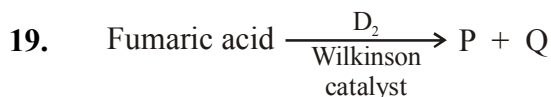
- (A) Pure optically active (B) Racemic mixture
(C) Diastereomeric mixture (D) Meso compound

18.



Correct statement regarding [P] is -

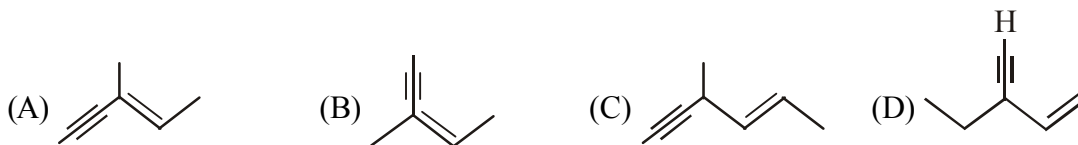
- (A) P is optically inactive (B) P is alkenyne
(C) P is diene (D) P is diyne



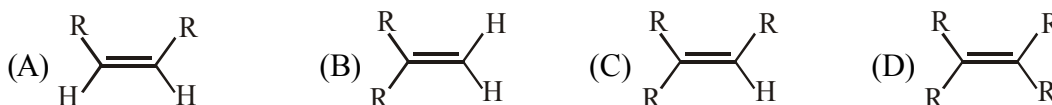
Product (P) and (Q) are -

- (A) Diastereomers (B) Enantiomers
(C) Identical (D) Constitutional isomers

20. The “enyne” that produces a chiral compound upon treatment with Lindlar catalyst-

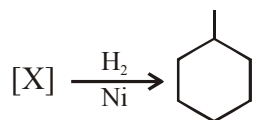


21. Which one of the following alkenes will react fastest with H_2 under catalytic hydrogenation condition ?



Integer Type :

22. Total number of alkene which on reaction give following alkane -

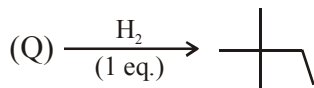
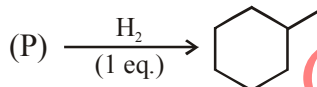
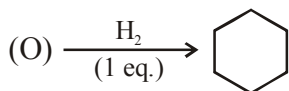
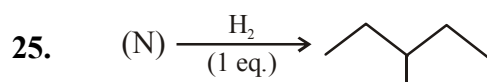


23. How many alkene will give isopentane by catalytic hydrogenation ?

24.
$$[X] \xrightarrow[\text{Pd-BaSO}_4]{\text{H}_2 \text{ (1 eq. per alkyne)}} [Y]$$

All alkynes having molecular formula C_5H_8 Number of isomeric products

Find the value of $(X + Y)$?



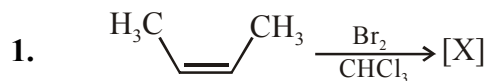
@Bohring_bot

26.
$$[X] \xrightarrow[\text{Ni}]{\text{H}_2} \text{trans-1,2-dideuteriocyclohexane}$$

Possible alkene (Exclusive product)

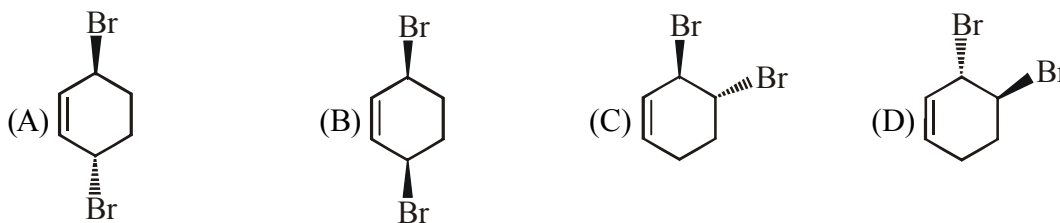
EXERCISE-C

One or more correct :-

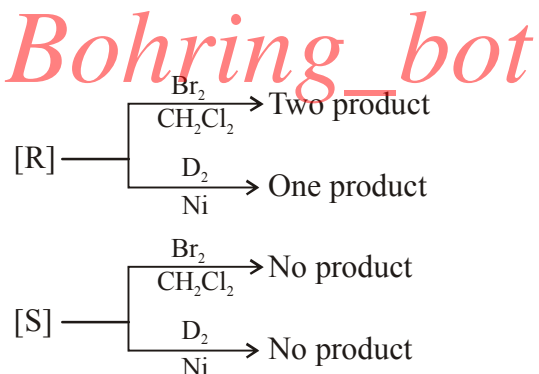


Which of the following statement(s) best describe the above reaction ?

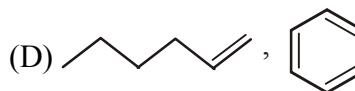
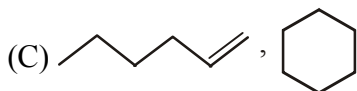
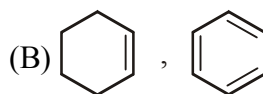
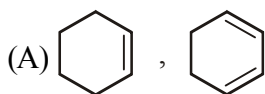
- (A) It is a stereospecific reaction (B) It is a stereoselective reaction
(C) It is electrophilic addition reaction (D) Reaction results in decoloration of bromine



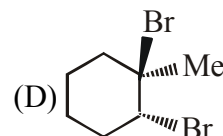
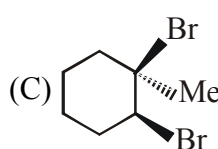
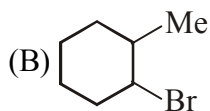
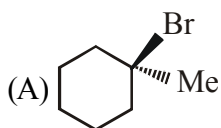
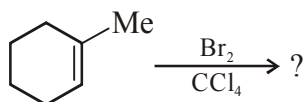
3. (R) and (S) consist of 6 carbon each. Results of bromine addition $\xrightarrow[\text{CHCl}_3]{\text{Br}_2}$ and deuterium addition ($\xrightarrow[\text{Ni}]{\text{D}_2}$) are as follows :



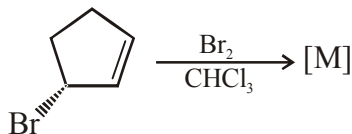
Which of the following pairs given below could be (R) & (S) respectively -



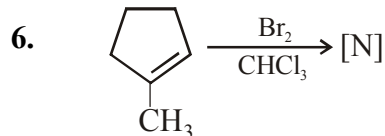
4. Identify the major product ?



5. Product of the following reaction can be best described as -

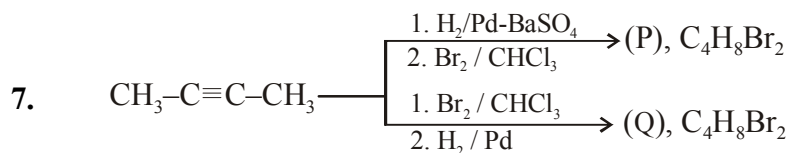


- (A) Meso product
(B) A pair of enantiomers
(C) Structural isomer
(D) A pair of diastereomers



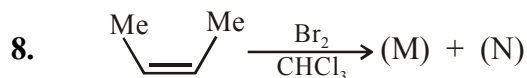
Stereochemistry of the products is are

- (A) Diastereomers
(B) Meso
(C) Racemic mixture
(D) Pure enantiomer

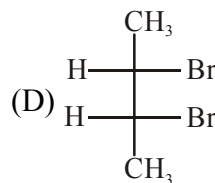
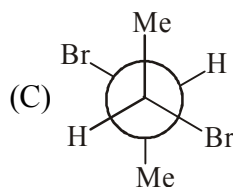
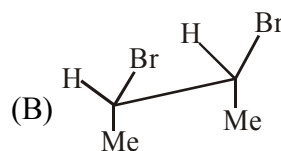
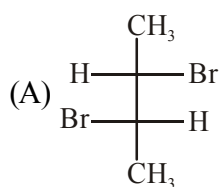


Relationship between (P) and (Q) is:

- (A) Diastereomers
(B) Identical
(C) Enantiomers
(D) Geometrical isomers



Identify (M) & (N) ?



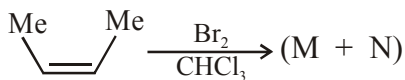
9. Consider the following reaction -



Identify the correct statement(s) ?

- (A) Br₂ molecule acts as a Lewis base
(B) Br₂ addition occurs with “anti” stereospecificity
(C) Intermediate bromonium ion acts as a Lewis acid
(D) Final product is a mixture of enantiomers

10.



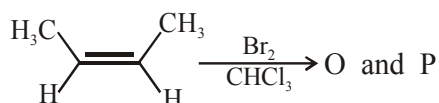
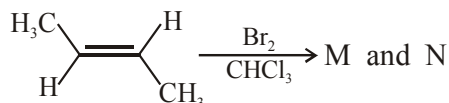
(A) M and O are enantiomers

(B) N and P are diastereomers

(C) O and P are identical

(D) M and P are optically active

11. The correct statements for the following addition reaction is(are) -



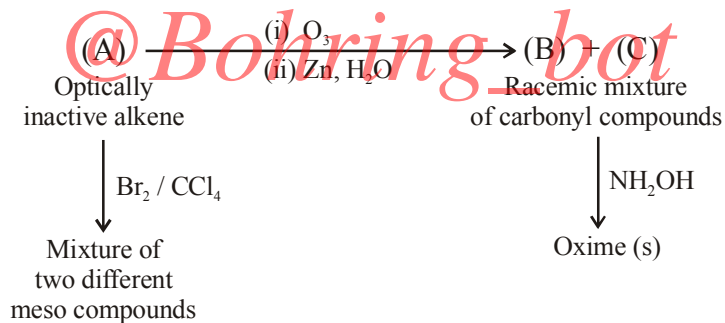
(A) (M and O) and (N and P) are two pairs of enantiomers

(B) Bromination proceed through trans addition in both the reaction

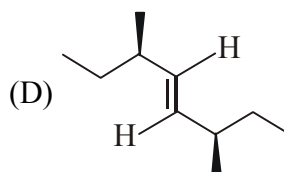
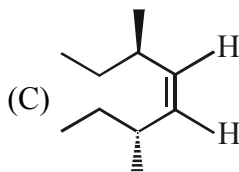
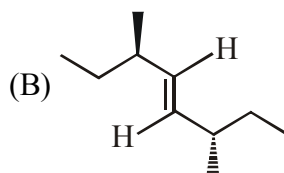
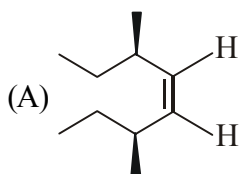
(C) O and P are identical molecules

(D) (M and O) and (N and P) are two pairs of diastereomers

12. In the given reaction sequence -

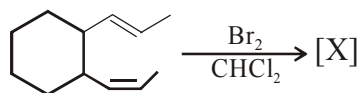


Alkene [A] is -



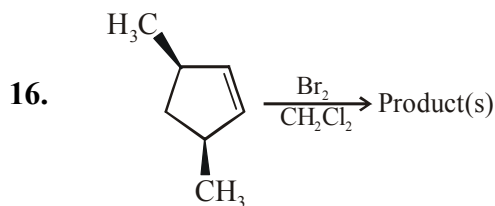
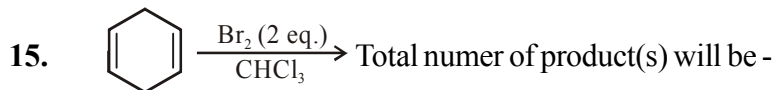
Integer Type :

13.

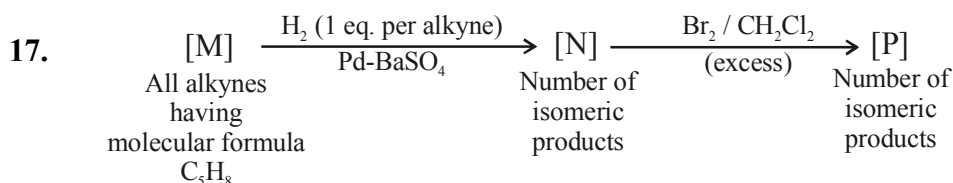


Find the number of product(s)

14. Isomeric alkenes of molecular formula C_5H_{10} , each react with $\text{Br}_2 / \text{CH}_2\text{Cl}_2$, and give “n” number of dibromo products. Out of ‘n’ products, find the number of optically active products.



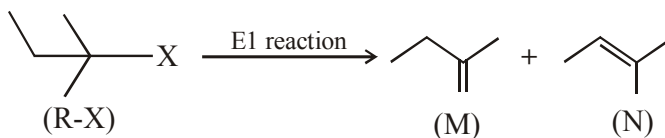
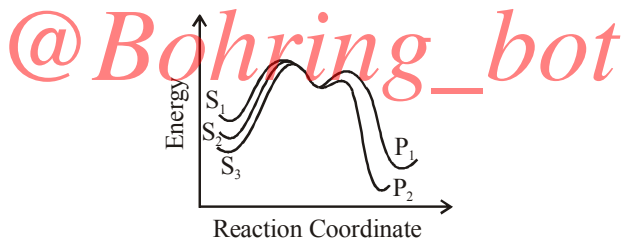
Find the number of optically active product(s).



Find the value of [P-N]

Passage-01

Consider the E1 reaction of tert-amyl halide from the energy profile given below.



18. In the above reaction, X = Cl, Br or I. Based on the graph, identify the alkyl halides (R-X) as S₁, S₂ and S₃

(A) S₁ = R-Cl, S₂ = R-Br and S₃ = R-I

(B) S₁ = R-I, S₂ = R-Br and S₃ = R-Cl

(C) S₁ = R-Cl, S₂ = R-I and S₃ = R-Br

(D) S₁ = R-I, S₂ = R-Cl and S₃ = R-Br

19. Identify product P₁ and its yield relative to P₂

(A) P₁ is M and is the major product

(B) P₁ is N and is the minor product

(C) P₁ is N and is the major product

(D) P₁ is M and is the minor product

20. Complete catalytic hydrogenation of naphthalene gives decaline (C₁₀H₁₈). The number of isomers of decaline formed and the total number of isomers of decaline possible are respectively.

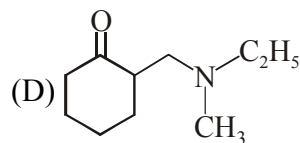
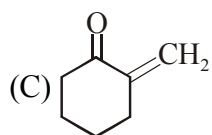
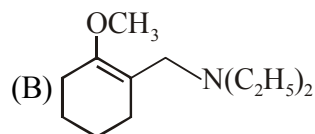
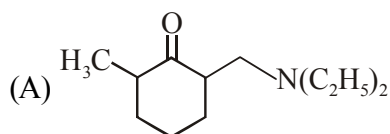
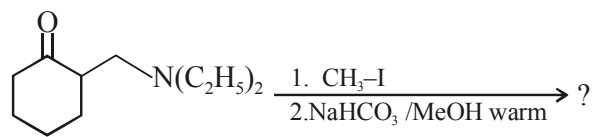
(A) 1, 2

(B) 2, 2

(C) 2, 4

(D) 3, 4

21. The major product obtained in the following reaction is -

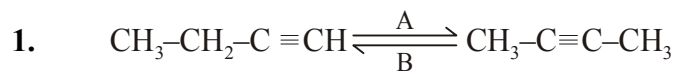


22. Hydration by oxymercuration of 3-methyl-1-butene gives 3-methyl-2-butanol. What is the observed rotation of this product, using a 20 % solution in ethanol in a 1 dm polarimeter cell ? (Assume the specific rotation of the pure (S)-enantiomer to be $+16^\circ$)

- (A) $+3.2^\circ$ (B) -3.2° (C) 180° (+ or -) (D) 0°

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EXERCISE-D



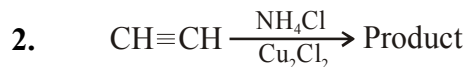
(A) and (B) are respectively -

(A) alc. KOH, NaNH_2

(B) NaNH_2 , alc. KOH

(C) NaNH_2 , Lindlar catalyst

(D) Lindlar catalyst, NaNH_2



Product is :

(A) $\text{Cu-C}\equiv\text{C-Cu}$

(B) $\text{CH}_2=\text{CH-C}\equiv\text{CH}$

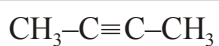
(C) $\text{CH}\equiv\text{C-Cu}$

(D) $\text{Cu-C}\equiv\text{C-NH}_4$

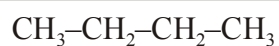
3. Which reagent is most useful for distinguishing compound (I) from rest of the compound -



(I)



(II)



(III)



(IV)

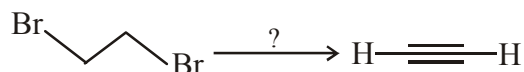
(A) Alkaline KMnO_4

(B) $\text{Br}_2 / \text{CHCl}_3$

(C) $\text{Br}_2 / \text{CH}_3\text{COOH}$

(D) Ammonical AgNO_3

4. The reagent(s) for the following conversion -



is/are :

(A) alc. KOH

(B) alc. KOH followed by NaNH_2

(C) aq. KOH followed by NaNH_2

(D) $\text{Zn} / \text{CH}_3\text{OH}$

5. Cyclohexene is best prepared from cyclohexanol by which of the following -

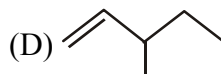
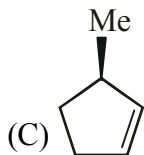
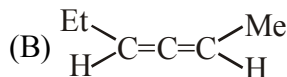
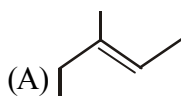
(A) Conc. H_2SO_4

(B) Conc. $\text{HCl} / \text{ZnCl}_2$

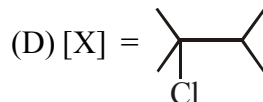
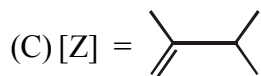
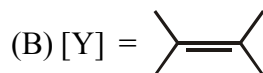
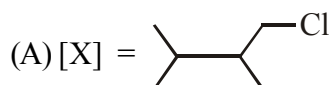
(C) Conc. HCl

(D) Conc. HBr

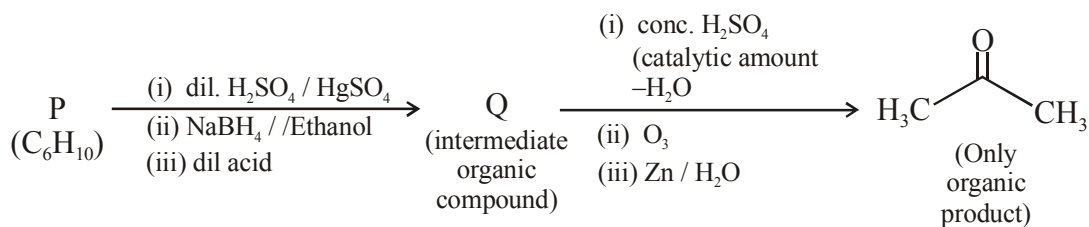
6. $\text{M}(\text{C}_6\text{H}_{12})$, an optically active hydrocarbon which on catalytic hydrogenation gives an optically inactive compound C_6H_{14} . [M] is :



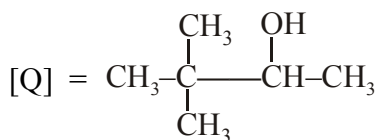
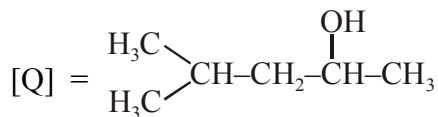
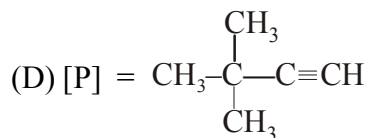
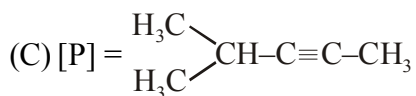
7. An alkyl halide [X], $C_6H_{13}Cl$, on treatment with potassium tert-butoxide gives two isomeric alkenes (Y) and (Z), C_6H_{12} . Both alkenes on hydrogenation give 2,3-dimethyl butane. Predict the structure of [X], [Y] and [Z].



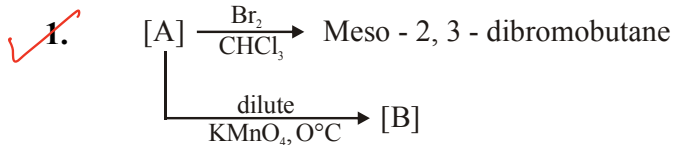
✓ 8.



Identify the correct match -

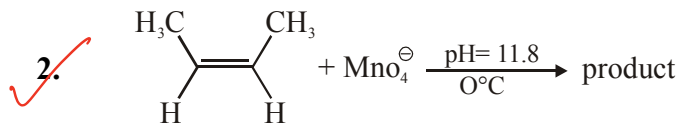


EXERCISE-E



Product (s) [B] is (are) :-

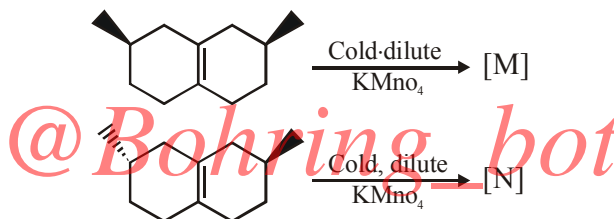
- (A) Enantiomers (B) Diastereomers (C) Identical (D) Meso



Identify the correct statement (s) regarding product :-

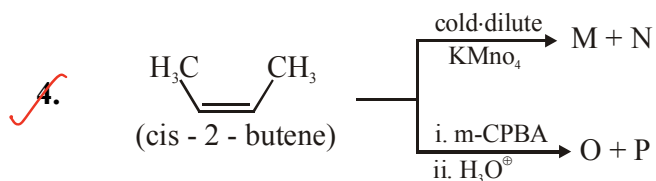
- (A) It has C_2 -axis of symmetry (B) It has plane of symmetry
 (C) It is optically active (D) It has non zero dipole moment.

3. Consider the given reactions :-



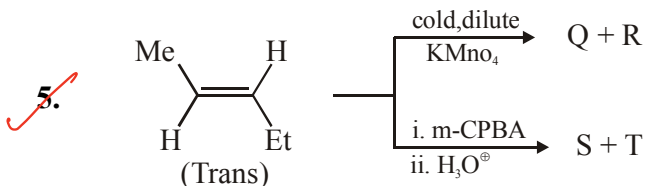
Find the sum of (M + N) :-

- (A) 2 (B) 3 (C) 4 (D) 5



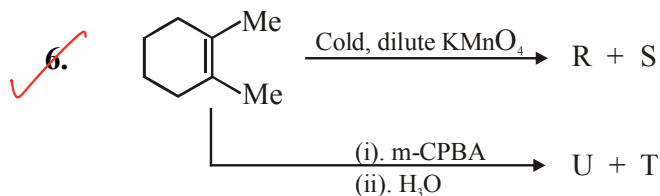
Identify the correct statement (s) :-

- (A) M and N are enantiomers. (B) N and P are diastereomers
 (C) O and P are identical (D) M and P are enantiomers.



Identify the correct statement (s) :-

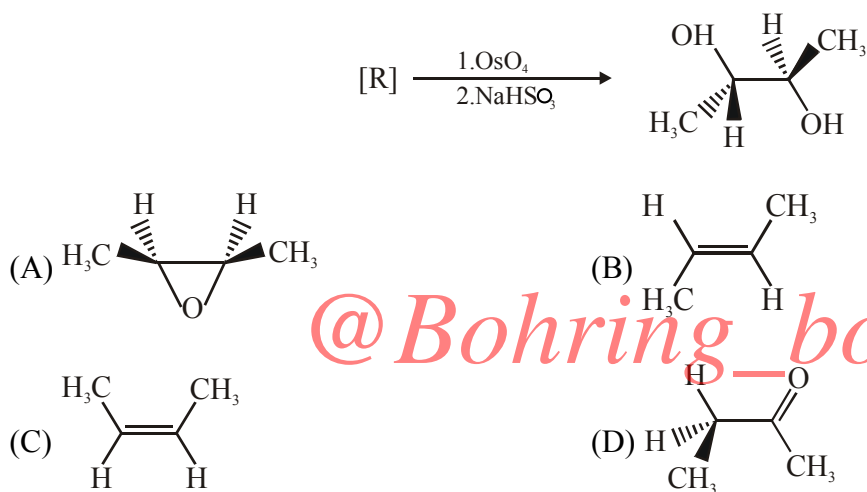
- (A) Q and R are optically active (B) R and S are diastereomers
 (C) R has axis of symmetry (D) Q and T are identical



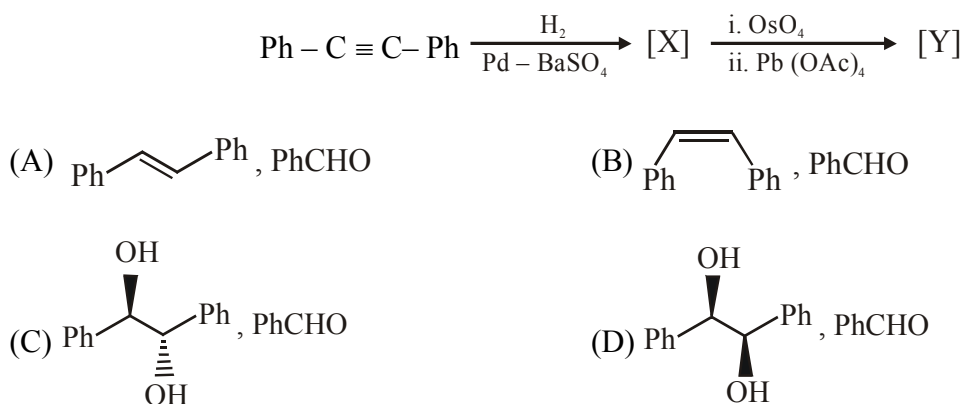
Identify the correct statement (s) :-

- (A) It is stereospecific reaction.
- (B) R and S are superimposable mirror image.
- (C) Formation of U and T is Racemisation.
- (D) U and R are non - superimposable, non-mirror image

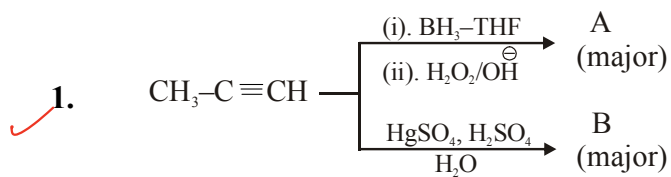
7. Identify the reactant that would give the product shown in following reaction :-



8. Identify the product (X,Y) in the following sequence :-

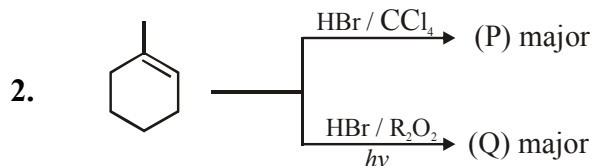


EXERCISE-F



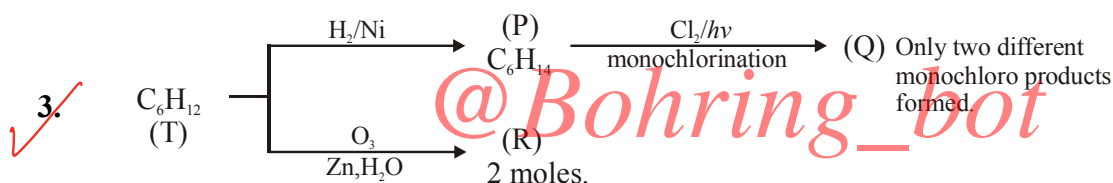
Relationship between A and B is :-

- (A) Tautomers (B) Functional isomers
(C) Position isomers (D) Identical

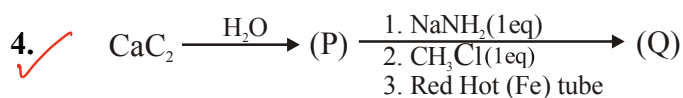
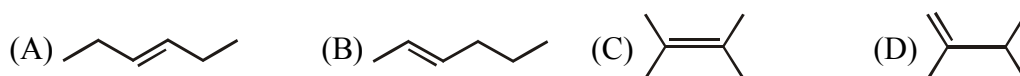


Correct relationship between (P) and (Q) is :-

- (A) Chain isomers (B) Position isomers
(C) Geometrical isomers (D) Identical

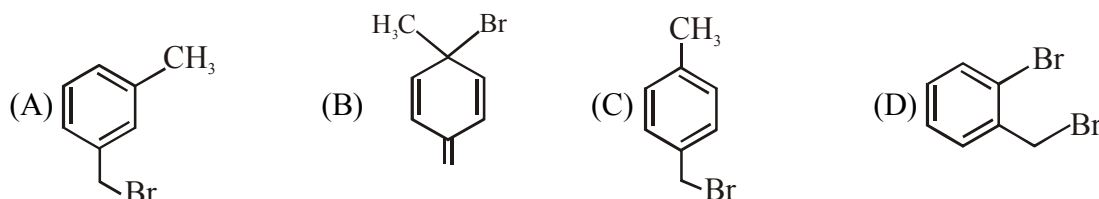
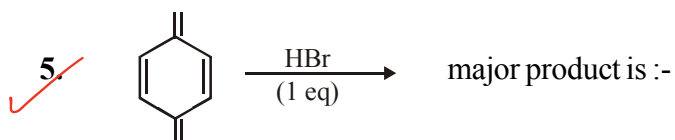


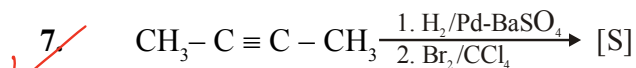
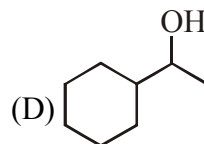
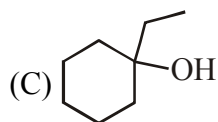
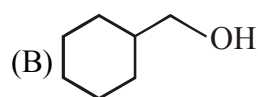
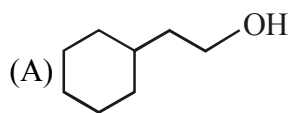
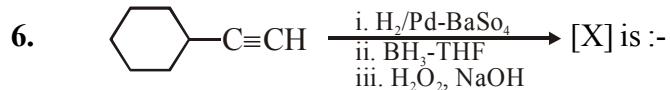
Identify reactant (T), on the basis of above reaction sequences :-



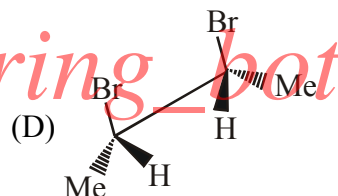
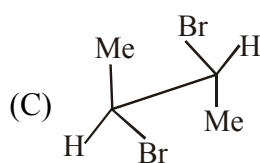
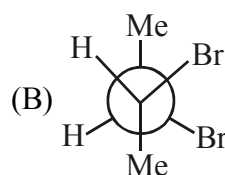
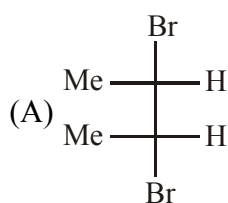
Identify (Q) :-

- (A) 1,3,5 - trimethyl benzene (B) Toluene
(C) m - Xylene (D) Mesitylene

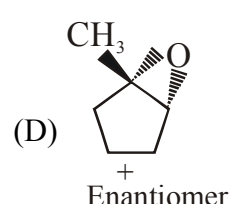
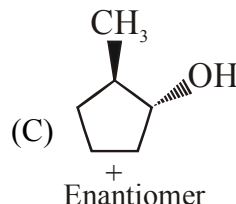
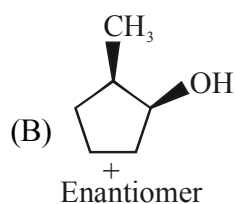
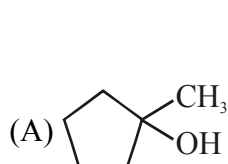
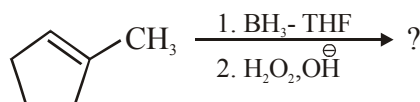




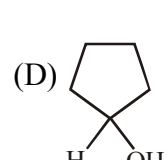
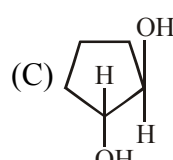
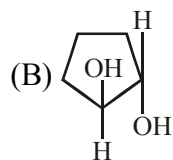
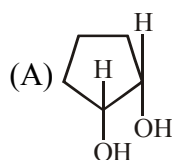
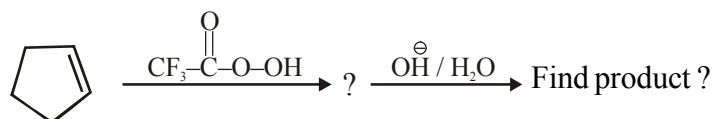
Product [S] is (are) :-



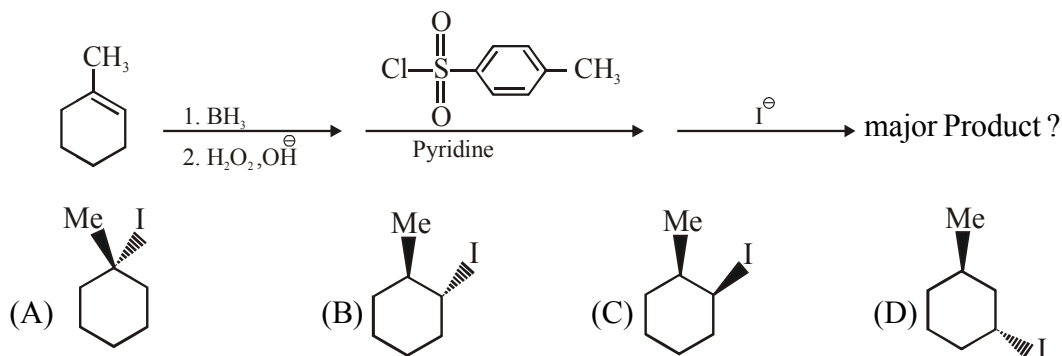
8. Which product (s) would you expect to obtain from the following sequence of reactions ?



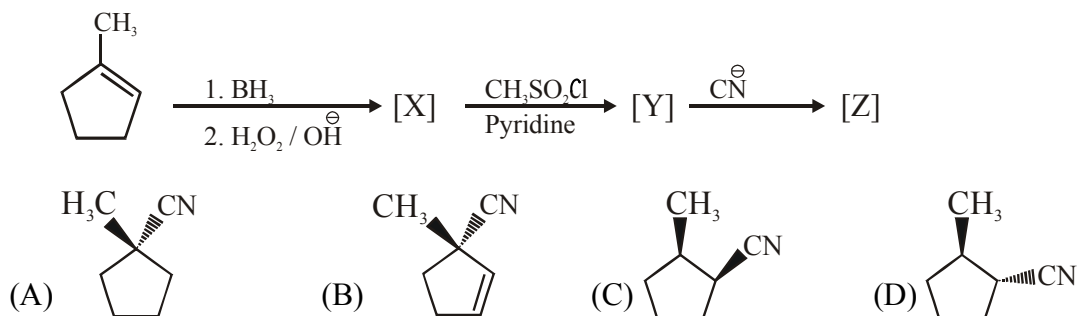
9. What would be the final product ?



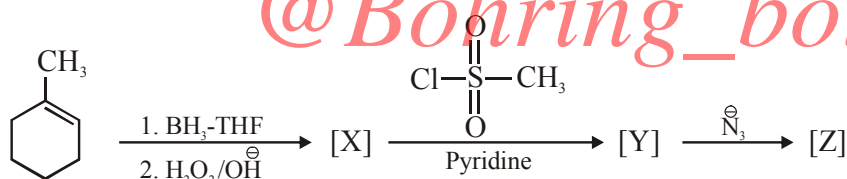
10. Choose the major product of the following reaction sequence :-



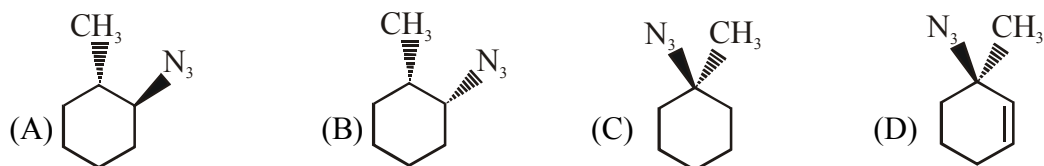
11. Choose the major product (Z) of the following reaction :-



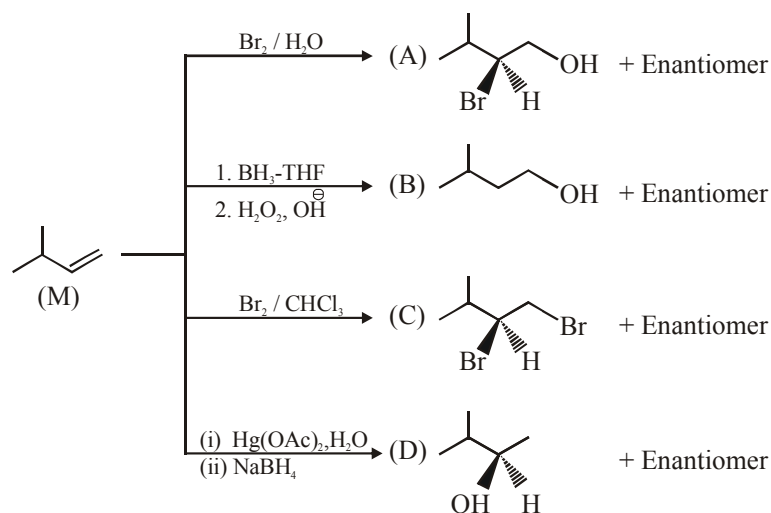
12. Consider the following reaction sequence :-



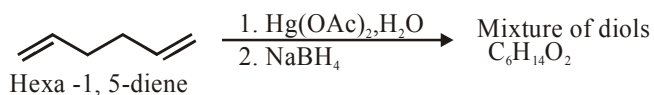
Choose the major product [Z], In each case assume that the enantiomer is formed as well.



13. Which one of the following is incorrect ?



14. The compound hexa - 1,5 - diene was subjected to oxymercuration and a mixture of diols of formula $C_6H_{12}O_2$ was formed -

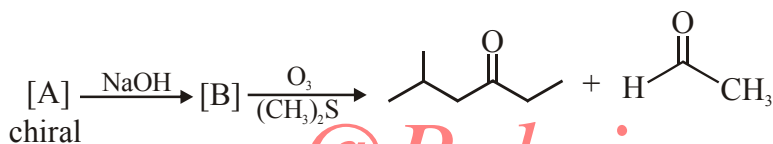


How many stereoisomers would you expect there to be :

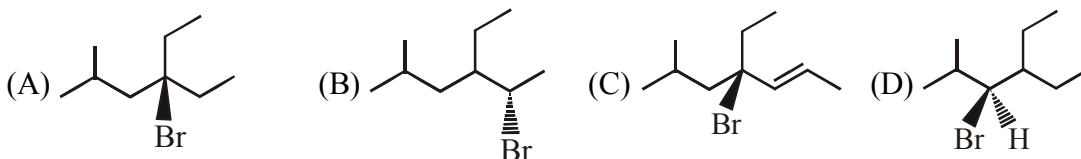
- (A) 1 (B) 2 (C) 3 (D) 4
15. In above Question no. 14 :-

If the diol synthesis shown in question 14 was carried out using hydroboration $\left(\begin{matrix} 1. \text{BH}_3\text{-THF} \\ 2. \text{H}_2\text{O}_2, \text{OH}^- \end{matrix} \right)$ instead of oxymercuration, how many stereoisomers would you expect there to be ?

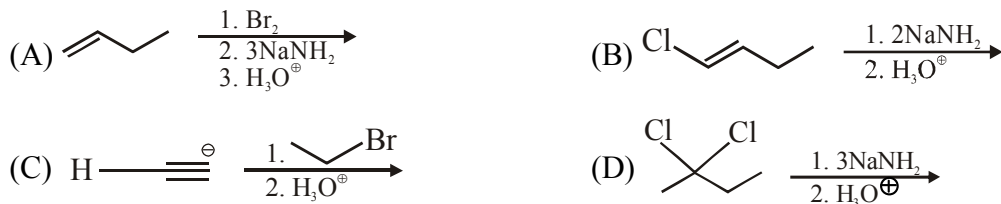
- (A) 1 (B) 2 (C) 3 (D) 0
16. A chiral molecule (A), is treated with NaOH to give a product (B). Compound (B) was subjected to ozonolysis and the following products were obtained :



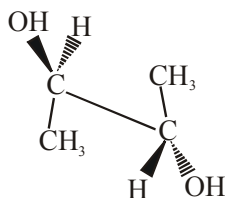
Which one of the following is molecules [A] ?



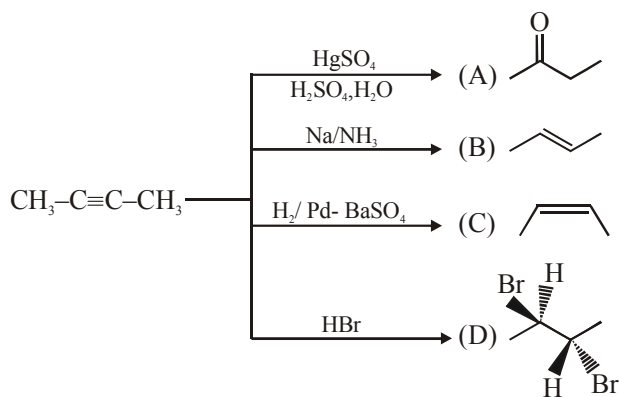
17. Choose the reaction that could be used to synthesize 1-butyne ?



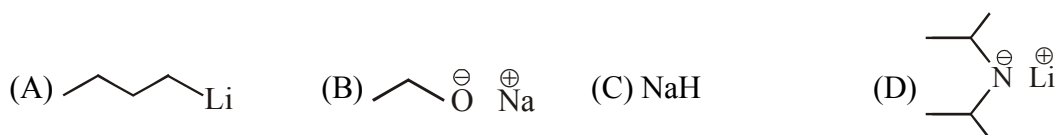
18. Choose the reaction that would “NOT” produce, as the major product, the compound shown on the right (2R, 3S)- butane-2, 3-diol



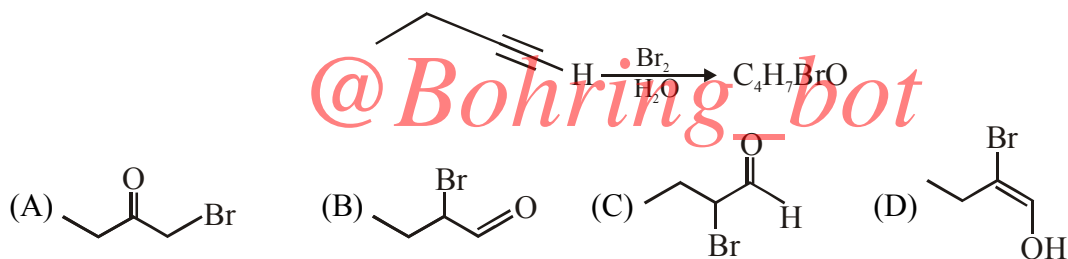
19. Choose the incorrect reaction :-



20. Choose the species that could not be used as the base in the following reaction :-



21. Choose the most likely product from among the following compounds :-



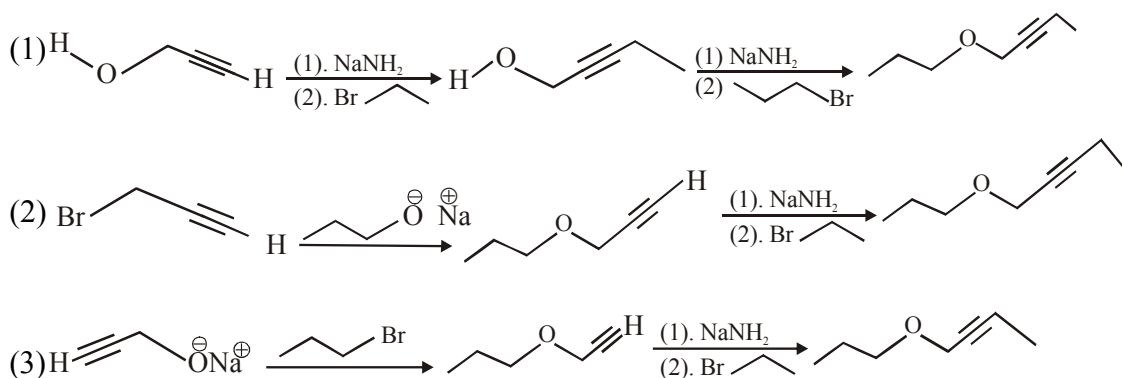
22. The NBS bromination of the compound shown below gives a mixture of products.



Choose the compound that should be the main product of this reaction :-

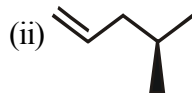
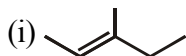


23. Which of the following scheme will be proceed to give the compound indicated as the major product :



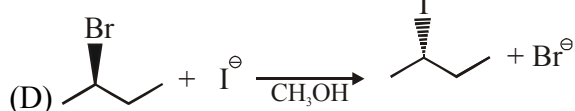
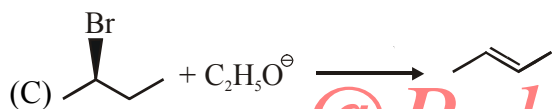
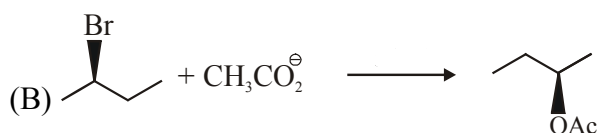
(A) 1 Only (B) 1 and 2 (C) 2 and 3 (D) All

24. Choose the incorrect statement about the following alkenes :-



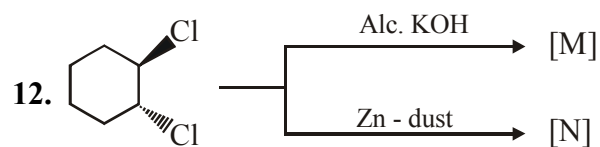
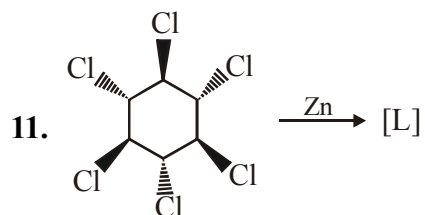
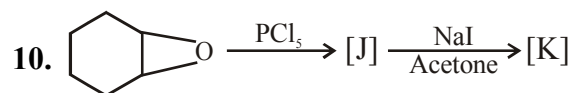
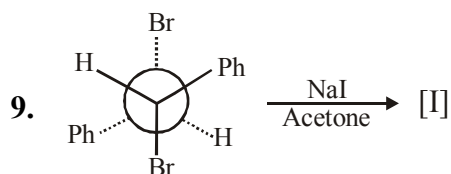
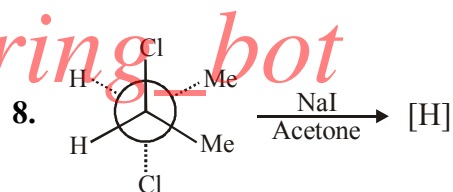
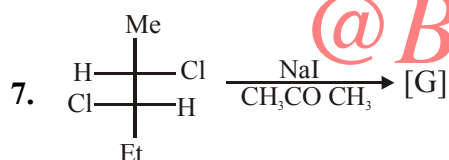
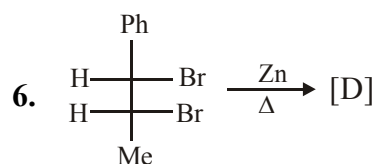
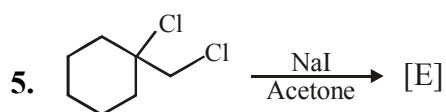
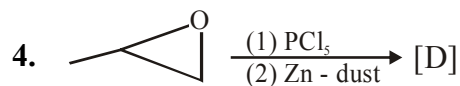
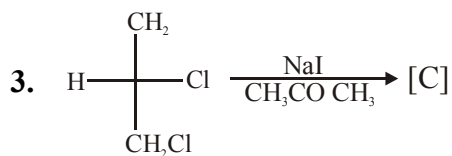
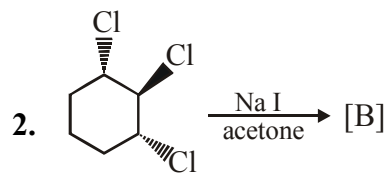
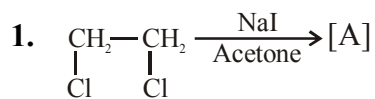
- (A) Hydrogenation of achiral alkene (i) will give chiral products
(B) Hydrogenation of chiral alkene (ii) will give a chiral product
(C) Hydrogenation of alkene (ii) will yield more heat than will hydrogenation of alkene (i)
(D) The two alkenes (i) and (ii) are configurational isomers

25. Choose the answer that incorrectly shows the major organic product (s) of the reaction shown :-

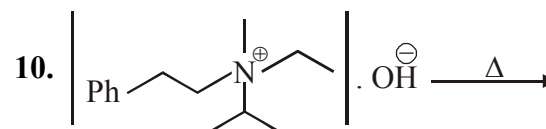
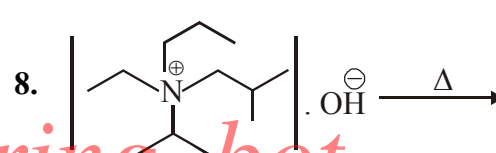
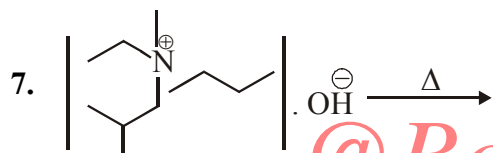
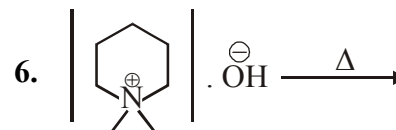
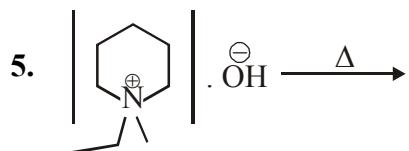
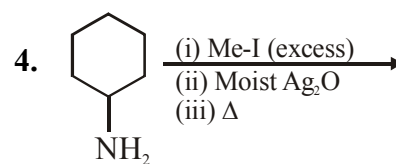
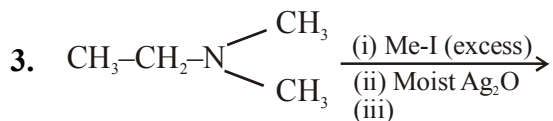
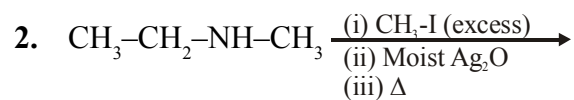
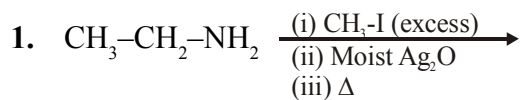


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EXERCISE - G

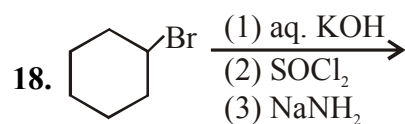
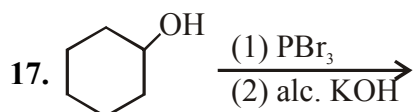
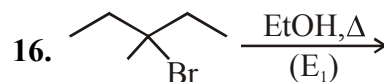
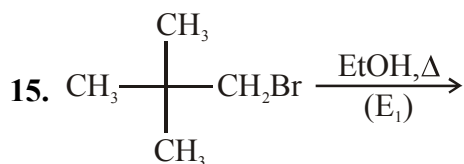
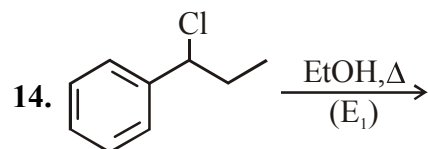
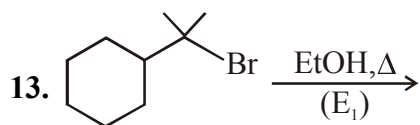
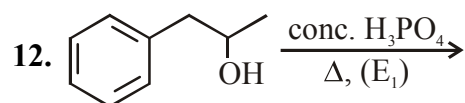
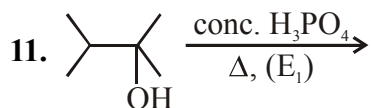
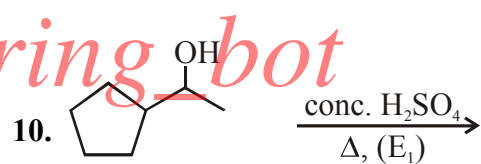
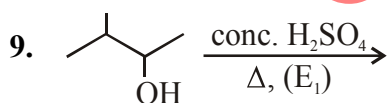
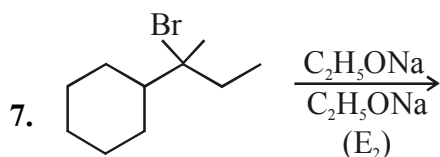
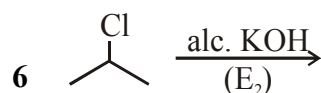
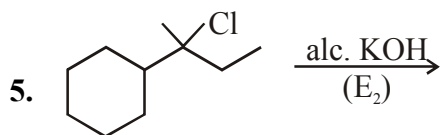
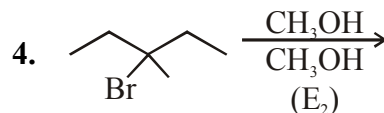
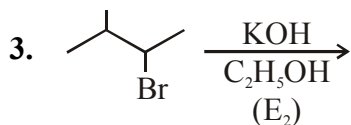
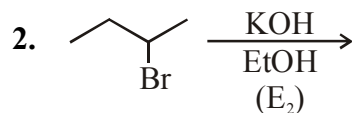
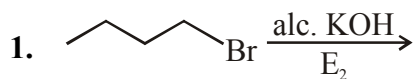


EXERCISE - H

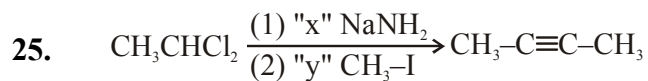
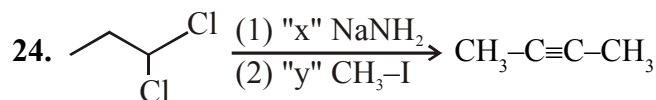
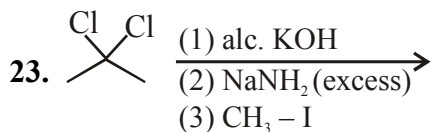
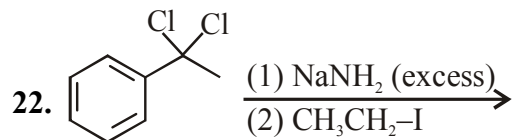
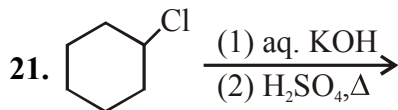
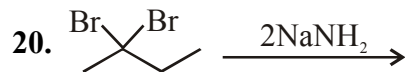
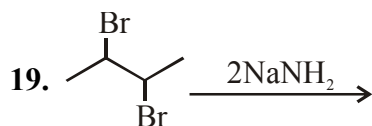


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EXERCISE-I



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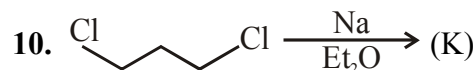
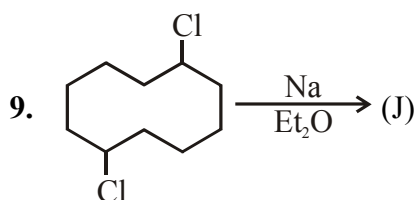
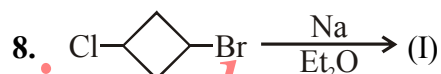
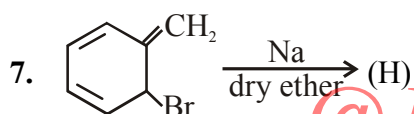
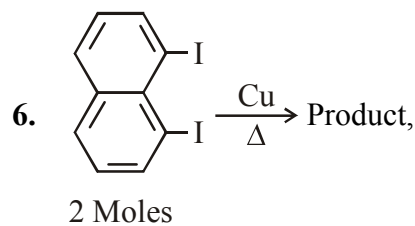
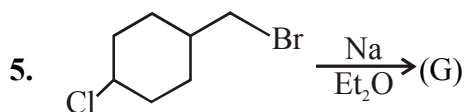
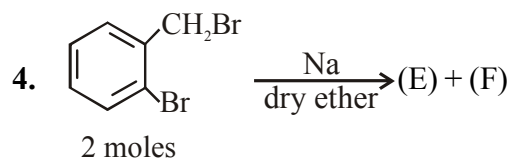
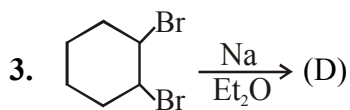
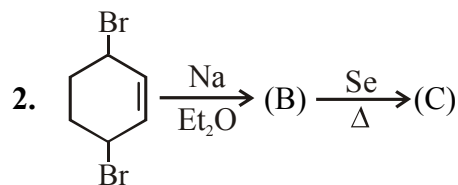
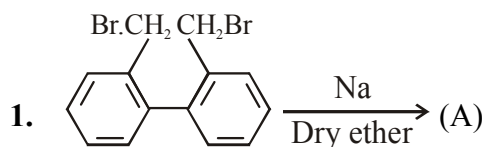


Find the value of (x) and (y)

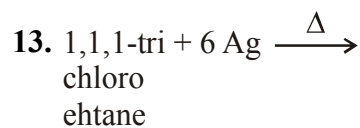
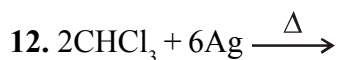
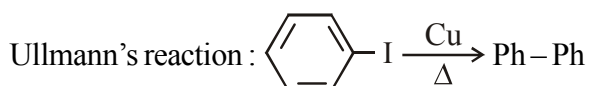
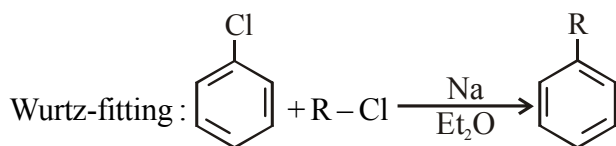
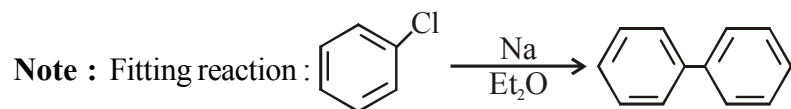
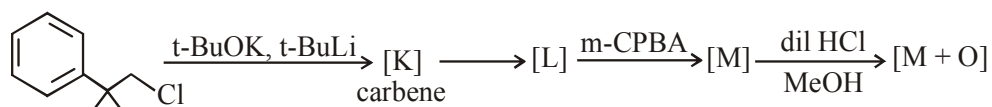
Find the value of "x" and "y".

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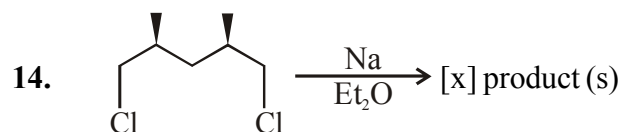
EXERCISE (J)



11. Complete the following reaction sequence and write structure of [K] to [O] -



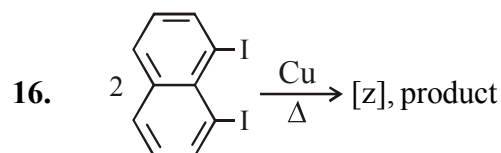
Integers :



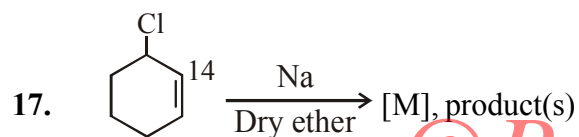
Find the number of optically inactive product(s).



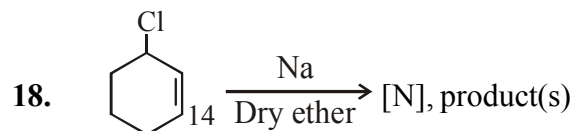
Find the number of optically active product(s).



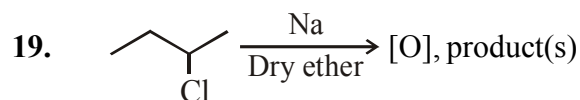
Find the number of resonating structure(s) possible for [z].



Find the number of product(s) possible. (excluding stereomers)

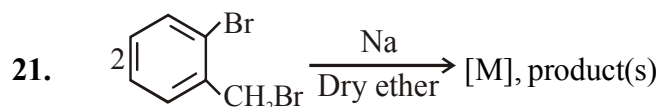
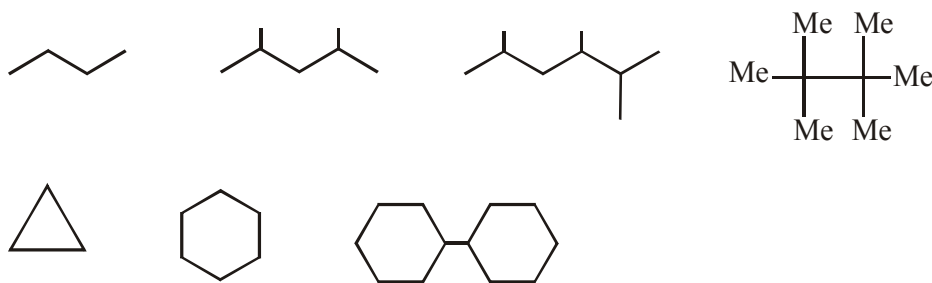


Find the number of product(s) possible (excluding stereomers)

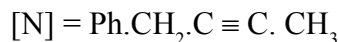
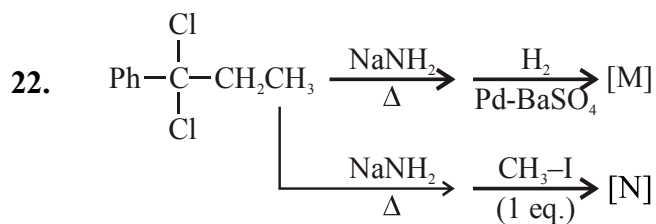


Find the total number of product(s).

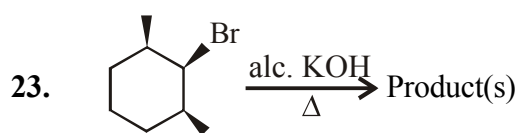
20. Number of compounds which can be obtained in good yield by wurtz reaction using a single reactant –



Find the value of [M]

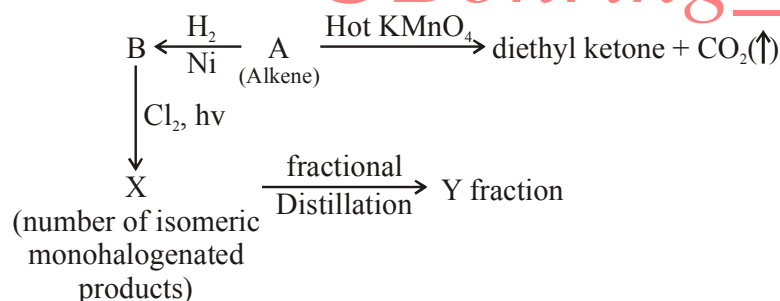


Number of equivalents of NaNH_2 used in the formation of $[\text{M}]$ and $[\text{N}]$ are x and y respectively. Find the sum of $(x + y)$.

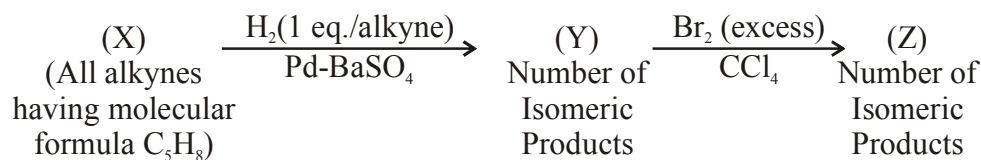


Number of E_2 alkene products (including stereoisomers) is/are :-

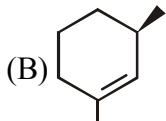
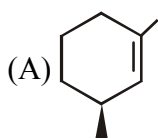
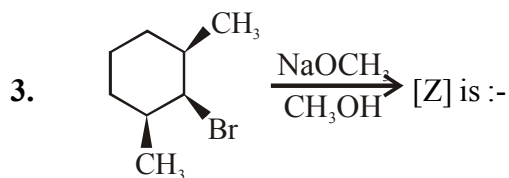
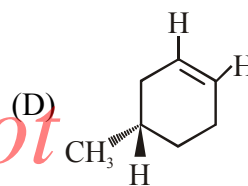
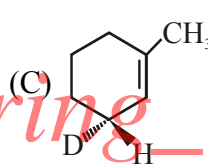
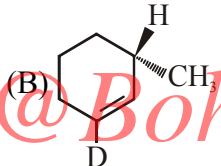
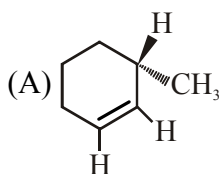
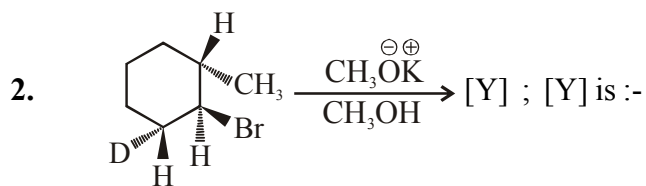
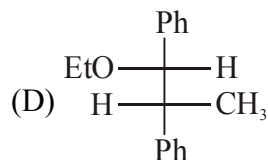
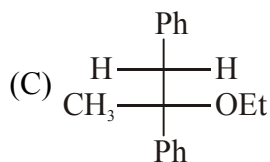
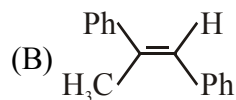
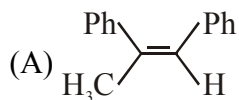
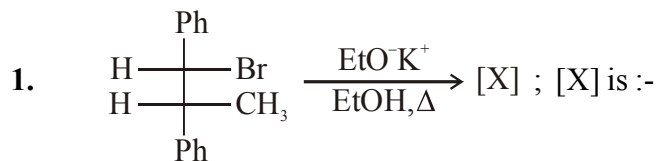
24. Identify magnitude of $X - Y$



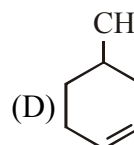
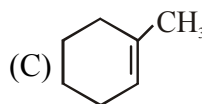
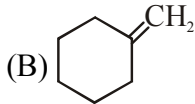
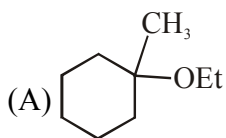
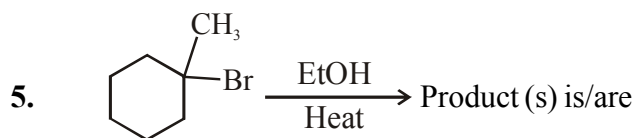
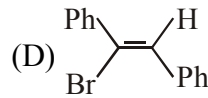
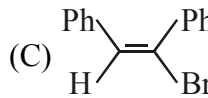
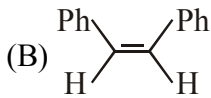
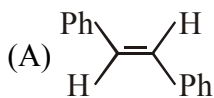
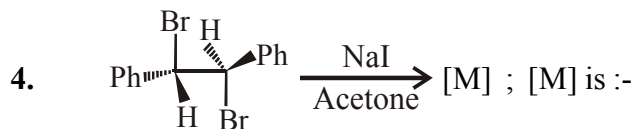
25. What is the value of $Y + Z$

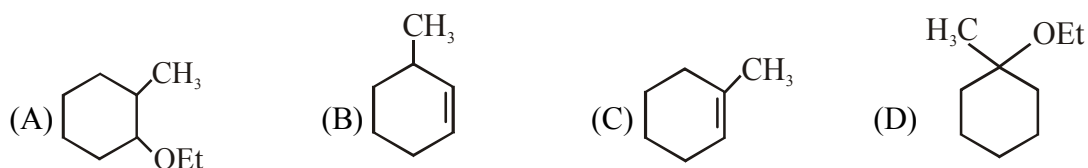
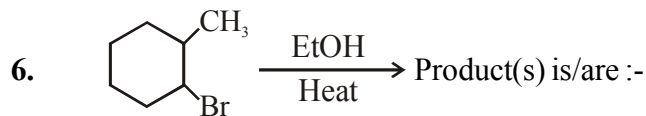


EXERCISE-K

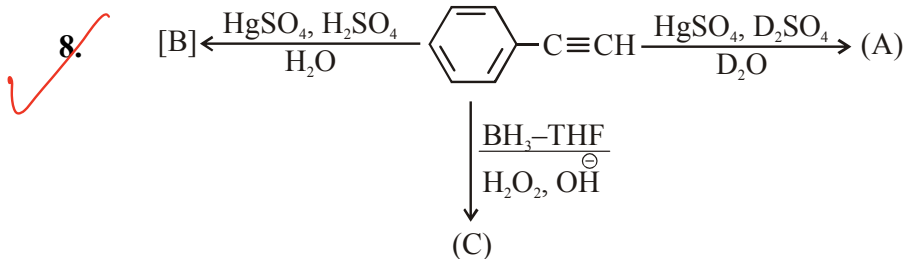
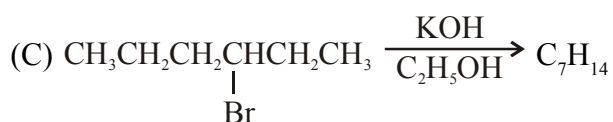
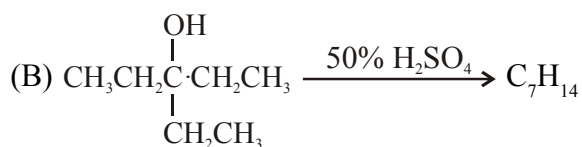


(C) Racemic mixture (D) No reaction





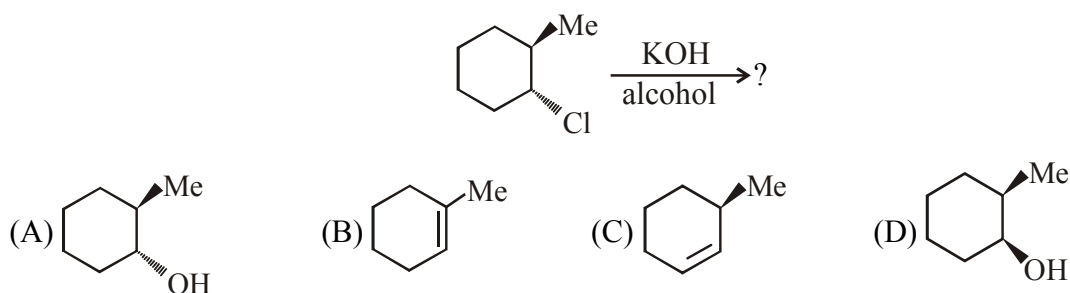
7. Which reaction would yield a single alkene product?



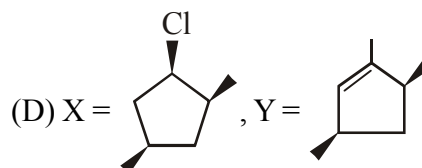
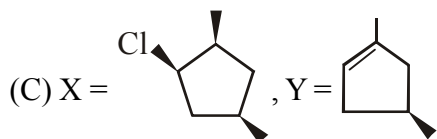
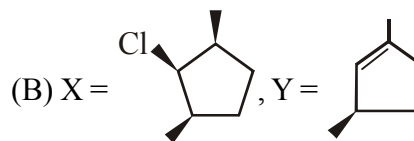
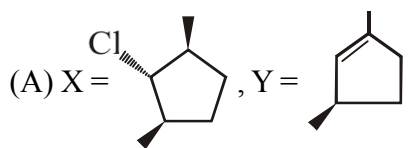
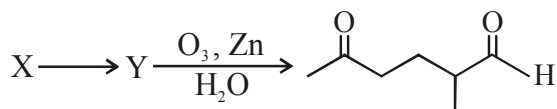
Which of the following statement(s) is(are) correct -

- (A) Product (B) on reaction with aq. bleaching power give sweet smelling colorless liquid.
- (B) Product (C) do not give fehling test.
- (C) Blue color of fehling solution remains blue with (A)
- (D) Product (A) give 2,4-DNP test.

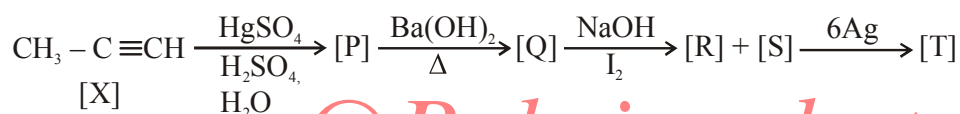
9. The product obtained in the following reaction is



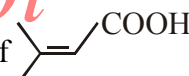
10. Compound X, $C_7H_{13}Cl$ is reacted with sodium ethoxide and gives an elimination product Y, C_7H_{12} . Treatment of ozone followed by Zinc and water gives the compound below. X and Y are respectively.



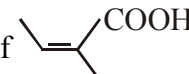
11. Correct option regarding following reaction sequence –



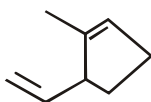
(A) 'T' is homologue of 'X'

(B) 'R' is sodium salt of 

(C) 'P' can produce 'S' with NaOI

(D) 'R' is sodium salt of 

12. The IUPAC name of the following compound is :-



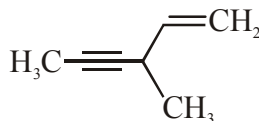
(A) 1-methyl-2-vinylcyclopentene

(B) 1-vinyl-2-methylcyclopent-2-ene

(C) 1-methyl-5-vinylcyclopentene

(D) 2-methyl-3-vinylcyclopentene

13. The IUPAC name for the following compound is :-



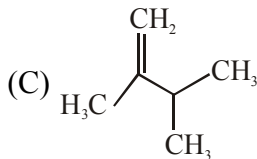
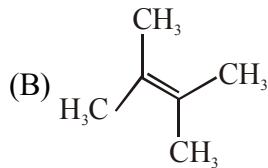
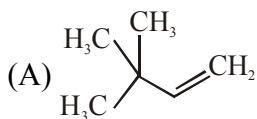
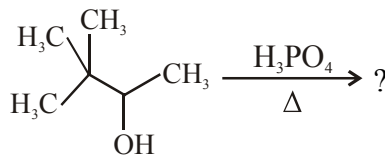
(A) 4-vinyl-2-pentyne

(B) 4-methylhex-2-en-5-yne

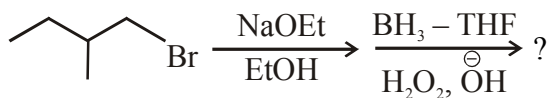
(C) 3-methylhex-4-yn-1-ene

(D) 3-methylhex-1-en-4-yne

14. The major product obtained in the following reaction is :-



15. The product obtained in the following transformation



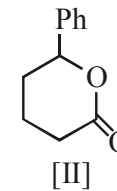
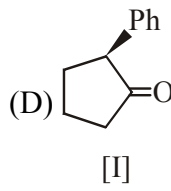
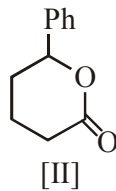
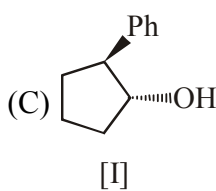
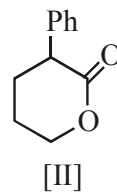
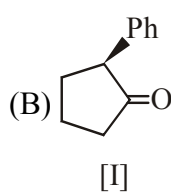
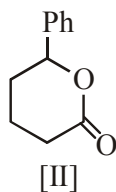
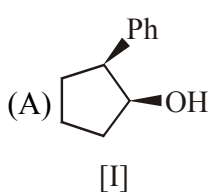
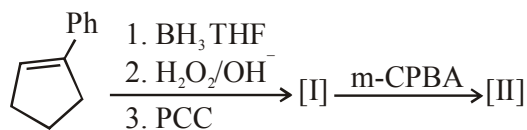
(A) 2 - methyl - 1 - butanol

(B) 2 - methyl - 2 - butanol

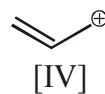
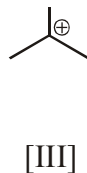
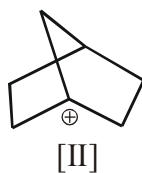
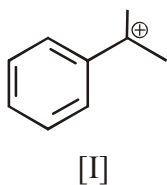
(C) 3 - methyl - 1 - butanol

(D) 3 - methyl - 2 - butanol

16. The products formed in the following reaction are



17. The correct order of stability for the following

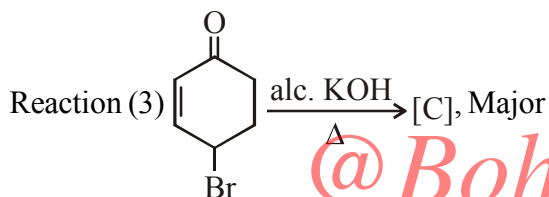
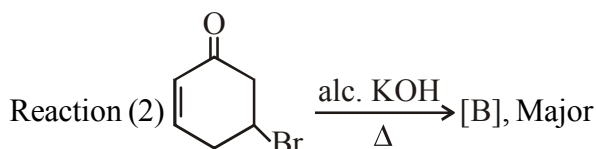
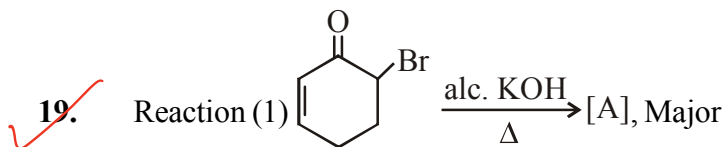
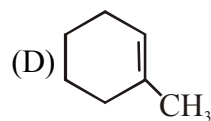
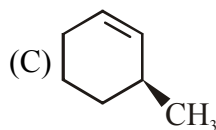
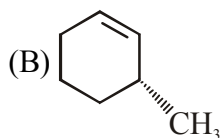
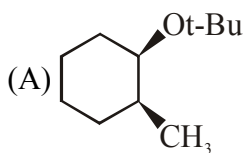
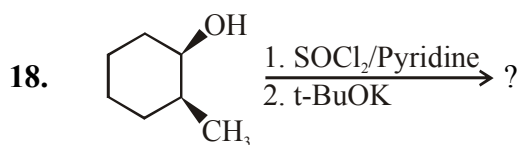


(A) $\text{I} < \text{III} < \text{IV} < \text{II}$

(B) $\text{III} < \text{II} < \text{IV} < \text{I}$

(C) $\text{II} < \text{IV} < \text{III} < \text{I}$

(D) $\text{IV} < \text{III} < \text{I} < \text{II}$



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Correct statement regarding product obtained in above reactions :-

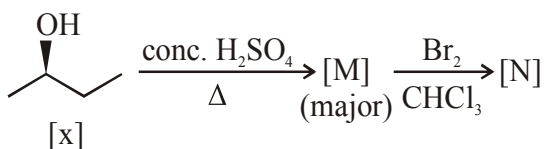
(A) A = B but C is different

(B) A = C but B is different

(C) B = C but A is different

(D) A = B = C, all products are identical.

20. ✓ Compound [X] is an optically active alcohol.



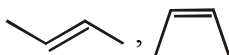
(A) optical activity of [X] retains indefinitely in contact with aqueous base.

(B) N posses axis of symmetry.

(C) monobromination of N give two products only.

(D) most stable conformer of N is having Non-zero dipole moment.

21. Which of the following reactants would give exactly the same products from both (cis) and (trans)-2-butene?



(A) Br₂

(B) PhCO₃H

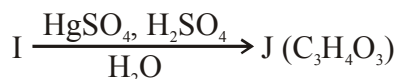
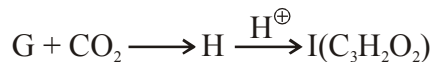
(C) (i) BH₃ – THF ; (ii) H₂O₂, NaOH

(D) OsO₄, NaHSO₃

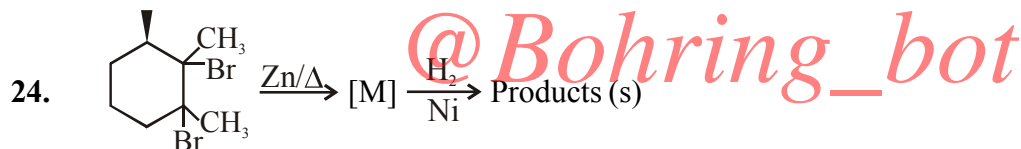
22. Compound [x] reacts with diborane followed by alkaline Hydrogen peroxide to form compound [y]. [y] on reaction with a mixture of sodium bromide in sulphuric acid followed by bromobenzene and sodium in ether gives n-pentyl benzene. Compound [x] is :-



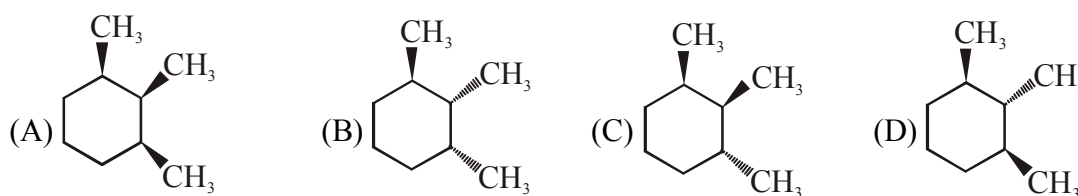
23. Acetylene + $\text{CH}_3\text{MgBr} \longrightarrow \text{G} + \text{CH}_4$



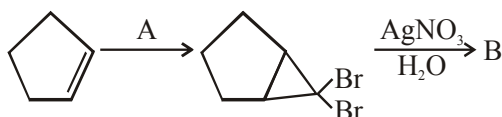
- (A) (M) is oxallic acid (B) (J) is pyruvic acid ($\text{CH}_3 - \overset{\text{O}}{\parallel} \text{C} - \text{COOH}$)
 (C) (M) on heating give acetic acid
 (D) In (I), all atoms can be in same plane in one of the possible conformation

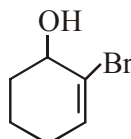
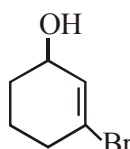
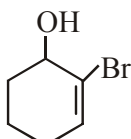
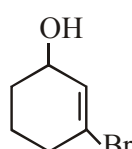


Identify the product(s)



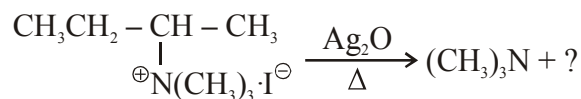
25. The reagent [A] required, and the major product [B] formed in the following reaction sequence are



- (A) $\text{A} = \text{CH}_2\text{Br}_2$ and BuO^-K^+ , $\text{B} =$  (B) $\text{A} = \text{CH}_2\text{Br}_2$ and BuO^-K^+ , $\text{B} =$ 
 (C) $\text{A} = \text{CHBr}_3$ and BuO^-K^+ , $\text{B} =$  (D) $\text{A} = \text{CHBr}_3$ and BuO^-K^+ , $\text{B} =$ 

EXERCISE-L

1. The major product in the reaction given below is :-



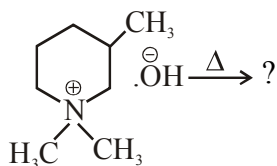
(A) trans - 2 - butene

(B) cis - 2 - butene

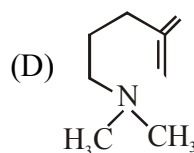
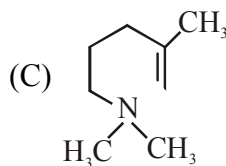
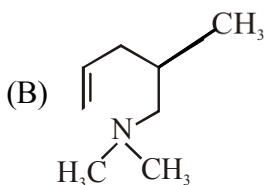
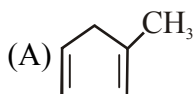
(C) 1 - butene

(D) 2 - iodobutane

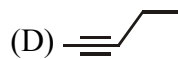
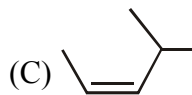
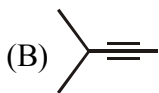
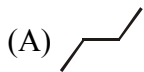
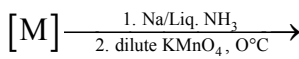
2.



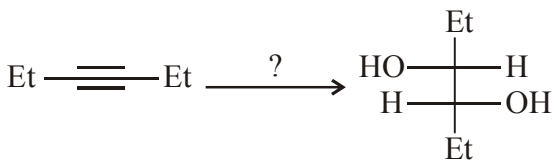
The major product in the above reaction will be :-



3.



4.



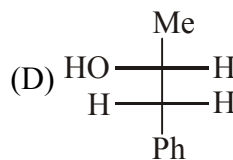
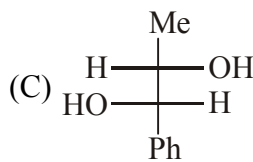
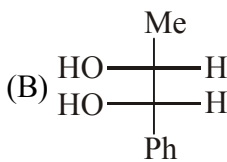
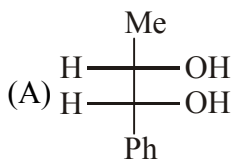
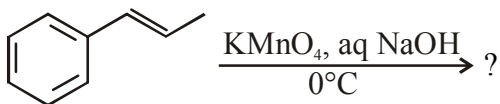
(A) (i) Na/NH_3 (ii) $\text{CH}_3\text{CO}_3\text{H}$ (iii) H_3O^+

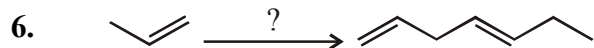
(B) (i) H_2/Pd (ii) $\text{KMnO}_4/\text{aq. NaOH}/0^\circ\text{C}$

(C) (i) $\text{H}_2/\text{Lindlar catalyst}$ (ii) $\text{KMnO}_4/\text{aq. NaOH}/0^\circ\text{C}$

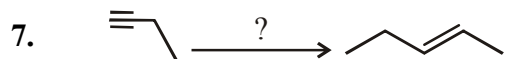
(D) (i) $\text{H}_2/\text{Lindlar catalyst}$ (ii) $\text{CH}_3\text{CO}_3\text{H}$ (iii) H_3O^+

5.

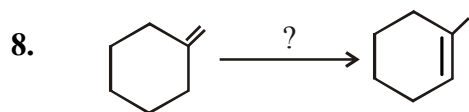




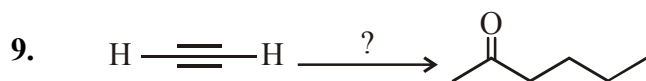
- (A) (i) NBS, Heat (ii) 1-Butyne, NaNH_2 (iii) H_2 , Pd-BaSO₄
 (B) (i) HBr, Peroxide (ii) 2-Butyne, NaNH_2 (iii) Na/NH₃ (Liq)
 (C) (i) NBS, Heat (ii) 1-Butyne, NaNH_2 (iii) Na/NH₃ (Liq)
 (D) (i) NBS, Heat (ii) 1-Butyne, NaNH_2



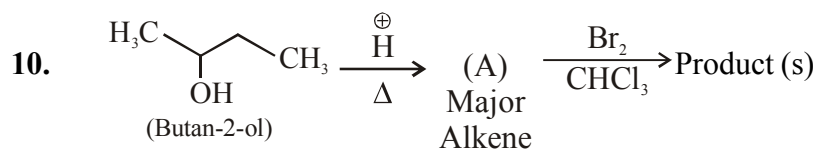
- (A) (i) NaNH_2 (ii) CH_3Br
 (B) (i) NaNH_2 (ii) $\text{CH}_3\text{CH}_2\text{Br}$
 (C) (i) NaNH_2 (ii) CH_3Br (iii) Na/NH₃ (Liq)
 (D) (i) NaNH_2 (ii) $\text{CH}_3\text{CH}_2\text{Br}$ (iii) Na/NH₃ (Liq)



- (A) H_2SO_4
 (B) (i) HBr (ii) $\text{KOC}(\text{CH}_3)_3/\text{C}(\text{CH}_3)_3\text{CO}_2\text{H}/\text{Heat}$
 (C) (i) HBr/Peroxides (ii) KOH/EtOH/Heat
 (D) (i) Br_2 (ii) KOH/EtOH/Heat



- (A) (i) NaNH_2 (ii) 1-Bromobutane (iii) O_3 then Zn/ H_2O
 (B) (i) NaNH_2 (ii) 1-Bromobutane (iii) BH_3 -THF (iv) aq. H_2O_2 , NaOH
 (C) (i) O_3 then H_2O (ii) NaNH_2 (iii) 1-Bromobutane
 (D) (i) NaNH_2 (ii) 1-Bromobutane (iii) HgSO_4 , H_2SO_4 , H_2O



Product (s) is (are) :-

- (A) Meso compound (B) Enantiomeric mixture
 (C) Diastereomers (D) Structural isomers

HYDROCARBON ANSWER KEY

EXERCISE-B

1	2	3	4	5	6	7	8	9	10
C	B	ABD	D	AC	ACD	CD	BCD	B	ACD
11	12	13	14	15	16	17	18	19	20
B	ABC	2	6	7	C	A	B	B	C
21	22	23	24	25	26				
A	6	3	6	13	4				

EXERCISE-C

1	2	3	4	5	6	7	8	9	10
ABCD	A	B	D	AD	C	B/C	A	BCD	BC
11	12	13	14	15	16	17	18	19	20
BD	B	16	12	3	2	3	B	D	A
21	22								
C	D								

EXERCISE-D

1	2	3	4	5	6	7	8		
A	B	D	B	A	D	B C D	D		

EXERCISE-E

1	2	3	4	5	6	7	8		
A	BD	B	B	AB	ABD	C	B		

EXERCISE-F

1	2	3	4	5	6	7	8	9	10
B	B	C	AD	C	A	B	C	BC	C
11	12	13	14	15	16	17	18	19	20
C	B	AB	C	D	B	ABCD	A	D	B
21	22	23	24	25					
A	B	C	ABD	B					

EXERCISE-J

14	15	16	17	18	19	20	21	22	23
1	1	9	1	3	3	4	2	5	2
24	25								
3	9								

EXERCISE-K

1	2	3	4	5	6	7	8	9	10
A	A	ABC	A	ABC	ABCD	B	ACD	C	B
11	12	13	14	15	16	17	18	19	20
ABC	C	D	B	A	D	C	C	D	A
21	22	23	24	25					
C	D	CD	ABC	C					

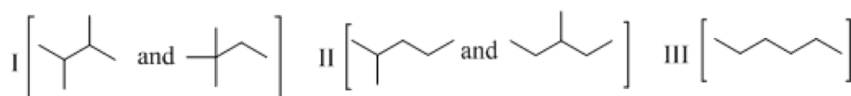
EXERCISE-L

1	2	3	4	5	6	7	8	9	10
C	B	B	D	C	C	C	A	D	A

Previous years lit JEE Questions

22. Isomers of hexane, based on their branching, can be divided into three distinct classes as shown in the figure.

[Figure]



The correct order of their boiling point is

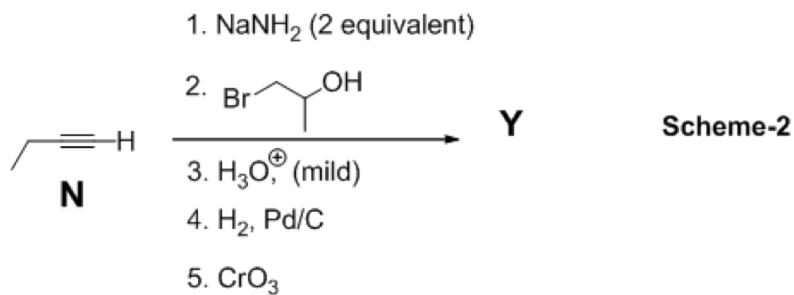
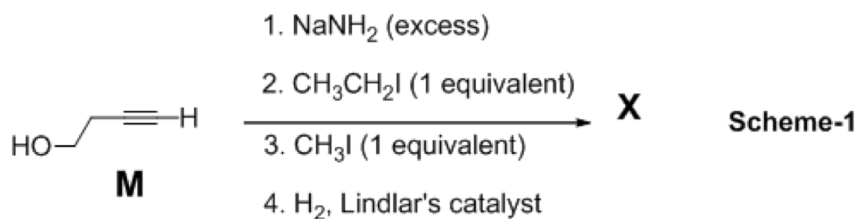
- (A) I > II > III
(B) III > II > I
(C) II > III > I
(D) III > I > II

Answer (B)

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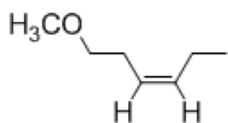
Paragraph For Question 31 and 32

Schemes 1 and 2 describe sequential transformation of alkynes **M** and **N**. Consider only the **major products** formed in each step for both the schemes.

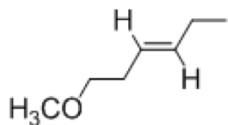


31. The product **X** is

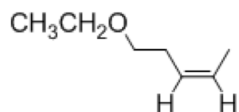
(A)



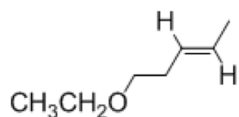
(B)



(C)



(D)



Answer (A)

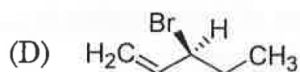
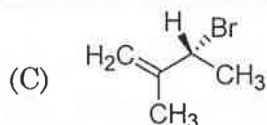
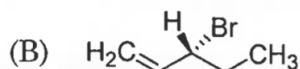
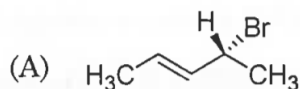
@Bohring_bot

32. The correct statement with respect to product **Y** is

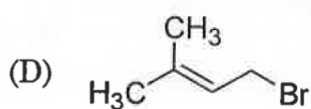
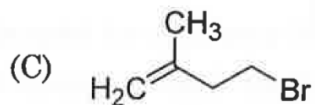
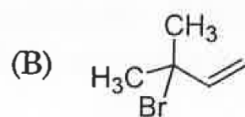
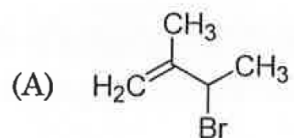
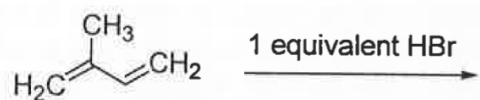
- (A) It gives a positive Tollens test and is a functional isomer of **X**.
(B) It gives a positive Tollens test and is a geometrical isomer of **X**.
(C) It gives a positive iodoform test and is a functional isomer of **X**.
(D) It gives a positive iodoform test and is a geometrical isomer of **X**.

Answer (C)

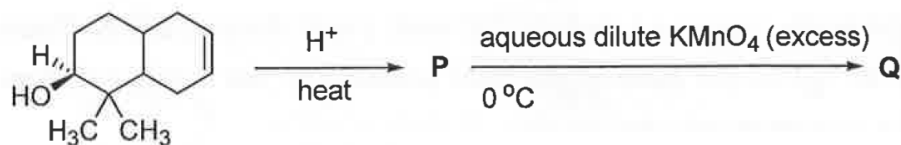
Q.29 Compound(s) that on hydrogenation produce(s) optically inactive compound(s) is(are)



Q.31 In the following reaction, the major product is



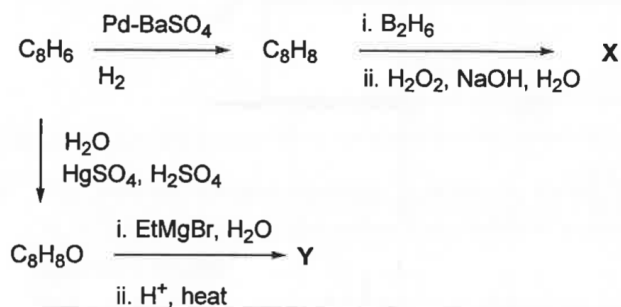
Q.21 The number of hydroxyl group(s) in **Q** is



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PARAGRAPH 1

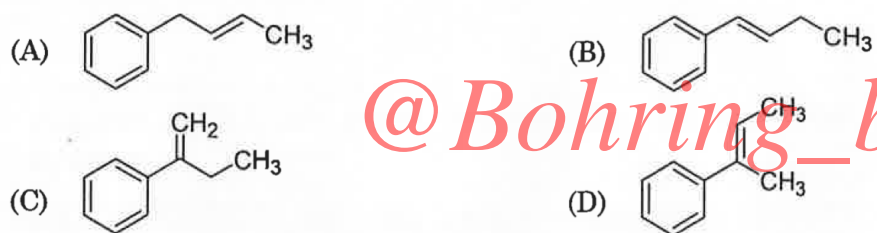
In the following reactions



Q.37 Compound **X** is

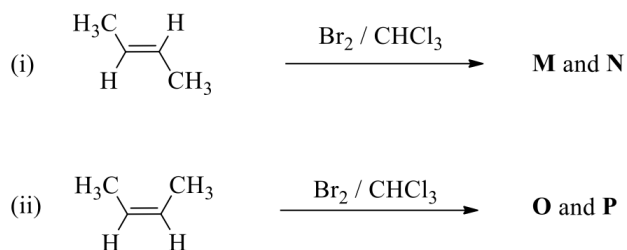


Q.38 The major compound **Y** is



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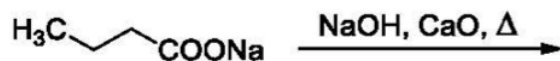
Q.25 The correct statement(s) for the following addition reactions is(are)



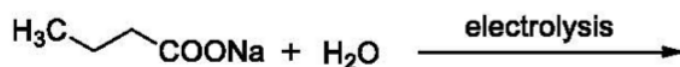
- [A] **O** and **P** are identical molecules
- [B] (**M** and **O**) and (**N** and **P**) are two pairs of diastereomers
- [C] (**M** and **O**) and (**N** and **P**) are two pairs of enantiomers
- [D] Bromination proceeds through *trans*-addition in both the reactions

Q.5 Which of the following reactions produce(s) propane as a major product?

(A)



(B)



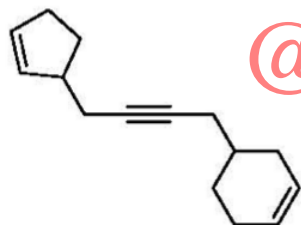
(C)



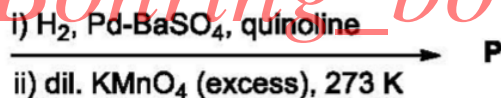
(D)



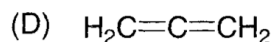
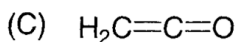
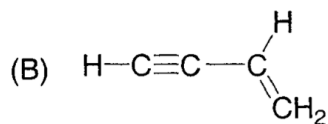
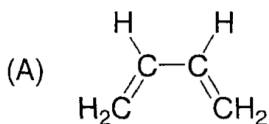
Q.13 Total number of hydroxyl groups present in a molecule of the major product P is ____



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11. Amongst the given options, the compound(s) in which all the atoms are in one plane in all the possible conformations (if any), is (are)



ANSWER: BC

22. The total number of alkenes possible by dehydrobromination of 3-bromo-3-cyclopentylhexane using alcoholic KOH is

ANSWER: 5

3. The synthesis of 3-octyne is achieved by adding a bromoalkane into a mixture of sodium amide and an alkyne. The bromoalkane and alkyne respectively are

A) $\text{BrCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{C}\equiv\text{CH}$

B) $\text{BrCH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{CH}_2\text{C}\equiv\text{CH}$

C) $\text{BrCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{C}\equiv\text{CH}$

D) $\text{BrCH}_2\text{CH}_2\text{CH}_2\text{CH}_3$ and $\text{CH}_3\text{CH}_2\text{C}\equiv\text{CH}$

ANSWER: D

8. The bond energy (in **kcal mol⁻¹**) of a C–C single bond is approximately

A) 1

B) 10

C) 100

D) 1000

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ANSWER KEY

EXERCISE-B

1. C	2. B	3. A	4. D	5. ACD	6. ACD	7. ABCD
8. BCD	9. B	10. ACD	11. B	12. ABC	13. 2	14. 7
15. 7	16. AC	17. B	18. B	19. B	20. C	21. A
22. 6	23. 3	24. 6	25. 20	26. 4		

EXERCISE-C

1. ABCD	2. A	3. B	4. D	5. AD	6. C	7. B C
8. A	9. BCD	10. BC	11. BD	12. B	13. 8	14. 12
15. 3	16. 2	17. 3	18. B	19. D	20. A	21. C
22. D						

EXERCISE-D

1. A	2. B	3. D	4. B	5. A	6. D	7. C
8. B						

EXERCISE-E

1. A	2. BD	3. B	4. B	5. AB	6. ABD	7. C
8. B						

EXERCISE-F

1. B	2. B	3. C	4. AD	5. C	6. A	7. A
8.	9. BC	10. C	11. C	12. B	13. B	14. C
15. A	16. B	17. D	18. A	19. D	20. B	21. A
22. B	23. C	24. ABD	25. A			

EXERCISE-J

14. 1	15. 1	16. 9	17. 1	18. 3	19. 3	20. 5
21. 2	22. 7	23. 2	24. 3	25. 9		

EXERCISE-K

1. A	2. A	3. D	4. A	5. A	6. A	7. B
8. ACD	9. C	10. D	11. ABC	12. C	13. D	14. B
15. A	16. D	17. C	18. D	19. D	20. A	21. C
22. D	23. CD	24. A	25. A			

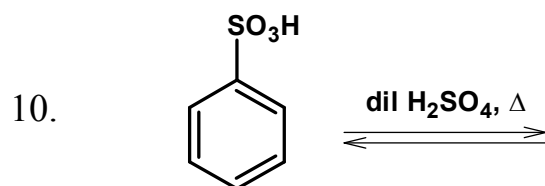
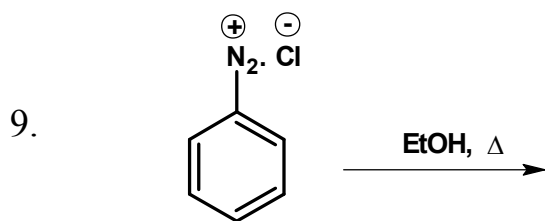
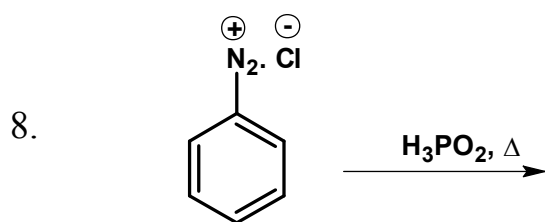
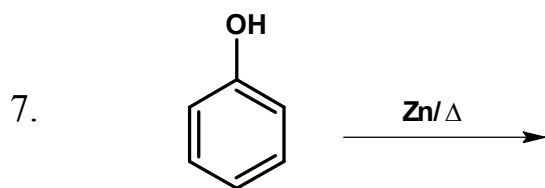
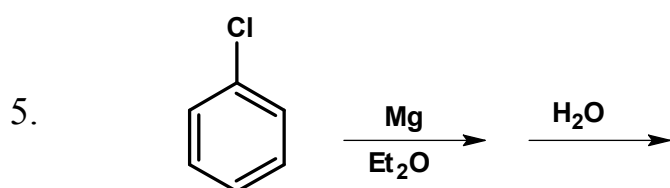
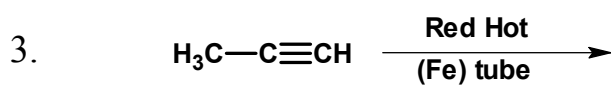
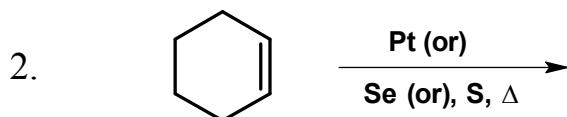
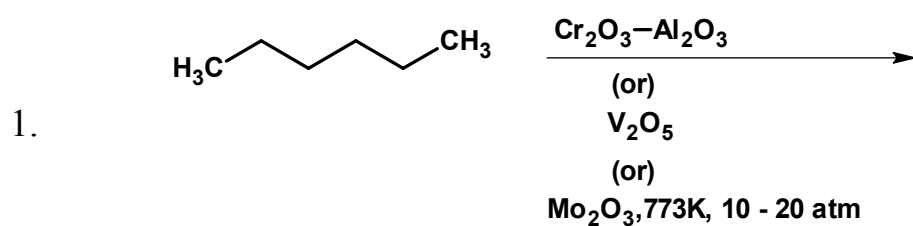
EXERCISE-L

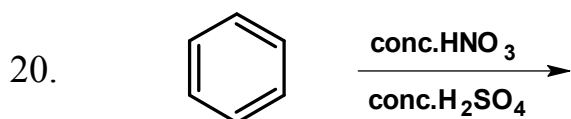
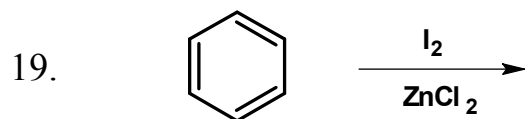
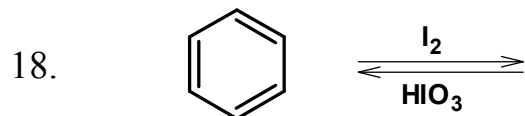
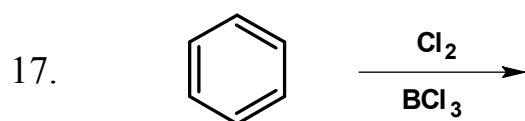
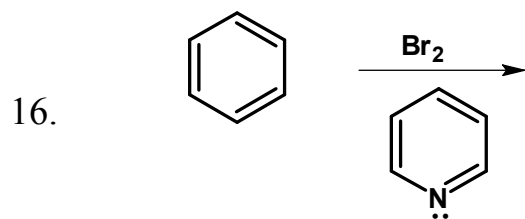
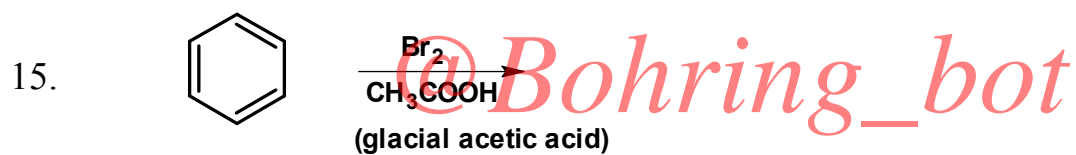
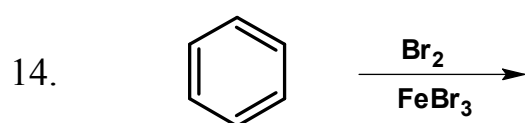
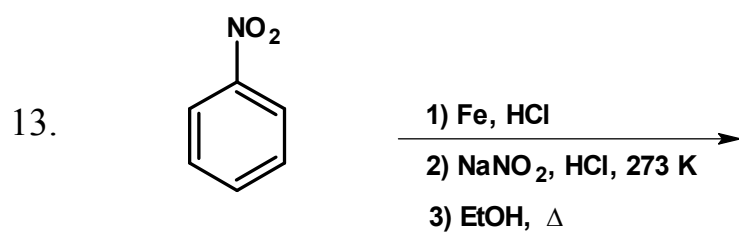
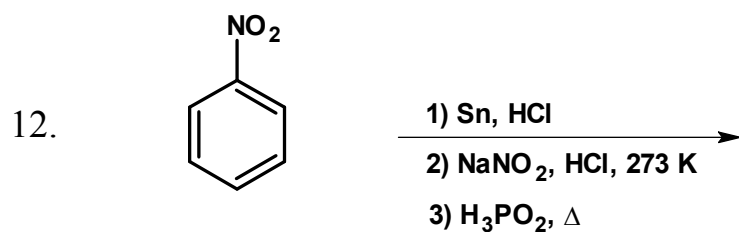
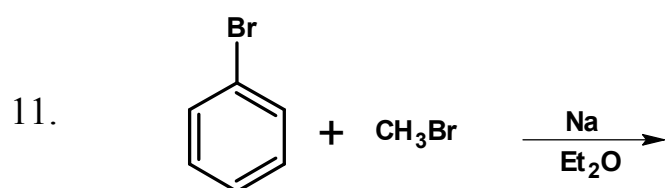
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8. A	9. D	10. A				

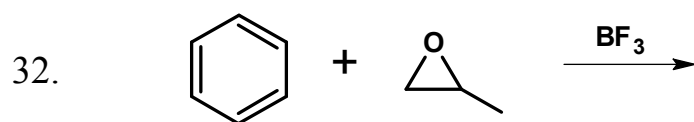
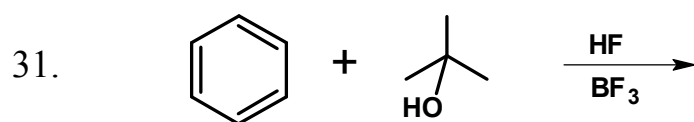
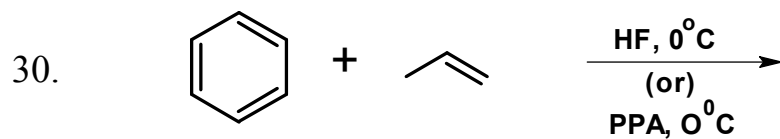
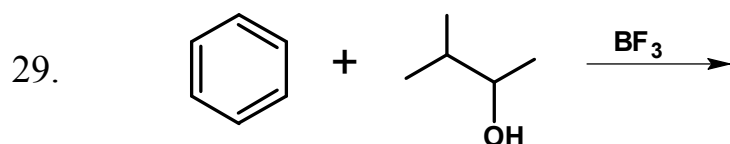
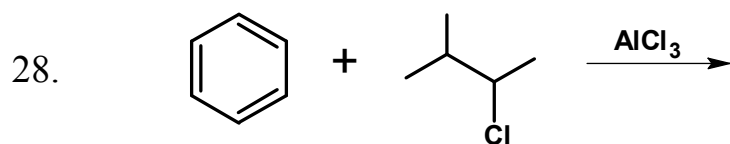
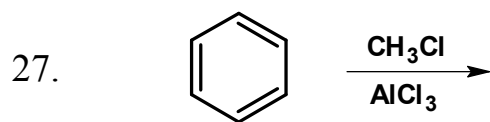
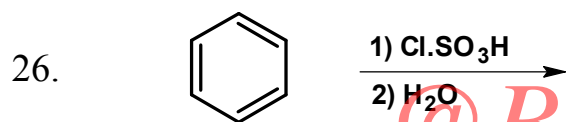
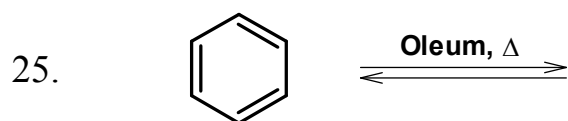
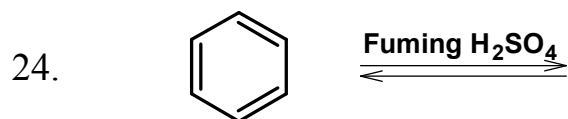
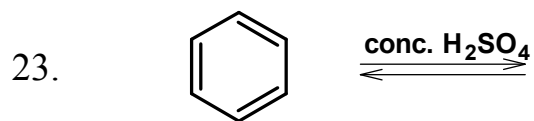
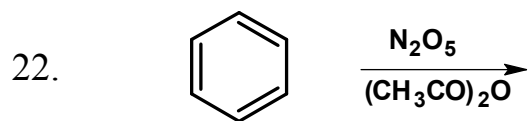
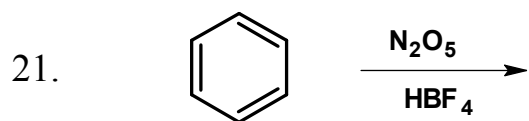
Handout- benzene

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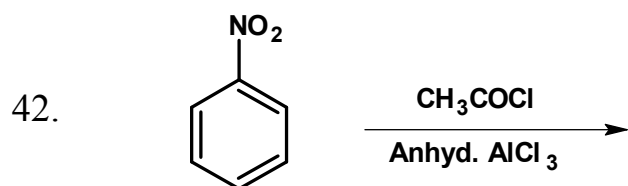
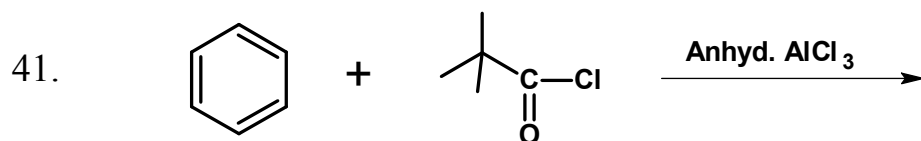
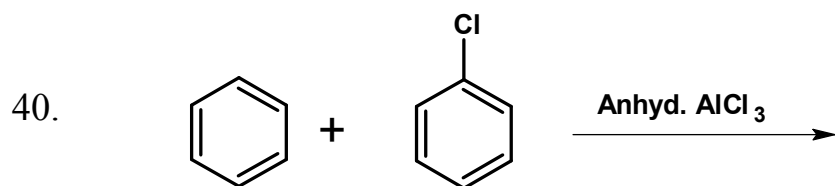
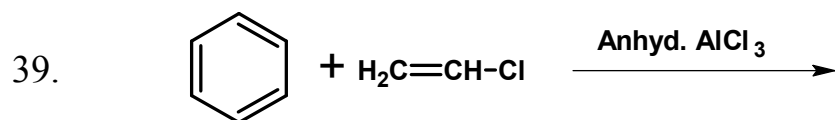
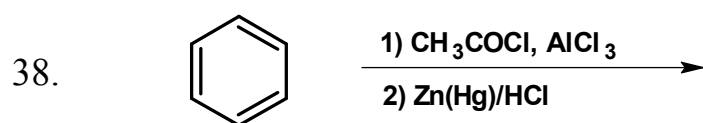
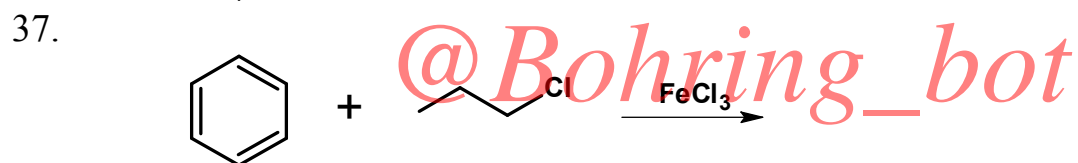
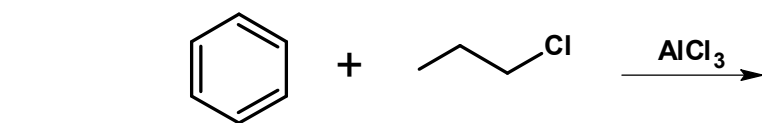
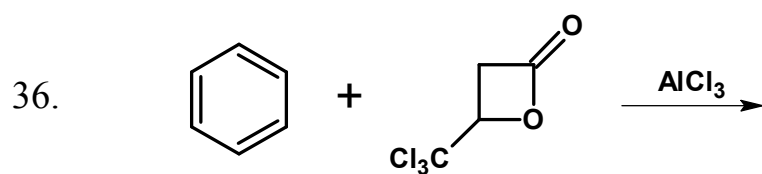
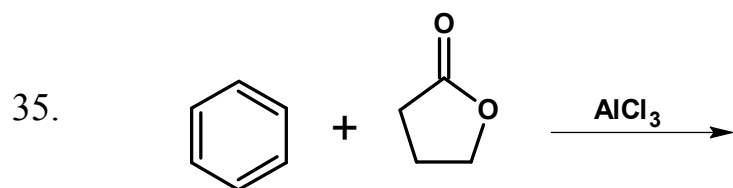
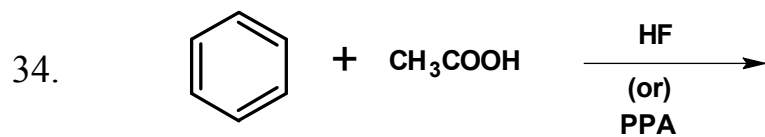
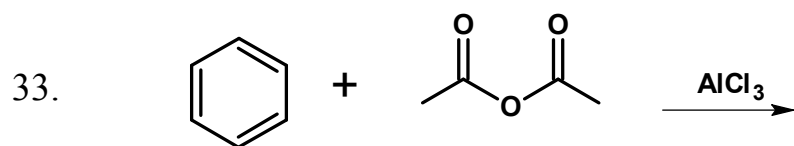
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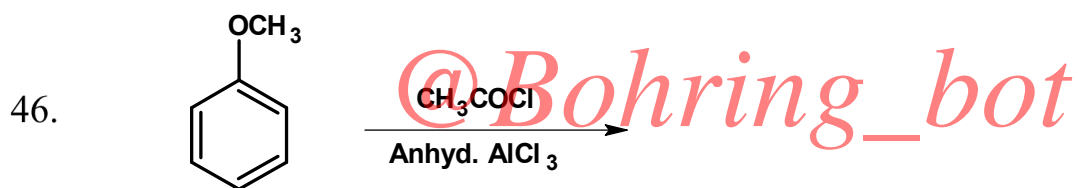
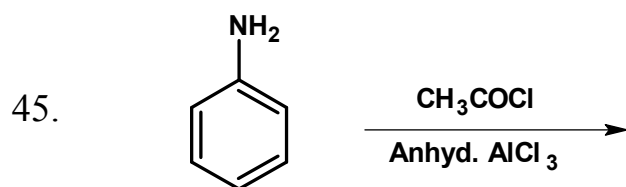
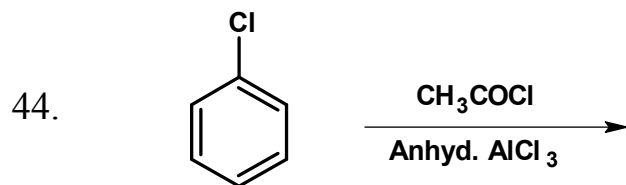
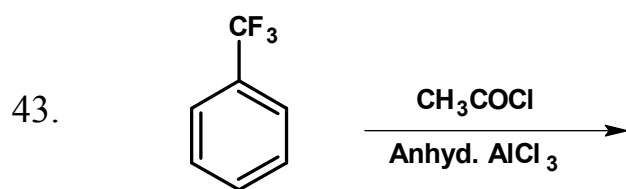






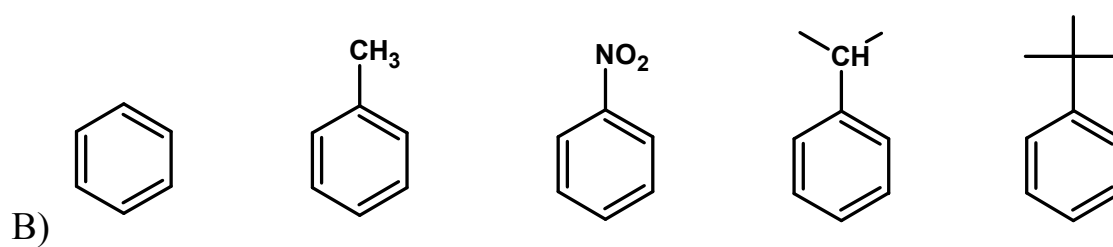
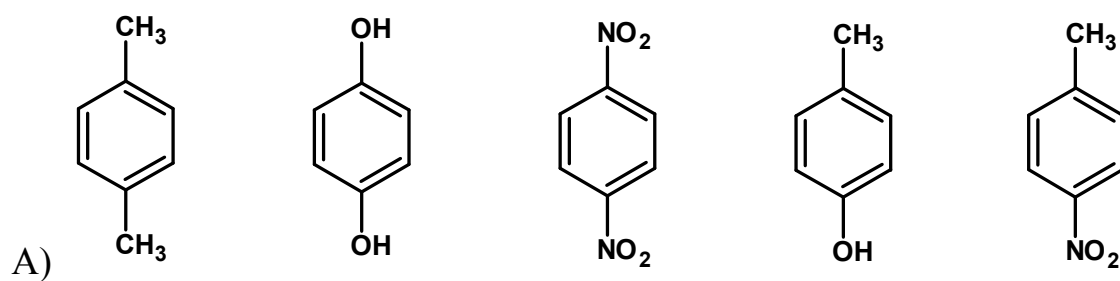
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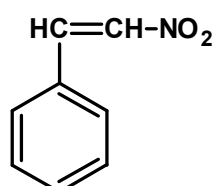
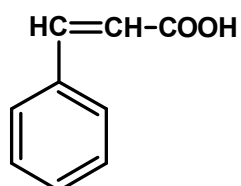
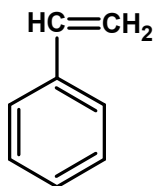
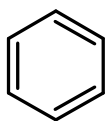


LEVEL-2

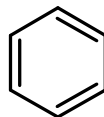
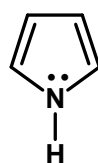
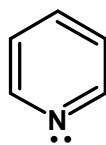
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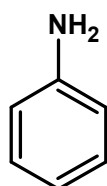
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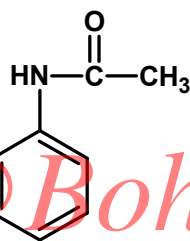
D)



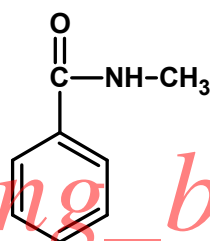
F)



1

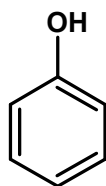


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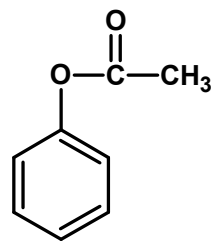


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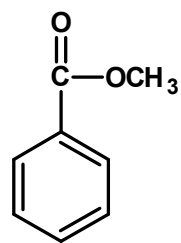
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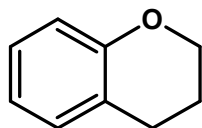


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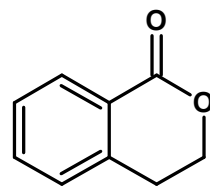


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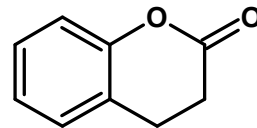
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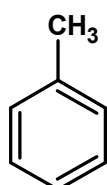


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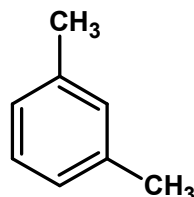


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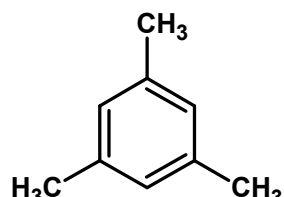
I)



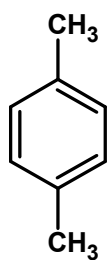
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2

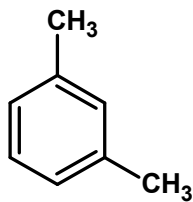


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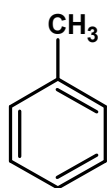


J)

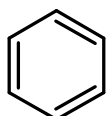
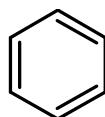
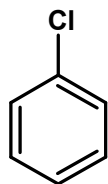
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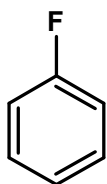
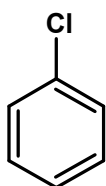
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K)



L)

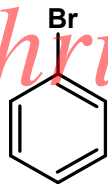


M)

1



2



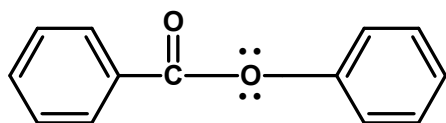
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4

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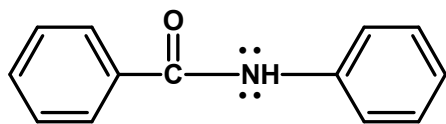
2. Which phenyl ring readily attacked by E^+ ?



1)

(A)

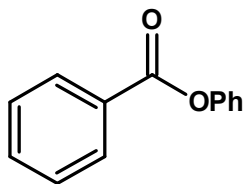
(B)



2)

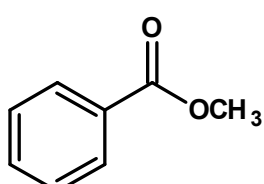
(A)

(B)



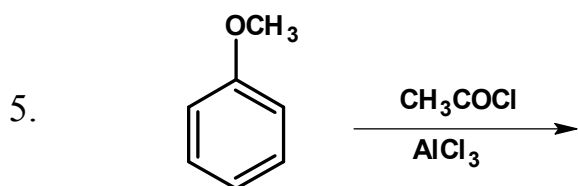
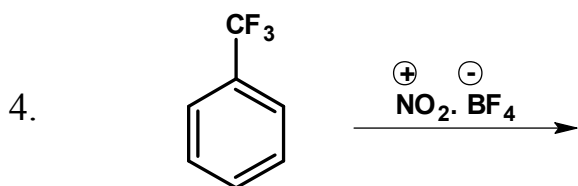
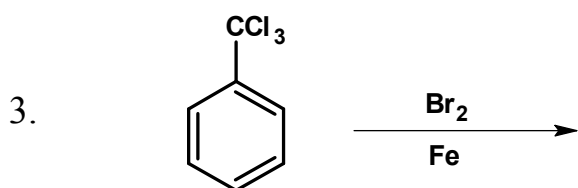
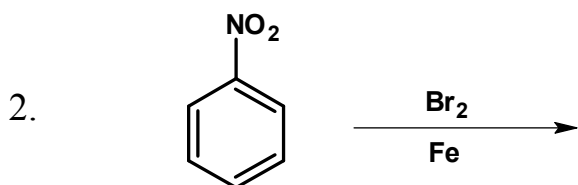
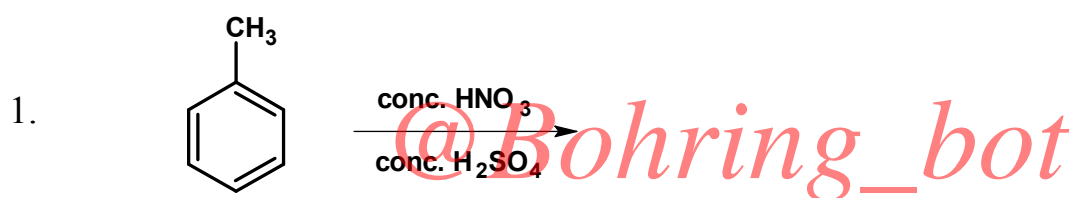
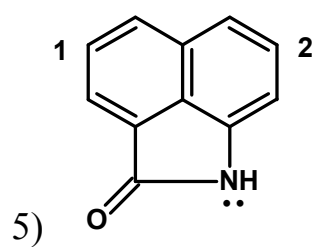
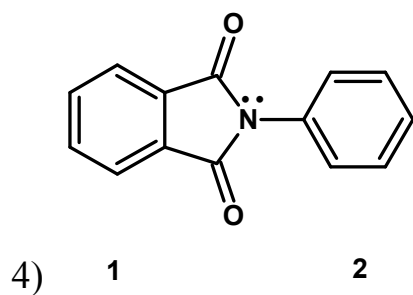
3)

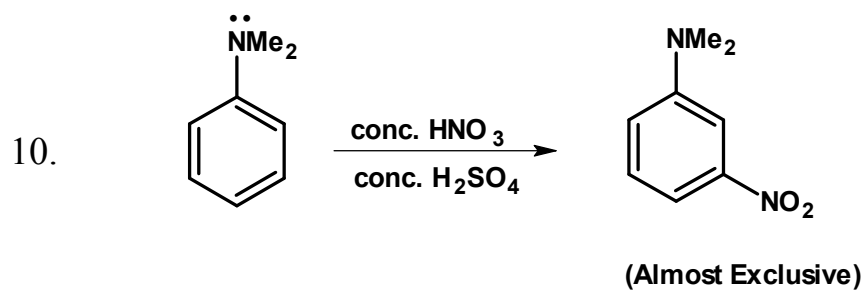
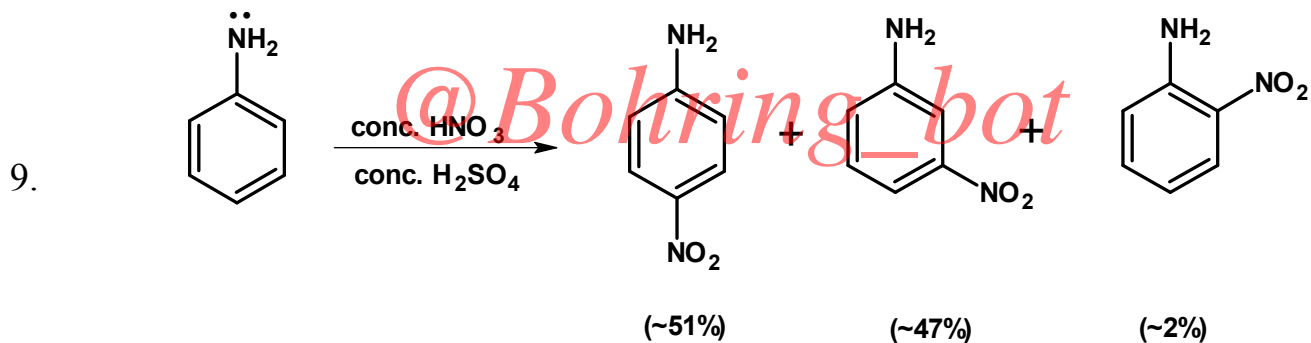
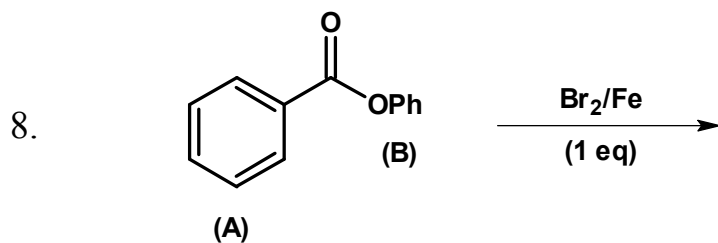
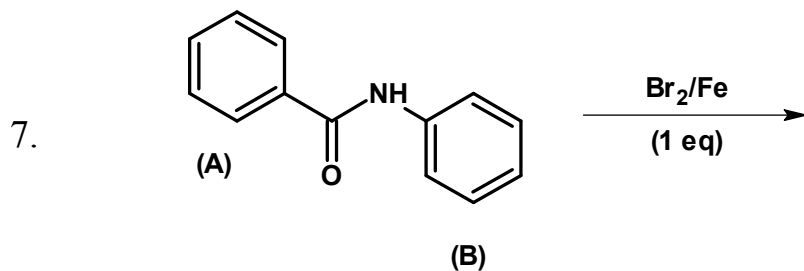
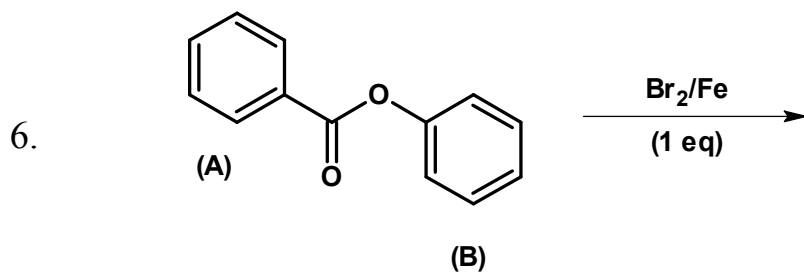
(A)



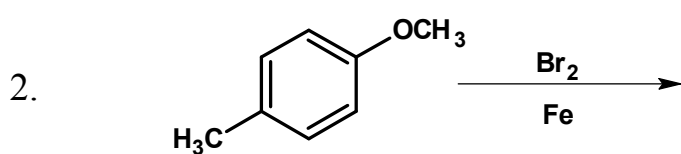
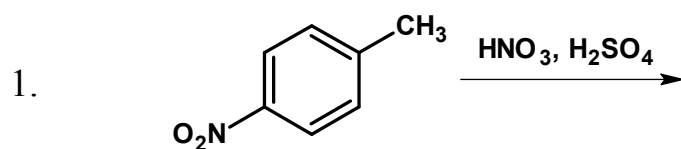
(B)

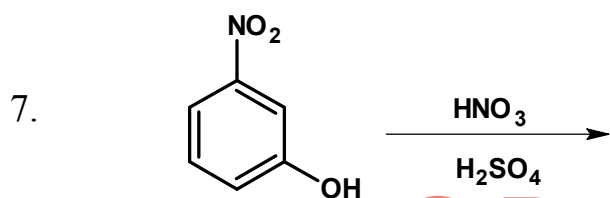
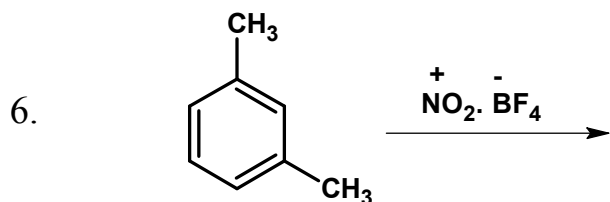
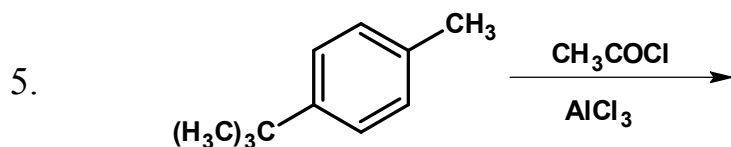
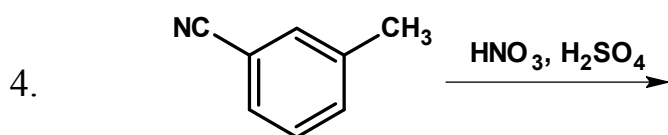
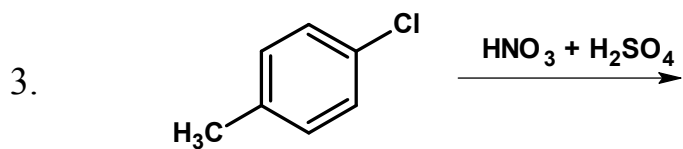
Which is more readily attacked by $E^{\oplus}$ out of two



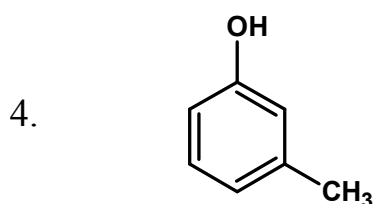
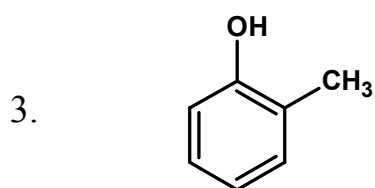
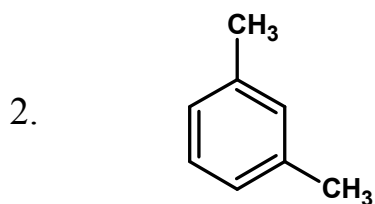
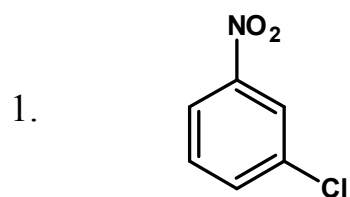


3. Write the answers?

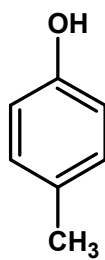




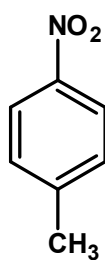
4. Identify the orientation for E^+ ? *@Bohring_bot*



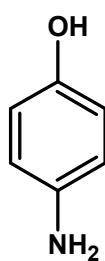
5.



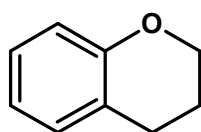
6.



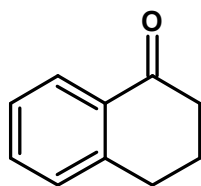
7.



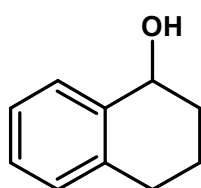
8.



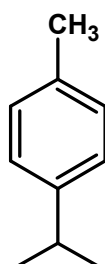
9.



10.

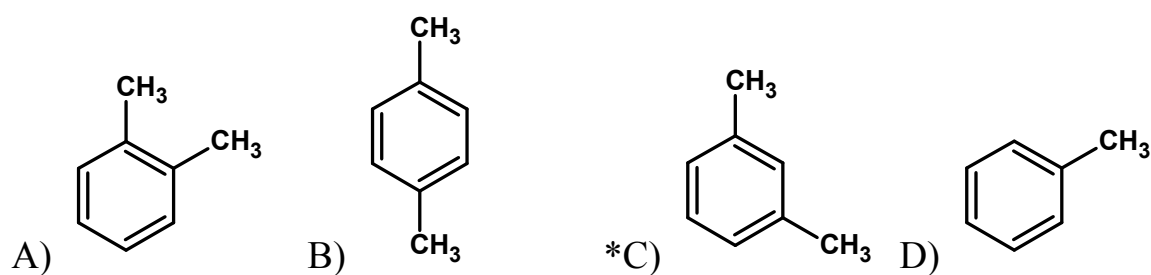


11.



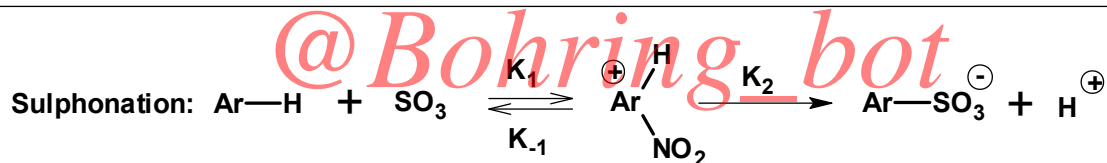
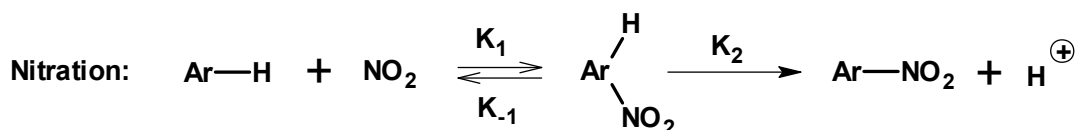
@Bohring_bot

5. Which is most reactive?



LEVEL-3

1. Which of the following statement related to Nitration & sulfonation of Benzene is/are correct?



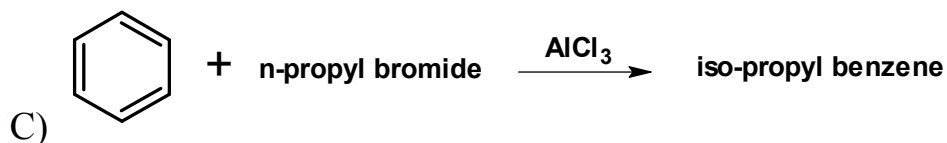
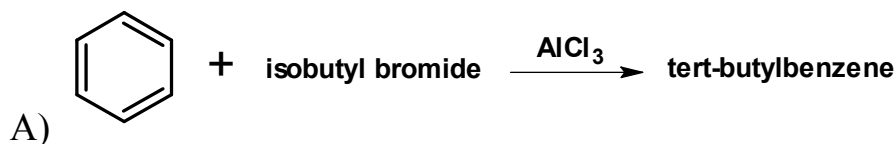
A) For Nitration, $K_2 \gg K_{-1}$

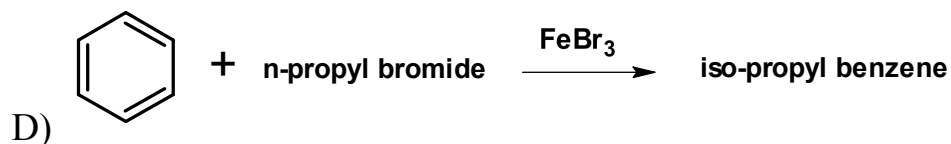
B) For Sulphonation $K_2 \gg K_{-1}$

C) For Nitration $K_2 \sim K_1$

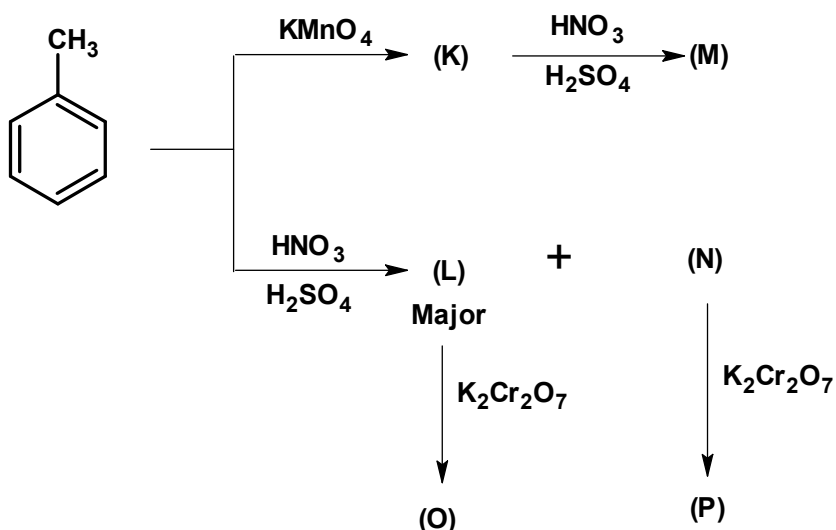
D) For Sulphonation $K_2 \sim K_{-1}$

2. Find the correct match(s) given below?





3.

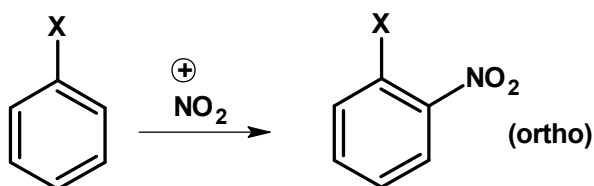


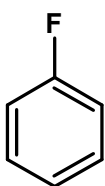
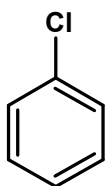
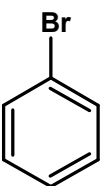
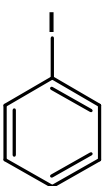
- A) (L) & (N) are position isomers B) (L) & (P) are position isomers
 C) (O) & (N) are position isomers D) (M) & (P) are position isomers

4. Identify the correct statement(s)

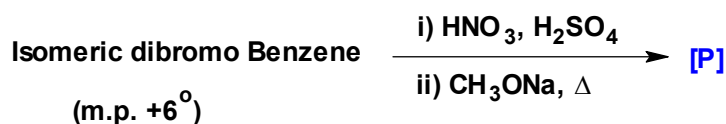
- A) Aniline is one million times as reactive as benzene
 B) Nitrobenzene is one million times as reactive as benzene
 C) Aniline is one million times as reactive as benzene
 D) Nitrobenzene is one million times as reactive as benzene.

5. On nitration of Halobenzene, which will give ortho product in more % among all?



- A)  B)  C)  D) 

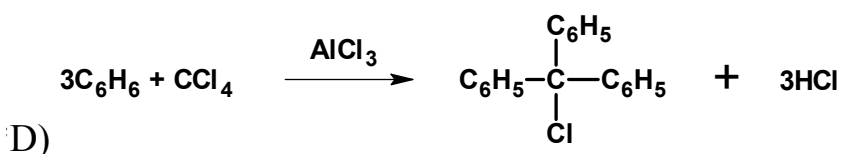
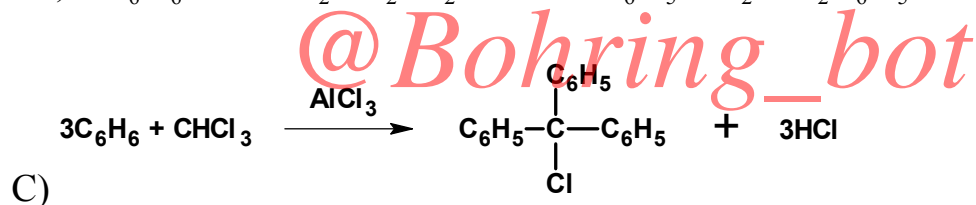
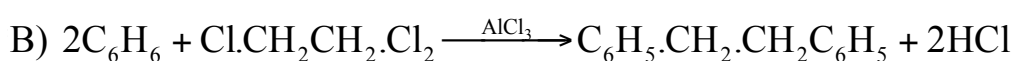
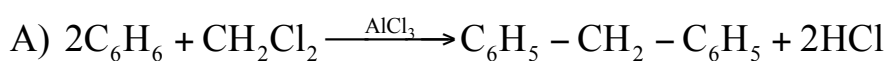
6. There are three bottles containing the three isomeric dibromo benzenes, they have the melting points $+87^\circ$, $+6^\circ$ and -7° . By a great deal of work, we prepare six dibromonitrobenzene ($\text{C}_6\text{H}_5\text{Br}_2\text{NO}_2$) and find that one is related to dibromobenzene of m.p. $+87^\circ$, two to the isomer of m.p. $+6^\circ$, and three isomer of m.p. -7° .



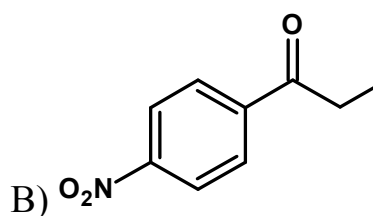
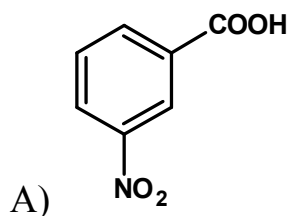
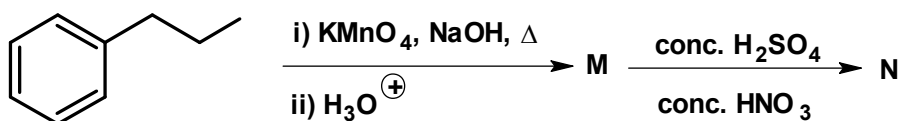
Find the locant of $-\text{Br}$ as per the IUPAC?

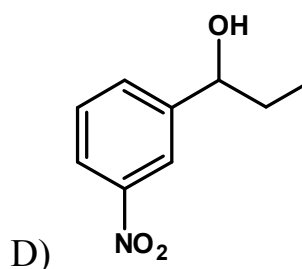
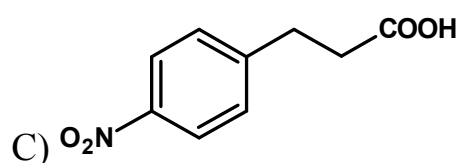
- A) 3 B) 2 C) 1 D) 4

7. Among the given reactions, which reactant(s) and product are correctly matched



8. The product 'N' of the following is



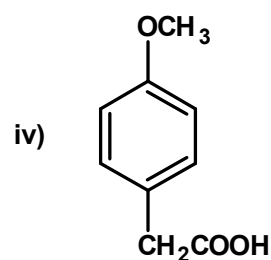
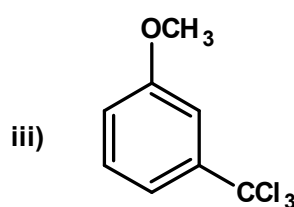
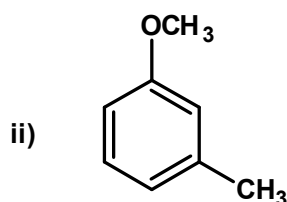
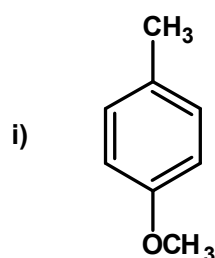


9. i) chlorobenzene is mononitrated to 'M'
 ii) nitrobenzene is mono-chlorinated to 'N'
 iii) anisole is mono-nitrated to 'P'
 iv) 2-nitrochlorobenzene is mononitrated to 'Q'

out of M, N, P, Q, the compound that undergoes reaction with NaOH Fastest is

- A) M B) N C) P D) Q

10. The order of reactivity of the following compounds in electrophilic monochlorination at the most favorable position is

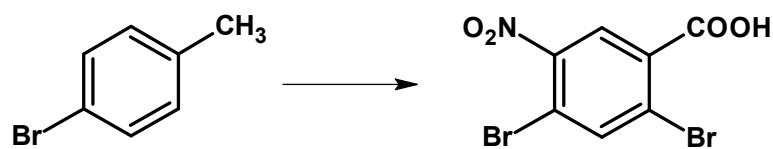


- A) I < II < IV < III B) III < IV < I < II
 C) IV < III < II < I D) III < II < IV < I

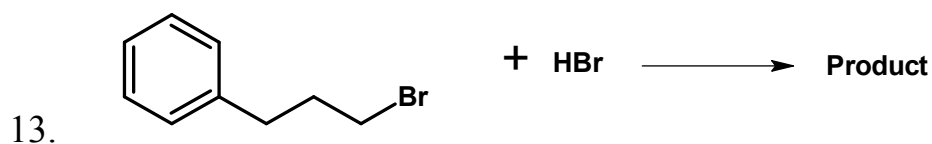
11. The reaction of 1-phenylpropane with limited amount of chlorine in presence of light gives mainly

- A) 4-chloropropyl benzene B) 1-chloro-1-phenyl propane
 C) 3-chloro-1-phenyl propane D) 2-chloro-1-phenyl propane

12. The best sequence of reactions for the following conversions is



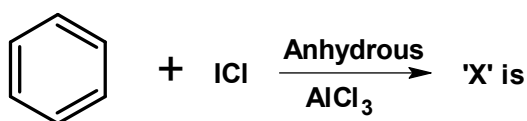
- A) i) 1mol Br₂ / FeBr₃ ii) KMnO₄, Heat iii) HNO₃ + H₂SO₄
- B) i) HNO₃ + H₂SO₄ ii) 1mol Br₂ / FeBr₃ iii) KMnO₄, Heat
- C) i) KMnO₄, Heat ii) HNO₃ + H₂SO₄ iii) 1mol Br₂ / FeBr₃
- D) i) 1mol Br₂ / FeBr₃ ii) HNO₃ + H₂SO₄ iii) KMnO₄, Heat



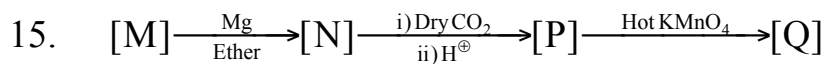
The product in above reaction is

- A)
- B)
- C)
- D) This reaction cannot take place

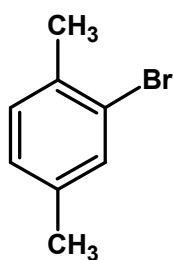
14. The compound X in the reaction



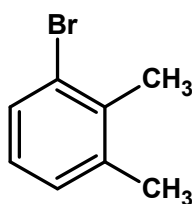
- A)
- B)
- C)
- D)



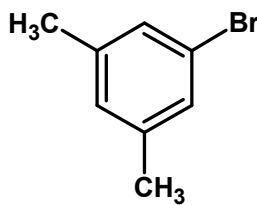
The two isomeric compounds which will give the same tricarboxylic acid after the above sequence of reactions are



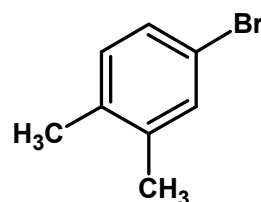
(I)



(II)



(III)



(IV)

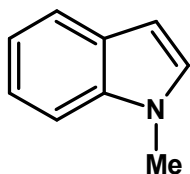
A) I, II

B) III, IV

C) I, IV

D) II, III

16. The correct order of rate of electrophilic aromatic substitution for the following given compound is



(I)



(II)



(III)

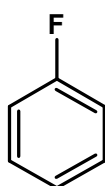
A) I > II > III

B) II > I > III

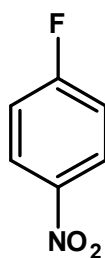
C) III > II > I

D) I > III > II

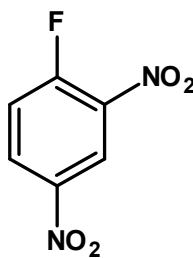
17. The order of reactivity of the following compounds towards displacement by methylamine is



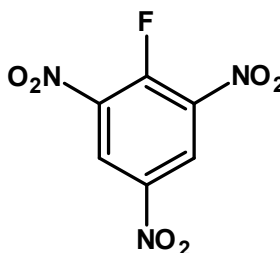
(I)



(II)



(III)



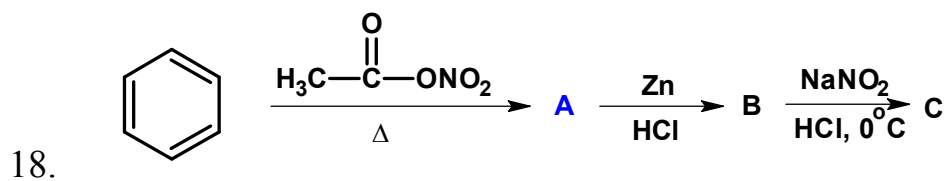
(IV)

A) I > II > III > IV

B) IV > III > II > I

C) II > III > IV > I

D) IV > II > III > I



Which statement(s) is/are true

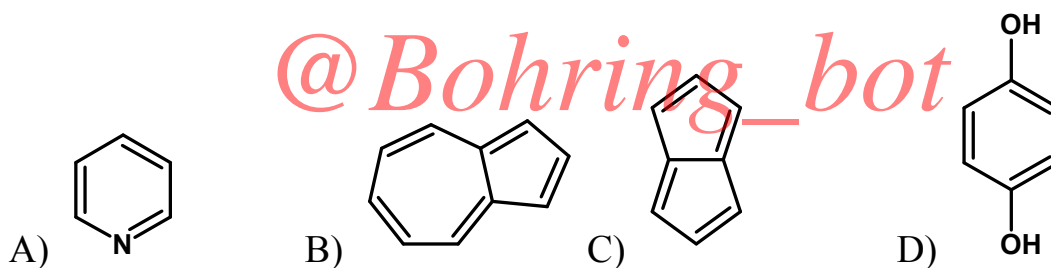
A) Compound C with NaBF_4 / Δ produces fluorobenzene

B) Compound B give 51% meta product on Nitration

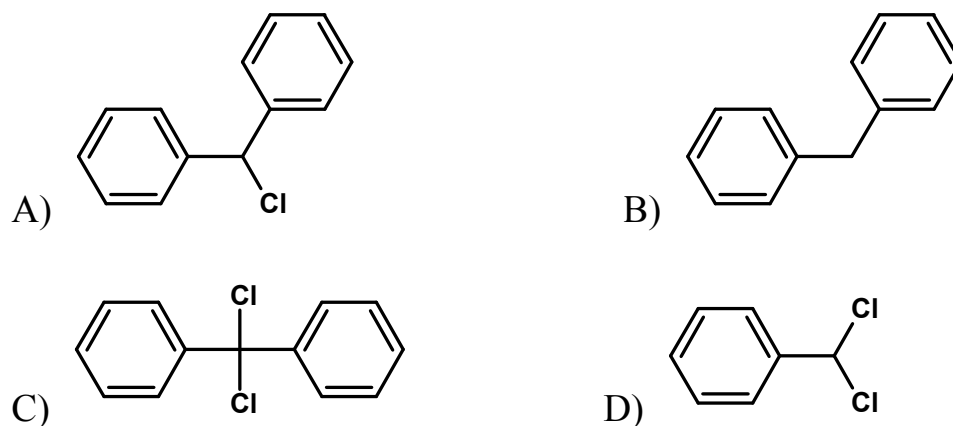
C) Compound A give meta-nitrotoluene on reaction with CH_3Cl in presence of anhydrous AlCl_3 .

D) Compound C give chlorobenzene with CuCl / HCl

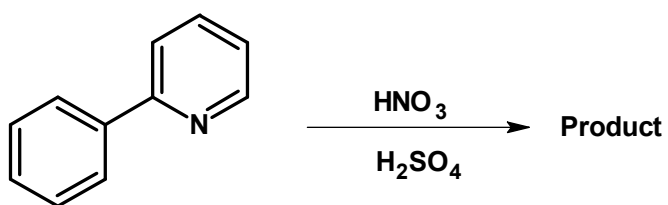
19. Polar ($\mu \neq 0$) compound is /are



20. Which of the following is obtained as the product when excess of benzene reacts with CH_2Cl_2 in presence of anhydrous AlCl_3

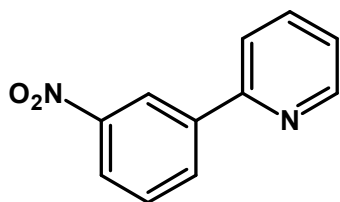


21. Which of the following statement is/are correct about the product



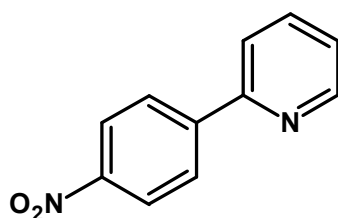
The product is

A)



The product is

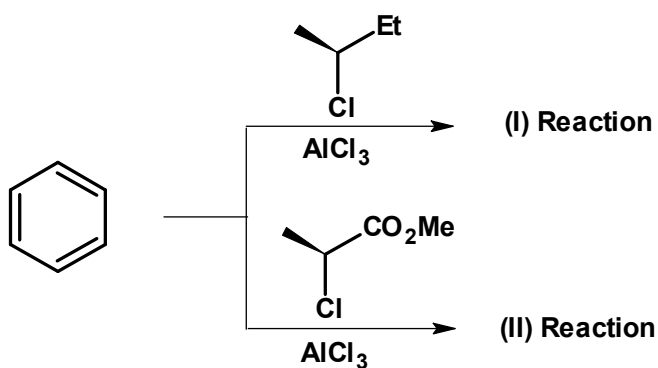
B)



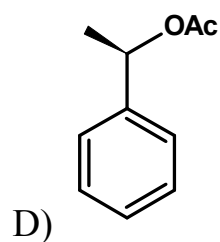
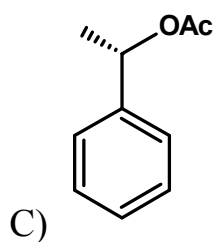
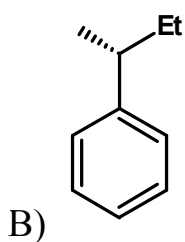
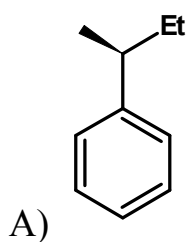
C) Pyridine act as a deactivating and o/p directing

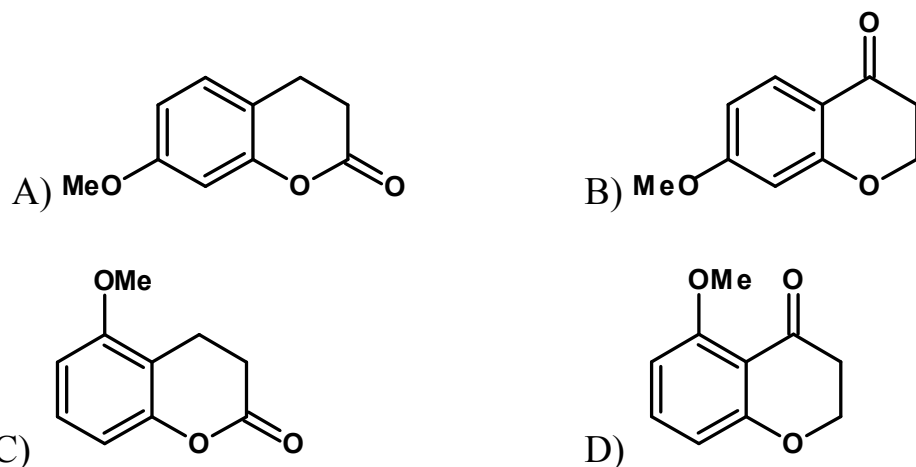
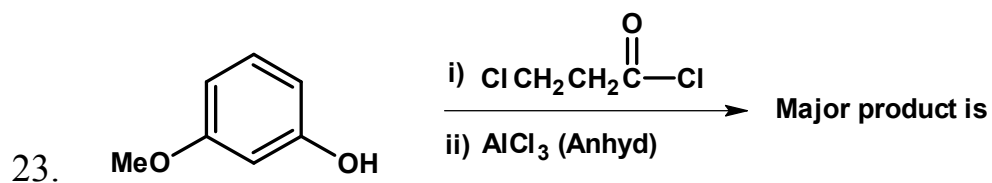
D) Pyridine act as an activating and m-directing

22. Identify the product(s) in given reaction I, II



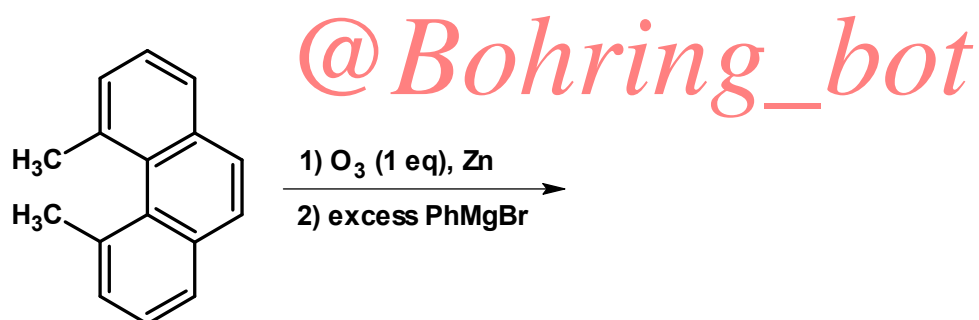
(OMs = Mesityl)



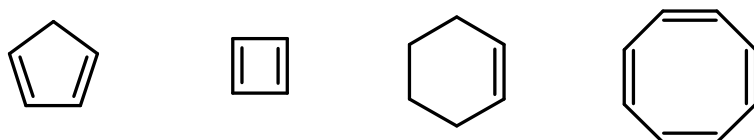


INTEGER

1. The number of possible stereoisomers obtained in the following reaction is

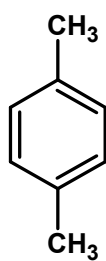


2. How many compounds can react with Na. metal

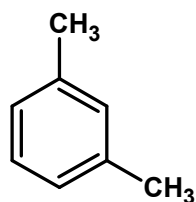


3. How many xylene(s) can give 1, 3-dibromo product as one of the possible product (as per IUPAC nomenclature) on reaction with $\xrightarrow[\text{FeBr}_3]{\text{Br}_2}$.

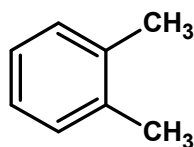
(Hint: Consider methyl gp, CH_3 – as o/p directing & use benzene as suffix, not toluene)



(K)



(L)

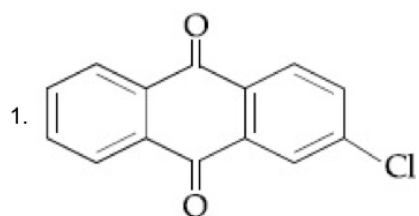
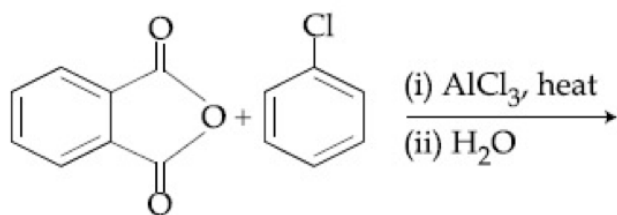


(M)

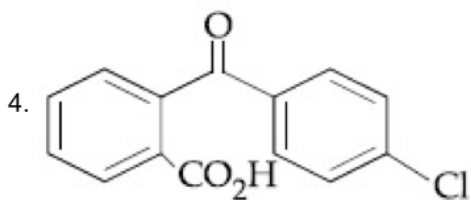
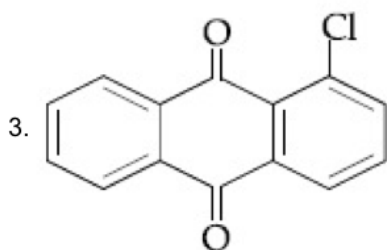
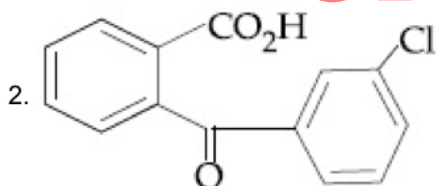
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JEE mains

The major product of the following reaction is :



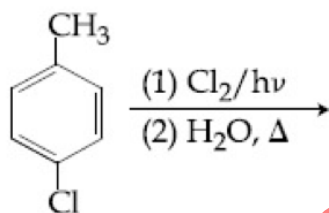
@Bohring_bot



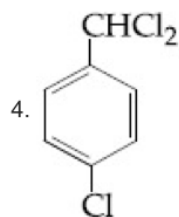
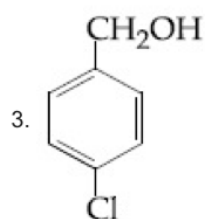
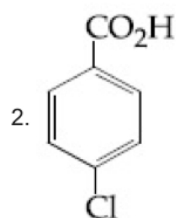
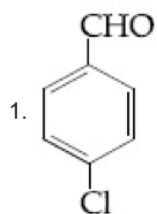
Polysubstitution is a major drawback in :

1. Friedel Craft's alkylation
2. Reimer Tiemann reaction
3. Acetylation of aniline
4. Friedel Craft's acylation

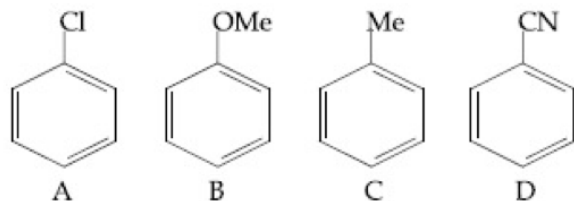
The major product of the following reaction is :



@Bohring_bot

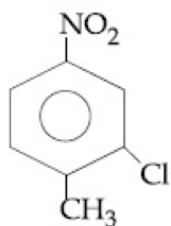


The increasing order of reactivity of the following compounds towards aromatic electrophilic substitution reaction is :



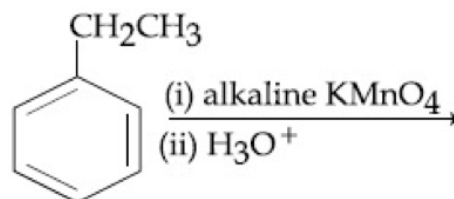
1. $D < A < C < B$
2. $B < C < A < D$
3. $A < B < C < D$
4. $D < B < A < C$

The correct IUPAC name of the following compound is :



1. 5-chloro-4-methyl-1-nitrobenzene
2. 2-chloro-1-methyl-4-nitrobenzene
3. 3-chloro-4-methyl-1-nitrobenzene
4. 2-methyl-5-nitro-1-chlorobenzene

. The major product of the following reaction is :

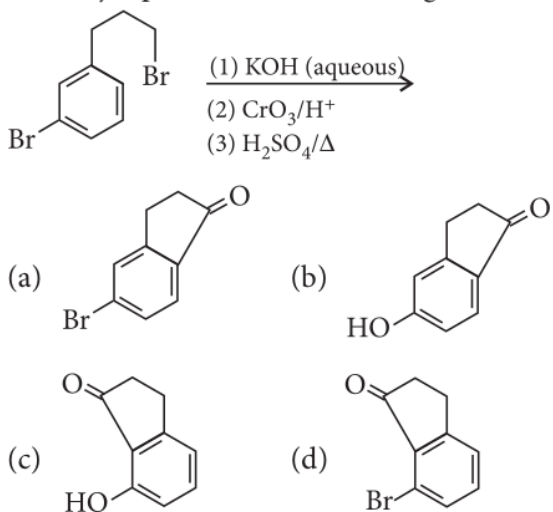


- 1.
- 2.
- 3.
- 4.

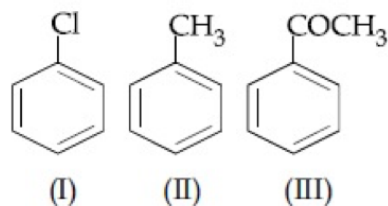
p-Hydroxybenzophenone upon reaction with bromine in carbon tetrachloride gives :

- 1.
- 2.
- 3.
- 4.

The major product of the following reaction is

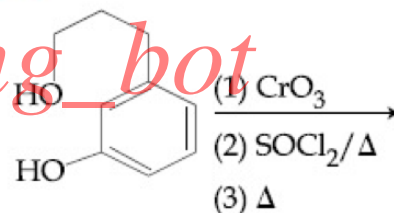


The increasing order of the reactivity of the following compounds towards electrophilic aromatic substitution reactions is :



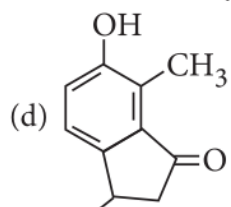
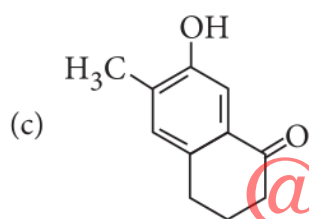
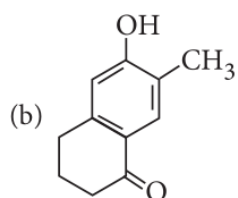
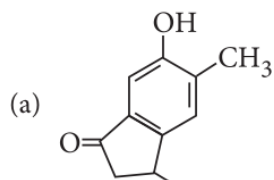
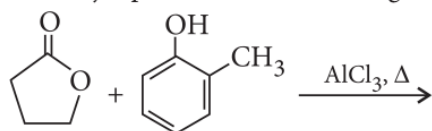
1. II < I < III
2. III < II < I
3. III < I < II
4. I < III < II

The major product of the following reaction is :

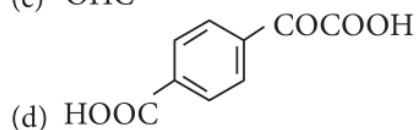
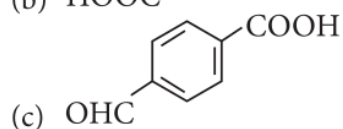
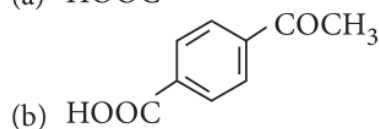
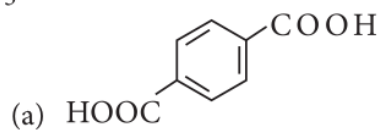
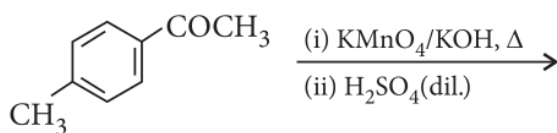


- 1.
- 2.
- 3.
- 4.

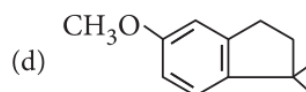
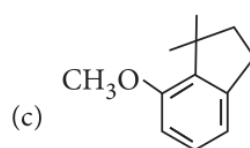
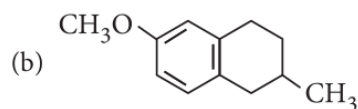
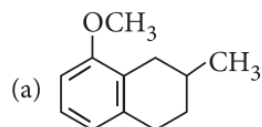
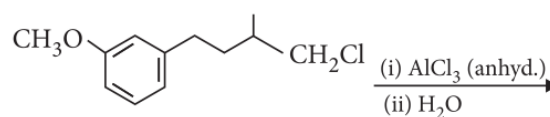
The major product of the following reaction is



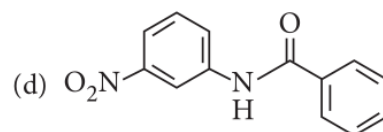
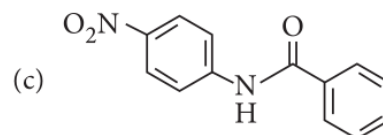
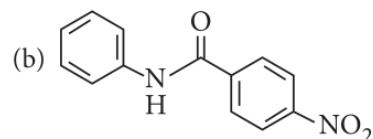
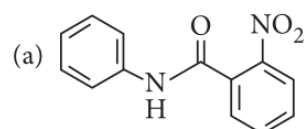
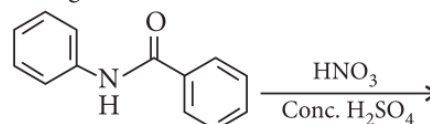
The major product of the following reaction is



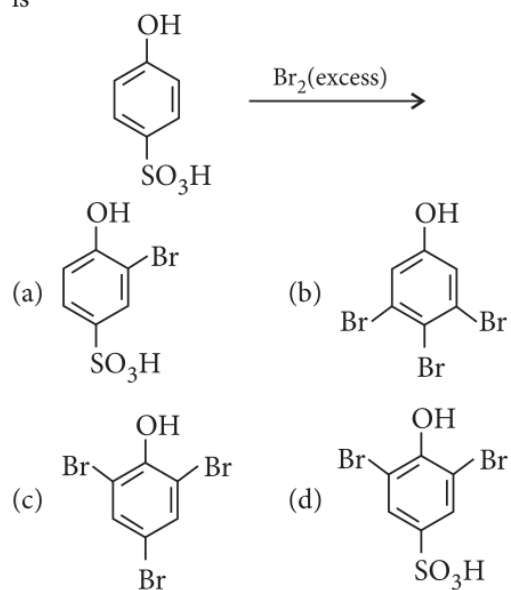
The major product of the following reaction is



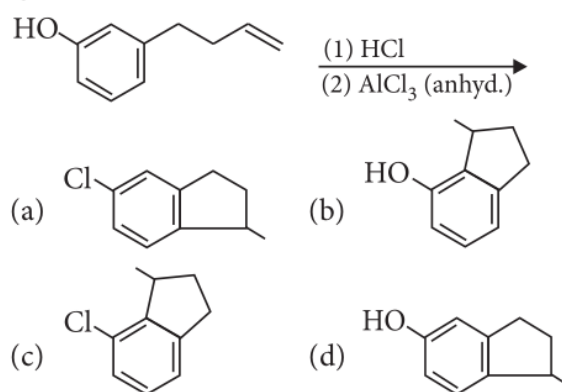
What will be the major product in the following mononitration reaction?



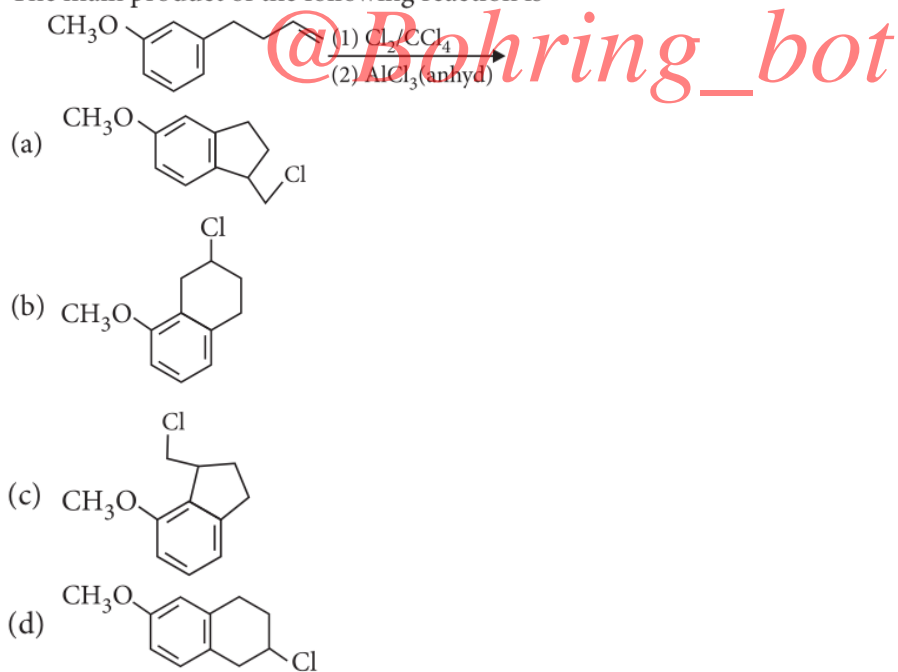
The major product of the following reaction is



The major product of the following reaction is



The main product of the following reaction is



ORGANIC CHEMISTRY

IIT JEE

@Bohring_bot

BENZENE

PREPARED BY: ASHISH SIR

SYNOPSIS

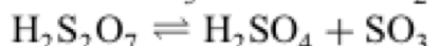
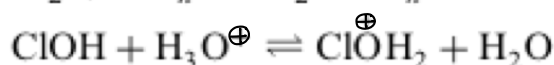
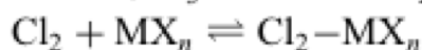
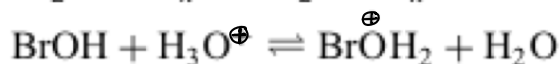
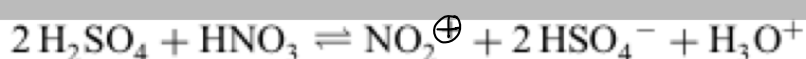
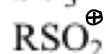
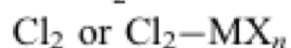
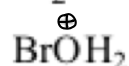
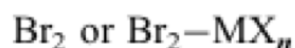
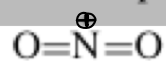
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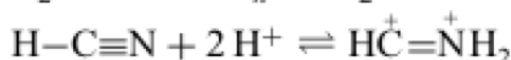
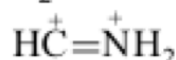
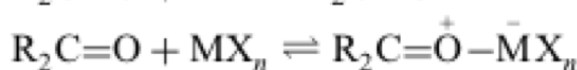
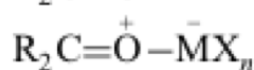
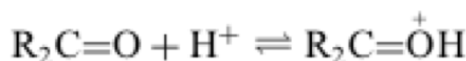
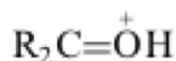
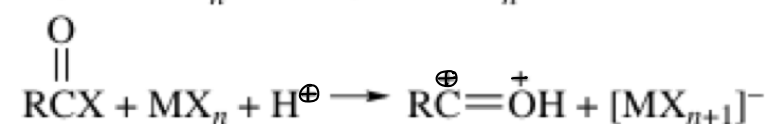
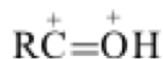
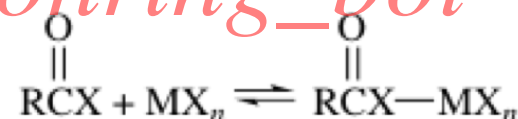
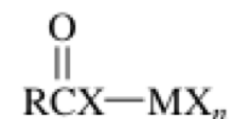
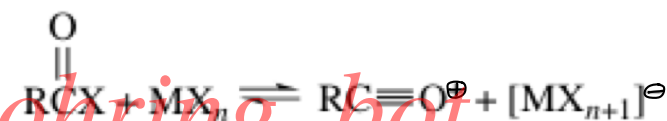
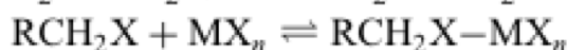
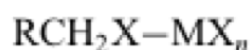
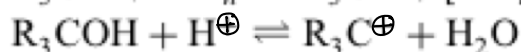
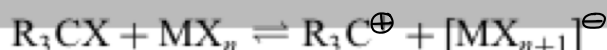
(Electrophiles)

(Mode of generation)

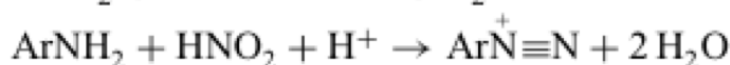
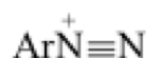
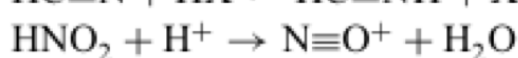
A. Electrophiles capable of substituting both activated and deactivated aromatic rings



B. Electrophiles capable of substituting activated but not deactivated aromatic rings



C. Electrophiles capable of substituting only strongly activated aromatic rings

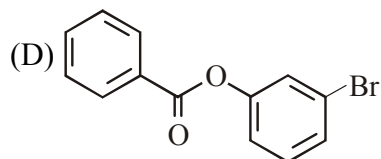
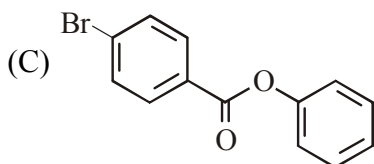
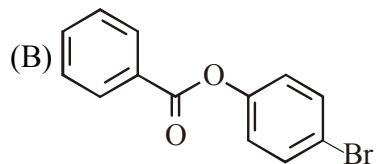
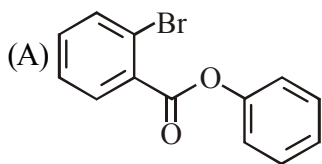


BENZENE

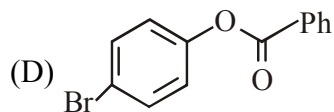
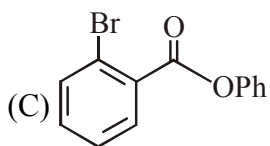
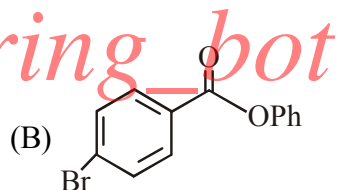
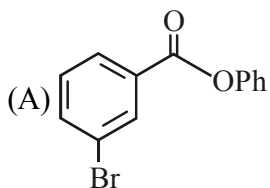
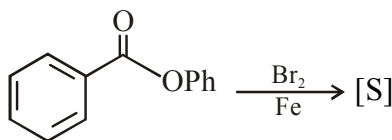
EXERCIS-A

One or more correct :

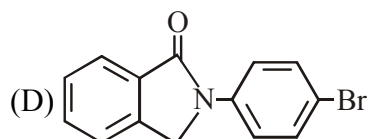
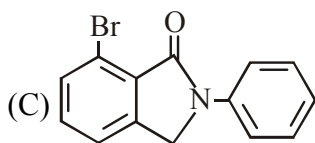
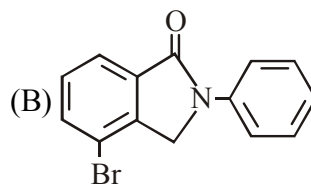
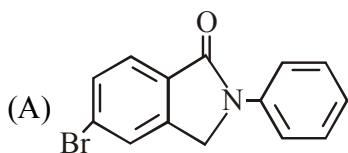
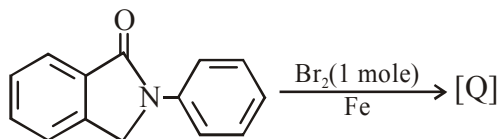
1. The major product formed on monobromination of phenyl benzoate is :-



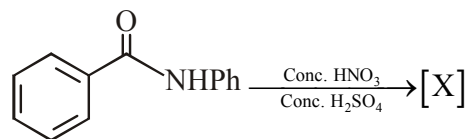
2. Identify the major product of given reaction :



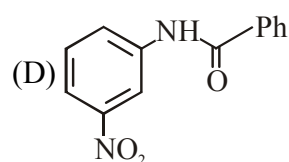
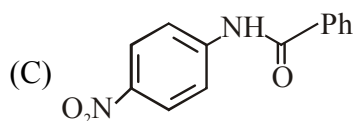
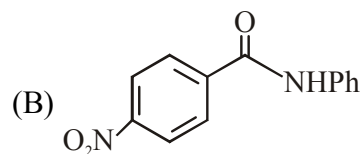
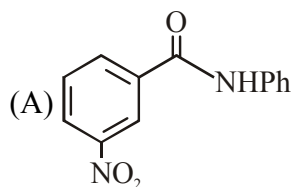
3. The major product in the given reaction is :-



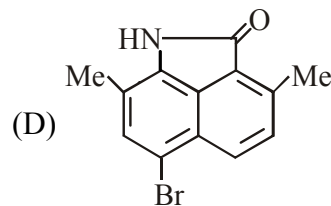
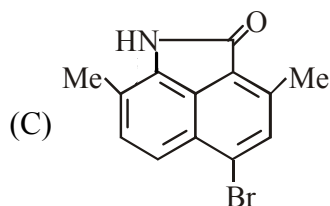
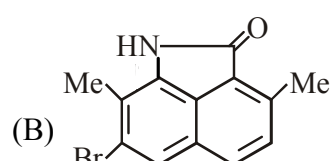
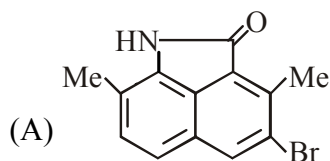
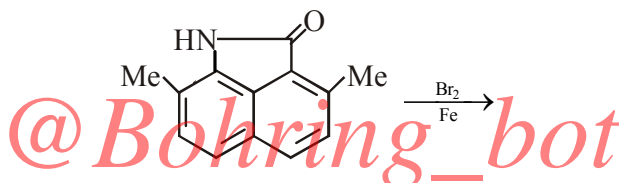
4. In the following reaction :-



the structure of the major product $[X]$ is :



5. Identify the major product in following reaction :-



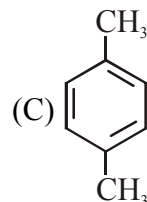
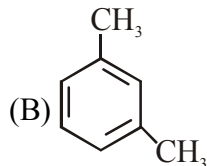
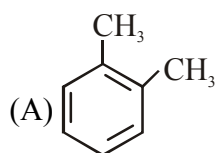
6. Identify the correct statement :-

(A) Nitration of benzene : $K_{\text{C}_6\text{H}_6} > K_{\text{C}_6\text{D}_6} > K_{\text{C}_6\text{T}_6}$ (B) Sulphonation of benzene : $K_{\text{C}_6\text{H}_6} > K_{\text{C}_6\text{D}_6} > K_{\text{C}_6\text{T}_6}$

(C) Nitration of benzene : $K_{\text{C}_6\text{H}_6} = K_{\text{C}_6\text{D}_6} = K_{\text{C}_6\text{T}_6}$ (D) Sulphonation of benzene : $K_{\text{C}_6\text{T}_6} > K_{\text{C}_6\text{D}_6} > K_{\text{C}_6\text{H}_6}$

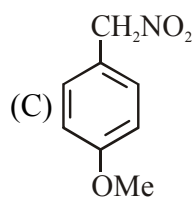
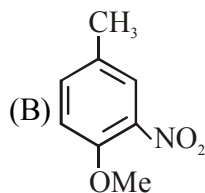
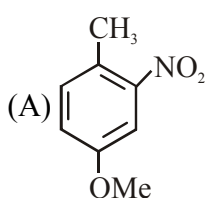
7. Ring Nitration of dimethyl benzene results in the formation of only one nitro dimethyl benzene as product.

Identify the structure of dimethyl benzene :-



(D) None

8. If p-methoxy toluene is nitrated, the major product is :-



(D) No reaction

9. Which of the following is most reactive towards chlorination ($\text{Cl}_2/\text{AlCl}_3$) :-

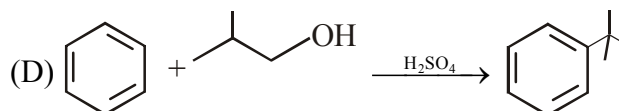
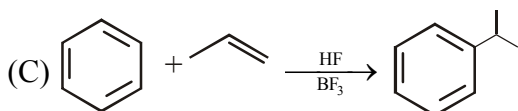
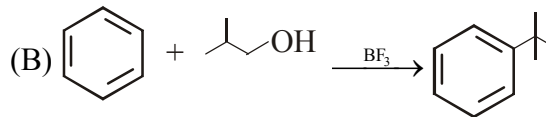
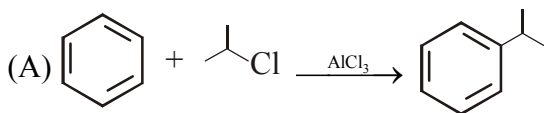
(A) m-xylene

(B) o-xylene

(C) p-xylene

(D) Toluene

10. Identify the correct match :-

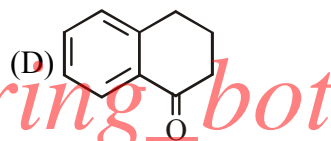
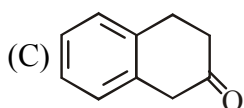


11. Find the correct product :-

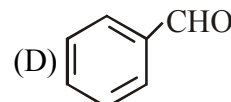
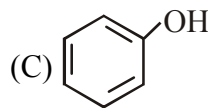
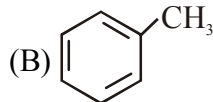
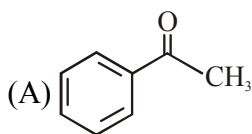
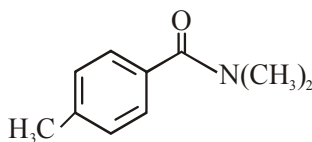


(A) $\text{Ph}-\text{CH}_2-\text{CH}=\text{CH}_2$

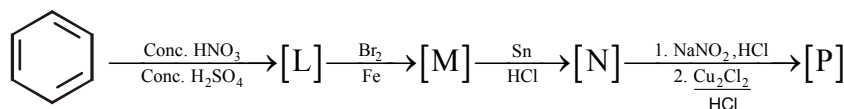
(B) $\text{Ph}-\text{CH}=\text{CH}-\text{CH}_3$



12. (Q) $\xrightarrow[5. \text{HN}(\text{CH}_3)_2]{1. \text{Br}_2/\text{FeBr}_3, 2. \text{Mg/Ether}, 3. \text{CO}_2 \text{ then } \text{H}_3\text{O}^+, 4. \text{SOCl}_2}$



13. The product [P] obtained through the following sequence of reaction is :-



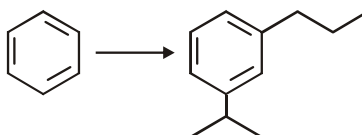
(A) 3-Chloroaniline

(B) 4-Bromochloro benzene

(C) 3-Bromochloro benzene

(D) 3-Bromo aniline

14. The sequence of three steps involved in the following conversion is



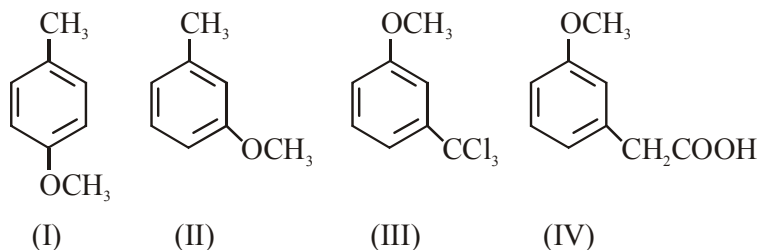
(A) (i) Friedel-Crafts alkylation; (ii) Reduction; (iii) Friedel-Crafts acylation

(B) (i) Friedel-Crafts acylation; (ii) Friedel-Crafts alkylation; (iii) Reduction

(C) (i) Friedel-Crafts acylation; (ii) Reduction; (iii) Friedel-Crafts alkylation

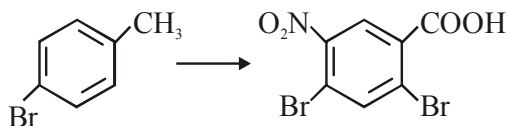
(D) (i) Friedel-Crafts alkylation; (ii) Friedel-Crafts acylation; (iii) Reduction

15. The order of reactivity of following compounds in electrophilic monochlorination is :-



- (A) $I < II < IV < III$ (B) $III < IV < I < II$ (C) $IV < III < II < I$ (D) $III < II < IV < I$

16. The best sequence of reaction for the following conversion is



- (A) (i) 1 mol $Br_2/FeBr_3$ (ii) $KMnO_4$, heat (iii) $HNO_3 + H_2SO_4$
 (B) (i) $HNO_3 + H_2SO_4$ (ii) 1 mol $Br_2/FeBr_3$ (iii) $KMnO_4$, heat
 (C) (i) $KMnO_4$, heat (ii) $HNO_3 + H_2SO_4$ (iii) 1 mol $Br_2/FeBr_3$
 (D) (i) 1 mol $Br_2/FeBr_3$ (ii) $HNO_3 + H_2SO_4$ (iii) $KMnO_4$, heat

17. Which of the following is a difference between benzene and cyclooctatetraene

- (A) Benzene reacts rapidly with bromine and cyclooctatetraene does not.
 (B) Benzene contains sp^2 -hybridized carbon atoms and cyclooctatetraene does not.
 (C) Benzene has the formula C_nH_n and cyclooctatetraene does not.
 (D) Benzene is planar and cyclooctatetraene is not.

18. Among the following statement about the nitration of aromatic compounds, the false one is :-

- (A) Nitration is an electrophilic substitution.
 (B) The rate of nitration of benzene is almost the same as that of Hexa deutero benzene.
 (C) The rate of nitration of chloro benzene is greater than that of Benzene.
 (D) The nitration of Benzene is very much faster than that of Hexa deutero benzene.

19. In the reaction :



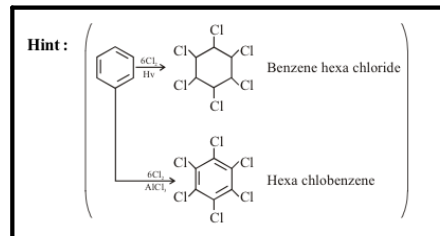
The major product obtained is :-

- (A)
- (B)
- (C)
- (D)

20. What is the final product after the following reaction has gone to completion?



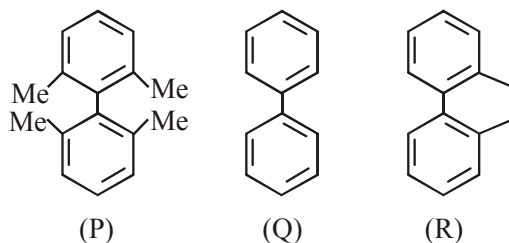
- (A) Benzene
(B) Benzene hexachloride
(C) Chlorobenzene
(D) Hexachlorobenzene



21. For electrophilic substitution in benzene derivatives which one of the following type of substituents is unknown?

- (A) Activating and m-directing
(B) Deactivating and m-directing
(C) Activating and o,p-directing
(D) Deactivating and o,p-direction

22. Arrange the following compounds in decreasing order of reactivity towards electrophiles?

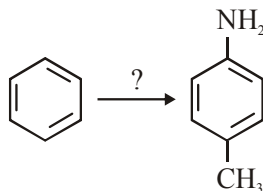


- (A) $P > Q > R$ (B) $Q > P > R$ (C) $Q > R > P$ (D) $R > Q > P$

23. Which of the following compounds has the highest boiling point :-

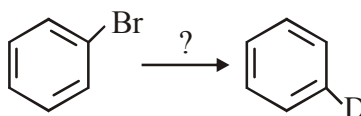
- (A) Mesitylene (B) Benzene (C) Toluene (D) Cyclo hexane

24. The sequence of reagents that will accomplish the synthesis of p-toluidine (4-amino toluene) from benzene is:-



- (A) (i) CH_3Cl , AlCl_3 (ii) HNO_3 , H_2SO_4 (iii) H_2
 (B) (i) CH_3Cl , AlCl_3 (ii) HNO_3 , H_2SO_4 (iii) Fe/HCl (iv) NaOH
 (C) (i) HNO_3 , H_2SO_4 (ii) Fe/HCl (iii) NaOH (iv) CH_3Cl , AlCl_3
 (D) (i) HNO_3 , H_2SO_4 (ii) CH_3Cl , AlCl_3 (iii) Fe/HCl (iv) NaOH

25. Which reagent(s) is (are) not suitable for the following transformation :-

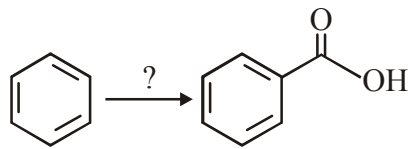


- (A) n-BuLi followed by D_2O (B) Li Metal followed by D_2O
 (C) $(\text{CH}_3)_2\text{CuLi}$ followed by D_2O (D) Mg followed by D_2O

26. Heating benzene in a large excess of 80% D_2SO_4 in D_2O results in what product?

- (A) $\text{C}_6\text{H}_5\text{SO}_3\text{D}$ (B) $\text{C}_6\text{H}_5\text{OH}$ (C) $\text{C}_6\text{H}_5\text{D}$ (D) C_6D_6

27. Which reagent sequence is best for given conversion :-

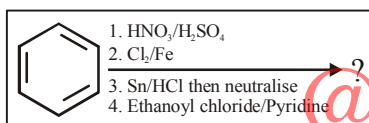


- (A) (i) $\text{CH}_3\text{COCl}/\text{AlCl}_3$ (ii) LiAlH_4 then H_3O^+ (B) (i) $\text{CH}_3\text{COCl}/\text{AlCl}_3$ (ii) Cold H_2O
 (C) (i) $\text{CH}_3\text{Cl}/\text{AlCl}_3$ (ii) $\text{KMnO}_4/\text{H}_3\text{O}^+/\text{Heat}$ (D) (i) Mg/THF (ii) CO_2 (iii) H_3O^+

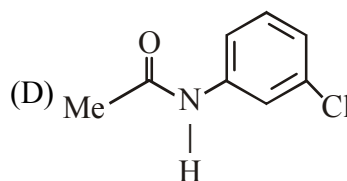
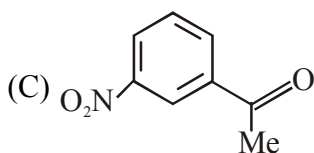
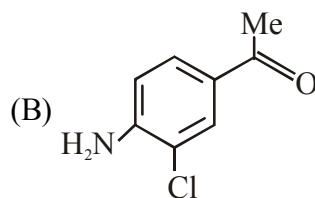
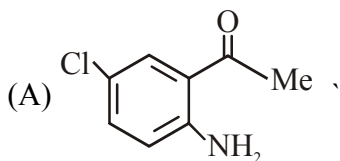
28. Which of the following statements is TRUE regarding the electrical conductivity of HCl in aqueous solution, HCl as a gas and HCl in benzene?

- (A) All three conduct electricity because HCl is a strong acid
 (B) HCl in aqueous solution only conducts electricity because it is ionized but not HCl gas and HCl in benzene
 (C) HCl in benzene is not conducting because benzene is a non-polar solvent but HCl gas, HCl in aqueous solution conduct electricity
 (D) HCl in aqueous solution and HCl in benzene are conducting because they are solution but not HCl gas

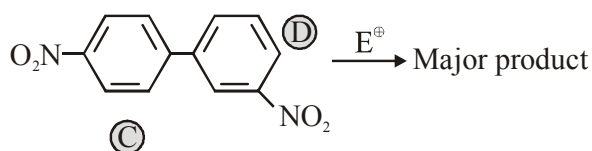
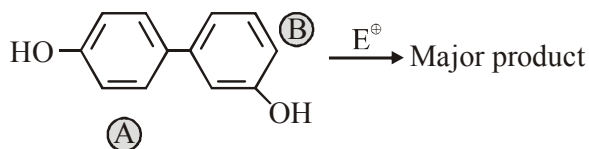
29.



Identify the product –



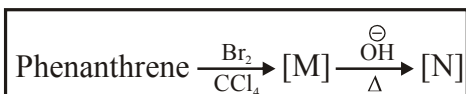
30.



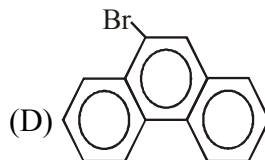
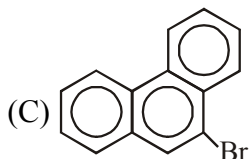
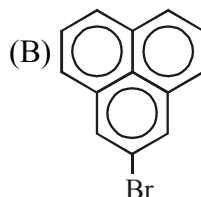
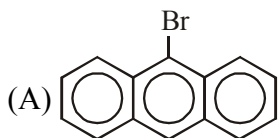
- (A) In Ist case, E^+ attack on ring A. (B) In IInd case, E^+ attack on ring C.
 (C) In IInd case, E^+ attack on ring D. (D) In Ist case, E^+ attack on ring B.

EXERCISE-B

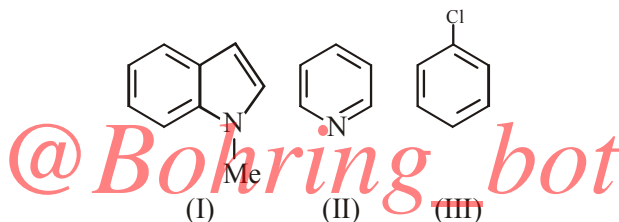
1.



[N] will be :-



2. The correct order of rate of electrophilic aromatic substitution for the following given compound is :-



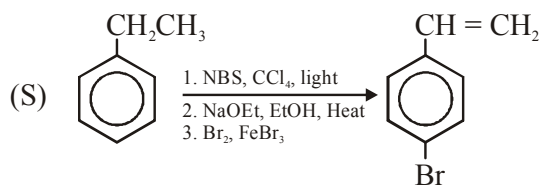
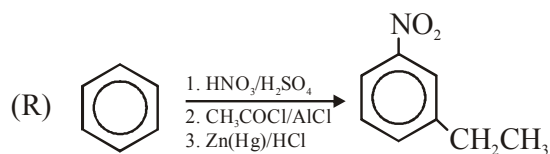
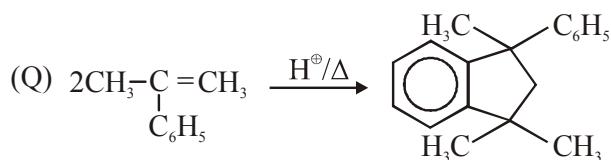
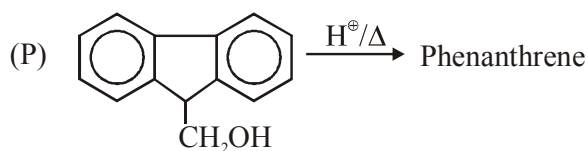
(A) I > II > III

(B) II > I > III

(C) III > II > I

(D) I > III > II

3. Which of the following match of reactant, product is/are correct?



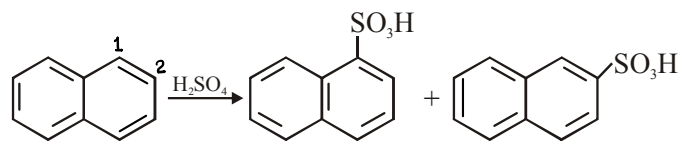
(A) P

(B) Q

(C) R

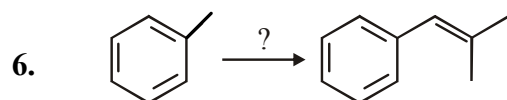
(D) S.

5. The sulfonation of naphthalene was carried out at two different temperature, 80°C and 160°C. Both temperature produced the same two products, except the relative distribution of products were different, as shown below.



The 1-sulfonation product is favoured at low temperatures because :

- (A) High temperatures cause carbocation rearrangements
 (B) Low temperatures direct to the 1-position of polycyclic aromatics
 (C) There is steric crowding around the SO₃H group
 (D) The cation intermediate is more stable.



- (A) i. HCl, peroxides ii. P(Ph)₃ iii. n-BuLi iv. 2-propanone
 (B) i. NBS, CHCl₃ ii. P(Ph)₃ iii. n-BuLi iv. 2-propanone
 (C) i. NBS, CHCl₃ ii. Mg iii. 2-bromopropane
 (D) i. NBS, CHCl₃ ii. Mg iii. 2-bromopropane iv. H₂SO₄, heat

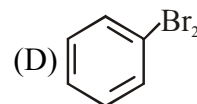
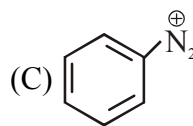
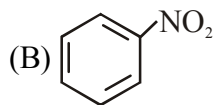
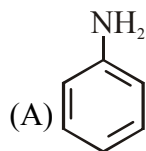
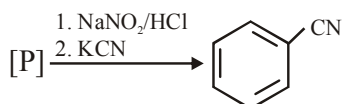
7. Consider the following statements.

1. chlorobenzene is mononitrated to [M].
2. nitrobenzene is monochlorinated to [N].
3. anisole is mononitrated to [P].
4. 2-nitrochloro benzene is mononitrated to [Q].

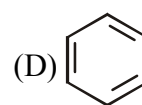
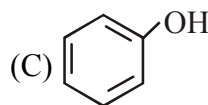
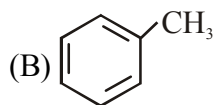
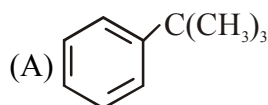
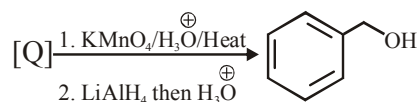
Out of M, N, P and Q, the compound that undergoes reaction with aq. NaOH fastest is :-

- (A) M (B) N (C) P (D) Q

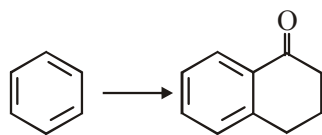
8. Choose the starting material from the selection provided, that gives the product via the transformation shown:-



9. Choose the starting material from the selection provided, that gives the product via transformation shown :-

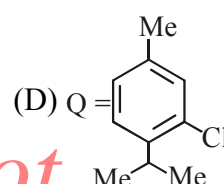
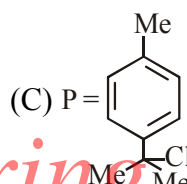
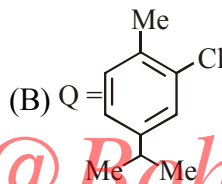
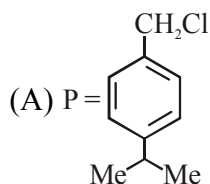
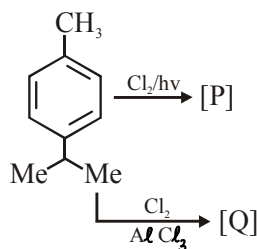


10. The appropriate reagents required for carrying out the following transformation are :

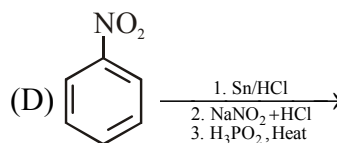
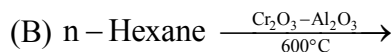
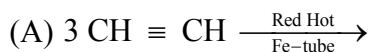


- (A) (i) succinic anhydride, AlCl_3 ; (ii) Zn/Hg , HCl ; (iii) H_2SO_4
 (B) (i) maleic anhydride, AlCl_3 ; (ii) $\text{H}_2\text{N-NH}_2$, KOH ; (iii) H_2SO_4
 (C) (i) succinic anhydride, FeCl_3 ; (ii) LiAlH_4 ; (iii) H_2SO_4
 (D) (i) phthalic anhydride, $\text{F}_3\text{B} \cdot \text{OEt}_2$; (ii) $\text{HS}(\text{CH}_2)_2\text{SH}$, H^+ ; (iii) Raney Ni; (iv) polyphosphoric acid

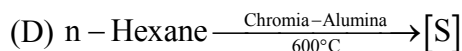
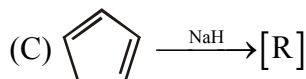
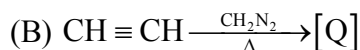
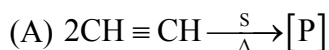
11. Identify the major product of both respectively :-



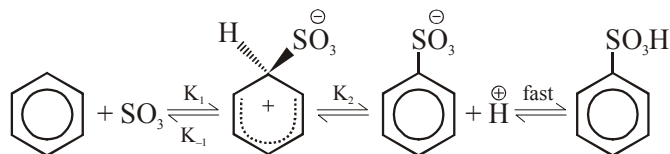
12. Which of the following preparation method(s) for Benzene is (are) correct :-



13. Among the following reactions given, aromatic product is (are) possible in –



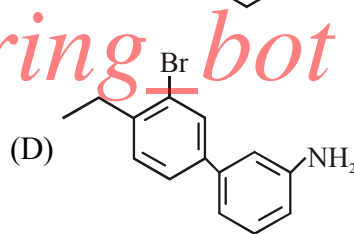
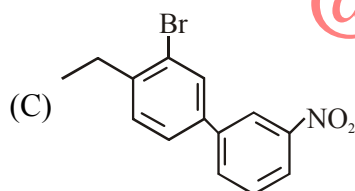
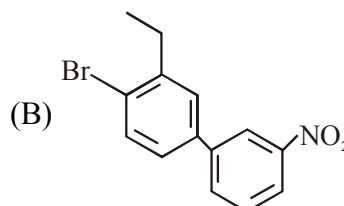
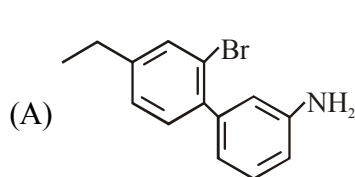
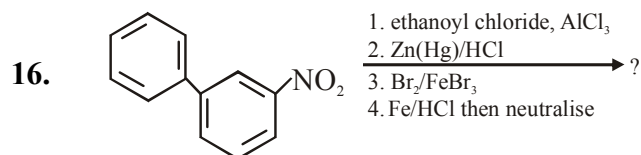
14. Sulphonation, like iodination, is reversible and is believed to take place in concentrated sulphuric acid via the following pathway :-



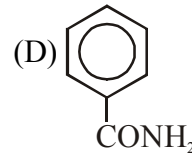
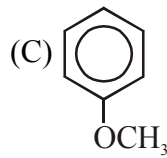
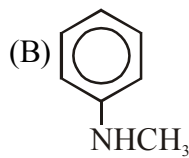
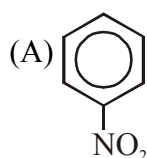
Find the correct statement(s) for above given reaction :-

- (A) Benzene is sulphonated fairly slowly by hot concentrated H_2SO_4 but rapidly by oleum.
 (B) In oleum, the σ -complex believed to undergo protonation of SO_3^- before undergoing C-H fission.
 (C) Kinetic isotopic effect exhibit because $K_{-1} \gg K_2$.
 (D) Rate of sulphonation reaction = $\text{C}_6\text{H}_6 > \text{C}_6\text{D}_6 > \text{C}_6\text{T}_6$

15. Which of the following statement(s) regarding friedal craft alkylation is(are) correct?
- (A) The action of $\text{Me}_3\text{C}.\text{CH}_2\text{Cl}/\text{AlCl}_3$ on benzene yields almost wholly the rearranged product $\text{Ph}.\text{CMe}_2.\text{CH}_2\text{Me}$.
- (B) Intial electrophile complex being polarized enough to allow the rearrangement of $[\text{Me}_3\text{C}.\text{CH}_2]^\delta+ \cdots \text{Cl} \cdots \overset{\delta-}{\text{Al}}\text{Cl}_3$ to the more stable $[\text{Me}_2\text{C}.\text{CH}_2\text{Me}]^\delta+ \cdots \text{Cl} \cdots \overset{\delta-}{\text{Al}}\text{Cl}_3$.
- (C) By contrast $\text{Me}_3\text{C}.\text{CH}_2\text{Cl}/\text{FeCl}_3$, on benzene is found to yield almost wholly unrearranged product $\text{PhCH}_2.\text{CMe}_3$.
- (D) The % of rearranged product from a given halide & Lewis acid being less at lower temperature as compared to that of high temperature.



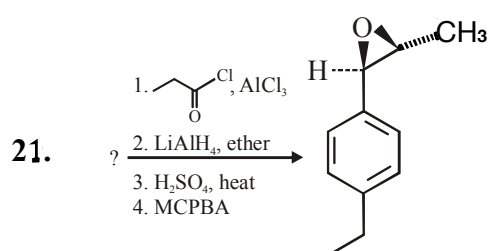
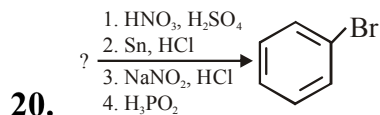
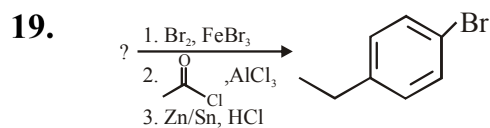
17. Which of these reacts most rapidly with $\text{CH}_3\text{I}/\text{AlCl}_3$?



18. Which of the following statements best describes a key step in the mechanism of the reaction between benzene and bromine in presence of FeBr_3 ?
- (A) A bromide ion attacks benzene in the slow step.
- (B) FeBr_3 forms a π -complex with benzene.
- (C) A complex of FeBr_3 and Br_2 reacts with benzene.
- (D) In a concerted process, Br_2 attack benzene, displacing a proton and producing bromo benzene.

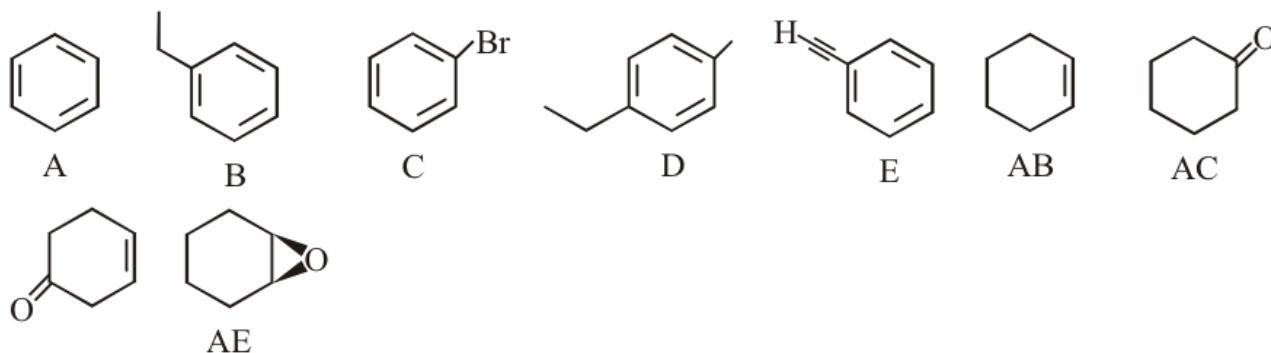
NOTE :

For each of the questions 18-20 choose the starting material from the selection provided that gives the product indicated via the transformations shown.



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List of Possible Answer for Questions 19-21

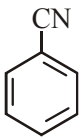
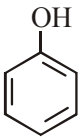
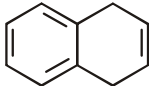
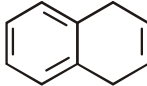
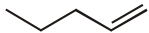


Answer = 19 = A, 20 = C, 21 = B

EXERCISE-C

Part-I

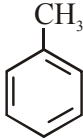
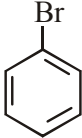
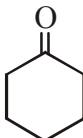
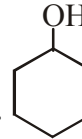
For each of the question 1 to 7 identify the major product obtained from each of the reaction sequences shown by selecting from the list of possible products provided.

1. 
 $\xrightarrow[4. \text{CH}_3\text{OH}]{1. \text{Br}_2, \text{FeBr}_3, 2. \text{H}_3\text{O}^+, \text{heat}, 3. \text{SOCl}_2}$?
2. 
 $\xrightarrow[5. \text{CuBr}]{1. \text{CH}_3\text{COCl}, \text{Et}_3\text{N}, 2. \text{HNO}_3, \text{H}_2\text{SO}_4, 3. \text{Zn/HCl}, 4. \text{NaNO}_2, \text{H}_2\text{SO}_4}$?
3. 
 $\xrightarrow[4. \text{CH}_3\text{Br}]{1. \text{BH}_3, \text{H}_2\text{O}_2, \text{OH}^-, 2. \text{Na}_2\text{Cr}_2\text{O}_7, 3. \text{LDA, THF, } -78^\circ\text{C}}$?
4. 
 $\xrightarrow[3. \text{Br}_2, \text{FeBr}_3]{1. \text{hot KMnO}_4, 2. \text{P}_2\text{O}_5}$?
5. 
 $\xrightarrow[3. \text{NaBH}_4, \text{MeOH}]{1. \text{NaOH, heat}, 2. \text{CH}_2=\text{CH}-\text{CHO, heat}}$?
6. 
 $\xrightarrow[4. \text{H}_2\text{O}]{1. \text{H}_2, \text{Pd}, 2. \text{Br}_2, \text{hv}, 3. \text{Mg, ether}}$?
7. 
 $\xrightarrow[3. \text{NaOMe, MeOH; then and MeBr}]{1. \text{BH}_3, \text{H}_2\text{O}_2, \text{OH}^-, 2. \text{PCC}}$?

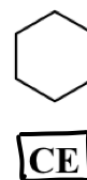
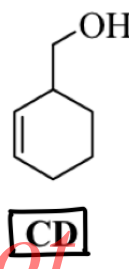
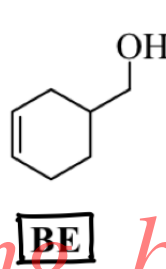
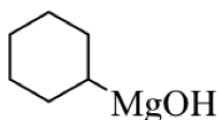
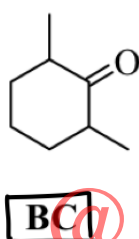
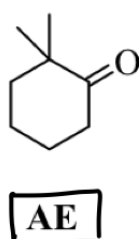
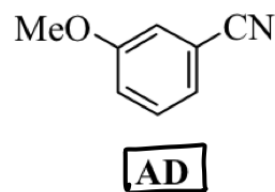
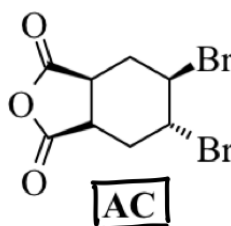
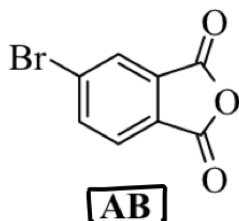
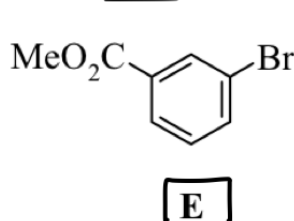
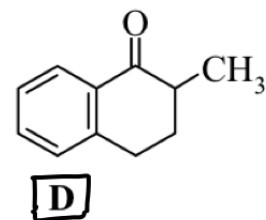
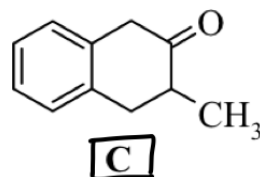
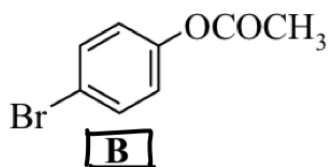
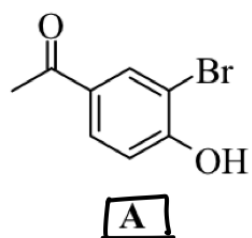
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Part-II

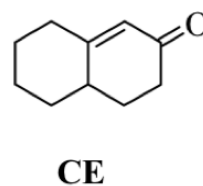
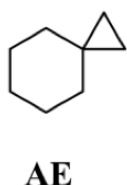
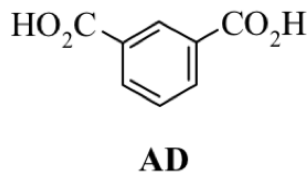
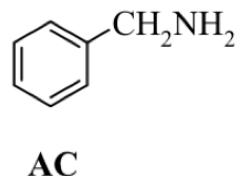
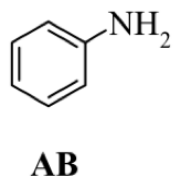
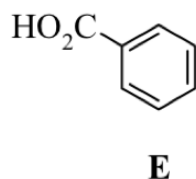
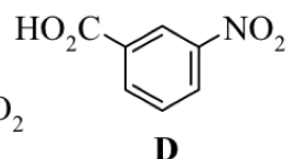
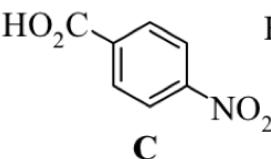
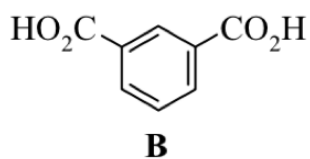
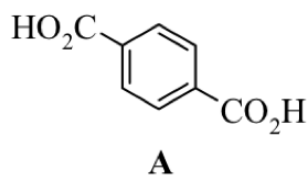
For each of the question 8 to 13 identify the major product obtained from each of the reaction sequences shown by selecting from the list of possible products provided.

8. 
 $\xrightarrow[(3) \text{NH}_2\text{NH}_2]{(1) \text{Br}_2/\text{hv}, (2) \text{Phthalimide K}^+\text{acetone}}$?
9. 
 $\xrightarrow[4. \text{H}_3\text{O}^+]{1. \text{Br}_2/\text{FeBr}_3, 2. \text{Mg/ether}, 3. \text{CO}_2}$?
10. 
 $\xrightarrow[2. \text{K}_2\text{Cr}_2\text{O}_7/\text{H}_3\text{O}^+/\Delta]{1. \text{HNO}_3/\text{H}_2\text{SO}_4}$?
11. 
 $\xrightarrow[5. \text{H}_3\text{O}^+/\Delta]{1. \text{HNO}_3/\text{H}_2\text{SO}_4, 2. \text{Sn/HCl}, 3. \text{NaNO}_2/\text{HCl}, 4. \text{Cu}_2(\text{CN})_2}$?
12. 
 $\xrightarrow[\Delta]{1. \text{KOH}, 2. \text{CH}_2=\text{CH}-\text{CO}-\text{CH}_3, 5\% \text{ NaOH}}$?
13. 
 $\xrightarrow[3. \text{CH}_2\text{I}_2/\text{Zn}]{1. \text{K}_2\text{Cr}_2\text{O}_7, \text{H}_3\text{O}^+, 2. \text{CH}_2=\text{PPh}_3}$?

List of possible answers for Questions 1 to 7



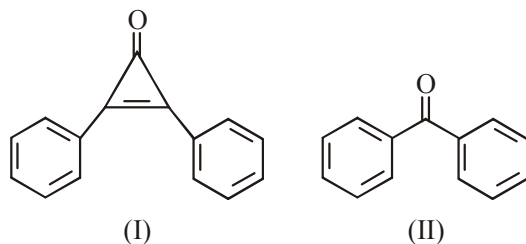
List of possible answers for Questions 8-13



14. Find out the correct statement(s) about aromaticity

(A) In cyclobutadiene two unpaired π electrons are present in π -non bonding molecular orbitals which are responsible for destabilisation of cyclobutadiene.

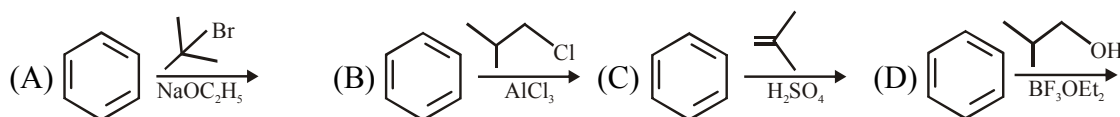
(B) Dipole moment of I structure is more than II structure in the following



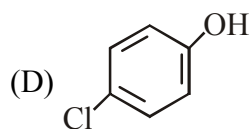
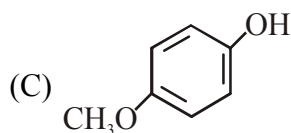
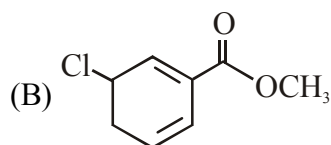
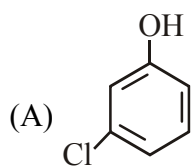
(C) Cycloheptatrienyl bromide is less soluble in H_2O than in non polar solvent.

(D) Due to antiaromatic nature, [4]-annulene is less stable than [6]-annulene.

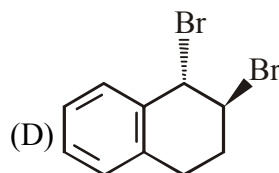
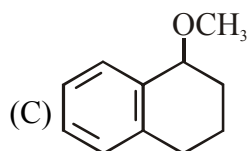
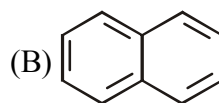
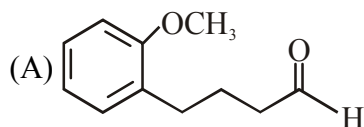
15. Among the following, reaction(s) which gives(give) tert-butyl benzene as the major product is(are)

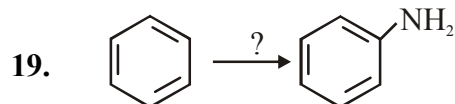
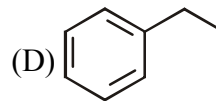
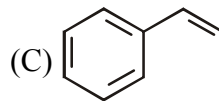
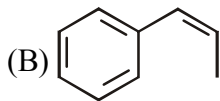
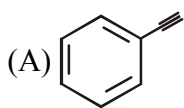
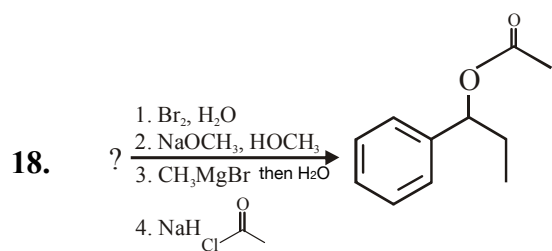


16. $\xrightarrow{\begin{matrix} 1. AlCl_3, \text{CH}_3COCl \\ 2. CH_3CO_3H \\ 3. AlCl_3, Cl_2 \\ 4. NaOCH_3 \\ 5. \text{acidic workup} \end{matrix}}$? *@Bohring_bot*



17. $\xrightarrow{\begin{matrix} 1. FeCl_3 \\ 2. LiAlH_4 \\ 3. H_2SO_4 \\ 4. NBS, UV \text{ light} \\ 5. NaOCH_3, \text{ heat} \end{matrix}}$?



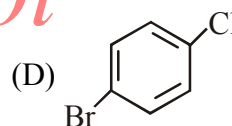
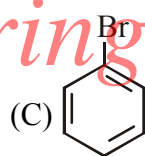
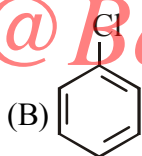
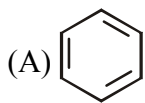
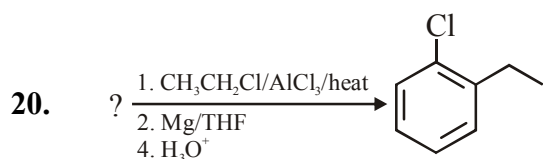


(A) i. NaNO_2/HCl ii. Sn/HCl iii. NaOH

(B) i. $\text{HNO}_3/\text{H}_2\text{SO}_4$ ii. Sn/HCl iii. NaOH

(C) i. NaNO_2/HCl ii. H_2/Pd

(D) i. Br_2/Fe ii. NH_3



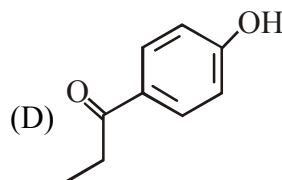
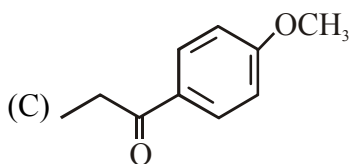
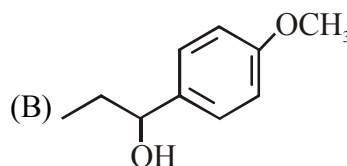
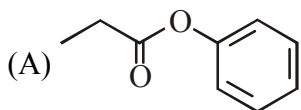
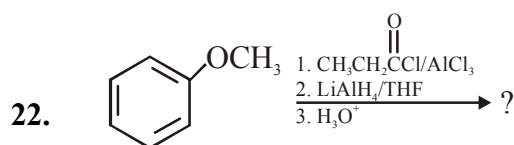
21. When phenyl ethanoate is alkylated with chloroethane/ AlCl_3 , the para product is the major product. This is because :

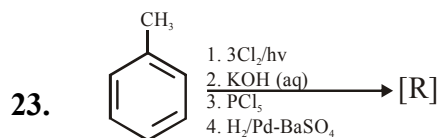
(A) Alkylation usually occurs at the *para* position.

(B) The $-\text{CO}_2\text{CH}_3$ group is an activator and an *ortho/para* director.

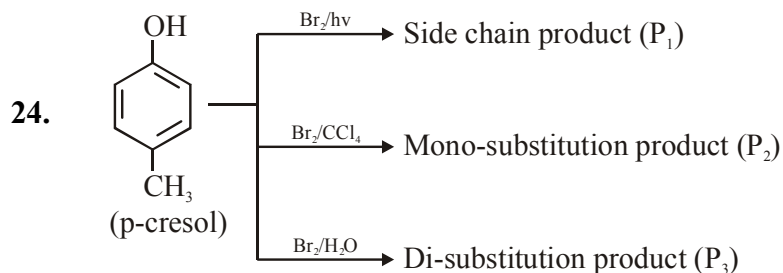
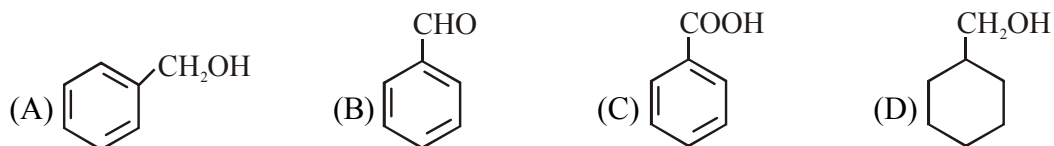
(C) The $-\text{Et}$ group is weakly activating and an *ortho/para* director.

(D) The $-\text{O}_2\text{CCH}_3$ group is an activator and an *ortho/para* director.



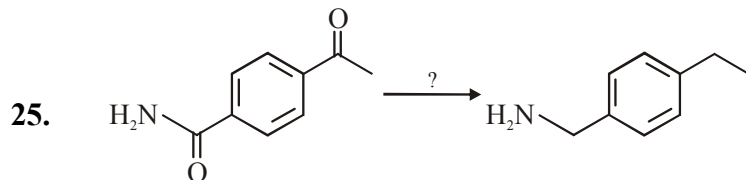


[R] will be :-

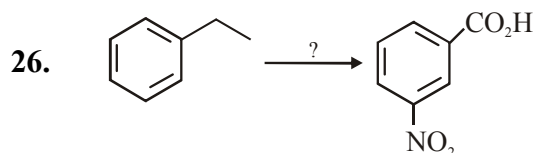


The observed pattern of substitution can be explained by :-

- (A) Free-radical substitution
(B) Electrophilic aromatic substitution
(C) Solvent effect on phenolic group
(D) Electronic effect of (CH_3) group

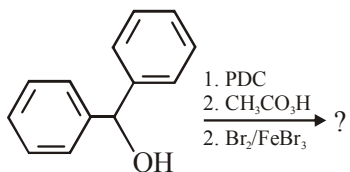


- (A) $\text{H}_2\text{NNH}_2/\text{NaOH/heat}$
(B) $\text{LiAlH}_4/\text{THF}$ then H_3O^+
(C) NaBH_4
(D) i. $\text{HOCH}_2\text{CH}_2\text{OH}/\text{TsOH}$, ii. $\text{LiAlH}_4/\text{THF}$ then H_3O^+ iii. $\text{H}_3\text{O}^+/\text{heat}$
(E) i. Zn/Hg/HCl , ii. $\text{LiAlH}_4/\text{THF}$ then H_3O^+

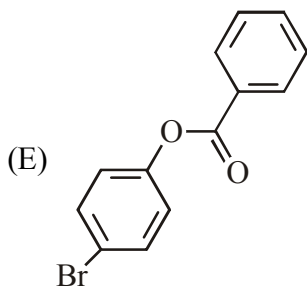
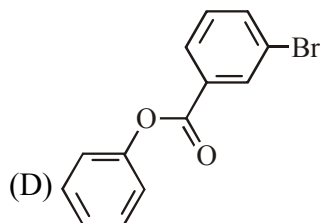
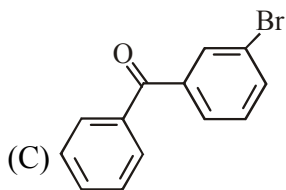
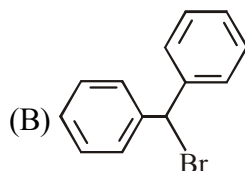
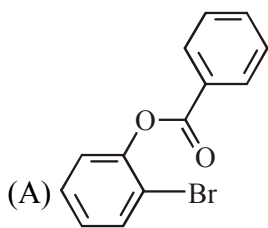


- (A) i. $\text{HNO}_3/\text{H}_2\text{SO}_4$; ii. $\text{KMnO}_4/\text{H}_3\text{O}^+/\text{heat}$
(B) i. $\text{KMnO}_4/\text{H}_3\text{O}^+/\text{heat}$; ii. $\text{NaNO}_2/\text{H}_2\text{SO}_4$
(C) i. $\text{KMnO}_4/\text{H}_3\text{O}^+/\text{heat}$; ii. $\text{HNO}_3/\text{H}_2\text{SO}_4$
(D) i. $\text{NaNO}_2/\text{H}_2\text{SO}_4$; ii. $\text{KMnO}_4/\text{H}_3\text{O}^+/\text{heat}$

27.

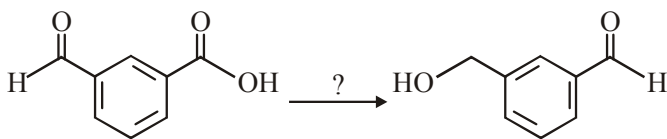


Major product



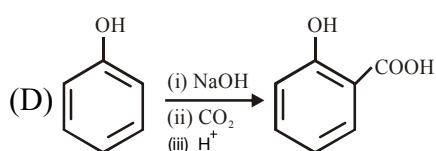
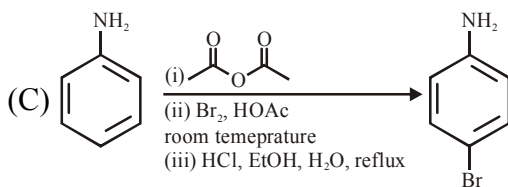
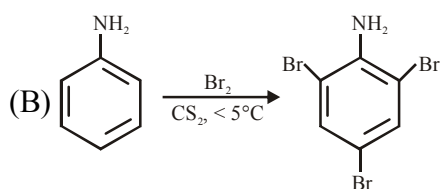
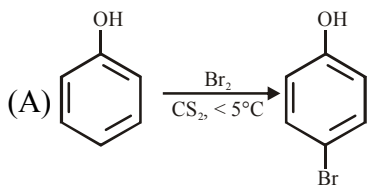
@Bohring_bot

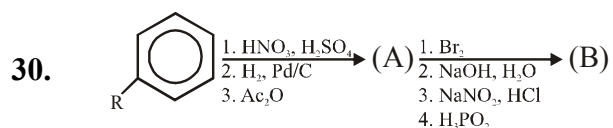
28.

(A) i. LiAlH_4 then H^+ workup ii. 1,2-ethanediol/ TsOH iii. aq. H_2SO_4 , heat(B) i. aq. H_2SO_4 , heat ii. 1,2-ethanediol/ TsOH iii. LiAlH_4 then H^+ workup(C) i. excess LiAlH_4 then H^+ workup ii. excess $\text{PDC}/\text{CH}_2\text{Cl}_2$ (D) i. 1,2-ethanediol/ TsOH ii. LiAlH_4 then H^+ workup iii. aq. H_2SO_4 , heat(E) i. 1,2-ethanediol/ TsOH ii. NaBH_4 iii. KOH , heat

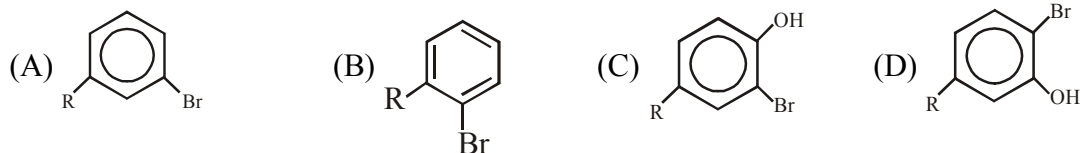
29.

Correct major product(s) with appropriate condition

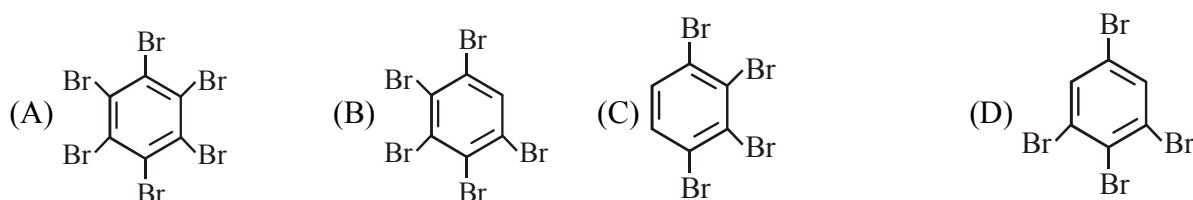
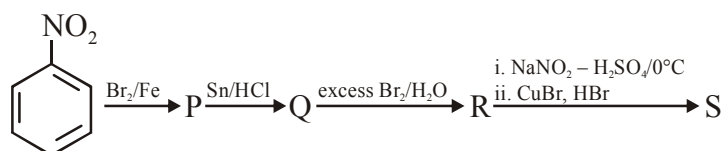




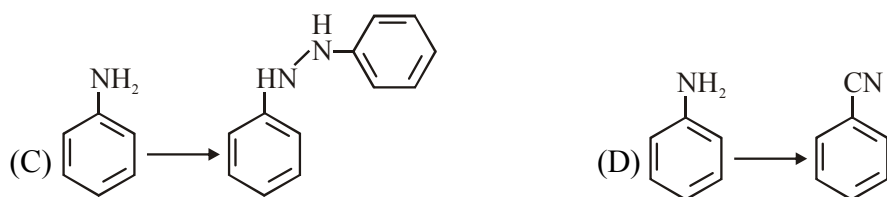
Product (B) will be



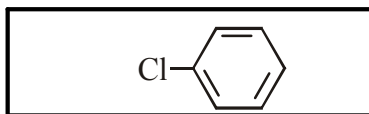
31. The major product 'S' of the following reaction sequence is :



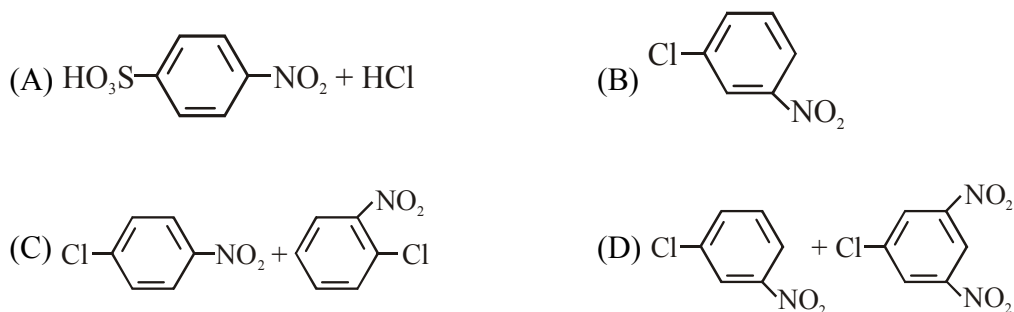
32. Identify Sandmeyer reaction from the following

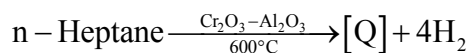
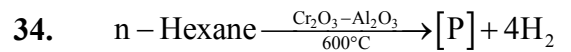


33.

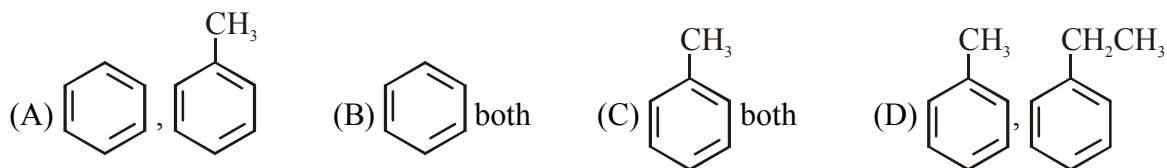


Nitration of chlorobenzene with a mixture of nitric and sulfuric acids yields which of the following as the major product (or) products?

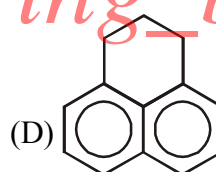
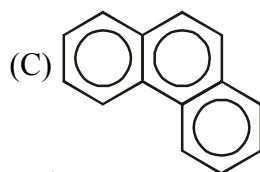
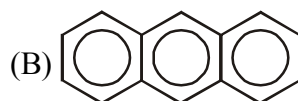
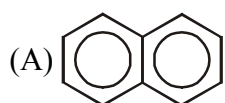
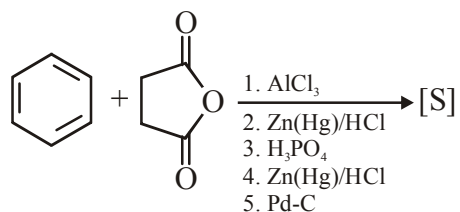




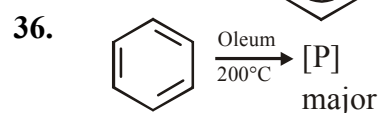
(P) & (Q) are respectively –



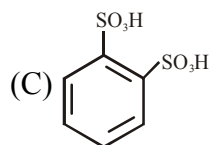
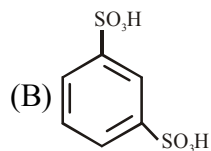
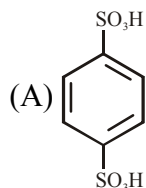
35. Find the product of the below reaction sequence is :-



@Bohring_bot

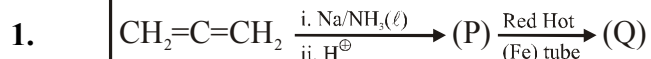


Major product [P] is :-



(D) None

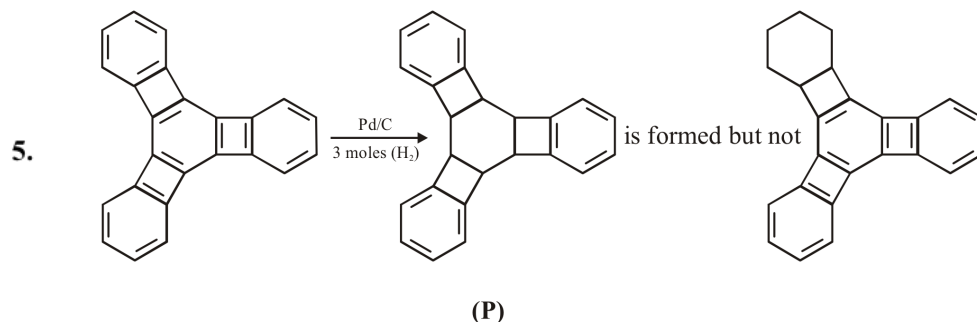
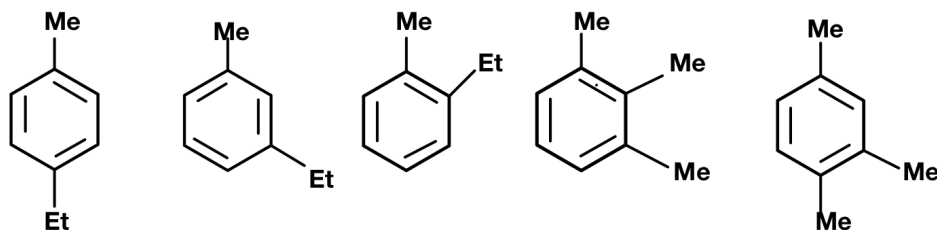
EXERCISE-D



Compound (P) can give white ppt with ammonical silver nitrate, and compound (Q) is thermodynamically more stable among its position isomers, now maximum number of atoms in (Q) that are in one plane in any possible conformation are "X". then find (X – 3Y)?

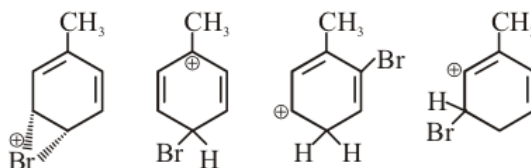
[Where "Y" is number of position isomers of (Q)]

2. How many diene-yne molecule(s) are possible (without branching but including stereoisomers) for molecular formula C_6H_6 ?
3. Among the following statements on the nitration of aromatic compounds, find the number of false statement(s)?
- The rate of benzene is almost the same as that of hexadeuterobenzene
 - Loss of aromaticity when $\text{NO}_2^{\oplus}$ attacks at ortho and para position and not at Meta position
 - Nitrobenzene prepared from benzene by using a mixture of conc. HNO_3 and conc. H_2SO_4 in which HNO_3 act as a base
 - Nitration is an electrophilic substitution reaction.
4. Out of the following given compounds, on nitration, how many can give "2-nitro substituted aromatic" product according to IUPAC nomenclature.



How many stereoisomers are possible for product P.

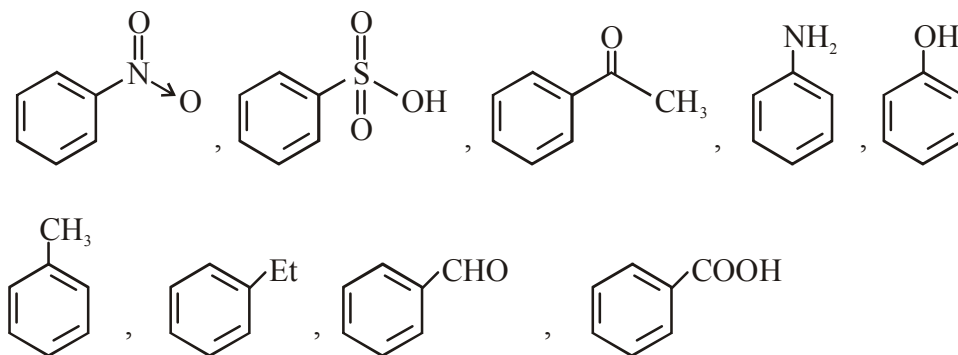
6. How many out of the following is (are) intermediate(s) in the bromination reaction of toluene



structure-1
is WRONG.

$\text{Ph}-\overset{\oplus}{\text{C}}\text{H}-\text{CH}_2\text{Br}$
cyclic bromonium
ring opens if
stabilised by
Resonance.

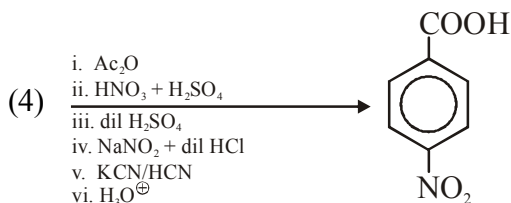
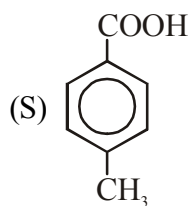
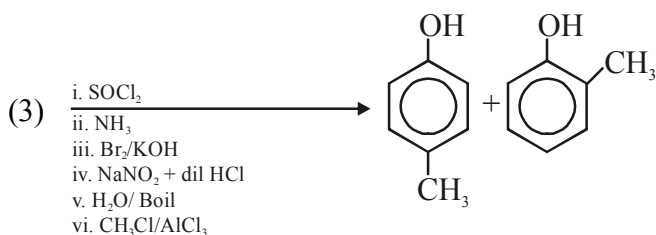
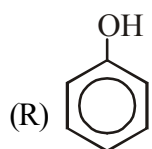
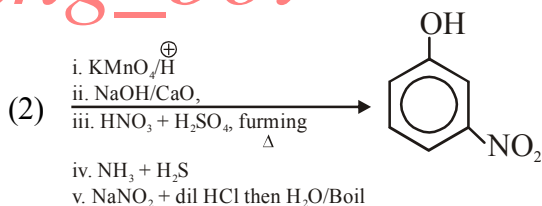
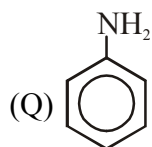
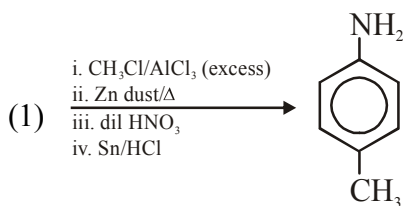
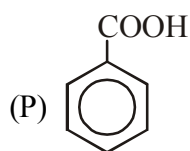
7. Number of compounds that will not give Friedel craft reaction are :



8. n -octane $\xrightarrow[600^\circ\text{C}]{\text{Cr}_2\text{O}_3-\text{Al}_2\text{O}_3}$ [X] Number of isomeric benzene derivatives. Find the value [X] is :-

9. List I

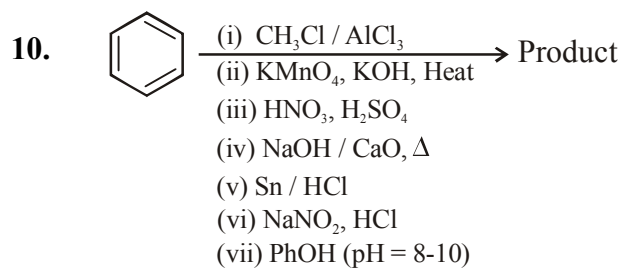
List-II



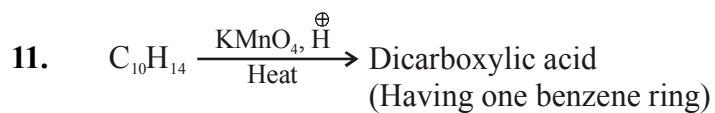
NOTE : If your answer is

P	Q	R	S
1	2	3	4

, give your answer as (1,2,3,4)

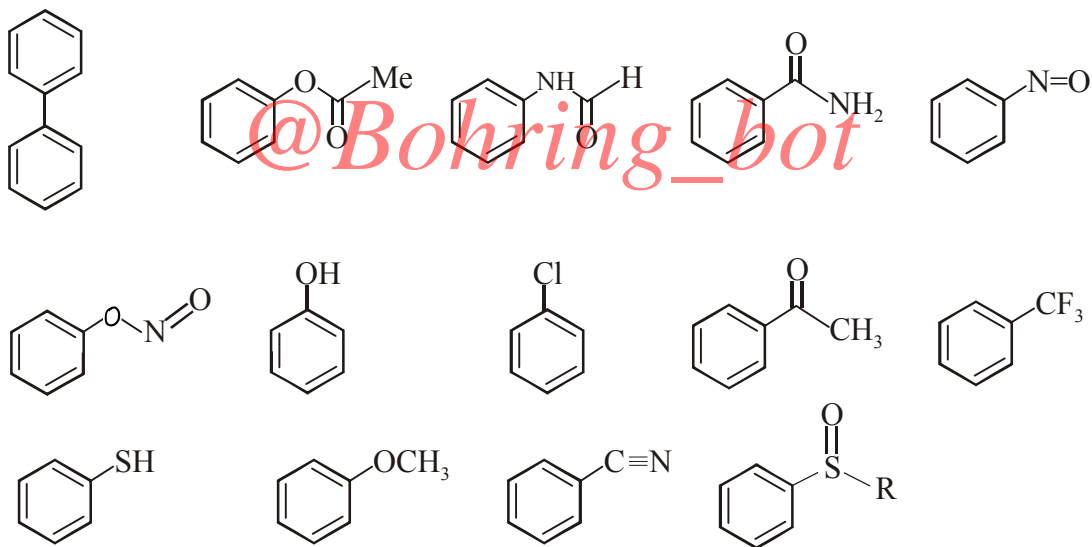


[X] = Total number of lone pair present in product.

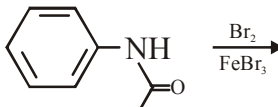
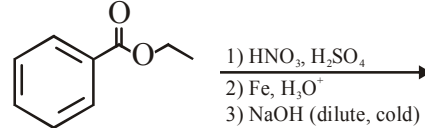
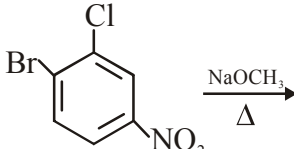
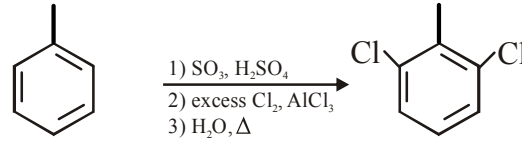
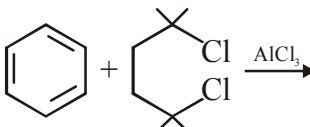
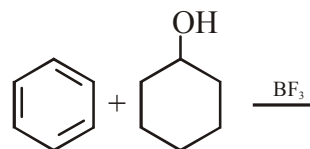
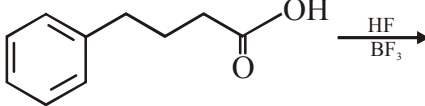
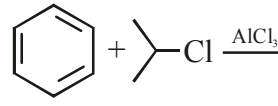
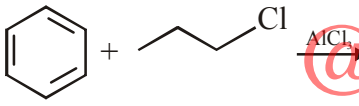
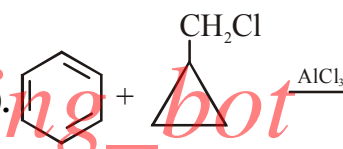
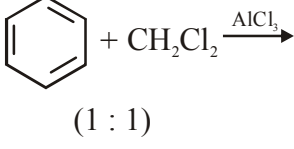
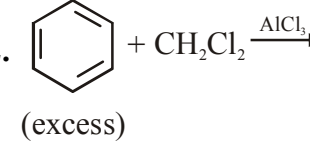
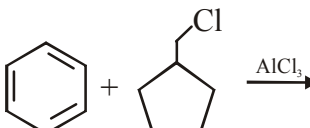
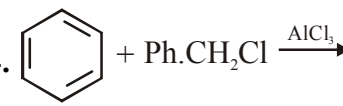
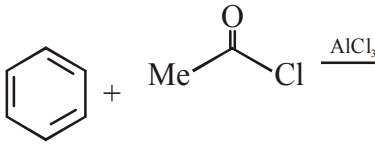
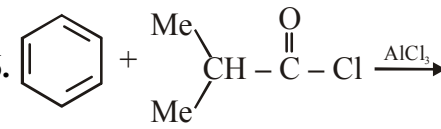
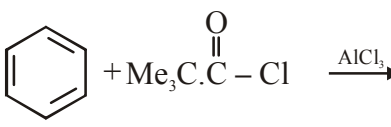
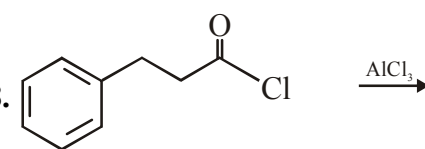
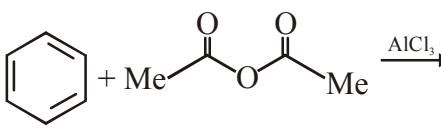
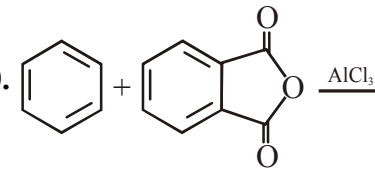


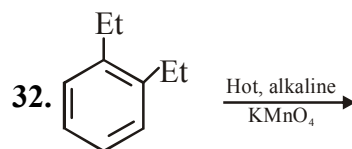
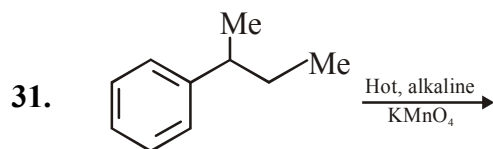
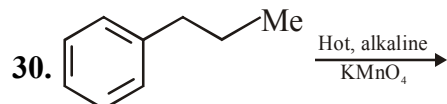
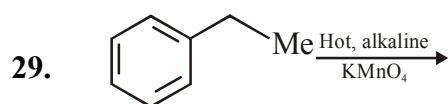
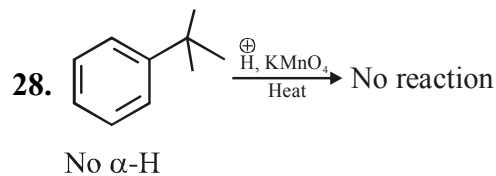
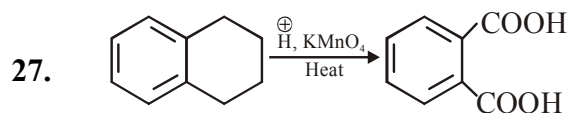
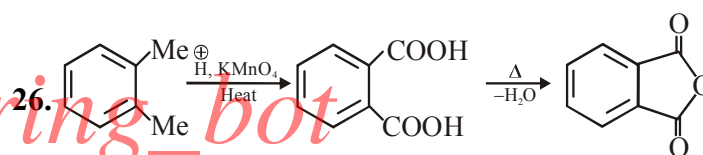
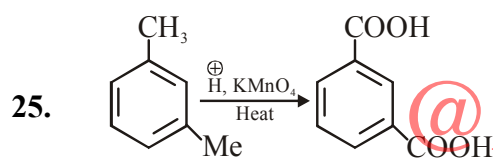
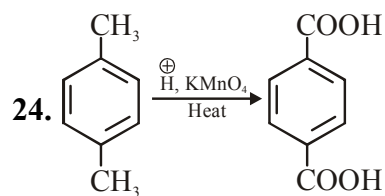
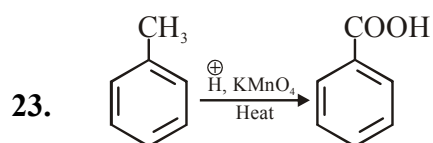
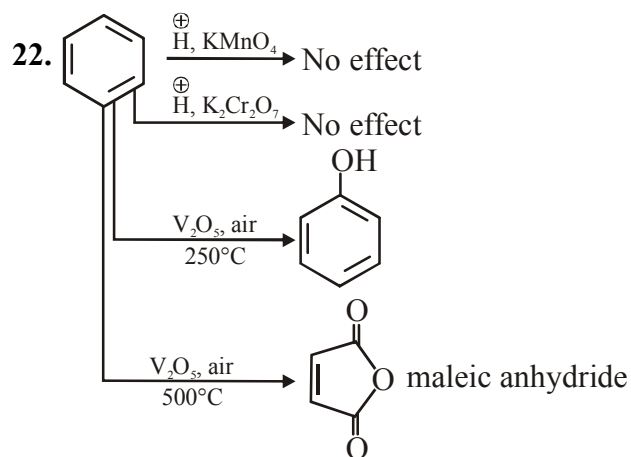
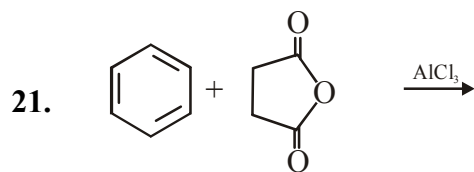
How many isomers (excluding stereoisomers) of C₁₀H₁₄ gives dicarboxylic acid when react with hot acidic KMnO₄. (Di carboxylic acid having one benzene ring)

12. Number of compounds that will give ortho/para as major product is (are) :

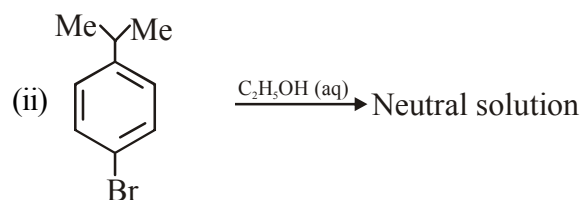
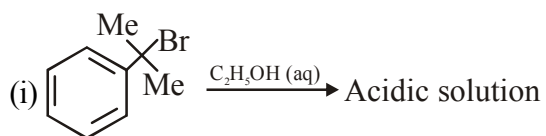


EXERCISE-F

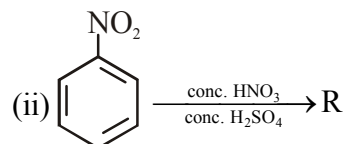
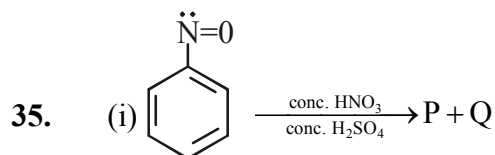
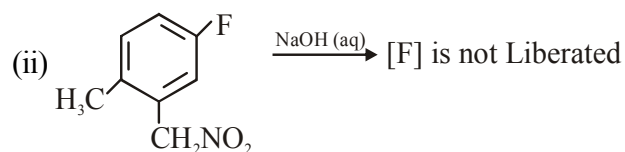
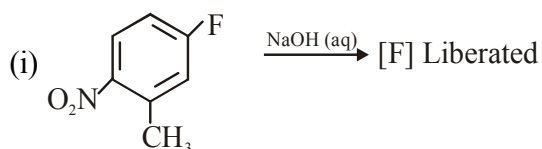
1.  CC(=O)Nc1ccccc1.BrBr>>[FeBr3]
2.  CCOC(=O)c1ccccc1.O=[N+]([O-])O.O=S(=O)(O)O>>[Fe].[H3O+].[Na]OH
(1) HNO₃, H₂SO₄
(2) Fe, H₃O⁺
(3) NaOH (dilute, cold)
3.  Brc1cc(Cl)cc([N+](=O)[O-])cc1.CO[Na]>>[Delta]
4.  Cc1ccccc1.O=S(=O)(O)O>>[SO3].[H2SO4].ClCl.Cl[Al](Cl)Cl.O.[Delta]
(1) SO₃, H₂SO₄
(2) excess Cl₂, AlCl₃
(3) H₂O, Δ
5.  C1(C)C(Cl)CC(Cl)CC1.c1ccccc1>>[AlCl3]
6.  C1CCCCC1O.c1ccccc1>>[BF3]
7.  OC(=O)CCCC1=CC=CC=C1>>[HF].[BF3]
8.  CC(C)Cl.c1ccccc1>>[AlCl3]
9.  CCCCl.c1ccccc1>>[AlCl3]
10.  C1CC1CCl.c1ccccc1>>[AlCl3]
11.  CCCl.c1ccccc1>>[AlCl3]
(1 : 1)
12.  CCCl.c1ccccc1>>[AlCl3]
(excess)
13.  C1CCCC1CCl.c1ccccc1>>[AlCl3]
14.  c1ccccc1CCl.c1ccccc1>>[AlCl3]
15.  CC(=O)Cl.c1ccccc1>>[AlCl3]
16.  CC(C)C(=O)Cl.c1ccccc1>>[AlCl3]
17.  CC(C)(C)C(=O)Cl.c1ccccc1>>[AlCl3]
18.  CC(=O)CC(=O)Cl.c1ccccc1>>[AlCl3]
19.  CC(=O)OC(=O)C.c1ccccc1>>[AlCl3]
20.  O=C1OC(=O)c2ccccc12.c1ccccc1>>[AlCl3]



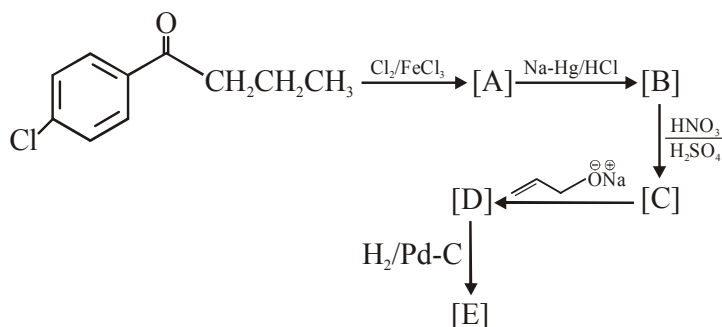
33. Give reasons :



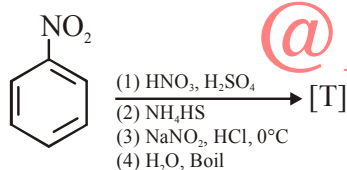
34. Give reasons :



36. Write the structure of products A,B,C,D and E in the following scheme :-

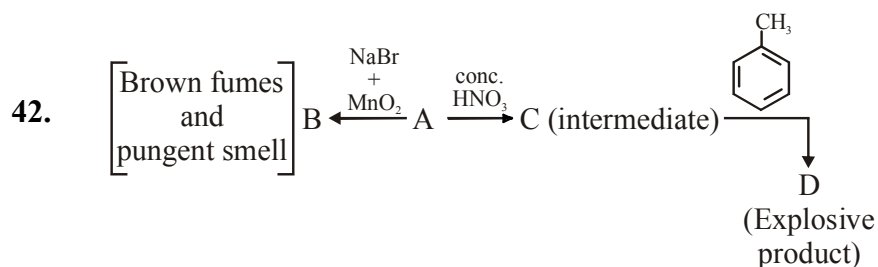
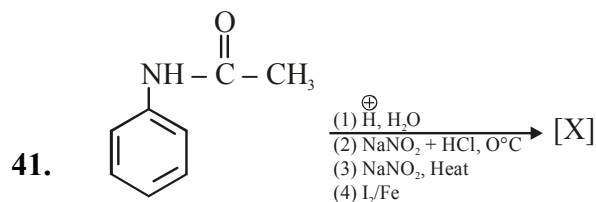
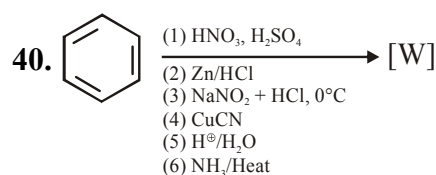
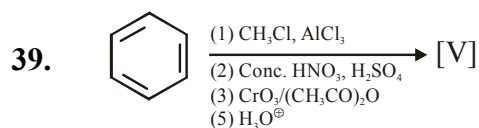
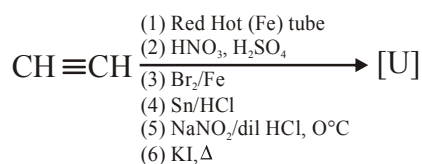


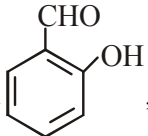
37. Write the product of given reaction sequences :-

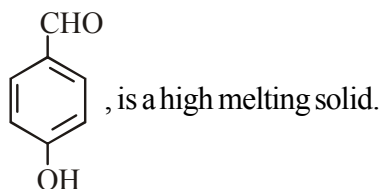


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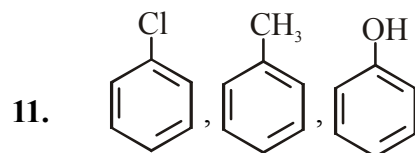
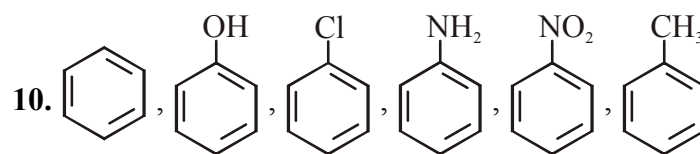
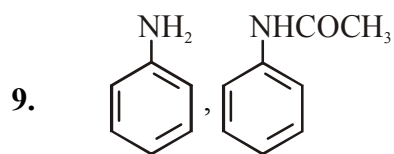
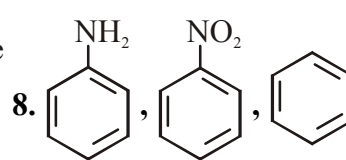
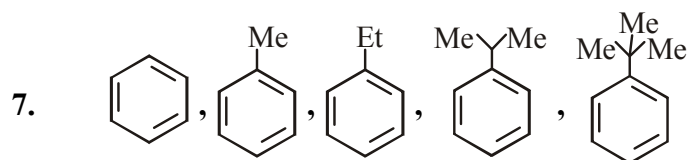
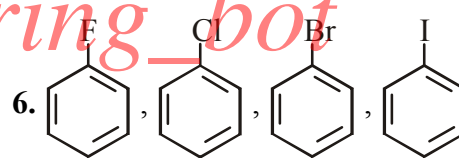
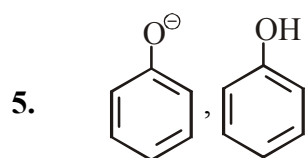
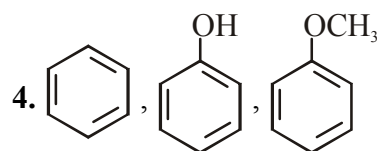
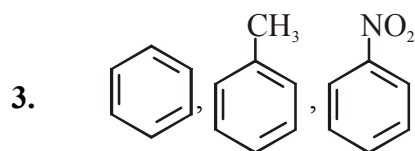
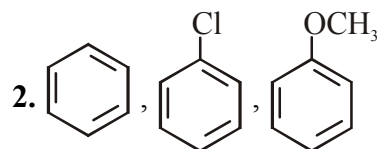
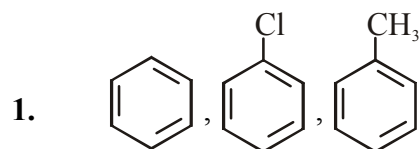
38. Write the structure of product for given reaction sequences :-



43. Explain , (ortho hydroxy benzaldehyde) is liquid at room temperature while (p-hydroxy benzaldehyde)



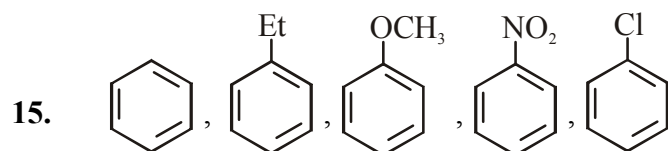
Rate of Electrophilic Aromatic Substitution



12. Benzene, Mesitylene, Toluene, m-xylene, p-xylene

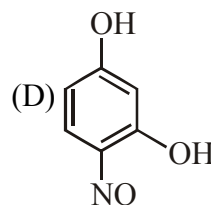
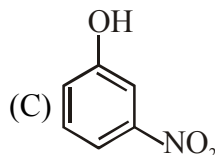
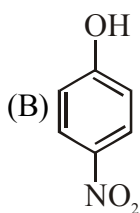
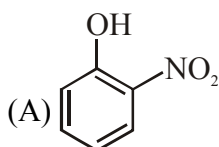
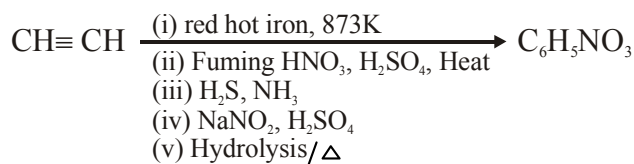
13. Acetanilide, Acetophenone, Aniline, Benzene

14. Terephthalic acid, Toluene, p-Toluic acid, p-xylene

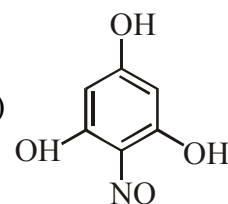
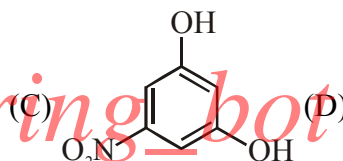
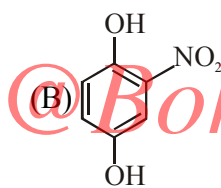
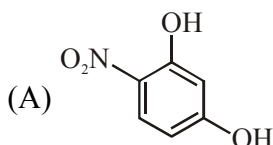
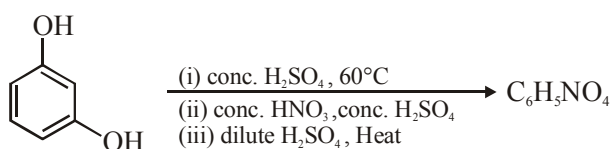


EXERCISE-G

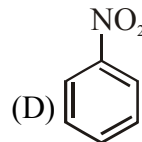
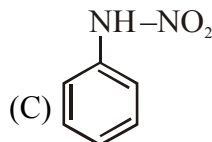
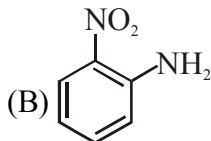
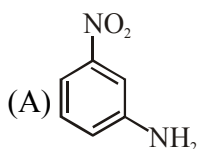
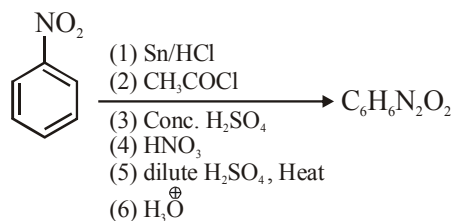
1. Major product of the given reaction is :-



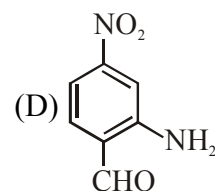
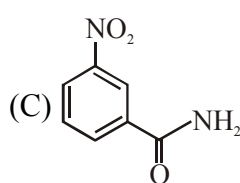
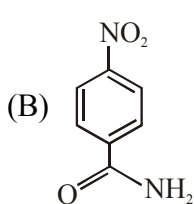
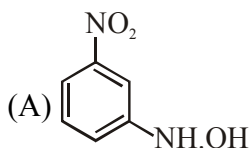
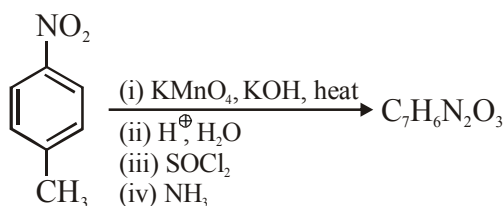
2. Major product of the given reaction is :-

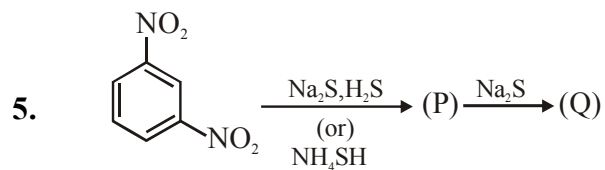


3. Major product of the given reaction is :-

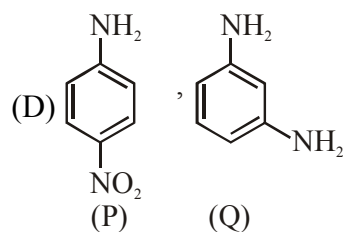
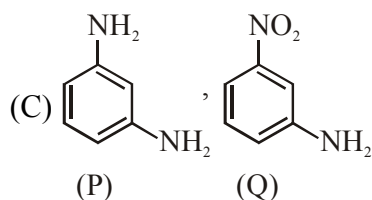
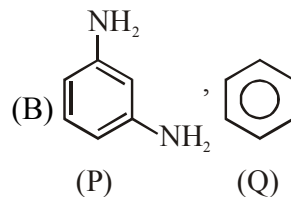
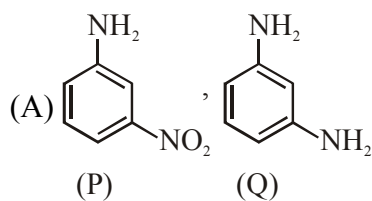


4. Major product of the given reaction is :-

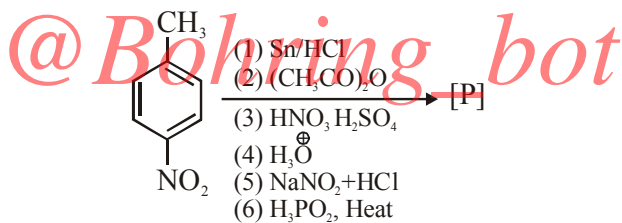




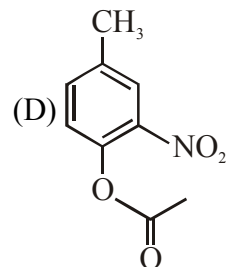
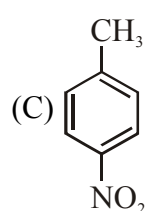
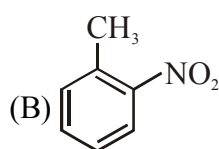
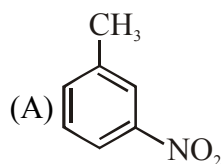
(P) and (Q) are respectively :-



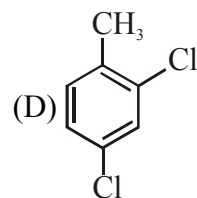
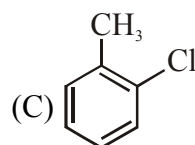
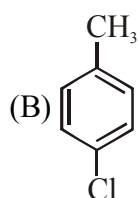
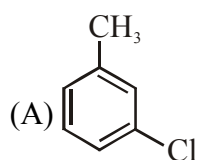
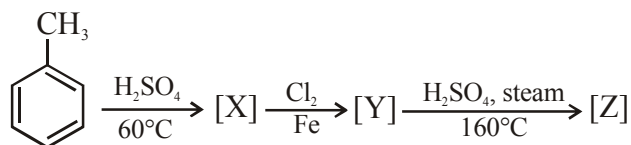
6. Find product of given reaction sequences :-



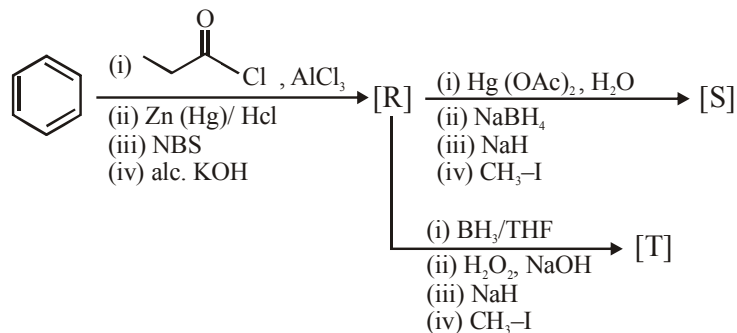
[P] will be :-



7. Product [Z] is :-



8. Relationship between final products [G] & [H] is :-



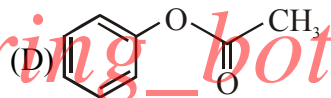
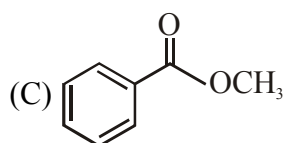
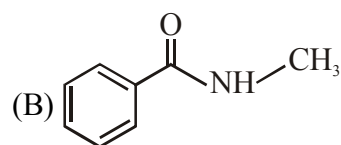
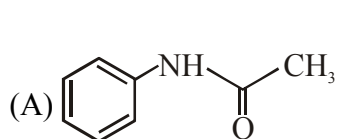
(A) Position isomers

(B) Geometrical isomers

(C) Optical isomers

(D) Chain isomers

9. Which of the following is ortho/para directing ?



10. Of the species PhSH , Ph-S(=O)-R , $\text{Ph-S(=O)}_2\text{-R}$ and $\text{Ph-S(=O)}_2\text{-OR}$, the meta substituted product is obtained from :-

(A) PhSH

(B) Ph-S(=O)-R

(C) $\text{Ph-S(=O)}_2\text{-R}$

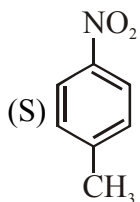
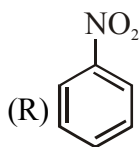
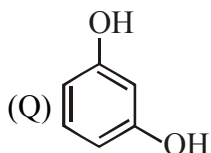
(D) $\text{Ph-S(=O)}_2\text{-OR}$

EXERCISE-H

1. Match the four starting materials (P, Q, R, S) given in List-I with the corresponding reaction schemes (I, II, III, IV) provided in List-II and select the correct answer using the code given below the list :-

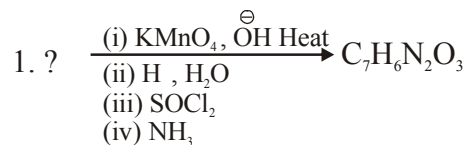
List-I

(P) $\text{CH}\equiv\text{CH}$

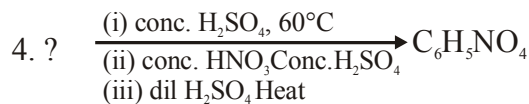
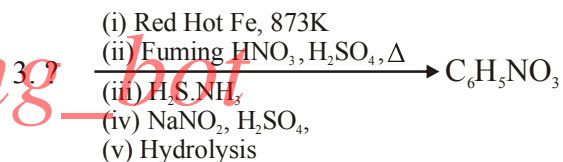
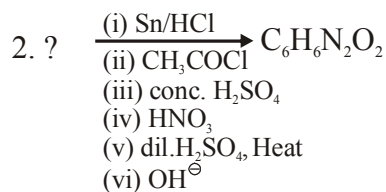


List-II

Scheme-I



Scheme-II



Codes :

	[P]	[Q]	[R]	[S]
(A)	1	4	2	3
(B)	3	1	4	2
(C)	3	4	2	1
(D)	4	1	3	2

ANSWER KEY

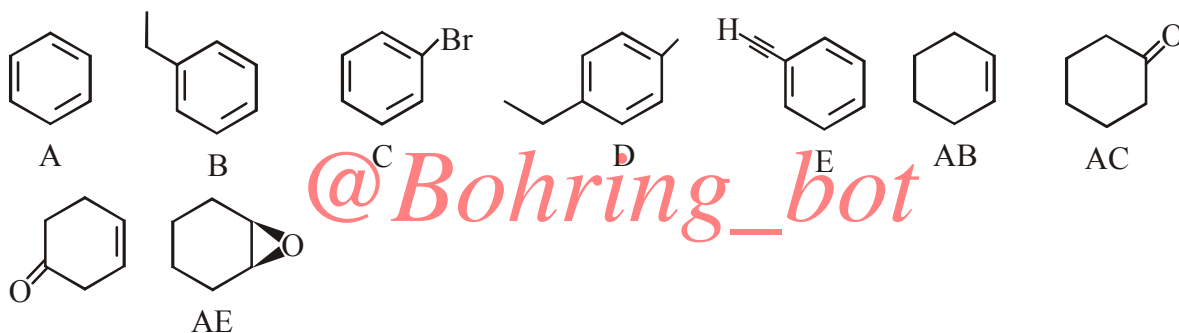
EXERCISE-A

1. (B)	2. (D)	3. (D)	4. (C)	5. (D)
6. (B,C)	7. (C)	8. (B)	9. (A)	10. (A,B,C,D)
11. (D)	12. (B)	13. (C)	14. (B)	15. (B)
16. (A)	17. (D)	18. (C,D)	19. (D)	20. (B)
21. (A)	22. (D)	23. (A)	24. (B)	25. (A,C)
26. (D)	27. (C)	28. (B)	29. (D)	30. (B,D)

EXERCISE-B

1. (C,D)	2. (D)	3. (A,B)	5. (D)	6. (B)
7. (D)	8. (A)	9. (B)	10. (A)	11. (C)(B)
12. (A,B,C,D)	13. (A,B,D,C)	14. (A,B,C,D)	15. (A,B,C,D)	16. (D)
17. (C)	18. (C)			

List of Possible Answer for Questions 19-21

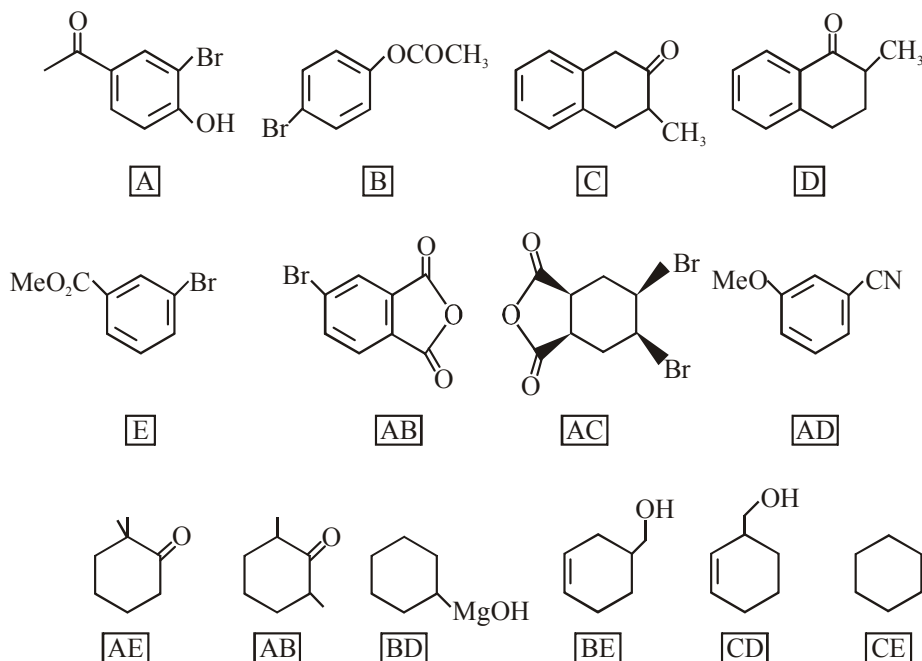


Answer = 19 = A, 20 = C, 21 = B

EXERCISE-C

PART-I

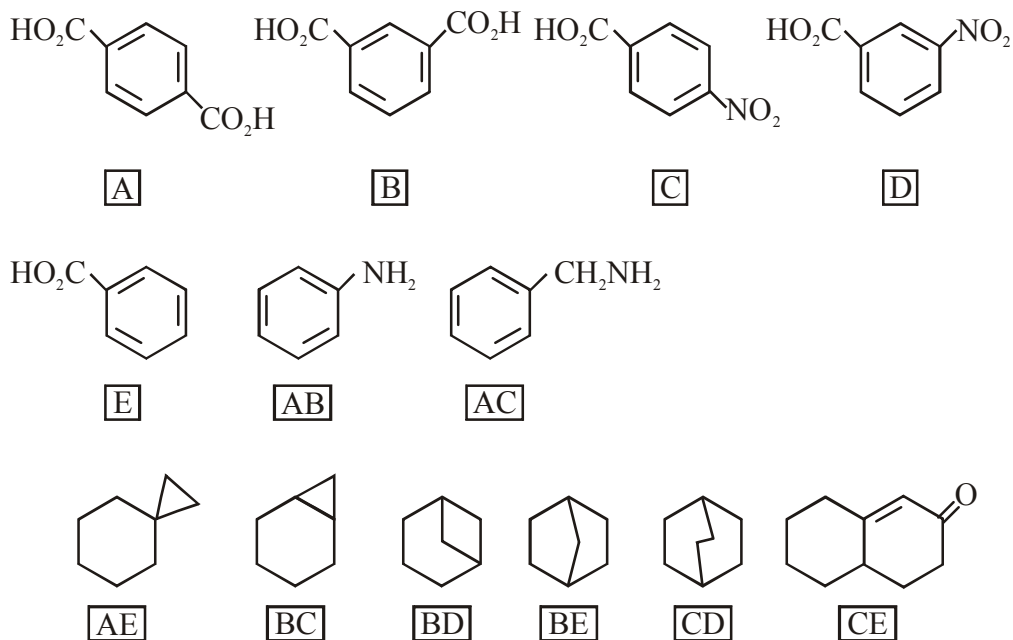
List of Possible Answers for Questions 1 to 7



Ans. 1 – E, 2 – B, 3 – C, 4 – AB, 5 – BE, 6 – CE, 7 – AE

PART-II

List of Possible Answer for Question 08-13



Ans. 8. – AC, 9. – A, 10. – C, 11. – E, 12. – CE, 13. – AE

- | | | | | |
|---------------|-----------|---------|---------|---------|
| 14. (B,D) | 15. (BCD) | 16. (D) | 17. (B) | 18. (C) |
| 19. (B) | 20. (D) | 21. (D) | 22. (B) | 23. (B) |
| 24. (A,B,C) | 25. (E) | 26. (C) | 27. (E) | 28. (D) |
| 29. (A,B,C,D) | 30. (A) | 31. (B) | 32. (D) | 33. (C) |
| 34. (A) | 35. (A) | 36. (B) | | |

EXERCISE-D

- | | | | | |
|---------|----------|--------|-----------|---------|
| 1. (9) | 2. (7) | 3. (1) | 4. (1) | 5. (2) |
| 6. (1) | 7. (6) | 8. (4) | 9. (3412) | 10. (4) |
| 11. (9) | 12. (10) | | | |

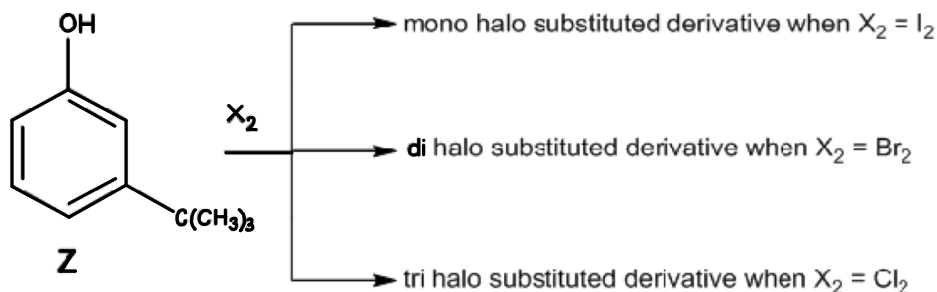
EXERCISE - G

- | | | | | |
|--------|--------|--------|----------|-----------|
| 1. (C) | 2. (A) | 3. (B) | 4. (B) | 5. (A) |
| 6. (A) | 7. (C) | 8. (A) | 9. (A,D) | 10. (C,D) |

EXERCISE - H

- | |
|--------|
| 1. (C) |
|--------|

22. The reactivity of compound **Z** with different halogens under appropriate conditions is given below:



The observed pattern of electrophilic substitution can be explained by

- (A) the steric effect of the halogen
 (B) the steric effect of the *tert*-butyl group
 (C) the electronic effect of the phenolic group
 (D) the electronic effect of the *tert*-butyl group

Answer: (A), (B) and (C)

38. Match the four starting materials (**P**, **Q**, **R**, **S**) given in **List I** with the corresponding reaction schemes (**I**, **II**, **III**, **IV**) provided in **List II** and select the correct answer using the code given below the lists.

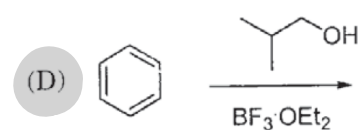
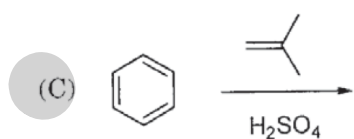
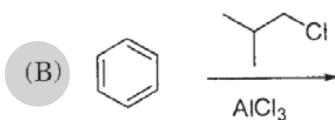
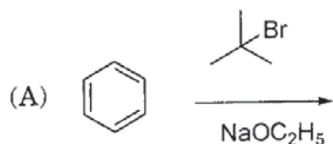
List-I	List-II
(P) $H-C\equiv C-H$	1. Scheme I (i) $KMnO_4, HO^{\ominus}$, heat (ii) $H^{\oplus}$, H_2O ? $\xrightarrow{(iii) SOCl_2 (iv) NH_3}$ $C_7H_6N_2O_3$
(Q)	2. Scheme II (i) Sn/HCl (ii) CH_3COCl (iii) conc. H_2SO_4 ? $\xrightarrow{(iv) HNO_3 (v) \text{dil. } H_2SO_4, \text{ heat } (vi) HO^{\ominus}_4}$ $C_6H_6N_2O_2$
(R)	3. Scheme III (i) red hot iron, 873 K (ii) fuming HNO_3, H_2SO_4 , heat ? $\xrightarrow{(iii) H_2S.NH_3 (iv) NaNO_2, H_2SO_4 (v) \text{hydrolysis}}$ $C_6H_5NO_3$
(S)	4. Scheme IV (i) conc. H_2SO_4 , 60 °C ? $\xrightarrow{(ii) \text{conc. } HNO_3, \text{ conc. } H_2SO_4 (iii) \text{dil. } H_2SO_4, \text{ heat}}$ $C_6H_5NO_4$

Codes:

	P	Q	R	S
(A)	1	4	2	3
(B)	3	1	4	2
(C)	3	4	2	1
(D)	4	1	3	2

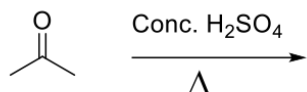
Answer (C)

Q.32 Among the following, reaction(s) which gives(give) *tert*-butyl benzene as the major product is(are)

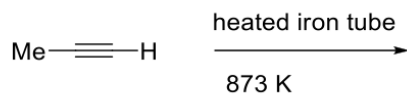


Q.5 The reaction(s) leading to the formation of 1,3,5-trimethylbenzene is (are)

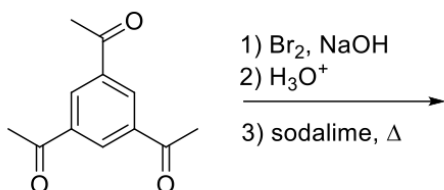
(A)



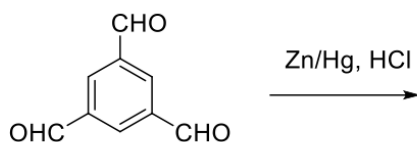
(B)



(C)



(D)

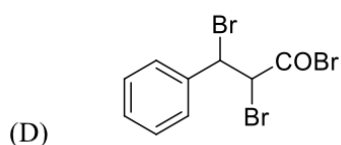
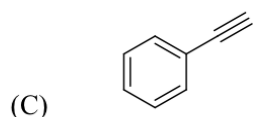
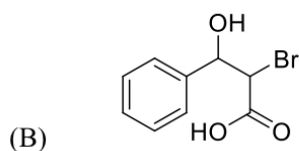
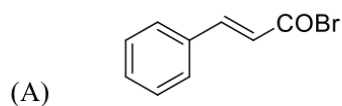


PARAGRAPH "X"

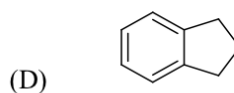
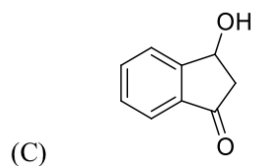
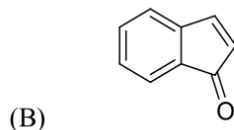
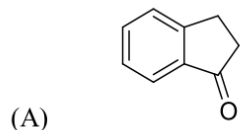
Treatment of benzene with CO/HCl in the presence of anhydrous $\text{AlCl}_3/\text{CuCl}$ followed by reaction with $\text{Ac}_2\text{O}/\text{NaOAc}$ gives compound **X** as the major product. Compound **X** upon reaction with $\text{Br}_2/\text{Na}_2\text{CO}_3$, followed by heating at 473 K with moist KOH furnishes **Y** as the major product. Reaction of **X** with $\text{H}_2/\text{Pd-C}$, followed by H_3PO_4 treatment gives **Z** as the major product.

(There are two questions based on PARAGRAPH "X", the question given below is one of them)

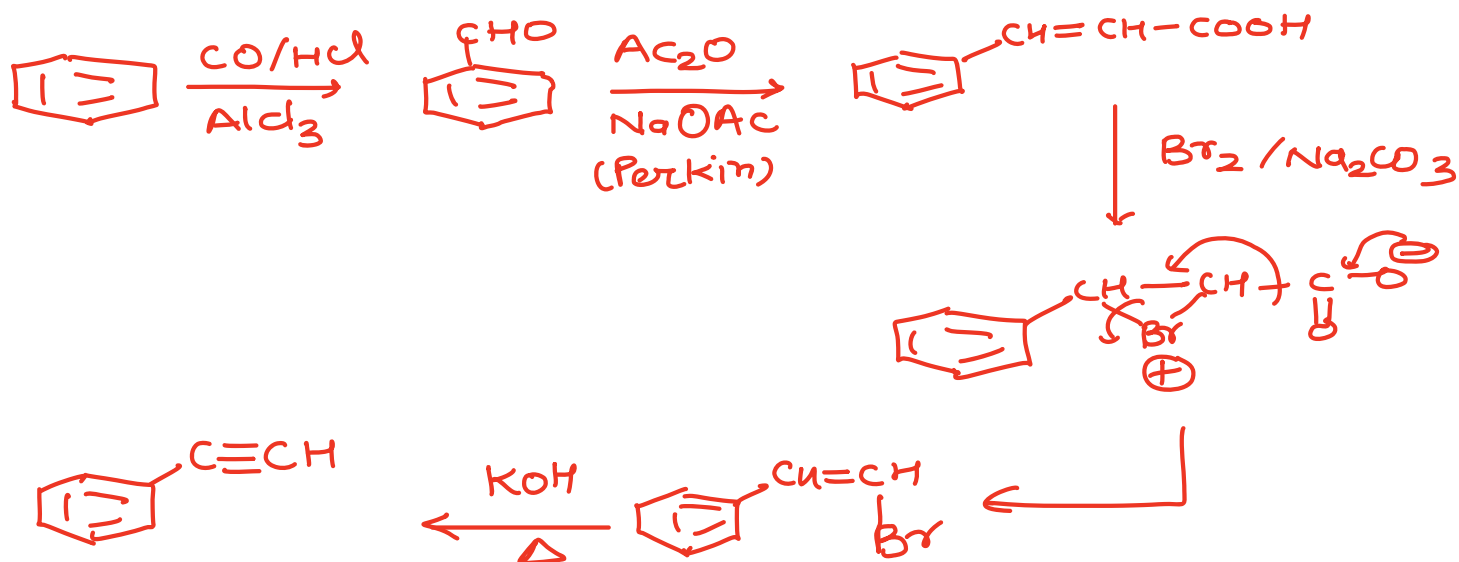
Q.15 The compound **Y** is



Q.16 The compound **Z** is

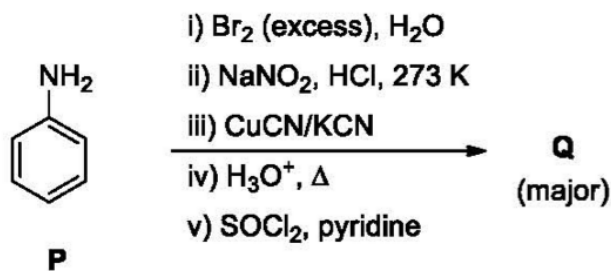


@Bohring_bot

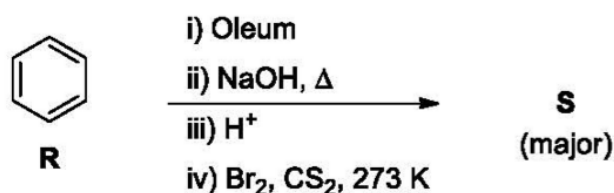


Q.18 Schemes 1 and 2 describe the conversion of P to Q and R to S, respectively. Scheme 3 describes the synthesis of T from Q and S. The total number of Br atoms in a molecule of T is _____

Scheme 1:



Scheme 2:



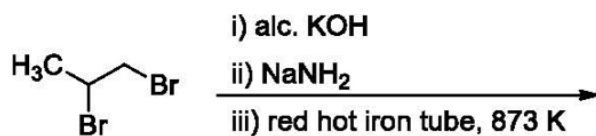
Scheme 3:



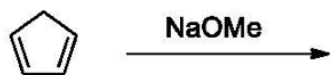
@Bohring_bot

Q.6 Choose the correct option(s) that give(s) an aromatic compound as the major product

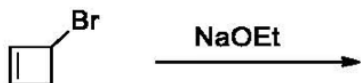
(A)



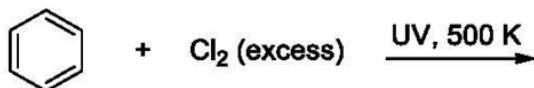
(B)



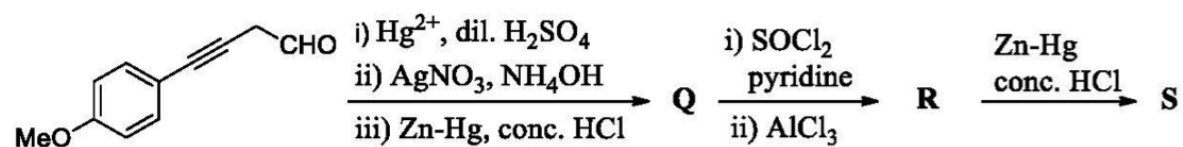
(C)



(D)

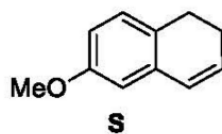
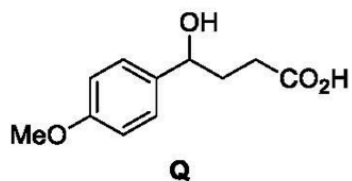


Q.7 Choose the correct option(s) for the following reaction sequence

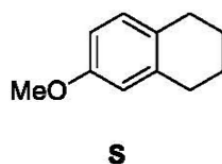
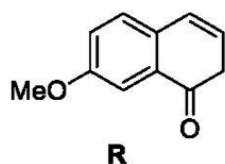


Consider Q, R and S as major products

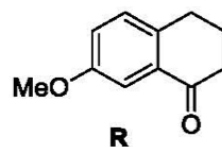
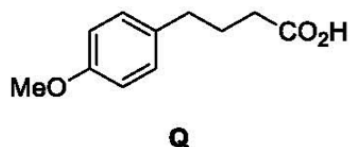
(A)



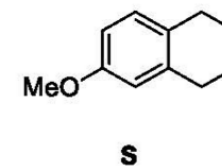
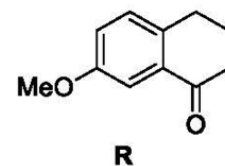
(B)



(C)



(D)



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